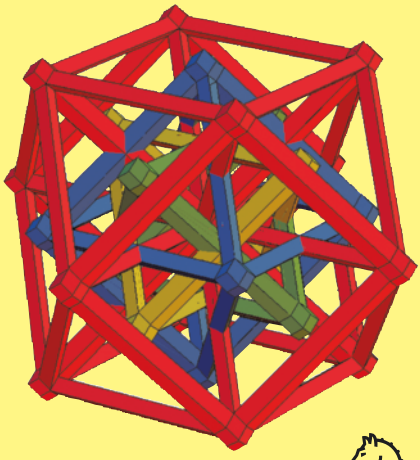
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Yvette Kosmann-Schwarzbach
Translated by Stephanie Frank Singer

Groups and Symmetries

From Finite Groups to Lie Groups

Second Edition



Springer

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Cover illustration: The figure on the front cover represents weights for the fundamental representation of sl(4). Courtesy of Adrian Ocneanu.

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Sophus Lie (1842–1899), around 1865, at the end of his studies at the University of Christiana (Oslo), approximately seven years before his first work on continuous groups, later known as "Lie groups".

(Photo Frederik Klem/Joronn Vogt, with the kind permission of Joronn Vogt and Arild Stubhaug)

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Introduction

Johnny walked to a corner of the room,
faced the wall, and started mumbling to himself.
After a while he turned around and said:
"You need the theory of group characters".
Whereupon von Neumann went to Issai Schur,
obtained reprints of two of his papers
and gave them to Wigner.[1]

Symmetries of geometric figures, crystals and other macroscopic physical items have been observed and analyzed for centuries. In modern terms, the symmetries of an object form a group. The concept of a group emerged slowly in the middle of the nineteenth century. Afterwards, though, the study of groups accelerated markedly. Due to the work of Sophus Lie (1842–1899), Georg Frobenius (1849–1917), Wilhelm Killing (1847–1923), Élie Cartan (1869–1951), Issai Schur (1875–1941), Hermann Weyl (1885–1955) and many, many others, group theory expanded enormously. Applications to quantum mechanics and elementary particle theory were developed throughout the twentieth century. If this history interests you, read the introduction to *Group Theory and Physics* by Shlomo Sternberg (2003), and consult the books by Charles Curtis (1999), Thomas Hawkins (2000) and Armand Borel (2001), or the essays and transcriptions of round-tables between physicists and mathematicians in the volume *Symmetries in Physics*.[2]

[1] A. Pais, *The Genius of Science*, Oxford University Press, Oxford, 2000, p. 335. This scene took place in 1926 in Berlin. "Johnny" was the mathematician and physicist John von Neumann (1903–1957). The papers by I. Schur were probably "Neue Begründung der Theorie der Gruppencharaktere" and his joint paper with F. G. Frobenius, "Über die reellen Darstellungen der endlichen Gruppen", which had appeared in 1906 in the *Sitzungsberichte Preussischen Akademie der Wissenschaften zu Berlin*, or his 1908 article, "Über die Darstellung der symmetrischen Gruppe durch lineare homogene Substitutionen", published in the same journal.

[2] M. G. Doncel, A. Hermann, L. Michel and A. Pais, eds., *Symmetries in Physics (1600–1980)*, Seminari d'Història de les Ciencies, Universitat Autònoma de Barcelona, Bellaterra, 1987.

Sophus Lie wrote in an 1877 letter to the mathematician Adolph Mayer[3] that he had "created group theory" in January 1873. He was certainly referring to the groups that he called "continuous" and that have since become known as "Lie groups" and in French as "groupes de Lie".[4] Lie was trying to extend the application of groups from the study of algebraic equations, where Évariste Galois had introduced them, to the study of differential equations. Inspired by the work of Camille Jordan and by a collaboration with Felix Klein, Lie published an article titled *Über Gruppen von Transformationen* in 1874. As early as 1871, the idea of an infinitesimal generator of a one-parameter group of transformations had already appeared in hiswork;[5] the set of infinitesimal generators of one-parameter subgroups of a continuous group forms what today is called a "Lie algebra".[6] Between 1888 and 1890, Killing established a classification of semi-simple Lie algebras over the field of complex numbers which was corrected and completed by Cartan in his 1894 thesis. Cartan would succeed in classifying Lie algebras over the field of real numbers, a more difficult problem, in 1914.

Frobenius, in response to a question that had been posed by Dirichlet,[7] invented the theory of characters of finite groups in 1896, and Schur, beginning in 1901, developed the representation theory of finite and infinite groups. Their joint work, *Über die reellen Darstellungen der endlichen Gruppen* was published in the proceedings of the Berlin Academy in 1906. The theory of characters was used by William Burnside in the second edition of his treatise *Theory of Groups of Finite Order* which appeared in 1911.

It was Eugene Wigner and Weyl who showed the pertinence of group theory, particularly group representations, for the quantum mechanics being developed in the 1920's by Werner Heisenberg and Paul Dirac. Wigner was the first to introduce group theory into physics in two articles in the *Zeitschrift für Physik* in 1927. His book *Gruppentheorie und ihre Anwendung auf die Quantenmechanik der Atomspektren*, which appeared in 1931, was followed in 1939 by a fundamental article in the *Annals of Mathematics* where

[3] Mayer (1839–1908) was a professor at the University of Heidelberg whose work on the calculus of variations and partial differential equations was close to that of Lie.

[4] The term "groupes de Lie" first appeared in French in 1893 in the thesis of Lie's student Arthur Tresse. See p. 58.

[5] This point of view was essential to Emmy Noether's 1918 article in the *Göttinger Nachrichten* establishing the relationship between symmetries of a variational problem and conservation laws. About ten years later, Noether published a very important article situating the representation theory of finite groups and of algebras in the context of non-commutative rings.

[6] This name was suggested much later by Weyl in his 1933–1934 lectures at the Institute for Advanced Study in Princeton.

[7] Peter Gustav Lejeune Dirichlet (1805–1859) taught in Paris, Breslau (now Wrocław), Berlin, and finally Göttingen. His works on number theory and on analysis are equally fundamental.

he determined the representations of the Poincaré group. At the end of 1927, Weyl published an article in the *Zeitschrift für Physik* in which he introduced what is now called "Weyl quantization", and the first edition of his *Gruppentheorie und Quantenmechanik* came out in 1928. In 1932 Bartel van der Waerden's *Die gruppentheoretische Methode in der Quantenmechanik* appeared. These three books marked the beginning of the use of group theory in theoretical physics, which continues to this day.

After Heisenberg had solved a two-electron problem related to the spectrum of the helium atom, Wigner became interested in the more-than-two identical particles problem. He first solved the case of three or four elements, then went to consult von Neumann on how to extend his already very complicated computations to the case of an arbitrary number of electrons. In 1983, he recalled,[8]

> As to representation theory: I realized that that there must be such a theory, but I had no knowledge of it. Dr. von Neumann to whom I presented the problem (and I presented to him the representations of the permutation groups of three and four elements because those I could establish by long hand calculations) gave me a reprint of the article of Frobenius and Schur [1906]. And that was wonderful!

One may consider group representation theory as a vast generalization of Fourier analysis. It has been developed continuously and has, since the middle of the twentieth century, seen innumerable applications to differential geometry, ergodic theory, probability theory, number theory, the theory of automorphic forms and dynamical systems, as well as to physics, chemistry, molecular biology and signal processing. Currently it is the basis of several branches of mathematics and physics. Some of the more recent and very active domains of applications of Lie theory are geometric mechanics, geometric deep learning, Lie group machine learning, neurogeometry, pattern recognition, wavelet transforms, and many other fields of information theory and computer science.

***

The goal of this course is to introduce and illustrate the basic notions of finite group theory and, more generally, compact topological groups, to introduce the notions of Lie algebra and Lie group, at least in the case of linear Lie groups (closed subgroups of the linear groups), and to study the Lie groups SO(3) and SU(2) in detail, then the Lie group SU(3) and its application to the theory of quarks, and to present an introduction to the theory of spinors. The idea of a group representation, an action on a vector space by linear transformations, plays the most fundamental role in this study.

In Chapter 1, we recall some general facts about groups and group actions, and we give examples of finite and infinite groups.

[8] Round Table on the Evolution of Symmetries, chaired by Louis Michel, in Doncel *et al.*, p. 633.

To study the representations of finite groups, we exploit the properties of the characters of these representations, that is, the traces of the linear transformations that define the representation in question. This is done in Chapter 2, where we define irreducible representations and intertwining operators. We state Schur's lemma which, though simple, will be essential to the proofs of important consequences such as the orthogonality of the characters of irreducible representations. We study the regular representation. A brief section will introduce induced representations.

Chapter 3 extends certain results proved for finite groups to compact groups, using the existence of an invariant measure on each compact group, the Haar measure. In this chapter, some topological and analytic results are stated without proof and references to the literature are supplied. A table summarizes the properties of representations of compact groups.

The beginning of Chapter 4 consists of an introduction to the general notion of a Lie algebra and a review of exponentiation of matrices. Next we study the Lie algebra of a linear Lie group, that is, the set of infinitesimal generators of one-parameter subgroups, with the operation of matrix commutation, and we show the relationship between the representations of a Lie group and the representations of its Lie algebra.

In Chapter 5 we study the Lie groups SO(3) and SU(2), proving the fundamental property that SU(2) is the universal cover of the rotation group SO(3).

In Chapter 6 we determine all the irreducible representations, first of $\mathfrak{sl}(2,\mathbb{C})$, and then of SU(2) and SO(3). These results are as important for the general theory of group representations as they are for quantum mechanics.

Chapter 7 consists of a study of spherical harmonics which appear in the representation theory of the group of rotations of Euclidean space of three dimensions. This is an introduction to the theory of special functions.

In Chapter 8 we begin the study of irreducible representations of SU(3) by considering examples, and we show that quark theory appears as a consequence of the mathematical properties of this group. In this chapter we introduce the roots and weights which are, more generally, the essential tools in the representation theory of a large class of Lie algebras, the semi-simple Lie algebras.

Chapter 9 deals with Clifford algebras, spin groups and the theory of spinors which constitutes another topic of great importance in theoretical physics.

For each chapter we give references to various sources for the material treated in the text, as well as for further study.

An original feature of this book, new in this second edition, is the publication of a section entitled "Topics in History", in each chapter, before the list of exercises. These pages are notes on the history of the material treated in this book with an emphasis on the content of the chapter they follow. The style varies from chapter to chapter, sometimes providing a didactic story of the development of the subject matter, sometimes relating a collection of

varied facts, mostly biographical, and remarks, and sometimes sketching an anecdotal scientific story.

At the end of each chapter, there are exercises of various degrees of difficulty, many of which introduce additional definitions and results.

The nine chapters of the book are followed by a series of problems with detailed solutions which constitute an application of many of the results proved in the text and also introduce several more advanced topics.

We have used many notations and concepts from the physics literature in the hope that this book will be useful to students of the physical sciences as well as to students of mathematics. Except for Chapters 3, 8, and 9, complete proofs of all results are given. The symbol □ denotes the end of a proof.

The book ends with a bibliography and an index.

***

The eight chapters that constituted the first edition of this book arose from a course taught for several years to third-year students majoring in mathematics or physics at the École Polytechnique, in Palaiseau near Paris. This course, roughly corresponding to an advanced undergraduate or beginning graduate course, was taught in nine sessions of ninety-minute lectures followed by two-hour exercise classes. The problems that complement this texbook were drawn from the examinations that we administered at the end of the term.

***

This book is designed to serve as an introduction to more advanced texts and to encourage the reader to develop and use the ideas sketched here. The bibliography on the last pages contains some very classical texts, recent textbooks on diverse aspects of group and representation theory, books that will be useful as reference for results not proved here, and books that further develop harmonic anlysis or representation theory in various ways, chosen from an extremely vast literature, as well as many references to historically significant publications. Among these works, we have starred those that are most directly related to the matter treated in this text. Others, perhaps in spite of their titles, go "beyond an introduction". We hope that our readers will find instruction and pleasure in this book.

Paris, April 2022

Acknowledgments

This book is a revised and enlarged edition of *Groups and Symmetries. From Finite Groups to Lie Groups*, first published by Springer in 2010. That edition was a translation into English of the text that I had previously published in French, *Groupes et Symétries: Groupes finis, groupes et algèbres de Lie, représentations* (Éditions de l'École Polytechnique, 2004, 2006). I am very grateful to Dr. Stephanie Singer for her excellent translation from the French for that first English edition, of which much remains in this second edition, to Vitalyi Kushnirevitch for his expert reading of the proofs of the first edition, to Christophe Eckes who more recently advised me on some historical questions. To Ann Kostant and the editors of the series *Universitext* for first accepting this textbook for publication and then inviting me to prepare a second edition, I address many thanks, as well as to the staff of Springer, New York, in particular Elizabeth Loew, for their patience and their help in its realization. I also thank the institutions and colleagues who kindly allowed me to reproduce the portraits of mathematicians and physicists that illustrate this book.

I wish to reiterate my thanks to all those who helped me in the preparation of the French editions: Alain Guichardet, for his numerous comments on the notes that preceded this book, André Rougé for his clarification of the history of the eightfold way, my colleagues Nicole Berline and Pascale Harinck, as well as Jean-Paul Blaizot and Jean-Claude Tolédano, for their encouragement and useful feedback, and the students of the École Polytechnique from whose remarks I learned a great deal. I am most grateful to Shlomo Sternberg, whose book on group theory and physics inspired me greatly, and to Adrian Ocneanu, of Pennsylvania State University, who sent me diagrams illustrating the theory of roots of Lie algebras, one of which appears on the cover of *Groupes et Symétries*, as well as on both the first and the present English editions. To Pierre Cartier, who advised me on the new chapter introduced in this second edtion, I address my warmest thanks.

This new edition would not have appeared without the help of Bertram E. Schwarzbach, who revised all the new material. I am deeply thankful to him.

Chapter 1

General Facts About Groups



We briefly recall the main concepts in the theory of groups, give examples of finite and of infinite groups, and we define the notion of a group action.

1 Review of Definitions

A *group* is a set equipped with an associative composition law, containing an identity element and such that each element has an inverse. The identity element, also called the unit element, is usually denoted by e, or by 1 or, in the case of a matrix group, by I.

A group is called *commutative* or *Abelian* if its composition law is commutative. In this case the composition is generally denoted by $+$ and the identity element is generally denoted by 0.

We denote by $|X|$ the cardinality of a finite set X. The *order of a finite group*, G, is the number $|G|$ of elements of the group.

An element $g \in G$ is said to be of order n ($n \geq 1$) if n is the smallest integer such that

$$g^n = e.$$

Example. The rotation around the origin of the plane through an angle $\frac{2\pi}{n}$ is an element of order n in the group of rotations of the plane.

A *subgroup* H of a group G is a subset of G satisfying the conditions:

(i) $e \in H$,

(ii) if $g \in H$, then $g^{-1} \in H$,

(iii) if $g \in H$ and $g' \in H$, then $gg' \in H$.

The *subgroup generated* by a subset of a group G is the smallest subgroup of G containing the subset.

Let H be a subgroup of a group G. The *left H-coset* of g in G is the set

$$gH = \{gh \mid h \in H\}.$$

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The set of left H-cosets of G is the quotient set denoted by G/H. The *right H-coset* of g in G is the set

$$Hg = \{hg \mid h \in H\}.$$

The set of right H-cosets of G is the quotient set denoted by $H\backslash G$. The group G is the union of the left (resp., right) H-cosets, each of which has $|H|$ elements. We deduce that the order of H divides the order of G, and that the number of left H-cosets equals the number of right H-cosets. We can now state the following result.

Theorem 1.1 (Lagrange's Theorem). *Let G be a finite group and let H be a subgroup of G. Then $|H|$ divides $|G|$ and*

$$|G|/|H| = |H\backslash G| = |G/H|.$$

When H is a subgroup of G, the integer $|G|/|H|$ is called the *index* of H in G.

A *morphism of groups* (or *group morphism*), $\varphi : G_1 \to G_2$, is a mapping of a group G_1 into a group G_2 such that for every $g, g' \in G_1$,

$$\varphi(gg') = \varphi(g)\varphi(g'),$$

which implies that $\varphi(e_1) = e_2$, where e_i is the identity element of G_i $(i = 1, 2)$, and $\varphi(g^{-1}) = (\varphi(g))^{-1}$, for each $g \in G$.

Remark. Note that most authors use the term *homomorphism* instead of *morphism*.

An *isomorphism of groups* is a bijection that is also a morphism. Its inverse is then also an isomorphism.

An *automorphism of a group* is an isomorphism of a group onto itself. In particular, for each $g \in G$, the endomorphism $\mathcal{C}_g$ of G defined by

$$\mathcal{C}_g : h \mapsto ghg^{-1}, \text{ for all } h \in G,$$

is an automorphism. It is called the *inner automorphism* or the *conjugation* defined by g.

A subgroup of G is called *normal* or *invariant* if it is stable under conjugation by every element of G.

If a subgroup H of G is normal, the set of right H-cosets of G is isomorphic to the set of left H-cosets, and the set of H-cosets is then a group called the *quotient group* of G by H, and denoted by G/H. The projection $p : G \to G/H$, defined by $g \mapsto gH$, is a morphism of groups.

The kernel of a morphism of groups, $\varphi : G_1 \to G_2$, is a normal subgroup of G_1. Conversely, if H is a normal subgroup of a group G, then it is the kernel of a group morphism, the projection $p : G \to G/H$.

The *center* of G is, by definition, the set $\{h \in G \mid \forall g \in G, hg = gh\}$. The center of G is a normal subgroup of G.

If G_1 and G_2 are groups, their *direct product* is the product $G_1 \times G_2$ with group law,

$$(g_1, g_2).(g_1', g_2') = (g_1 g_1', g_2 g_2').$$

The notion of a *semi-direct product* of groups is introduced in Exercise 1.4.

2 Examples of Finite Groups

2.1 Cyclic Groups

A group is called *cyclic* if it is generated by a single element. Every cyclic group is Abelian.

For n a positive integer, the set of integer multiples of n is denoted by $n\mathbb{Z}$. It is a subgroup of the Abelian group $\mathbb{Z}$, in fact it is a normal subgroup since all subgroups of an Abelian group are normal.

The following groups are isomorphic and are cyclic groups of order n:

- $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$. For $n = 2$, the group $\mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$ is denoted additively by $\{0, 1\}$ or multiplicatively by $\{1, -1\}$.
- The group of rotations of the plane around O, through angles $\frac{2k\pi}{n}$, $0 \leq k \leq n-1$, under composition.
- The group of complex numbers, $\{e^{\frac{2ik\pi}{n}} \mid 0 \leq k \leq n-1\}$, under multiplication.
- The subgroup $\{1, g, g^2, \ldots, g^{n-1}\}$ if g is an element of order n in a group G.

2.2 Symmetric Groups

The permutation group of a set of cardinality n is denoted by $\mathfrak{S}_n$ and called the *symmetric group on n elements*. The order of $\mathfrak{S}_n$ is $n!$.

Every element of $\mathfrak{S}_n$ can be written as a product of transpositions. To every element $\sigma \in \mathfrak{S}_n$, we associate the number 1 or -1 according to the parity of the number of transpositions, which parity is independent of the decomposition. This number is denoted by $(-1)^\sigma$ and called the *signature* of σ. The map $\sigma \in \mathfrak{S}_n \mapsto (-1)^\sigma \in \mathbb{Z}_2$ is a group morphism, called the *signature morphism*.

The *alternating group* $\mathfrak{A}_n$ is the kernel of the signature morphism. If $n \geq 2$, it is a normal subgroup of index 2 of $\mathfrak{S}_n$.

2.3 Dihedral Groups

The *dihedral group* $D_{(n)}$ is the group of rotations and symmetries of the plane preserving a regular n-sided polygon ($n \geq 3$). It is a subgroup of order $2n$ of $\mathfrak{S}_n$. Some authors use the notation D_{2n} for this group. (See Exercise 1.6.)

2.4 Crystallographic Groups

We denote by $\mathrm{O}(3)$ the group of linear isometries of $\mathbb{R}^3$, and by $\mathrm{SO}(3)$ the group of rotations of $\mathbb{R}^3$, which is the kernel of the determinant map from $\mathrm{O}(3)$ to $\mathbb{R}$, and thus a normal subgroup of $\mathrm{O}(3)$. For each regular polyhedron – tetrahedron, cube, octahedron, icosahedron (fullerene), dodecahedron –, we define the corresponding group of direct symmetries (resp., group of symmetries), that is, the subgroup of $\mathrm{SO}(3)$ (resp., subgroup of $\mathrm{O}(3)$) leaving the solid globally invariant. These are finite groups, and they are known as *crystallographic groups*. The first is of index 2 in the second.

	$\mathrm{SO}(3)$	$\mathrm{O}(3)$
tetrahedron	$\mathfrak{A}_4$ order 12	$\mathfrak{S}_4$ order 24
cube	$\mathfrak{S}_4$ order 24	$\mathfrak{S}_4 \times \mathbb{Z}_2$ order 48
octahedron	$\mathfrak{S}_4$ order 24	$\mathfrak{S}_4 \times \mathbb{Z}_2$ order 48
icosahedron	$\mathfrak{A}_5$ order 60	$\mathfrak{A}_5 \times \mathbb{Z}_2$ order 120
dodecahedron	$\mathfrak{A}_5$ order 60	$\mathfrak{A}_5 \times \mathbb{Z}_2$ order 120

The classification of all finite subgroups of $\mathrm{SO}(3)$ and of $\mathrm{O}(3)$, which is well known, is of great importance in physics, in particular in crystallography.

3 Examples of Infinite Groups

Among the groups with an infinite number of elements, there are *discrete groups* such as, for example, the Abelian group $\mathbb{Z}$. But we are mainly interested in the non-discrete groups called "continuous groups", of which we now give examples.

Let $\mathbb{K}$ denote the field $\mathbb{R}$ or $\mathbb{C}$. We denote by $\mathrm{GL}(n, \mathbb{K})$ the group of linear isomorphisms of $\mathbb{K}^n$, called the *linear group* in dimension n. This is the group of invertible $n \times n$ matrices with coefficients in $\mathbb{K}$, of which we shall consider several subgroups. We denote the *transpose* of a matrix A by ${}^tA$. The *complex conjugate* of a complex number or matrix is denoted by a bar over that number or matrix. We denote the *determinant* of an endomorphism u of a finite-dimensional $\mathbb{K}$-vector space by $\det u$ or by $\det A$ if A is the matrix of u in a basis.

- Special linear group over $\mathbb{K}$,

$$\mathrm{SL}(n, \mathbb{K}) = \{A \in \mathrm{GL}(n, \mathbb{K}) \mid \det A = 1\}.$$

- Orthogonal group over $\mathbb{K}$,

$$\mathrm{O}(n, \mathbb{K}) = \{A \in \mathrm{GL}(n, \mathbb{K}) \mid A\, {}^tA = I\}.$$

In particular, we call the real orthogonal group the *orthogonal group*, and we denote it simply by $\mathrm{O}(n)$.

- Special orthogonal group over $\mathbb{K}$,

$$\mathrm{SO}(n, \mathbb{K}) = \{A \in \mathrm{O}(n, \mathbb{K}) \mid \det A = 1\}.$$

In particular, we call the real special orthogonal group the *special orthogonal group*, and we denote it simply by $\mathrm{SO}(n)$.

More generally, if p and q are nonnegative integers such that $p+q=n$, we denote by J_{pq} the diagonal matrix with p diagonal elements equal to 1 followed by q diagonal elements equal to -1, and we set

$$\mathrm{O}(p,q) = \{A \in \mathrm{GL}(n,\mathbb{R}) \mid A\ J_{pq}\ {}^tA = J_{pq}\}$$

and

$$\mathrm{SO}(p,q) = \{A \in \mathrm{O}(p,q) \mid \det A = 1\}.$$

In this notation, $\mathrm{O}(p,0) = \mathrm{O}(0,p) = \mathrm{O}(p)$ and $\mathrm{SO}(p,0) = \mathrm{SO}(0,p) = \mathrm{SO}(p)$.

In particular $\mathrm{O}(3,1)$, which is the group of isometries of Minkowski space-time, is called the *Lorentz group*. (In some publications, it is the canonically isomorphic group $\mathrm{O}(1,3)$ which is called the Lorentz group.)

- Unitary group

$$\mathrm{U}(n) = \{A \in \mathrm{GL}(n,\mathbb{C}) \mid A\ {}^t\bar{A} = I\}.$$

The matrix ${}^t\bar{A}$ is the adjoint of the matrix A, also denoted by A^*.

- Special unitary group

$$\mathrm{SU}(n) = \{A \in \mathrm{U}(n) \mid \det A = 1\}.$$

Definition 3.1. *A* topological group *is a group G that is a Hausdorff topological space such that composition $(g,g') \mapsto gg'$ is a continuous map from $G \times G$ into G and inversion $g \mapsto g^{-1}$ is a continuous map of G into G.*

The linear group $\mathrm{GL}(n,\mathbb{K})$ with its usual topology, as an open subset of $\mathbb{K}^{n^2}$, is a locally compact topological group and each of the groups listed above is a closed subgroup of a linear group. The groups $\mathrm{O}(n)$ and $\mathrm{U}(n)$, as well as $\mathrm{SO}(n)$ and $\mathrm{SU}(n)$ are compact.

Below we give the definition of real and complex Lie groups, which brings in the notion of *manifolds*, an abstraction of the notion of a surface in a Cartesian space covered by overlapping patches of coordinate domains.

A *real Lie group* of real dimension $N \in \mathbb{N}$, or simply a *Lie group of dimension N*, is a group that is a smooth real manifold of dimension N such that composition and inversion are smooth maps.

A *complex Lie group* of complex dimension $N \in \mathbb{N}$ is a group that is a complex-analytic manifold of complex dimension N such that composition and inversion are analytic maps. Every complex Lie group of complex dimension N is a real Lie group of real dimension $2N$.

All closed subgroups of $\mathrm{GL}(n,\mathbb{K})$ are real Lie groups. This is a theorem that we state without proof. The closed subgroups of $\mathrm{GL}(n,\mathbb{K})$ are called *linear Lie groups*. They are the only ones we shall consider, and we shall call them simply *Lie groups*.

All of the examples in this section are examples of Lie groups. In this text we study the Lie groups SO(3) and SU(2) in detail (Chapters 5 and 6), and we give introductions to SU(3) (Chapter 8) and to the spin groups (Chapter 9).

Remark. The group $\mathrm{GL}(n,\mathbb{C})$ and some of its subgroups are also complex Lie groups. This is the case for $\mathrm{SL}(n,\mathbb{C})$, $\mathrm{O}(n,\mathbb{C})$ and $\mathrm{SO}(n,\mathbb{C})$, but not for $\mathrm{U}(n)$, nor for $\mathrm{SU}(n)$, which are real Lie groups only.

4 Group Actions and Conjugacy Classes

Definition 4.1. *Let G be a group and let M be a set. A* group action *(or simply an* action*) of G on M is a mapping $\alpha : G \times M \to M$, denoted by $(g,m) \mapsto g \cdot m$, such that for every $m \in M, e \cdot m = m$ and for every g and $g' \in G, g \cdot (g' \cdot m) = (gg') \cdot m$. If there is an action of G on M, we say that G acts on M.*

In other words, α defines a group morphism of G into the group of bijections of M onto itself.

If G is a topological group and if M is a topological space, we shall assume that the action α of G on M is continuous. If G is a Lie group and if M is a smooth manifold, we shall similarly assume that α is smooth.

The *orbit* of $m \in M$ under the action of G is the set $\{g \cdot m \mid g \in G\}$. The orbits define a partition of M.

Examples.

- The trivial action of G on a set M consists of sending each element of the group to the identity mapping of M on itself. In this case, the orbits are the points of M.
- Under the action of $G = \mathrm{O}(2)$ on the unit sphere $M = S^2 \subset \mathbb{R}^3$ by rotations around the axis Oz, the orbit of $m \in S^2$ is a point if m is the north pole or the south pole, and is otherwise a circle with the axis Oz (a one-dimensional submanifold).
- The action of G on G by left (resp., right) translation is the action

$$(g,m) \in G \times G \mapsto gm \in G$$

(resp., $(g,m) \in G \times G \mapsto mg^{-1} \in G$). In other words, we associate to $g \in G$ the left (resp., right) translation l_g (resp., $r_{g^{-1}}$) in G, thus defining a morphism of G into the bijections of G onto G (but not into the automorphisms of G).

We remark that the left coset, gH, of $g \in G$ relative to a subgroup H of G is the orbit of g under the action of $H \subset G$ acting by right translation.

- The action of G on G by conjugation, denoted by $(g,h) \mapsto \mathcal{C}_g(h)$ and defined by

$$\mathcal{C}_g(h) = ghg^{-1},$$

is an action of G on itself by automorphisms. The orbit of $h \in G$,

$$C_h = \{ghg^{-1} \mid g \in G\},$$

is called the *conjugacy class* of h.

Examples.

- The conjugacy class of e is $\{e\}$.
- If G is Abelian, the conjugacy class of $g \in G$ is $\{g\}$. This is the case, for example, for SO(2).
- In $\mathfrak{S}_n$, the number of conjugacy classes is the number of *partitions* of n. (See Exercise 1.3.)
- In $\mathrm{GL}(n, \mathbb{K})$, the conjugacy class of a matrix A is $\{PAP^{-1} \mid P \in \mathrm{GL}(n, \mathbb{K})\}$, the set of all matrices *similar* to A.
- In SO(3), two rotations are conjugate by a rotation if and only if their angles are equal or opposite. The conjugacy classes are in bijective correspondence with the interval $[0, \pi]$.

In fact, by a direct orthonormal change of basis, any rotation can be written as $g(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$, with $\theta \in [0, 2\pi)$. From the equality $g(-\theta) = g_0 g(\theta) g_0^{-1}$, where $g_0 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, it is easy to see that $g(\theta')$ is conjugate to $g(\theta)$ if and only if $\theta' = \theta$ or $\theta' = -\theta$.

References

For the classification of the finite subgroups of SO(3), see Sternberg (2003), Artin (2014), or Ledermann–Weir (1996). Sternberg also analyzes the subgroups of O(3) and identifies, for each subgroup, a crystal possessing that symmetry.

Basic notions in topology such as discrete spaces, Hausdorff spaces, compact and locally compact spaces, as well as connected spaces, are recalled in many books on representation theory. See, *e.g.*, Stillwell (2008) or Naĭmark–Štern (1982).

One can find an introduction to the theory of differentiable manifolds and Lie groups in Sagle–Walde (1973), Rossmann (2011), or M. J. D. Hamilton (2017), and, with applications to mechanics, in Marsden–Ratiu (2011). For the study of the differential geometry of Lie groups as manifolds see, for example, Warner (1983).

Topics in History

Permuting roots of polynomials

It is generally accepted that Lagrange's 1770 study of polynomials, "Reflections upon the algebraic resolution of equations", was subsequently an inspiration for

the first papers that introduced the concept of a group. Joseph-Louis Lagrange was born Giuseppe Lodovico Lagrangia in 1736, of French ancestors, in Turin which was, at the time, the capital of the kingdom of Sardinia, and died in Paris in 1813. He was not yet twenty years old when he corresponded with the great Leonhard Euler (1707–1783) whom he succeeded at the age of 30 as Director of Mathematics at the Berlin Academy. Some twenty years later, in 1787, he accepted a position at the Académie des Sciences in Paris where he had no teaching duties. When the École Polytechnique was created in 1794 as a school for military engineers, he was appointed professor while retaining his position at the Academy.

What is today called "Lagrange's theorem" is a generalization of what he proved in 1771: the number of distinct polynomials obtained by permuting the n variables in a polynomial of degree n divides $n!$. An interpretation of this result in terms of subgroups of a group was published nearly a century later, in a memoir by the French mathematician, Camille Jordan (1838–1922).

A gifted student

At the École Polytechnique Lagrange taught analysis and mechanics to college-level students. The following anecdote is fact, not legend. Among the students in the class of the already famous professor Lagrange in the year 1799 there was a young student named Siméon-Denis Poisson who was then only 18 but was to become very famous himself. One day, Poisson had an idea for a better proof of the binomial theorem than the one that Lagrange had presented in his lecture. He consulted the other students who agreed that his proof was indeed correct and sent a respectful note to the professor explaining his method. At the next lecture, the great Lagrange explained his student's proof and credited him for it! Poisson, who died in 1840, proved to be an important mathematician and mathematical physicist in his own right, becoming a lecturer, then a professor at the École Polytechnique in 1806 and, in 1809, he was named to the chair of mechanics when the Faculty of Science of the University of Paris, which was and still is known as the Sorbonne, was created.

Emergence of the concept of group

The notion of "permutation" can already be found around 150 BCE in the mathematics of the Jaina in India. In the West, the first idea of how to compose permutations arose at the beginning of the 19th century in the work of the less well known but nonetheless very important mathematician Paolo Ruffini (1765–1822) who was working in the Italian city of Modena. The notion of a group was made explicit, first by Évariste Galois (1811–1832), in his study of algebraic equations in a memoir of 1831 that in fact remained unpublished until 1846, long after he had died in a duel on May 30, 1832. Then, Augustin Cauchy (1789–1857), who had already published two articles on substitutions in the *Journal de l'École Polytechnique* in 1815, wrote a thorough study of permutations in successive notes to the Académie des Sciences, followed by a

book-length study entitled "Mémoire sur les arrangements que l'on peut former avec des lettres données" (On the permutations of given letters) in 1844.

It was the English mathematician Arthur Cayley (1821–1895), the author of 966 mathematical papers, who formulated a definition of an abstract group – as opposed to a group of transformations acting upon a space – in 1854, in an 8-page article in the *Philosophical Magazine*: "On the theory of groups, as depending on the symbolic equation $\theta^n = 1$":

> A set of symbols, $1, \alpha, \beta...$ all of them different, and such that the product of any two of them (no matter in what order), or the product of any one of them into itself, belongs to the set, is said to be a group,

with a footnote that specified: "The idea of a group as applied to permutations or substitutions is due to Galois." In fact, he was tacitly assuming associativity and a unit element so this was not yet a formal definition of infinite groups but he formulated the main axiom of their definition. He also suggested presenting the results of the multiplications in a table.

Later in the nineteenth century, the notion of a group was to be found in the "Traité des substitutions" that Jordan published in 1870, then in the lecture notes of Heinrich Weber (1842–1913) for his course on number theory at the University of Königsberg in 1880 and in his later, influential textbook, *Lehrbuch der Algebra*, which appeared in 1903. It was in 1930 and 1931 that Bartel van der Waerden (1903–1996), the Dutch mathematician who had been a student of Emmy Noether[1] in Göttingen, published his *Moderne Algebra*, in two volumes, which renewed the subject and was a precursor of the modern texts.

From finite to continuous groups

As the theory of finite groups progressed, that of continuous groups emerged in the study of geometry in the late eighteenth century and led, after 1870, to the fundamental work of Sophus Lie (1842–1899) about whom more in the next chapters.

Who was Sophus Lie?

Lie was a Norwegian mathematician who worked in the capital city, Christiania, present-day Oslo, and later in Leipzig in Germany. Many of his publications appeared in the little-known Norwegian journal, *Forhandlinger Christiania*. All of his articles were written in German, except for his doctoral dissertation which appeared in Norwegian in 1871 and was published in German translation in volume 1 of his *Collected Works*, which comprise 7 volumes, the publication of which began in 1922, long after his death at the age of 56. His health had failed him and he had become more and more isolated and at odds with the world at large. But his work endures.

[1] See below p. 34 for more on Noether.

Exercises

Exercise 1.1 *Application of Lagrange's Theorem.*

Show that in a finite group G of order n, for each element $a \in G$, $a^n = e$. Conclude that every morphism of G into the group $\mathrm{GL}(1, \mathbb{C}) = \mathbb{C}^*$ takes values in the group of n-th roots of unity.

Exercise 1.2 *Center of* $\mathfrak{S}_n$.

Show that, for $n \geq 3$, the only element in the center of the symmetric group $\mathfrak{S}_n$ is the identity.

Exercise 1.3 *Conjugacy classes of* $\mathfrak{S}_n$.

An element in $\mathfrak{S}_n$ is called a *cycle* of length k $(1 \leq k \leq n)$ if it is a circular permutation of k elements that leaves the other $n - k$ elements invariant.

(a) Show that each permutation is the composition of disjoint cycles. Is this decomposition unique?

(b) Show that in $\mathfrak{S}_n$ two elements are conjugate if and only if they have the same decomposition into cycles. Deduce that the number of conjugacy classes of $\mathfrak{S}_n$ is the number of partitions of the integer n, that is, of finite sequences of integers, $\lambda_1, \lambda_2, \ldots, \lambda_\ell$, such that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0$, and $\sum_{i=1}^{\ell} \lambda_i = n$.

Exercise 1.4 *Semi-direct products of groups.*

If a group H acts on a group N by group automorphisms, we define a multiplication on $G = N \times H$ by

$$(n, h)(n', h') = (n(h \cdot n'), hh').$$

(a) Show that, equipped with this multiplication, G is a group. This group, denoted by $N \rtimes H$, is called the *semi-direct product* of N and H. What is the inverse of (n, h) in G? Show that N, identified with the subset $N \times \{1\}$ of $N \times H$, is a normal subgroup of $N \rtimes H$.

(b) Show that the group $\mathfrak{S}_3$ is the semi-direct product of the alternating group $\mathfrak{A}_3$ and $\mathbb{Z}_2$. Is the group $\mathfrak{S}_n$ a semi-direct product of the alternating group $\mathfrak{A}_n$ and $\mathbb{Z}_2$, for $n \geq 3$?

(c) Find the standard definition of the Galilean group and of the Poincaré group in any textbook. Show that each of these groups is a semi-direct product.

Exercise 1.5 *Group of isometries of the plane.*

(a) Show that the group $\mathbb{Z}_2 = \{1, -1\}$ acts by group automorphisms on $\mathrm{SO}(2)$ via $\varepsilon \cdot g = g^\varepsilon$, where $\varepsilon = \pm 1$. Show that $\mathrm{O}(2)$ is the semi-direct product $\mathrm{SO}(2) \rtimes \mathbb{Z}_2$.

(b) Determine the conjugacy classes of SO(2) and of O(2).

Exercise 1.6 *The dihedral group* $D_{(n)}$.

Let $D_{(n)}$ be the dihedral group which is of order $2n$, *i.e.*, the group of rotations and symmetries of the plane that leave invariant a regular n-sided polygon ($n \geq 3$).

(a) Show that $D_{(n)} = \Gamma_n \rtimes \mathbb{Z}_2$, where Γ_n is the cyclic group of order n and the group $\mathbb{Z}_2 = \{1, -1\}$ acts on Γ_n by $\varepsilon \cdot g = g^{\varepsilon}$, where $\varepsilon = \pm 1$.

(b) Show that $D_{(n)}$ is isomorphic to $\mathfrak{S}_n$ if and only if $n = 3$.

(c) The *quaternion group* is the set $\{\pm 1, \pm i, \pm j, \pm k\}$ with the group multiplication law $i^2 = j^2 = k^2 = -1$, $ij = -ji = k$, $jk = -kj = i$, $ki = -ik = j$. Show that $D_{(4)}$ is not isomorphic to the quaternion group.



Ferdinand Georg Frobenius (1849–1917), who created the representation theory of finite groups around 1896, was a professor in Berlin and Zurich. His collected works were published in 1968.

(Collection of the Mittag-Leffler Institute, Swedish Royal Academy of Science)

Chapter 2

Representations of Finite Groups



In mathematics and physics, the notion of a group representation is fundamental. The idea is to study the different ways that groups can act on vector spaces by linear transformations. In this chapter, unless otherwise indicated, we shall consider only representations of finite groups in *complex, finite-dimensional* vector spaces.

1 Representations

1.1 General Facts

Let G be a finite group. If E is a vector space over $\mathbb{K}$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, we denote by $\mathrm{GL}(E)$ the group of $\mathbb{K}$-linear isomorphisms of E. (The group $\mathrm{GL}(E)$ is not finite unless $E = \{0\}$.)

Definition 1.1. *A* representation *of a group G in a finite-dimensional complex vector space E is a morphism of groups, $\rho : G \to \mathrm{GL}(E)$.*

Thus, for every $g, g' \in G$,

$$\rho(gg') = \rho(g)\rho(g'), \quad \rho(g^{-1}) = (\rho(g))^{-1}, \quad \rho(e) = \mathrm{Id}_E.$$

The vector space E is called the *support* of the representation, and the dimension of E is called the *dimension* of the representation. We denote such a representation by (E, ρ), or simply ρ, or simply E, if no confusion is possible. If in particular $E = \mathbb{C}^n$, we say that the representation is a *matrix representation* of dimension n.

The *fundamental representation* of a subgroup G of $\mathrm{GL}(E)$ is the representation of G on E defined by the canonical injection of G into $\mathrm{GL}(E)$.

Any representation such that $\rho(g) = \mathrm{Id}_E$ for each $g \in G$ is called a *trivial representation.*

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Y. Kosmann-Schwarzbach, *Groups and Symmetries*, Universitext,
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Example. Here is a first example of a representation of a non-Abelian group. Let $t \in \mathfrak{S}_3$ be the transposition $123 \mapsto 132$ and c the cyclic permutation $123 \mapsto 231$ that generate $\mathfrak{S}_3$. We set $j = e^{\frac{2i\pi}{3}}$. Then $j^3 = 1$ and $j^2 + j + 1 = 0$. We can represent $\mathfrak{S}_3$ on $\mathbb{C}^2$ by defining

$$\rho(e) = I, \quad \rho(t) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho(c) = \begin{pmatrix} j & 0 \\ 0 & j^2 \end{pmatrix}.$$

Definition 1.2. *Let $(\ |\ )$ be a scalar product on E. We say that the representation ρ is* unitary *if $\rho(g)$ is unitary for every g, that is, if*

$$\forall g \in G, \forall x, y \in E, \ (\rho(g)x \mid \rho(g)y) = (x \mid y).$$

A representation (E, ρ) is called *unitarizable* if there is a scalar product on E such that ρ is unitary.

In order to prove the following theorem, as well as many other propositions, we shall use a fundamental property:

Lemma 1.3. *Let G be a finite group. For every function φ on G taking values in a vector space,*

$$\forall g \in G, \quad \sum_{h \in G} \varphi(gh) = \sum_{h \in G} \varphi(hg) = \sum_{k \in G} \varphi(k). \tag{1.1}$$

Proof. In fact, once g is chosen, every element of G can be written uniquely as gh (resp., as hg), where $h \in G$. □

Theorem 1.4. *Every representation of a finite group is unitarizable.*

Proof. Let (E, ρ) be a representation of a finite group G, and let $(\ |\ )$ be a scalar product on E. We consider

$$(x \mid y)' = \frac{1}{|G|} \sum_{g \in G} (\rho(g)x \mid \rho(g)y),$$

which is a scalar product on E. In fact, suppose that $(x \mid x)' = 0$, that is, $\sum_{g \in G} (\rho(g)x \mid \rho(g)x) = 0$. Then, for each $g \in G$, $(\rho(g)x \mid \rho(g)x) = 0$ and, in particular, $(x \mid x) = 0$, whence $x = 0$.

This scalar product on E is invariant under ρ. In fact,

$$\begin{aligned} (\rho(g)x \mid \rho(g)y)' &= \frac{1}{|G|} \sum_{h \in G} (\rho(h)\rho(g)x \mid \rho(h)\rho(g)y) \\ &= \frac{1}{|G|} \sum_{h \in G} (\rho(hg)x \mid \rho(hg)y) = (x \mid y)', \end{aligned}$$

where we have used the fundamental equation (1.1), which holds for any function φ on G. Thus ρ is a unitary representation of G on $(E, (\ |\ )')$. □

1.2 Irreducible Representations

Let (E, ρ) be a representation of G. A vector subspace $F \subset E$ is called *invariant* or *stable* under ρ (or under G, if the name of the representation is understood) if, for every $g \in G$, $\rho(g)F \subset F$. Since F is finite-dimensional, the condition $\rho(g)F \subset F$ implies $\rho(g)F = F$. We can therefore speak of the representation ρ restricted to F which is a representation of G on F. We denote it by $\rho\,|_F$. Such a representation restricted to an invariant subspace is also called a *subrepresentation.*

Definition 1.5. *A representation (E, ρ) of G is called* irreducible *if $E \neq \{0\}$ and if the only vector subspaces of E invariant under ρ are $\{0\}$ and E itself.*

Example. The representation of dimension 2 of $\mathfrak{S}_3$ defined in the preceding Example is irreducible, since the eigenspaces of $\rho(t)$ and $\rho(c)$ have trivial intersection.

Proposition 1.6. *Every irreducible representation of a finite group is finite-dimensional.*

Proof. Let (E, ρ) be an irreducible representation of a finite group G and let $x \in E$. Because the subset $\{\rho(g)x \mid g \in G\}$ is finite, it generates a finite-dimensional vector subspace of E. If $x \neq 0$, this vector subspace of E is not equal to $\{0\}$. Because this subspace is invariant under ρ, it coincides with E, which is thus finite-dimensional. □

1.3 Direct Sum of Representations

Definition 1.7. *Let (E_1, ρ_1) and (E_2, ρ_2) be representations of G. Then*

$$(E_1 \oplus E_2, \rho_1 \oplus \rho_2),$$

where $(\rho_1 \oplus \rho_2)(g)(x_1, x_2) = (\rho_1(g)(x_1), \rho_2(g)(x_2))$, for $g \in G$, $x_1 \in E_1$, $x_2 \in E_2$, is a representation of G called the direct sum *of the representations (E_1, ρ_1) and (E_2, ρ_2).*

Clearly a direct sum of representations of positive dimensions cannot be irreducible, even if the summands are irreducible. For matrix representations ρ_1 and ρ_2, the matrices of the direct sum of the representations ρ_1 and ρ_2 are block-diagonal matrices,

$$\begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix}.$$

More generally, if m is a positive integer, we can use induction to define the direct sum of m representations, $\rho_1 \oplus \cdots \oplus \rho_m$. If (E, ρ) is a representation of G we denote by $m\rho$ the representation $\rho \oplus \ldots \oplus \rho$ (direct sum of m terms) on the vector space $E \oplus \cdots \oplus E$ (m terms).

A representation is called *completely reducible* if it is a direct sum of irreducible representations. In particular, any irreducible representation is completely reducible.

Lemma 1.8. *Let ρ be a unitary representation of G on $(E, (\ \mid\ ))$. If $F \subset E$ is invariant under ρ, then $F^{\perp} = \{y \in E \mid \forall x \in F,\ (x \mid y) = 0\}$ is also invariant under ρ.*

Proof. Let $y \in F^{\perp}$. Then, for every $g \in G$ and $x \in F$, $(x \mid \rho(g)y) = (\rho(g^{-1})x \mid y) = 0$, because F is invariant under ρ. Thus $\rho(g)y \in F^{\perp}$. □

Theorem 1.9. (Maschke's Theorem). *Every finite-dimensional representation of a finite group is completely reducible.*

Proof. Let (E, ρ) be a representation of a finite group G. By Theorem 1.4, one may suppose this representation to be unitary. If ρ is not irreducible, let F be a vector subspace of E invariant under ρ such that $F \neq \{0\}$ and $F \neq E$. Then, $E = F \oplus F^{\perp}$, where F (by hypothesis) and $F^{\perp}$ (by Lemma 1.8) are invariant under ρ, and $\dim F < \dim E$, $\dim F^{\perp} < \dim E$. By induction on the dimension of E, we obtain the desired result. □

In fact, this theorem is true under more general conditions. (See the study of compact groups in Chapter 3.)

1.4 Intertwining Operators and Schur's Lemma

Recall that G is a finite group and that by "representation" we mean a representation in a finite-dimensional, complex vector space.

Definition 1.10. *Let (E_1, ρ_1) and (E_2, ρ_2) be representations of G. We say that a linear map, $T : E_1 \to E_2$,* intertwines *ρ_1 and ρ_2 if*

$$\forall g \in G,\ \rho_2(g) \circ T = T \circ \rho_1(g),$$

in which case T is called an intertwining operator *for ρ_1 and ρ_2.*

The definition can be expressed in the commutativity of the following diagram for each $g \in G$:

$$\begin{array}{ccc} E_1 & \xrightarrow{T} & E_2 \\ {\scriptstyle \rho_1(g)}\downarrow & & \downarrow{\scriptstyle \rho_2(g)} \\ E_1 & \xrightarrow{T} & E_2 \end{array}$$

The following expressions are often used to express the same property:

- T is equivariant under ρ_1 and ρ_2,
- T is a morphism of G-vector spaces,
- T is a G-morphism,
- $T \in \mathrm{Hom}_G(E_1, E_2)$.

If $E_1 = E_2 = E$ and if $\rho_1 = \rho_2 = \rho$, an intertwining operator for ρ_1 and ρ_2 is just an operator that *commutes* with ρ.

Definition 1.11. *The representations ρ_1 and ρ_2 are called* equivalent *if there is a* bijective *intertwining operator for ρ_1 and ρ_2.*

If T is such a bijective intertwining operator,

$$\forall g \in G,\ \rho_2(g) = T \circ \rho_1(g) \circ T^{-1}.$$

The existence of an intertwining operator is an equivalence relation on representations, which leads to the notion of an *equivalence class of representations*. We let $\sim$ denote this equivalence relation.

Two representations (E_1, ρ_1) and (E_2, ρ_2) are equivalent if and only if there is a basis B_1 of E_1 and a basis B_2 of E_2 such that, for every $g \in G$, the matrix of $\rho_1(g)$ in the basis B_1 is equal to the matrix of $\rho_2(g)$ in the basis B_2. In particular, if the representations (E_1, ρ_1) and (E_2, ρ_2) are equivalent, then E_1 is isomorphic to E_2.

For matrix representations, we thus obtain *similar matrices*: if $E_1 = E_2 = \mathbb{C}^n$, and if ρ_1 and ρ_2 are equivalent, then the matrices $\rho_1(g)$ and $\rho_2(g)$ are similar via the same similarity matrix for every g.

If ρ_0 is an n-dimensional representation of G on E, the choice of a basis (e_i) of E determines a matrix representation $(\mathbb{C}^n, \rho)$; by changing to basis (e'_i) via a matrix T, one obtains the equivalent representation $(\mathbb{C}^n, \rho')$,

$$\rho'(g) = T \circ \rho(g) \circ T^{-1}.$$

Lemma 1.12. *If T intertwines ρ_1 and ρ_2, the kernel of T, $\operatorname{Ker} T$, is invariant under ρ_1, and the image of T, $\operatorname{Im} T$, is invariant under ρ_2.*

Proof. If $x \in E_1$ and $Tx = 0$, then $T(\rho_1(g)x) = \rho_2(g)(Tx) = 0$. Thus $\operatorname{Ker} T$ is a subspace of E_1 invariant under ρ_1.

Let $y \in \operatorname{Im} T$. There exists $x \in E_1$ such that $y = Tx$. Then $\rho_2(g)y = \rho_2(g)(Tx) = T(\rho_1(g)x)$, and hence $\operatorname{Im} T$ is a subspace of E_2 invariant under ρ_2. □

Lemma 1.13. *If T commutes with ρ, each eigenspace of T is invariant under ρ.*

Proof. In fact, if $Tx = \lambda x, \lambda \in \mathbb{C}$, then $T(\rho(g)x) = \lambda\rho(g)x$. Thus the eigenspace of T corresponding to the eigenvalue λ is invariant under ρ. □

Theorem 1.14. (Schur's Lemma). *Let T be an operator intertwining irreducible representations (E_1, ρ_1) and (E_2, ρ_2) of a finite group, G.*

- *If ρ_1 and ρ_2 are not equivalent, then $T = 0$.*
- *If $E_1 = E_2 = E$ and $\rho_1 = \rho_2 = \rho$, then T is a scalar multiple of the identity of E.*

Proof. If ρ_1 and ρ_2 are not equivalent, T is not bijective. Hence either $\operatorname{Ker} T \neq \{0\}$, or $\operatorname{Im} T \neq E_2$. By Lemma 1.12, $\operatorname{Ker} T$ is invariant under ρ_1. Because ρ_1 is irreducible, if $\operatorname{Ker} T \neq \{0\}$, then $\operatorname{Ker} T = E_1$, hence $T = 0$. By Lemma 1.12, $\operatorname{Im} T$ is invariant under ρ_2. Because ρ_2 is irreducible, if $\operatorname{Im} T \neq E_2$, then $\operatorname{Im} T = \{0\}$, and hence $T = 0$.

If $E_1 = E_2 = E$ and $\rho_1 = \rho_2 = \rho$, then, for every $g \in G, \rho(g) \circ T = T \circ \rho(g)$, and T commutes with the representation ρ. Let λ be an eigenvalue of T, which must exist because T is an endomorphism of E, a vector space over $\mathbb{C}$, and let E_λ be the eigenspace associated to λ. By Lemma 1.13, E_λ is invariant under ρ. By hypothesis $E_\lambda \neq \{0\}$, therefore, since ρ is irreducible, $E_\lambda = E$, which means that $T = \lambda \operatorname{Id}_E$. We remark that the proof of the second part of the theorem uses the hypothesis that the vector space of the representation is a complex vector space. □

Conversely, if each operator commuting with the representation ρ is a scalar multiple of the identity, then ρ is irreducible. In fact, if ρ were not irreducible, the projection onto a non-trivial invariant subspace would be a non-scalar operator commuting with ρ.

Remark. Lemma 1.13 has very important consequences in quantum mechanics. The symmetry operators of a system represented by a Hamiltonian $\widehat{H}$ (an operator acting on a Hilbert space) are precisely the operators that commute with $\widehat{H}$. For each energy level, that is, for each eigenvalue of the Hamiltonian, there is a corresponding eigenspace. By this lemma, each eigenspace is the support of a representation of the group of symmetries of the system. *Wigner's principle* then states that, for each energy level, the corresponding representation is an irreducible representation of the full symmetry group of the system. The dimension of the representation corresponding to the given energy level is called the degree of *degeneracy* of the energy level.

2 Characters and Orthogonality Relations

2.1 Matrix Coefficients

We shall denote by $\mathcal{F}(G)$, or sometimes by $\mathbb{C}[G]$, the vector space of functions on the finite group G taking values in $\mathbb{C}$. When this vector space is equipped with the scalar product defined below, we call the resulting finite-dimensional Hilbert space $L^2(G)$. (This definition will be extended to compact groups.)

We adopt the convention that a *scalar product* is antilinear in the first argument and linear in the second.

Definition 2.1. *On $L^2(G)$, the scalar product is defined by*

$$(f_1 \mid f_2) = \frac{1}{|G|} \sum_{g \in G} \overline{f_1(g)} f_2(g).$$

We shall be interested in the matrix coefficients of representations.

Definition 2.2. *If ρ is a representation of G on $\mathbb{C}^n$ then, for every ordered pair (i, j), $1 \leq i \leq n$, $1 \leq j \leq n$, the function $\rho_{ij} \in L^2(G)$ defined for each $g \in G$ to be the coefficient of the matrix $\rho(g)$ in the i-th row and the j-th column, $(\rho(g))_{ij} \in \mathbb{C}$, is called a* matrix coefficient *of ρ.*

For a representation ρ on a vector space E, we define the matrix coefficients ρ_{ij} relative to a basis (e_i) satisfying

$$\rho(g)e_j = \sum_i \rho_{ij}(g)e_i,$$

where i is the row index and j is the column index. If ρ is a unitary representation on a finite-dimensional Hilbert space, then

$$\rho(g^{-1}) = (\rho(g))^{-1} = {}^t\overline{(\rho(g))}.$$

Hence, in an orthonormal basis,

$$\rho_{ij}(g^{-1}) = \overline{\rho_{ji}(g)}$$

and, in particular, the diagonal coefficients of $\rho(g)$ and $\rho(g^{-1})$ are complex conjugates.

2.2 Characters of Representations and Orthogonality Relations

We denote by Tr the trace of an endomorphism.

Definition 2.3. *Let (E, ρ) be a representation of G. The* character *of ρ is the function χ_ρ on G taking complex values defined by*

$$\forall g \in G,\ \chi_\rho(g) = \mathrm{Tr}(\rho(g)).$$

Equivalent representations have the same character.
For a matrix representation of dimension n,

$$\chi_\rho(g) = \sum_{i=1}^{n} (\rho(g))_{ii}. \tag{2.1}$$

On each conjugacy class of G, the function χ_ρ is constant.

Definition 2.4. *A* class function *on G is a function constant on each conjugacy class.*

Thus characters of representations are class functions on the group.

Proposition 2.5. *The following are elementary properties of characters:*

- $\chi_\rho(e) = \dim \rho$.
- $\forall g \in G,\ \chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$.
- *The character of a direct sum of representations is the sum of the characters, $\chi_{\rho_1 \oplus \rho_2} = \chi_{\rho_1} + \chi_{\rho_2}$.*

Proof. The first property is a consequence of formula (2.1). To prove the second formula, we may assume that ρ is unitary in a certain scalar product and choose an orthonormal basis. The direct sum property is obvious. □

If (E_1, ρ_1) and (E_2, ρ_2) are representations of the same group G, we define their *tensor product* to be $(E_1 \otimes E_2, \rho_1 \otimes \rho_2)$, where

$$(\rho_1 \otimes \rho_2)(g) = \rho_1(g) \otimes \rho_2(g),$$

for each $g \in G$. (See Exercise 2.5 for a review of the relevant definitions.) The following is an important property of characters.

Proposition 2.6. *The character of a tensor product of representations is the product of the characters,*

$$\chi_{\rho_1 \otimes \rho_2} = \chi_{\rho_1}\ \chi_{\rho_2}. \tag{2.2}$$

Proof. The equality follows from the fact that the trace of a tensor product of matrices is the product of the traces. □

By Proposition 2.5, for representations ρ_1 and ρ_2 of G,

$$(\chi_{\rho_1} \mid \chi_{\rho_2}) = \frac{1}{|G|} \sum_{g \in G} \chi_{\rho_1}(g^{-1}) \chi_{\rho_2}(g). \tag{2.3}$$

We shall show that the characters of inequivalent irreducible representations are orthogonal and that the character of an irreducible representation is of norm 1.

Proposition 2.7. *Let (E_1, ρ_1) and (E_2, ρ_2) be representations of G and let $u : E_1 \to E_2$ be a linear map. Then the linear map, $T_u : E_1 \to E_2$, defined by*

$$T_u = \frac{1}{|G|} \sum_{g \in G} \rho_2(g)\ u\ \rho_1(g^{-1}), \tag{2.4}$$

intertwines ρ_1 and ρ_2.

Proof. We calculate

$$\begin{aligned} \rho_2(g)\ T_u &= \tfrac{1}{|G|} \sum_{h \in G} \rho_2(gh)\ u\ \rho_1(h^{-1}) \\ &= \tfrac{1}{|G|} \sum_{k \in G} \rho_2(k)\ u\ \rho_1(k^{-1}g), \end{aligned}$$

by the fundamental equation (1.1). Hence,

$$\rho_2(g)\ T_u = T_u\ \rho_1(g).$$

The operator T_u is thus an intertwining operator for ρ_1 and ρ_2. □

Proposition 2.8. *Let (E_1, ρ_1) and (E_2, ρ_2) be irreducible representations of G, let $u : E_1 \to E_2$ be a linear map, and define T_u by the equation (2.4).*

(i) *If* ρ_1 *and* ρ_2 *are not equivalent, then* $T_u = 0$.

(ii) *If* $E_1 = E_2 = E$ *and* $\rho_1 = \rho_2 = \rho$, *then*

$$T_u = \frac{\operatorname{Tr} u}{\dim E} \operatorname{Id}_E.$$

Proof. The first assertion is clear by Schur's lemma (Theorem 1.14). For the second, we need only calculate λ given that $T_u = \lambda \operatorname{Id}_E$. So $\operatorname{Tr} T_u = \frac{1}{|G|}\sum_{g\in G} \operatorname{Tr} u = \operatorname{Tr} u$, thus $\lambda = \frac{\operatorname{Tr} u}{\dim E}$. □

Proposition 2.9. *Let* (E_1, ρ_1) *and* (E_2, ρ_2) *be irreducible representations of* G. *We choose bases in* E_1 *and* E_2.

(i) *If* ρ_1 *and* ρ_2 *are not equivalent,*

$$\forall i, j, k, \ell, \quad \sum_{g\in G} (\rho_2(g))_{k\ell} (\rho_1(g^{-1}))_{ji} = 0.$$

(ii) *If* $E_1 = E_2 = E$ *and* $\rho_1 = \rho_2 = \rho$,

$$\frac{1}{|G|} \sum_{g\in G} (\rho(g))_{k\ell} (\rho(g^{-1}))_{ji} = \frac{1}{\dim E} \delta_{ki}\delta_{\ell j}.$$

Proof. We use a basis (e_j) of E_1, $1 \le j \le \dim E_1$, and a basis (f_ℓ) of E_2, $1 \le \ell \le \dim E_2$. For $u : E_1 \to E_2$, T_u is defined by (2.4). We have, for $1 \le i \le \dim E_1, 1 \le k \le \dim E_2$,

$$(T_u)_{ki} = \frac{1}{|G|} \sum_{g\in G} \sum_{m=1}^{\dim E_1} \sum_{p=1}^{\dim E_2} (\rho_2(g))_{kp}\, u_{pm}\, (\rho_1(g^{-1}))_{mi}.$$

Let us choose our linear map u to be the map $u_{(\ell j)} : E_1 \to E_2$ defined by $u_{(\ell j)}(e_k) = \delta_{jk} f_\ell$. Then

$$(u_{(\ell j)})_{pm} = \delta_{\ell p}\delta_{jm},$$

and, consequently,

$$(T_{u_{(\ell j)}})_{ki} = \frac{1}{|G|} \sum_{g\in G} (\rho_2(g))_{k\ell} (\rho_1(g^{-1}))_{ji}.$$

Next we apply Proposition 2.8. If ρ_1 and ρ_2 are not equivalent, then $T_{u_{(\ell j)}}$ is always zero, hence (i). If $E_1 = E_2 = E$ and $\rho_1 = \rho_2 = \rho$, then

$$\frac{1}{|G|} \sum_{g\in G} (\rho(g))_{k\ell} (\rho(g^{-1}))_{ji} = (T_{u_{(\ell j)}})_{ki} = \frac{\operatorname{Tr} u_{(\ell j)}}{\dim E} \delta_{ki} = \frac{\delta_{ki}\delta_{\ell j}}{\dim E},$$

which proves (ii). □

Corollary 2.10. *Let (E_1, ρ_1) and (E_2, ρ_2) be unitary irreducible representations of G. We choose orthonormal bases in E_1 and E_2.*

(i) *If ρ_1 and ρ_2 are not equivalent, then for every i, j, k, ℓ,*

$$((\rho_1)_{ij} \mid (\rho_2)_{k\ell}) = 0.$$

(ii) *If $E_1 = E_2 = E$ and $\rho_1 = \rho_2 = \rho$, then for every i, j, k, ℓ,*

$$(\rho_{ij} \mid \rho_{k\ell}) = \frac{1}{\dim E}\delta_{ik}\delta_{j\ell}.$$

Proof. In fact, if ρ_1 is unitary for a scalar product on E_1 and if the chosen basis in E_1 is orthonormal, then

$$\frac{1}{|G|}\sum_{g\in G}(\rho_2(g))_{k\ell}(\rho_1(g^{-1}))_{ji} = \frac{1}{|G|}\sum_{g\in G}(\rho_2(g))_{k\ell}\overline{(\rho_1(g))_{ij}} = ((\rho_1)_{ij} \mid (\rho_2)_{k\ell}).$$

Proposition 2.9 thus implies (i) and (ii). □

Theorem 2.11. (Orthogonality Relations).

(i) *If ρ_1 and ρ_2 are inequivalent irreducible representations of G, then*

$$(\chi_{\rho_1} \mid \chi_{\rho_2}) = 0.$$

(ii) *If ρ is an irreducible representation of G, then*

$$(\chi_\rho \mid \chi_\rho) = 1.$$

Proof. By the equality (2.3) and the preceding proposition, if ρ_1 and ρ_2 are inequivalent irreducible representations, then $(\chi_{\rho_1} \mid \chi_{\rho_2}) = 0$. If $\rho_1 = \rho_2 = \rho$, $\frac{1}{|G|}\sum_{g\in G}\rho(g)_{ii}\rho(g^{-1})_{jj} = \frac{\delta_{ij}}{\dim E}$, whence $(\chi_\rho \mid \chi_\rho) = 1$. □

We define the set of *irreducible characters* of G to be the set of characters of inequivalent irreducible representations of G. We write χ_{ρ_i} or even χ_i to denote the character of an irreducible representation ρ_i. The preceding results can be formulated as follows.

Theorem 2.12. *Let G be a finite group. The irreducible characters of G form an orthonormal set in $L^2(G)$.*

Corollary 2.13. *The inequivalent irreducible representations of a finite group G are finite in number.*

We shall denote by $\widehat{G}$ the set of equivalence classes of irreducible representations of G.

2.3 Character Table

This is the name given to the table whose columns correspond to conjugacy classes of a group and whose rows correspond to inequivalent irreducible representations of the group. At the intersection of the row and the column one writes the value of the character of the representation, evaluated on an element (any element) of the conjugacy class. Let N be the number of conjugacy classes of the group G. (In other words, N is the number of columns; we shall show that it is also the number of rows.) Let g_i be an element of G with conjugacy class C_{g_i}, $1 \le i \le N$, which consists of $|C_{g_i}|$ elements. Let ρ_k and ρ_ℓ be irreducible representations of G. Then

$$(\chi_{\rho_k} \mid \chi_{\rho_\ell}) = \frac{1}{|G|} \sum_{i=1}^{N} |C_{g_i}| \, \overline{\chi_{\rho_k}(g_i)} \, \chi_{\rho_\ell}(g_i) = \delta_{k\ell}.$$

This formula can be restated as the following result.

Proposition 2.14. *If the i-th column is given weight $|C_{g_i}|$, the rows of the character table are orthogonal and of norm $\sqrt{|G|}$.*

We write character tables in the following form:

	$\lvert C_{g_1}\rvert$	.....	$\lvert C_{g_N}\rvert$
	g_1	.....	g_N
...	...	.....	...
χ_{ρ_k}	$\chi_{\rho_k}(g_1)$	.....	$\chi_{\rho_k}(g_N)$
...	...	.....	...
χ_{ρ_ℓ}	$\chi_{\rho_\ell}(g_1)$	.....	$\chi_{\rho_\ell}(g_N)$
...	...	.....	...

2.4 Application to the Decomposition of Representations

We denote by $\rho_1, \ldots, \rho_N$ the inequivalent irreducible representations of G. (We shall see in Corollary 3.6 that this number N equals the number of conjugacy classes of G.) More precisely, we choose from each equivalence class of representations of G a representative that we denote by ρ_i.

In the equalities below, the equal sign denotes membership in the same equivalence class.

Theorem 2.15. *Let ρ be any representation of G and let χ_ρ be its character. Then*

$$\rho = \bigoplus_{i=1}^{N} m_i \rho_i,$$

where

$$m_i = (\chi_{\rho_i} \mid \chi_\rho).$$

Proof. We know by Theorem 1.9 that ρ is direct sum of irreducible representations. We can group the terms corresponding to the same equivalence class of irreducible representations, ρ_i, and we find $\rho = \oplus_{i=1}^{N} m_i \rho_i$, for some non-negative integers, m_i. One sees then that $\chi_\rho = \sum_{i=1}^{N} m_i \ \chi_{\rho_i}$, and hence by orthogonality $(\chi_{\rho_i} \mid \chi_\rho) = m_i \, (\chi_{\rho_i} \mid \chi_{\rho_i}) = m_i$. □

Definition 2.16. *If ρ admits the decomposition*

$$\rho = m_1\rho_1 \oplus m_2\rho_2 \oplus \cdots \oplus m_N\rho_N,$$

then the non-negative integer m_i is the multiplicity *of ρ_i in ρ, and $m_i\rho_i$ is the* isotypic component *of type ρ_i of ρ.*

Corollary 2.17. *The decomposition into isotypic components is unique up to order.*

Corollary 2.18. *Two representations with the same character are equivalent.*

By the previous theorem,

$$(\chi_\rho \mid \chi_\rho) = \sum_{i=1}^{N} m_i^2.$$

Hence

Theorem 2.19. (Irreducibility Criterion). *A representation ρ is irreducible if and only if $(\chi_\rho \mid \chi_\rho) = 1$.*

3 The Regular Representation

3.1 Definition

In general, if a group G acts on a set M, then G acts linearly on the space $\mathcal{F}(M)$ of functions on M taking values in $\mathbb{C}$ by $(g, f) \in G \times \mathcal{F}(M) \mapsto g \cdot f \in \mathcal{F}(M)$, where

$$\forall x \in M, \ (g \cdot f)(x) = f(g^{-1}x).$$

We can see immediately that this gives us a representation of G on $\mathcal{F}(M)$.

Take $M = G$, the group acting on itself by left multiplication. One obtains a representation R of G on $\mathcal{F}(G)$ called the *left regular representation* (or simply *regular representation*) of G. Thus, by definition,

$$\forall g, h \in G, (R(g)f)(h) = f(g^{-1}h).$$

In the same way one can define the *right regular representation* R', associated to the right action of G on itself, by $(R'(g)f)(h) = f(hg)$. The right and left regular representations are equivalent. For a finite group G the vector space $\mathcal{F}(G)$ of maps of G into $\mathbb{C}$ is finite-dimensional, of dimension $|G|$. The regular representation is thus of dimension $|G|$.

We introduce the basis $(\epsilon_g)_{g\in G}$ of $\mathcal{F}(G)$ defined by:

$$\epsilon_g : G \to \mathbb{C} \quad \begin{cases} \epsilon_g(g) & = 1, \\ \epsilon_g(h) & = 0, \text{ if } h \neq g. \end{cases}$$

The regular representation of G satisfies

$$\forall g, h \in G,\ R(g)(\epsilon_h) = \epsilon_{gh}.$$

In fact, for every $k \in G$, $(R(g)\epsilon_h)(k) = \epsilon_h(g^{-1}k)$, and $\epsilon_h(g^{-1}k) = 1$ if $k = gh$, while $\epsilon_h(g^{-1}k) = 0$ otherwise. (In the right regular representation $\epsilon_h \mapsto \epsilon_{hg^{-1}}$.)

Proposition 3.1. *On $L^2(G) = \mathcal{F}(G)$ with scalar product $(\ \mid\ )$, the regular representation is unitary.*

Proof. For f_1 and $f_2 \in L^2(G)$ we have, for every $g \in G$,

$$\begin{aligned}(R(g)f_1 \mid R(g)f_2) &= \frac{1}{|G|}\sum_{h\in G} \overline{(R(g)f_1)(h)}(R(g)f_2)(h) \\ &= \frac{1}{|G|}\sum_{h\in G} \overline{f_1(g^{-1}h)}f_2(g^{-1}h) \\ &= \frac{1}{|G|}\sum_{k\in G} \overline{f_1(k)}f_2(k) = (f_1 \mid f_2).\end{aligned}$$

The operator $R(g)$ is thus unitary for every $g \in G$. □

3.2 Character of the Regular Representation

On the one hand,

$$\chi_R(e) = \mathrm{Tr}(R(e)) = \dim \mathcal{F}(G) = |G|.$$

On the other hand, if $g \neq e$, then

$$\chi_R(g) = \mathrm{Tr}(R(g)) = 0$$

because in this case, for every $h \in G$, $R(g)\epsilon_h \neq \epsilon_h$.

The regular representation, R, is reducible because $\sum_{g\in G} \epsilon_g$ generates a vector subspace W of $\mathcal{F}(G)$ of dimension 1 that is invariant under R. In fact, for every $g \in G$, $R(g)(\sum_{h\in G} \epsilon_h) = \sum_{h\in G} \epsilon_{gh} = \sum_{k\in G} \epsilon_k$. Furthermore $R\mid_W$ is equivalent to the trivial representation since, for every $x \in W$, $R(g)(x) = x$. We shall show that in fact the regular representation contains each irreducible representation of G with multiplicity equal to its dimension.

Example. The regular representation of $\mathfrak{S}_3$ on $\mathbb{C}[\mathfrak{S}_3]$ is of dimension 6. It decomposes into the direct sum of the one-dimensional trivial representation, the one-dimensional sign representation and two copies of the two-dimensional irreducible representation studied in the Example of Section 1.

3.3 Isotypic Decomposition

We now use the notation introduced in Section 2.4.

Proposition 3.2. *The decomposition of the regular representation R of G into isotypic components is*

$$R = \bigoplus_{i=1}^{N} n_i \rho_i,$$

where ρ_i, i=1, ..., N, are the irreducible representations of G, and n_i= dim ρ_i.

Proof. We know that

$$\chi_R(g) = \begin{cases} |G| & \text{if } g = e, \\ 0 & \text{if } g \neq e, \end{cases}$$

and hence $(\chi_{\rho_i} \mid \chi_R) = \chi_{\rho_i}(e) = \dim \rho_i$. □

Theorem 3.3. *We have*

$$\sum_{i=1}^{N} (n_i)^2 = |G|,$$

where $n_i = \dim \rho_i$.

Proof. We have $|G| = \chi_R(e) = \sum_{i=1}^{N} n_i \chi_{\rho_i}(e) = \sum_{i=1}^{N} (n_i)^2$. □

The equality $\sum_{i=1}^{N} (n_i)^2 = |G|$ is often used, for example, in order to determine the dimension of a "missing" irreducible representation when one already knows $N - 1$ representations.

3.4 Basis of the Vector Space of Class Functions

The dimension of the vector space of class functions on G taking values in $\mathbb{C}$ is equal to the number of conjugacy classes of G. We shall show that this is also the number of equivalence classes of irreducible representations of G.

Let (E, ρ) be a representation of G, and let f be a function on G. We consider the endomorphism ρ_f of E defined by

$$\rho_f = \sum_{g \in G} f(g)\rho(g). \tag{3.1}$$

Thus, by definition, for every $x \in E$, $\rho_f(x) = \sum_{g \in G} f(g)\rho(g)(x)$.

Lemma 3.4. *The endomorphism ρ_f has the following properties.*

(i) *If f is a class function, ρ_f commutes with ρ.*
(ii) *If f is a class function and if ρ is irreducible,*

$$\rho_f = \frac{|G|(\overline{f} \mid \chi_\rho)}{\dim \rho} \mathrm{Id}_E .$$

Proof. For every function f, we have

$$\begin{aligned}\rho_f \circ \rho(g) &= \textstyle\sum_{h\in G} f(h)\rho(h)\rho(g) = \sum_{h\in G} f(h)\rho(hg)\\ &= \textstyle\sum_{k\in G} f(kg^{-1})\rho(k) = \sum_{h\in G} f(ghg^{-1})\rho(gh).\end{aligned}$$

If f is assumed to be a class function, we obtain

$$\rho_f \circ \rho(g) = \rho(g)\sum_{h\in G} f(h)\rho(h) = \rho(g)\circ\rho_f.$$

Let us prove (ii). Assume that f is a class function and that ρ is irreducible. Then by (i) and Schur's lemma (Theorem 1.14), there exists a scalar $\lambda\in\mathbb{C}$ such that $\rho_f = \lambda\,\mathrm{Id}_E$. On the other hand, $\operatorname{Tr}\rho_f = \sum_{g\in G} f(g)\operatorname{Tr}\rho(g) = \sum_{g\in G} f(g)\chi_\rho(g) = |G|(\overline{f}\mid\chi_\rho)$, from which the result follows. □

Theorem 3.5. *The irreducible characters form an orthonormal basis of the vector space of class functions.*

Proof. We know that the characters, $\chi_{\rho_1},\dots,\chi_{\rho_N}$, of inequivalent irreducible representations, $\rho_1,\dots,\rho_N$, of G form an orthonormal set in $L^2(G)$ (Theorem 2.12). Let us show that this set spans the vector subspace of class functions. Let f be a class function such that, for $1\le i\le N$, $(f|\chi_{\rho_i}) = 0$. We consider $(\rho_i)_{\overline{f}} = \sum_{g\in G}\overline{f}(g)\rho_i(g)$. By the previous lemma, $(\rho_i)_{\overline{f}} = 0$ and we deduce, by decomposition, that for any representation ρ we have $\rho_{\overline{f}} = 0$. In particular, $R_{\overline{f}} = 0$, where R is the regular representation. Thus,

$$0 = R_{\overline{f}}(\epsilon_g) = \sum_{h\in G}\overline{f}(h)R(h)(\epsilon_g) = \sum_{h\in G}\overline{f}(h)\epsilon_{hg},$$

for $g\in G$, and, in particular,

$$0 = R_{\overline{f}}(\epsilon_e) = \sum_{h\in G}\overline{f}(h)\epsilon_h = \overline{f},$$

so $f = 0$. □

Corollary 3.6. *The number of equivalence classes of irreducible representations of a finite group is equal to the number of conjugacy classes of that group.*

In other words, *the character table is square.*

Proposition 3.7. *The columns of the character table of a finite group G are orthogonal and of norm $\sqrt{\frac{|G|}{|C_g|}}$, where $|C_g|$ denotes the number of elements of the conjugacy class of g. Explicitly,*

$$\begin{cases}\sum_{i=1}^N \overline{\chi_{\rho_i}(g)}\chi_{\rho_i}(g') = 0, \text{ if } g \text{ and } g' \text{ are not conjugate},\\ \frac{1}{|G|}\sum_{i=1}^N \overline{\chi_{\rho_i}(g)}\chi_{\rho_i}(g) = \frac{1}{|C_g|}.\end{cases}$$

Proof. By Theorem 3.5, if f is a class function, then

$$f = \sum_{i=1}^{N} (\chi_{\rho_i} \mid f)\chi_{\rho_i}.$$

For $g \in G$, consider the class function f_g which takes the value 1 on g and the value 0 on every other conjugacy class of G. We have

$$\begin{aligned}(\chi_{\rho_i} \mid f_g) &= \frac{1}{|G|} \sum_{h \in G} \overline{\chi_{\rho_i}(h)} f_g(h) \\ &= \frac{|C_g|}{|G|} \overline{\chi_{\rho_i}(g)},\end{aligned}$$

and thus $f_g = \frac{|C_g|}{|G|} \sum_{i=1}^{N} \overline{\chi_{\rho_i}(g)} \chi_{\rho_i}$. In particular, if $g' \notin C_g$, then

$$0 = f_g(g') = \frac{|C_g|}{|G|} \sum_{i=1}^{N} \overline{\chi_{\rho_i}(g)} \chi_{\rho_i}(g'),$$

which proves the first formula, and hence the orthognality of the columns of the character table. On the other hand, $1 = f_g(g) = \frac{|C_g|}{|G|} \sum_{i=1}^{N} \overline{\chi_{\rho_i}(g)} \chi_{\rho_i}(g)$, which proves the second formula. □

In the particular case where $g = e$, Proposition 3.7 yields a new proof of the equation $\sum_{i=1}^{N} (\dim \rho_i)^2 = |G|$.

4 Projection Operators

We introduce the projection operators onto the isotypic components of the decomposition of the vector space of any representation. Let (E, ρ) be a representation of G and let $\rho = \oplus_{i=1}^{N} m_i \rho_i$ be the decomposition of ρ into isotypic components. The support of the isotypic component $m_i \rho_i$, is $m_i E_i = E_i \oplus \cdots \oplus E_i$ (m_i terms). We denote this vector subspace of E by V_i. We shall write

$$V_i = m_i E_i = \bigoplus_{j=1}^{m_i} E_{i,j},$$

where each $E_{i,j}$, $1 \leq j \leq m_i$, is equal to E_i. We thus have $E = \oplus_{i=1}^{N} V_i$.

Theorem 4.1. *For each i, $1 \leq i \leq N$, we set*

$$P_i = \frac{\dim \rho_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g).$$

Then

(i) *P_i is the projection of E onto V_i under the decomposition $E = \oplus_{i=1}^N V_i$.*
(ii) *$P_i P_j = \delta_{ij} P_i$, for $1 \le i \le N$, $1 \le j \le N$.*
(iii) *If ρ is unitary, P_i is Hermitian, that is ${}^t\overline{P_i} = P_i$.*

Proof. (i) Let us choose i_0, $1 \le i_0 \le N$, and show that $P_{i_0}\big|_{V_{i_0}} = \mathrm{Id}_{V_{i_0}}$, while if $i \ne i_0$, then $P_{i_0}\big|_{V_i} = 0$. Let $x = \sum_{i=1}^N x_i$, where $x_i \in V_i$, and let $x_i = \sum_{j=1}^{m_i} x_{i,j}$, where $x_{i,j} \in E_{i,j}$, whence $x = \sum_{i=1}^N \sum_{j=1}^{m_i} x_{i,j}$. Then

$$\begin{aligned} P_{i_0}(x) &= \frac{\dim \rho_{i_0}}{|G|} \sum_{i=1}^N \sum_{j=1}^{m_i} \sum_{g\in G} \overline{\chi_{i_0}(g)} \rho(g) x_{i,j} \\ &= \frac{\dim \rho_{i_0}}{|G|} \sum_{i=1}^N \sum_{j=1}^{m_i} \left(\sum_{g\in G} \overline{\chi_{i_0}(g)} \rho_i(g) \right) x_{i,j}. \end{aligned}$$

Because χ_{i_0} is a class function and ρ_i is irreducible, we may apply Lemma 3.4 and we obtain

$$\sum_{g\in G} \overline{\chi_{i_0}(g)} \rho_i(g) = (\rho_i)_{\overline{\chi_{i_0}}} = \frac{|G|}{\dim \rho_i} (\chi_{i_0} \mid \chi_i) \,\mathrm{Id}_{E_i} = \frac{|G|}{\dim \rho_{i_0}} \delta_{ii_0} \,\mathrm{Id}_{E_{i_0}},$$

which finally leads to

$$P_{i_0}(x) = \sum_{i=1}^N \sum_{j=1}^{m_i} \delta_{i_0 i} x_{i,j} = \sum_{j=1}^{m_{i_0}} x_{i_0,j} = x_{i_0}.$$

(ii) The equations $P_i P_j = 0$ if $i \ne j$ and $P_i^2 = P_i$ follow from (i).
(iii) If ρ is unitary, then

$$\begin{aligned} \frac{|G|}{\dim \rho_i} \, {}^t\overline{P_i} &= \sum_{g\in G} \chi_i(g) \, {}^t\overline{\rho(g)} = \sum_{g\in G} \chi_i(g) \rho(g^{-1}) \\ &= \sum_{g\in G} \chi_i(g^{-1}) \rho(g) = \sum_{g\in G} \overline{\chi_i(g)} \rho_i(g), \end{aligned}$$

which is equal to $\frac{|G|}{\dim \rho_i} P_i$, which proves (iii). □

The decomposition $E = \oplus_{i=1}^N V_i$ is unique up to order. On the other hand, the decomposition $V_i = \oplus_{j=1}^{m_i} E_{i,j}$ is not in general unique. For example, if $\rho = \mathrm{Id}_E$ and the dimension of E is greater than 1 then ρ can be written in an infinite number of ways as a direct sum of one-dimensional representations.

5 Induced Representations

Induction is an operation which associates to a representation of a subgroup H of a group G a representation of the group G itself.

5.1 Definition

Let G be a finite group and H a subgroup. Let (F, π) be a representation of H. We define the vector space

$$E = \{\varphi \ : G \to F \mid \forall h \in H, \varphi(gh) = \pi(h^{-1})\varphi(g)\}, \tag{5.1}$$

and a representation $\rho = \pi^{\uparrow G}$ of G on E by

$$\forall \varphi \in E, \quad (\rho(g_0)\varphi)(g) = \varphi(g_0^{-1}g), \tag{5.2}$$

for every $g_0 \in G$ and for every $g \in G$. We see that $\rho(g_0)\varphi$ lies in E because

$$(\rho(g_0)\varphi)(gh) = \varphi(g_0^{-1}gh) = \pi(h^{-1})\varphi(g_0^{-1}g) = \pi(h^{-1})\left((\rho(g_0)\varphi)(g)\right),$$

and on the other hand we see that $g \mapsto \rho(g)$ is a group morphism of G into $\mathrm{GL}(E)$.

Definition 5.1. *The representation $\rho = \pi^{\uparrow G}$ of G on E is called the representation of G* induced *from the representation π of the subgroup H of G.*

For example, if $H = \{e\}$ and if π is the trivial representation of H on $\mathbb{C}$, then the vector space E is equal to $\mathbb{C}[G]$ and the representation of G induced from π is the regular representation of G.

5.2 Geometric Interpretation

We can interpret the vector space E as the space of sections of a vector bundle. We consider the Cartesian product $G \times F$ and we introduce the equivalence relation

$$(g, x) \sim (gh, \pi(h^{-1})x), \ \forall h \in H. \tag{5.3}$$

Let $G \times_\pi F$ be the quotient of $G \times F$ by this equivalence relation, and let

$$q : G \times_\pi F \to G/H$$

be the projection that sends the equivalence class of (g, x) to gH. Note that this is well-defined because, if $(g', x') \sim (g, x)$, then $g' = gh$, for some $h \in H$. The inverse image under the projection q of any point in G/H is isomorphic to the vector space F. We call $G \times_\pi F$ a *vector bundle* over G/H with *fiber* F.

A *section* of the projection $q : G \times_\pi F \to G/H$ (or of the vector bundle $G \times_\pi F$) is, by definition, a map ψ from G/H to $G \times_\pi F$ such that $q \circ \psi = \mathrm{Id}_{G/H}$.

Proposition 5.2. *The support E of the induced representation $\pi^{\uparrow G}$ is the vector space of sections of the projection $q : G \times_\pi F \to G/H$.*

Proof. To $\varphi \in E$ and $g \in G$ we associate the equivalence class of $(g, \varphi(g))$. The result depends only on the class of g modulo H. In fact, if $g' = gh$, with $h \in H$, we obtain the equivalence class of $(gh, \varphi(gh))$ which is equal to the equivalence class of $(g, \pi(h)\varphi(gh)) = (g, \varphi(g))$, since $\varphi \in E$. Thus one defines

a section of $q : G \times_\pi F \to G/H$. On the other hand, to any given section of q, we may associate an element of E by considering the second component of the equivalence class associated to an element of G/H. As this construction is the inverse of the previous one, we have thus obtained an isomorphism of the space E of the induced representation onto the vector space of sections of the vector bundle $G \times_\pi F$. □

The notion of an induced representation can be defined more generally than just for finite groups, and has many important applications in mathematics and physics.

References

The representations of finite groups are the subject of Serre's book (1977), of which Part I is an exposition of fundamental results. Finite groups are studied in Sternberg (2003), Simon (1996), Artin (2014) and Ledermann–Weir (1996). See Etingof (2011) for a modern account including the representations of associative algebras. Also see the first chapters of Fulton–Harris (2004) which are followed by chapters on Lie algebra representations, or the textbook by James and Liebeck (2001) which stresses arithmetic. All of these works discuss induced representations. For applications to physics, see Ludwig–Falter (2012) or Tung (1985).

Tensor products of vector spaces are introduced in Exercise 2.5 below. (Also see Exercise 2.7.) For supplementary material on the tensor, exterior and symmetric algebras of a vector space, see Greub (1978), Warner (1983), Sternberg (2003) or Knapp (2002).

One can find a historical and mathematical analysis of the theory of representation theory in the book by C. Curtis (1999). Also see Hawkins (2000) and Rossmann (2011).

For an in-depth study of induced representations, see, *e.g.*, Goodman–Wallach (2009) or Gurarie (1992) which includes applications to physics.

Topics in History

At the end of the nineteenth century

The early developments of the theory of representations of finite groups took place in the last decade of the nineteenth century and are mainly due to William Burnside (1852–1927) and Heinrich Maschke (1853–1908).

Theory of groups of finite order

Burnside, who was active in England, published extensively on the groups of finite order, beginning in 1893 with a 12-page article in the *Proceedings of the London Mathematical Society* and continuing until 1911 when he published

the second edition of his 530 page treatise, *Theory of Groups of Finite Order*, which had first appeared in 1897. He had graduated from Cambridge University in 1875 where he remained as a researcher and a member of the rowing team. From 1885 on, the career he followed was in the Royal Naval College in Greenwich where he taught naval officers hydrodynamics and other fields of applied mathematics, while he published on finite groups and their representations. His impressive list of publications begins with a paper on elliptic functions published in 1883, continues with numerous fundamental articles on the representations of finite groups and ends with an article published the year of his death, "On a group of order 25920 and the projective transformations of a cubic surface". He published some 170 articles and notes on group theory as well as on analysis and mechanics and nine books. The two volumes of his collected articles were finally published in 2004 with biographical and editorial material, a total of more than 1600 pages.

Finite linear groups of substitutions

Maschke was born in Breslau in Prussia (now Wrocław, Poland) and died in Chicago. He studied at the universities of Heidelberg and Berlin, obtained his doctorate from the University of Göttingen, taught in high schools, worked with Felix Klein (1849–1925) in Göttingen for several months in 1886–1887, and studied electrotechnics before emigrating to America in 1891. He was soon appointed to a position at the University of Chicago where he became a full professor in 1907. After 1900, he published in English. The theorem that bears his name can be found in an article that he published in 1898 in the *Mathematische Annalen*: "Finite linear groups of substitutions in which some coefficients vanish are intransitive". He presented his result in a lecture on "Some general theorems concerning linear substitution groups of finite order" delivered at the fourth regular meeting of the Chicago Section of the American Mathematical Society held at the University of Chicago on Thursday and Friday, December 29 and 30, 1898, a meeting composed of thirteen communications, attended by twenty-six persons, a large number that indicated "the interest in the [...] development of mathematics" of the members of the section[1]. Maschke's later work was on differential geometry.

Frobenius and Schur

Ferdinand Georg Frobenius (1849–1917) was educated in Berlin. His theory of characters was created in a series of articles that were published in the proceedings of the Berlin Academy beginning in 1895 and are reprinted in his collected works which appeared in three volumes in 1968. They contain beautiful character tables, for example, for a subgroup of $\mathfrak{S}_{12}$ of order 12.11.10.9.8 with 15 irreducible representations, and, on a folding page, for a subgroup of $\mathfrak{S}_{24}$ of order 24.23.22.21.20.48 with 26 irreducible representations. These articles

[1] One of the mathematicians present was a woman, Professor Mary F. Winston, who had defended her thesis, prepared in Göttingen under the direction of Felix Klein, in 1897.

include: "On finite groups" in 1895, "On the characters of the symmetric group" in 1900 and, the following year, "On the characters of the alternating group". He published on the theory of finite groups until his last year.

Born in Mogilev in Belarus, then in the Russian empire, Issai Schur (1875–1941) completed his doctorate, "On a class of matrices which are similar to a given matrix", in Berlin in 1901, a study of the representations of the general linear group of invertible complex matrices. He held positions in Bonn and later in Berlin where he was named to a university chair in 1919, as the successor to Frobenius, and three years later he was elected to the Berlin Academy of Science, succeeding Frobenius. In Berlin, he supervised the theses of twenty-two doctoral students. His collected works were published in 1973. There is proof of the collaboration of Frobenius with his younger colleague Schur in the two articles which they published jointly in 1906 in the proceedings of the Berlin Academy.

Schur's lectures in Zurich and after

In 1936, Schur gave a course on "The algebraic foundations of the representation theory of groups" at the ETH (Eidgenössische Technische Hochschule) in Zurich. Why in Zurich? After the Nazis came to power in Germany in 1933, Schur was dismissed from his position because he was Jewish and, in 1935, he was obliged to tend his resignation. It is at that time, when he had not yet left Berlin, that some Swiss colleagues invited him to lecture. His lecture notes from 1936, "The algebraic foundations of the representation theory of groups", which were mimeographed, are considered to be the first written account of the theory of representations of finite groups. In 2004, Schur's lectures were edited with commentaries in the ETH Library Series.

A fact, a guess, and a question. A copy of this book, signed by the author, is listed in the catalogue of the main mathematics library in Paris but could not be found by the librarian anywhere in the library in 2021. To whom was it dedicated? If I had to guess, I would say "to Élie Cartan". I'll add another question: Who stole this book?

The end. In 1938, Schur was ill and in great danger in Nazi Germany. He was able to leave for Switzerland in 1939 and, from there, to travel to Palestine where he settled in Tel Aviv. He was invited to give a lecture at the university of Jerusalem – to the creation of which Einstein had been instrumental since 1923 – but his health failed him during his lecture. Back in Tel Aviv, he died soon after at the age of 66.

Schur and Brauer plan to read Wigner

Richard Brauer (1901–1977) was born in Berlin and studied at the university under Schur. In 1926, he completed his dissertation which dealt with the characters of the irreducible representations of the real orthogonal groups. He began teaching as an assistant at the University of Königsberg where he had a

successful career of very active research and teaching. He was a superb teacher, covering a wide range of courses.

We know from Brauer's own recollections, cited in the preface to his collected works that appeared three years after his death, that, shortly before Hitler seized power, Schur suggested to him that, together, they should write a book on the representation theory of groups that would include chapters on the work of Eugene Wigner[2] on the applications of representation theory to quantum mechanics. But this project could not be carried out because of the situation in Germany in 1933, Brauer was compelled to leave Germany because he too was Jewish. He was fortunate enough to secure a position in the United States, emigrated in November, and arrived at the University of Kentucky where he had to teach in an unfamiliar language, English. Later he held a variety of prestigious positions, first at the Institute for Advanced Study in Princeton, then at the University of Toronto, the University of Michigan, and finally at Harvard, where he had many Ph.D. students and he kept publishing, often in collaboration.

Brauer and Noether

Earlier, Brauer had collaborated with Emmy Noether (1882–1935), the great mathematician who was an author of innovative, important work on the representation theory of algebras, at a time when practically no other woman worked in mathematics. She was the daughter of the mathematician Max Noether (1844–1921), a renowned algebraist teaching at the University of Erlangen. Her 1907 dissertation was written under the guidance of her father's colleague, Paul Gordan (1837–1912). The name of the mathematician Gordan would become associated with that of Alfred Clebsch (1833–1872) for their work on the theory of invariants. In fact, the "Clebsch–Gordan formula" is a modern group-theoretic interpretation of the finite series expansion for binary forms, *i.e.,* polynomials in two variables, to be found in work developed independently, published by Gordan in 1871 and by Clebsch in 1872.

In 1915, when Noether was already the author of five papers on invariant theory and fields of rational functions, and eight years after her doctorate, she was called to the University of Göttingen to assist Klein and Hilbert[3] who were then in the process of acquainting themselves with Einstein's general theory of relativity. Her work of the years 1915–1918 resulted in the now famous article, "Invariant variational problems", which today is the basis of innumerable papers on symmetries and conservation laws. But she soon left this topic related to mathematical physics and returned to research in pure algebra, ring theory, on which she published landmark papers in one of the highly regarded journals, such as the *Mathematische Annalen* and the *Mathematische Zeitschrift*. Never appointed a professor in spite of fame that extended well beyond the University of Göttingen, she taught a small band of devoted students and published numerous articles in abstract algebra.

[2] On Wigner, see p. 115 below.
[3] On Hilbert, see p. 52 below.

Noether began a collaboration with Brauer in 1927. They published an important joint paper that same year on "The minimal splitting fields of irreducible representations" and, in 1931, an article on simple algebras, this time with a third author, Helmut Hasse (1898–1979), who was already a professor in Marburg, a university town in central Germany, and would become the director of the mathematical institute in Göttingen in 1934. In 1933, when the racial laws against the Jews took efect, Emmy Noether was forced to emigrate. An offer of a position at Bryn Mawr College in Pennsylvania arrived on time for her to leave Germany. She sailed to the United States where she could carry on her teaching and her mathematical activity at this women's college, while she also travelled once a week to give lectures and seminars at the Institute for Advanced Study in Princeton. She died, following an operation, two years after she had left Germany.

Exercises

Exercise 2.1 *The symmetric group* $\mathfrak{S}_3$.

We write c for the cyclic permutation (123) and t for the transposition (23). Show that $\{c, t\}$ generates $\mathfrak{S}_3$, and that $tc = c^2t$, $ct = tc^2$. Find the conjugacy classes of the group $\mathfrak{S}_3$.

Exercise 2.2 *Representations of* $\mathfrak{S}_3$.

(a) Find the one-dimensional representations of the group $\mathfrak{S}_3$.

(b) Let e_1, e_2, e_3 be the canonical basis of $\mathbb{C}^3$. For $g \in \mathfrak{S}_3$, we set $\sigma(g)e_i = e_{g(i)}$. Show that this defines a three-dimensional representation σ of $\mathfrak{S}_3$ and that $V = \{(z_1, z_2, z_3) \in \mathbb{C}^3 \mid z_1 + z_2 + z_3 = 0\}$ is invariant under σ. This representation is called the *permutation representation* of the symmetric group.
We denote by ρ the restriction to V of the representation σ.

(c) Show that there is a basis (u_1, u_2) of V such that $\rho(t)u_1 = u_2$, $\rho(t)u_2 = u_1$, $\rho(c)u_1 = ju_1$, $\rho(c)u_2 = j^2u_2$, where $j^2 + j + 1 = 0$. Is the representation ρ irreducible?

(d) Find the character table of $\mathfrak{S}_3$.

(e) Give a geometric interpretation of $\mathfrak{S}_3$ as a group of symmetries and a geometric interpretation of the representation ρ.

Exercise 2.3 *The symmetric group* $\mathfrak{S}_4$.

Find the conjugacy classes and character table of the symmetric group $\mathfrak{S}_4$.

Exercise 2.4 *The alternating group* $\mathfrak{A}_4$.

Find the character table of $\mathfrak{A}_4$. Which representations of $\mathfrak{A}_4$ are the restriction of a representation of $\mathfrak{S}_4$? Which representations of $\mathfrak{S}_4$ have an irreducible restriction to $\mathfrak{A}_4$? Which have a reducible restriction?

Exercise 2.5 *Tensor products of vector spaces and of representations.*

We denote the dual of a vector space E by E^*, and the duality pairing by $\langle\ ,\ \rangle$.

If E and F are finite-dimensional vector spaces over $\mathbb{K}$ (where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$), one can define the tensor product $E \otimes F$ as the vector space of bilinear maps from $E^* \times F^*$ to $\mathbb{K}$. For $x \in E$, $y \in F$, we define the element $x \otimes y \in E \otimes F$ by

$$(x \otimes y)(\xi, \eta) = \langle \xi, x\rangle\langle \eta, y\rangle,$$

for every $\xi \in E^*$, $\eta \in F^*$.

(a) Let $(e_1, \dots, e_n)$ be a basis of E and let $(f_1, \dots, f_p)$ be a basis of F. Show that $(e_i \otimes f_j)_{1\le i\le n, 1\le j\le p}$ is a basis of $E \otimes F$.

(b) An element of $E \otimes E$ is called a *contravariant tensor* (or simply a *tensor*) of order 2 on E. Every contravariant tensor of order 2 on E can be written $T = \sum_{i,j=1}^{n} T^{ij} e_i \otimes e_j$, where the T^{ij} are scalars, called the *components* of T in the basis (e_i). What are the components of T after a change of basis?

(c) We can associate to $\xi \otimes y \in E^* \otimes F$ the linear map u of E into F defined by $u(x) = \langle \xi, x\rangle y$, for $x \in E$. Show that this defines an isomorphism of $E^* \otimes F$ onto $\mathcal{L}(E, F)$, the vector space of linear maps of E into F.

(d) Show that if $u : E \to E$ and $v : F \to F$ are linear maps, there is a unique endomorphism $u \otimes v$ of $E \otimes F$ satisfying $(u \otimes v)(x \otimes y) = u(x) \otimes v(y)$ for each $x \in E$, $y \in F$. In $E \otimes F$, we choose the basis

$$(e_1 \otimes f_1, e_1 \otimes f_2, \dots, e_1 \otimes f_p, e_2 \otimes f_1, e_2 \otimes\ f_2, \dots, e_2 \otimes f_p, \dots, e_n \otimes f_1, \dots, e_n \otimes f_p).$$

Write the matrix of $u \otimes v$, where u (resp., v) is an endomorphism of E (resp., F) with matrix $A = (a_{ij})$ (resp., $B = (b_{ij})$) in the chosen bases.

(e) If (E_1, ρ_1) and (E_2, ρ_2) are representations of a group G, we set, for $g \in G$,

$$(\rho_1 \otimes \rho_2)(g) = \rho_1(g) \otimes \rho_2(g).$$

Show that this defines a representation $\rho_1 \otimes \rho_2$ of G on $E_1 \otimes E_2$. What can one say about the character of $\rho_1 \otimes \rho_2$? If ρ_1 and ρ_2 are irreducible, is the representation $\rho_1 \otimes \rho_2$ irreducible?

Exercise 2.6 *The dual representation.*

Let (E, π) be a representation of a group G. For $g \in G$, $\xi \in E^*$, $x \in E$, we set $\langle\ \pi^*(g)(\xi),\ x\ \rangle\ =\ \langle\ \xi,\ \pi(g^{-1})(x)\ \rangle$. (As in Exercise 2.5, E^* is the dual of E, and $\langle\ ,\ \rangle$ is the duality pairing.)

(a) Show that this defines a representation π^* of G on E^*. The representation π^* is called the *dual* (or *contragredient*) of π.

(b) Show that if (E, π) and (F, ρ) are representations of a group G, then $g.u = \rho(g) \circ u \circ \pi(g^{-1})$, for $u \in \mathcal{L}(E, F)$ and $g \in G$, defines a representation of G on $\mathcal{L}(E, F)$, equivalent to $\pi^* \otimes \rho$.

Exercise 2.7 *Exterior and symmetric powers.*

Let E be a finite-dimensional vector space, with basis $(e_1, \ldots, e_n)$. We denote by $\bigwedge^2 E$ (resp., S^2E) the vector subspace of $E \otimes E$ generated by $e_i \otimes e_j - e_j \otimes e_i$, $1 \leq i < j \leq n$ (resp., $e_i \otimes e_j + e_j \otimes e_i$, $1 \leq i \leq j \leq n$). These definitions are independent of the choice of basis and $E \otimes E = \bigwedge^2 E \oplus S^2E$. The space $\bigwedge^2 E$ is the *exterior* (or *antisymmetric*) *power* of degree 2 of E, and the space S^2E is the *symmetric power* of degree 2 of E.

(a) If (E, ρ) is a representation of a group G, $\bigwedge^2 E$ and S^2E are invariant under $\rho \otimes \rho$. We denote the restriction of $\rho \otimes \rho$ to $\bigwedge^2 E$ (resp., S^2E) by $\bigwedge^2 \rho$ (resp., $S^2\rho$). Suppose that G is finite. Show that the characters of these representations satisfy, for each $g \in G$,

$$\chi_{\wedge^2\rho}(g) = \frac{1}{2}\left((\chi_\rho(g))^2 - \chi_\rho(g^2)\right), \quad \chi_{S^2\rho}(g) = \frac{1}{2}\left((\chi_\rho(g))^2 + \chi_\rho(g^2)\right).$$

(b) If ρ is the two-dimensional irreducible representation of $\mathfrak{S}_3$, find $\chi_{\wedge^2\rho}$ and $\chi_{S^2\rho}$. Decompose $\rho \otimes \rho$ into a direct sum of irreducible representations.

Exercise 2.8 *Equivalence of the left and right regular representations.*

Show that the left and right regular representations of a finite group are equivalent.

Exercise 2.9 *Representations of Abelian and cyclic groups.*

(a) Show that every irreducible representation of a finite group is one-dimensional if and only if the group is Abelian.
(b) Find all the inequivalent irreducible representations of the cyclic group of order n.

Exercise 2.10 *An application of the orthogonality relations.*

Let ρ_i and ρ_j be irreducible representations of a finite group G. Let $\chi_i = \chi_{\rho_i}$ and $\chi_j = \chi_{\rho_j}$. Show that, for every $h \in G$,

$$\frac{1}{|G|} \cdot \sum_{g \in G} \chi_i(g)\chi_j(g^{-1}h) = \frac{1}{\dim \rho_i}\chi_i(h)\delta_{ij}.$$

Exercise 2.11 *Regular representation of* $\mathfrak{S}_3$.

Decompose the regular representation of $\mathfrak{S}_3$ into a direct sum of irreducible representations.

Find a basis of each one-dimensional invariant subspace and a projection onto the support of the representation 2ρ, where ρ is the irreducible representation of dimension 2.

Exercise 2.12 *Real and complexified representations.*

Let E be a vector space over $\mathbb{R}$, of dimension n. A morphism of a finite group G into $\mathrm{GL}(E)$ is called a *real representation* of G, of (real) dimension n.

We consider $E^{\mathbb{C}} = E \oplus iE = E \otimes \mathbb{C}$, a vector space over $\mathbb{C}$, of complex dimension n, called the *complexification* of E.

(a) Show that every real representation of G on E can be extended uniquely to a (complex) representation of G on $E^{\mathbb{C}}$. This representation is called the *complexification* of the real representation.
(b) Let the symmetric group $\mathfrak{S}_3$ act on $\mathbb{R}^2$ by rotation through angles of $\frac{2k\pi}{3}$ and reflection. Show that the complexification of this representation is equivalent to the irreducible representation of $\mathfrak{S}_3$ on $\mathbb{C}^2$.
(c) Let the cyclic group of order 3 act on $\mathbb{R}^2$ by rotations through angles of $\frac{2k\pi}{3}$. Is this real representation irreducible?
(d) Are all irreducible real representations of Abelian groups one-dimensional?

Exercise 2.13 *Representations of the dihedral group.*

(a) Show that if H is an Abelian subgroup of order p of a finite group G of order n, then every irreducible representation of G is of dimension $\leq \frac{n}{p}$.
(b) Conclude that, for every $n \geq 3$, every irreducible representation of the dihedral group $D_{(n)}$ is one- or two-dimensional.

Exercise 2.14 *Peter–Weyl theorem for finite groups.*

Let $\rho^1, \rho^2, \ldots, \rho^N$ be unitary representations of a finite group G, chosen from each equivalence class of irreducible representations.

Show that the matrix coefficients of the representations ρ^k, $k = 1, \ldots, N$, in orthonormal bases form an orthogonal basis of $L^2(G)$. Conclude that every function $f \in L^2(G)$ has a "Fourier series,"

$$f = \sum_{k=1}^{N} \sum_{i,j=1}^{\dim \rho_k} d_k (\rho_{ij}^k | f) \rho_{ij}^k,$$

where the d_k are integers.

Exercise 2.15 *Representation of* $\mathrm{GL}(2, \mathbb{C})$ *on the polynomials of degree* 2.

Let G be a group and let ρ be a representation of G on $V = \mathbb{C}^n$. Let $P^{(k)}(V)$ be the vector space of complex polynomials on V that are homogeneous of degree k.

(a) For $f \in P^{(k)}(V)$, we set $\rho^{(k)}(g)(f) = f \circ \rho(g^{-1})$. Show that this defines a representation $\rho^{(k)}$ of G on $P^{(k)}(V)$.
(b) Compare $\rho^{(1)}$ and the dual representation of ρ.
(c) Suppose that $G = \mathrm{GL}(2, \mathbb{C})$, $V = \mathbb{C}^2$ and ρ is the fundamental representation. Let $k = 2$. To the polynomial $f \in P^{(2)}(\mathbb{C}^2)$ defined by $f(x, y) = ax^2 + 2bxy + cy^2$, we associate the vector $v_f = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{C}^3$. Let $\tilde{\rho}$ denote the representation of $\mathrm{GL}(2, \mathbb{C})$ on $\mathbb{C}^3$ defined by $\rho^{(2)}$ and the isomorphism above. Find the dual of $\tilde{\rho}$. k

Exercise 2.16 *Convolution.*

Let G be a finite group and let $\mathbb{C}[G]$ be the *group algebra* of G, that is, the vector space $\mathcal{F}(G)$ with multiplication defined by $\epsilon_g \epsilon_{g'} = \epsilon_{gg'}$, for g and $g' \in G$, and extended by linearity.

(a) Show that the product of two functions $f_1, f_2 \in \mathbb{C}[G]$ is the *convolution product* $(f_1 * f_2)(g) = \sum_{h \in G} f_1(h) f_2(h^{-1}g)$.
(b) Let ρ be a representation of G and suppose $f \in \mathbb{C}[G]$. Set $\rho_f = \sum_{g \in G} f(g)\rho(g)$. Show that $\rho_{f_1 * f_2} = \rho_{f_1} \circ \rho_{f_2}$.
(c) Show that $f \in \mathbb{C}[G]$ is a class function if and only if f is in the center of the algebra $\mathbb{C}[G]$ equipped with convolution (that is, f commutes in the sense of convolution with every function on G).

Exercise 2.17 *On the map $f \mapsto \rho_f$.*

For every representation (E, ρ) of G, and each function f on G, consider the endomorphism ρ_f of E defined by

$$\rho_f = \sum_{g \in G} f(g)\rho(g).$$

(a) Let R be the regular representation of G. Consider $R_f(\epsilon_g)$, for $g \in G$. Show that $R_f(\epsilon_e) = f$. Is the map $f \in \mathbb{C}[G] \mapsto R_f \in \mathrm{End}(\mathbb{C}[G])$ injective?
(b) Let ρ_i and ρ_j be irreducible representations of G and let χ_i (resp., χ_j) be the character of ρ_i (resp., ρ_j). Find ρ_f for $\rho = \rho_j$ and $f = \overline{\chi_i}$.

Exercise 2.18 *Tensor products of representations of $\mathfrak{S}_3$.*

Let ρ be the irreducible representation of dimension 2 of the symmetric group $\mathfrak{S}_3$. We set $\rho = \rho^{\otimes 1}$ and by induction we define, for every integer $k \geq 2$,

$$\rho^{\otimes k} = \rho^{\otimes (k-1)} \otimes \rho.$$

(a) For each positive integer k, decompose $\rho^{\otimes k}$ into a direct sum of irreducible representations.
(b) Let $\mathfrak{A}_3 \subset \mathfrak{S}_3$ denote the alternating group. For each positive integer k, decompose the restriction of $\rho^{\otimes k}$ to $\mathfrak{A}_3$ into a direct sum of irreducible representations.



Issai Schur, born in 1875 in Moilev, Belorussia (then in the Russian Empire, today Belarus), professor in Bonn and later in Berlin, member of the Prussian Academy of Sciences, lost his university position in 1935. Forced to emigrate to Palestine, he died in Tel-Aviv in 1941. Schur is, with Frobenius, a founder of representation theory.

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Chapter 3

Representations of Compact Groups



In this chapter, we shall extend to compact topological groups many of the properties proved in the case of finite groups. Some properties will be stated without proof.

1 Compact Groups

Recall that a topological group is a group equipped with the structure of a Hausdorff topological space (for example, a subset of a normed vector space) such that multiplication and inversion are continuous maps. A topological space is locally compact if each point has a compact neighborhood. A topological group that is a compact (resp., locally compact) space is called a *compact group* (resp., *locally compact group*).

If E is a Banach space (*i.e.*, a complete normed vector space) over the real or complex field, we denote by $\mathcal{L}(E, E)$ the vector space of continuous linear maps of E into E, also called *endomorphisms* of E or *continuous linear operators* or *bounded operators* on E. We equip it with the standard norm on linear maps defined, for each continuous, linear $u : E \to E$, by

$$\|u\| = \sup_{\|x\| \leq 1} \|u(x)\|.$$

For each Banach space, E, we denote by $\mathrm{GL}(E) \subset \mathcal{L}(E, E)$ the group of isomorphisms of E, that is, bijective and bicontinuous endomorphisms of E. In order to show that the inverse of a continuous linear operator is continuous, it suffices to show that the operator is bijective. We consider $\mathrm{GL}(E)$ to be a topological subspace of the normed vector space $\mathcal{L}(E, E)$.

The unit ball in a normed vector space is compact if and only if the space is finite-dimensional. Thus each *closed and bounded* subset of $\mathrm{GL}(E)$, where E is a finite-dimensional vector space, is compact. For example $\mathrm{U}(n) \subset \mathrm{GL}(n, \mathbb{C})$ and $\mathrm{O}(n) \subset \mathrm{GL}(n, \mathbb{R})$ are compact. Similarly, $\mathrm{SU}(n)$ and $\mathrm{SO}(n)$ are compact.

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The Abelian group $\mathbb{R}$ with its usual metric is locally compact but not compact. The groups $\mathrm{GL}(n,\mathbb{C})$ and $\mathrm{SL}(n,\mathbb{C})$, $\mathrm{GL}(n,\mathbb{R})$ and $\mathrm{SL}(n,\mathbb{R})$ are locally compact but not compact.

2 Haar Measure

On a finite group G, we know that for each function $f \in \mathcal{F}(G)$ and for every $g \in G$,

$$\sum_{h\in G} f(h) = \sum_{h\in G} f(gh) = \sum_{h\in G} f(hg).$$

If we denote by l_g (resp., r_g) the left (resp., right) multiplication by $g \in G$, we have, by definition, $f(gh) = (f \circ l_g)(h)$ and $f(hg) = (f \circ r_g)(h)$. Consequently, the averaging operation,

$$M \ : f \mapsto M(f) = \frac{1}{|G|} \sum_{g\in G} f(g),$$

satisfies

- M is a positive linear form on $\mathcal{F}(G)$, that is, M takes non-negative values on real non-negative functions,
- M is both left- and right-invariant, that is,

$$\forall g \in G,\ M(f \circ l_g) = M(f \circ r_g) = M(f),$$

- $M(1) = 1$.

On any compact group there is a measure, the *Haar measure*, which has analogous properties. More generally, on a locally compact group, there are measures which are either left-invariant or right-invariant but, in general, not both.

Theorem 2.1. *Let G be a locally compact group.*

(*i*) *There exists a positive measure, μ, on G, finite on compact sets and not identically zero, which is left-invariant, i.e., such that, for each integrable function f and for every $h \in G$,*

$$\int_G f(hg)\mathrm{d}\mu(g) = \int_G f(g)\mathrm{d}\mu(g).$$

Such a measure is unique up to multiplication by a positive real number. If f is a continuous, non-negative function, then $\int_G f(g)\mathrm{d}\mu(g) = 0$ implies $f = 0$.

(*ii*) *If G is compact, there is a unique left-invariant measure μ such that $\int_G \mathrm{d}\mu(g) = 1$.*

(*iii*) *On a compact group, every left-invariant measure is right-invariant.*

Proof. (i) The proof of this result is beyond the scope of this book. See, *e.g.*, Loomis (1953).

(ii) If μ_0 is a left-invariant measure on a compact group G and if $\int_G \mathrm{d}\mu_0(g) = m$, we set $\mu = \frac{1}{m}\mu_0$ and μ is clearly the unique left-invariant measure such that $\int_G \mathrm{d}\mu(g) = 1$.

(iii) Let μ be a left-invariant measure on a locally compact group G. For any continuous f with compact support, we set $\mu(f) = \int_G f(g)\mathrm{d}\mu(g)$. Let $h \in G$ and consider $\mu_h(f) = \int_G f(gh)\mathrm{d}\mu(g)$, that is, $\mu_h(f) = \mu(f \circ r_h)$. Then,

$$\forall k \in G, \mu_h(f \circ l_k) = \int_G f(kgh)\mathrm{d}\mu(g) = \int_G f(gh)\mathrm{d}\mu(g) = \mu_h(f),$$

and hence, by the uniqueness of left-invariant measures up to a factor, there is a positive real number depending on h, $\Delta(h)$, satisfying

$$\mu_h(f) = \Delta(h)\mu(f).$$

If G is compact, the constant function 1 is integrable. We obtain $\mu_h(1) = \mu(1) = \Delta(h)\mu(1)$. Hence $\Delta = 1$ and μ is thus also right-invariant, *i.e.*,

$$\int_G f(gh)\mathrm{d}\mu(g) = \int_G f(g)\mathrm{d}\mu(g),$$

for every $h \in G$. □

Definition 2.2. *On a compact group, the unique left- and right-invariant measure of total mass* 1 *is called the* Haar measure.

On a locally compact group G, the function $\Delta : h \in G \mapsto \Delta(h) \in \mathbb{R}$ is called the *modular function* of G. It satisfies

$$\Delta(hh') = \Delta(h)\Delta(h'),$$

since $\Delta(hh')\mu(f) = \mu_{hh'}(f) = \mu(f \circ r_{hh'}) = \mu(f \circ r_{h'} \circ r_h) = \Delta(h)\mu(f \circ r_{h'}) = \Delta(h)\Delta(h')\mu(f)$. We say that a locally compact group G is *unimodular* if $\Delta = 1$. The preceding theorem says that if G is compact, then G is unimodular.

One often writes $\int_G f(g)\mathrm{d}g$ instead of $\int_G f(g)\mathrm{d}\mu(g)$. Thus, if G is compact, then for each measurable function f,

$$\forall h \in G, \int_G f(g)\mathrm{d}g = \int_G f(hg)\mathrm{d}g = \int_G f(gh)\mathrm{d}g,$$

and we require that μ satisfy the normalization condition $\int_G \mathrm{d}g = 1$.

Examples.

- On an Abelian group, each left-invariant measure is obviously also right-invariant.

- In the case of the locally compact Abelian group $\mathbb{R}$, every invariant measure is proportional to the Lebesgue measure.
- If $G = \mathrm{U}(1) = S^1 = \{e^{i\theta} | \theta \in \mathbb{R}/2\pi\mathbb{Z}\} = \{z \in \mathbb{C} \mid |z| = 1\}$, then

$$\mathrm{d}g = \frac{\mathrm{d}\theta}{2\pi} \quad \text{or} \quad \mathrm{d}g = \frac{\mathrm{d}z}{2i\pi z}.$$

- For the groups SU(2) and SO(3), see Exercises 5.4, 5.5 and 5.6 of Chapter 5.

3 Representations of Topological Groups and Schur's Lemma

All the Hilbert spaces we consider are over the field of *complex numbers*, and we assume each Hilbert space to be *separable*, that is, we assume it has a countable Hilbert space basis.

3.1 General Facts

Definition 3.1. *Let G be a topological group. A* continuous representation *(or simply a* representation*) of G is defined to be a Hilbert space E and a morphism of groups $\rho : G \to \mathrm{GL}(E)$ such that, for every $x \in E$,*

$$g \in G \mapsto \rho(g)x \in E$$

is a continuous mapping.

A sufficient condition for the continuity of the map $g \mapsto \rho(g)x$, for every $x \in E$, is that

$$\lim_{g \to e} \|\rho(g) - \mathrm{Id}_E\| = 0,$$

that is, ρ should be continuous as a map of G into $\mathrm{GL}(E)$ with the topology induced by the norm of $\mathcal{L}(E, E)$. If E is finite-dimensional, this sufficient condition is also necessary.

The dimension, finite or infinite, of E is called the *dimension* of ρ.

The *trivial representation* of G on a vector space E is defined by $\rho(g) = \mathrm{Id}_E$, for every $g \in G$.

Definition 3.2. *Let E be a complex Hilbert space. If $u \in \mathcal{L}(E, E)$, the* adjoint *u^* of u is defined by*

$$\forall x, y \in E, (ux \mid y) = (x \mid u^* y),$$

and an element $u \in \mathrm{GL}(E)$ is a unitary operator *if $uu^* = u^*u = \mathrm{Id}_E$. The group of unitary operators of E is denoted by $\mathrm{U}(E)$.*

In finite dimensions and in an orthonormal basis, a unitary operator is represented by a unitary matrix.

A representation ρ of G on E is called *unitary* if E is a complex Hilbert space and if, for every $g \in G$, $\rho(g)$ is a unitary operator. Then, for every $g \in G$, $x, y \in E$, $(\rho(g)x \mid \rho(g)y) = (x \mid y)$ and, in particular, $\|\rho(g)x\| = \|x\|$.

Remark. In the same way, one can define representations on real Hilbert spaces. In this case one speaks of *orthogonal representations.*

3.2 Coefficients of a Representation

Let E^* be the dual of E. For $x \in E$ and $\xi \in E^*$, we set

$$\rho_{x\xi}(g) = \langle \xi, \rho(g)x \rangle,$$

where $\langle\ ,\ \rangle$ denotes the duality pairing. In finite dimensions, given a basis (e_i) of E, and the dual basis, (e_i^*), in E^*, we recover the matrix coefficients ρ_{ij} defined in Chapter 2, because

$$\rho_{e_j e_i^*}(g) = \langle e_i^*, \rho(g)e_j \rangle = \rho_{ij}(g).$$

One can also consider, for $x, y \in E$,

$$\varphi_{xy}^{\rho}(g) = (x \mid \rho(g)y).$$

If ρ is unitary,

$$\varphi_{xy}^{\rho}(g^{-1}) = \overline{\varphi_{yx}^{\rho}(g)}.$$

In finite dimensions, for each basis (e_i) of E, we consider the coefficients

$$\varphi_{ij}^{\rho}(g) = (e_i \mid \rho(g)e_j).$$

If the basis (e_i) is orthonormal, then

$$\varphi_{ij}^{\rho} = \rho_{ij}.$$

3.3 Intertwining Operators

Definition 3.3. *Let (E_1, ρ_1) and (E_2, ρ_2) be unitary representations of G. An* intertwining operator *for ρ_1 and ρ_2 is defined to be any* continuous *linear mapping $T : E_1 \to E_2$ such that, for every $g \in G$, $\rho_2(g) \circ T = T \circ \rho_1(g)$.*

In particular, if (E, ρ) is a unitary representation of G, a continuous linear mapping, $T : E \to E$, intertwines ρ with itself if and only if it commutes with ρ.

Two representations (E_1, ρ_1) and (E_2, ρ_2) are *equivalent* if there is a bijective (and thus bicontinuous) intertwining operator T for ρ_1 and ρ_2. Then E_1 is isomorphic to E_2 and $\rho_2(g) = T \circ \rho_1(g) \circ T^{-1}$. Thus one defines the *equivalence class* of a representation.

Definition 3.4. *A representation (E, ρ) of G is called* irreducible *if $E \neq \{0\}$ and if no* closed *non-trivial vector subspace of E is invariant under ρ.*

For a closed vector subspace F invariant under ρ, the restriction of ρ to F is called a *sub-representation* of ρ.

Proposition 3.5. *If (E, ρ) is a unitary representation, and if $F \subset E$ is a closed vector subspace invariant under ρ, then the orthogonal complement $F^\perp$ of F is also a closed vector subspace invariant under ρ, and the orthogonal projections P_F on F and $P_{F^\perp}$ on $F^\perp$ commute with ρ.*

Proof. We know that for every vector subspace F of E, the orthogonal complement, $F^\perp$, of F is closed. By hypothesis, for every $x \in F, \rho(g)x \in F$. Let $y \in F^\perp$. Then for every $x \in F$, $(x \mid y) = 0$. Hence for every $x \in F$, $(x \mid \rho(g)y) = (\rho(g)^{-1}x \mid y) = 0$, and so $\rho(g)y \in F^\perp$. Furthermore,

$$\rho(g)y = \rho(g)(P_F y + P_{F^\perp} y) = \rho(g)P_F y + \rho(g)P_{F^\perp} y.$$

The first term belongs to F, by invariance of F. Similarly, the second term belongs to $F^\perp$. Therefore the first term is $P_F(\rho(g)y)$ and the second is $P_{F^\perp}(\rho(g)y)$. □

3.4 Operations on Representations

One can take the *Hilbert direct sum*, $(\hat{\oplus}E_i, \hat{\oplus}\rho_i)$, of a countable family of unitary representations (E_i, ρ_i) of a topological group G. In the vector space of the representation, each E_i is a closed subspace, the E_i are pairwise orthogonal and the direct sum is dense.

A representation is called *completely reducible* (or *semi-simple*) if it is the Hilbert direct sum of irreducible representations.

If ρ is an irreducible representation, we use the notation $m\rho$ to denote the *isotypic component* of type ρ of a completely reducible representation, where m can be an integer or ∞.

We define the tensor product $(E_1 \otimes E_2, \rho_1 \otimes \rho_2)$ of two representations (E_1, ρ_1) and (E_2, ρ_2) of G by setting, for $g \in G$, $x_1 \in E_1$ and $x_2 \in E_2$,

$$((\rho_1 \otimes \rho_2)(g))\,(x_1 \otimes x_2) = \rho_1(g)x_1 \otimes \rho_2(g)x_2.$$

If ρ_1 and ρ_2 are infinite-dimensional representations, we consider $E_1 \hat{\otimes} E_2$, the tensor product completed under the norm associated to the scalar product, which can be shown to be positive,

$$(x_1 \otimes x_2 \mid x_1' \otimes x_2') = (x_1 \mid x_1')(x_2 \mid x_2').$$

The Hilbert space $E_1 \hat{\otimes} E_2$ is called the *Hilbert tensor product* of E_1 and E_2.

Theorem 3.6. *Every* finite-dimensional *unitary representation of a topological group is completely reducible.*

The proof is the same as in the case of finite groups using Proposition 3.5.

There exist (non-unitary) representations that are reducible but not completely reducible.

3.5 Schur's Lemma

Theorem 3.7 (Schur's Lemma). *Let G be a topological group and let (E_1, ρ_1) and (E_2, ρ_2) be irreducible unitary representations of G. Let T be a continuous linear mapping of E_1 into E_2 that intertwines ρ_1 and ρ_2. Then either $T = 0$, or T is an isomorphism (and consequently ρ_1 and ρ_2 are equivalent) and T is unique up to a multiplicative constant.*

In summary, if $\rho_2(g) \circ T = T \circ \rho_1(g)$, for every $g \in G$, either $\rho_1 \nsim \rho_2$ and $T = 0$, or $\rho_1 \sim \rho_2$ and T is an isomorphism. The proof, which we omit, uses the spectral decomposition of Hermitian operators. See, *e.g.*, Gurarie (1992).

Corollary 3.8. *Let G be a topological group and let (E, ρ) be a unitary representation of G. The representation ρ is irreducible if and only if every endomorphism of E that commutes with ρ is a scalar multiple of the identity.*

Proof. If ρ is not irreducible, the projection onto a non-trivial closed invariant subspace is a non-scalar endomorphism of E that commutes with ρ. The converse is a consequence of Schur's lemma. □

Corollary 3.9. *The unitary irreducible representations of an Abelian group are one-dimensional.*

Proof. Let (E, ρ) be a unitary irreducible representation of an Abelian group G. Let g be an element of G. Then, for every $h \in G$, $\rho(g)\rho(h) = \rho(h)\rho(g)$, and thus $\rho(g)$ commutes with ρ. By Corollary 3.8 of Schur's lemma, G acts by scalar multiplication. Because the representation ρ is assumed irreducible, it must be one-dimensional. □

4 Representations of Compact Groups

In this section, we consider a topological group G which we assume to be *compact* and, as above, we designate by "representation" a continuous representation on a separable complex Hilbert space.

4.1 Complete Reducibility

Theorem 1.4 of Chapter 2 is generalized as follows.

Theorem 4.1. *Every representation of a compact group is unitarizable.*

Sketch of Proof. Let G be a compact group, and let (E, ρ) be a representation of G. We set, for $x, y \in E$,

$$(x \mid y)' = \int_G (\rho(g)x \mid \rho(g)y)\mathrm{d}g,$$

where $\mathrm{d}g$ is the Haar measure on G. This is indeed a scalar product because if $(x|x)' = 0$, then, by Theorem 2.1, $(\rho(g)x \mid \rho(g)x) = 0$ for every $g \in G$, and consequently $x = 0$. On the other hand,

$$(\rho(g)x \mid \rho(g)y)' = \int_G (\rho(hg)x \mid \rho(hg)y)\mathrm{d}h = \int_G (\rho(h)x \mid \rho(h)y)\mathrm{d}h = (x \mid y)'.$$

Thus $\rho(g)$ is unitary for $(\ |\ )'$. Next one shows that the norms associated to $(\ |\ )$ and $(\ |\ )'$ are equivalent, which implies that the representation remains continuous and that the representations ρ on the Hilbert spaces $(E, (\ |\ ))$ and $(E, (\ |\ )')$ are equivalent, which means that the identity mapping of E is bicontinuous and intertwines the representations. □

Combining this theorem with Theorem 3.6, we obtain the following corollary which generalizes Maschke's theorem (Theorem 1.9 of Chapter 2).

Corollary 4.2. *Every finite-dimensional representation of a compact group is completely reducible.*

A proof of the following result, which generalizes Proposition 1.6 of Chapter 2, may be found in, *e.g.*, Naĭmark and Štern (1982).

Theorem 4.3. *Every irreducible representation of a compact group is finite-dimensional.*

Remark. This theorem, as specified above, assumes representations to be continuous and representation spaces to be separable complex Hilbert spaces. The theorem is not true in all generality, but it is true for continuous representations taking values in certain topological vector spaces more general than Hilbert spaces.

4.2 Orthogonality Relations

We define a scalar product on the vector space of continuous complex-valued functions on G by

$$(f_1 \mid f_2) = \int_G \overline{f_1(g)} f_2(g) \mathrm{d}g,$$

where $\mathrm{d}g$ is the Haar measure. We denote by $L^2(G)$ the Hilbert space obtained by completing this pre-Hilbert space under the norm defined by the scalar product. This is the Hilbert space of *square-integrable functions* on G. (Strictly speaking this vector space consists of equivalence classes of square-integrable functions under the equivalence relation of equality almost everywhere.)

We know that the irreducible representations of G are finite-dimensional. The orthogonality relations of characters of irreducible representations of finite groups extend to the compact case.

Theorem 4.4. *Let G be a compact group and let (E_1, ρ_1) and (E_2, ρ_2) be irreducible unitary representations of G. For every $x_1, y_1 \in E_1$ and $x_2, y_2 \in E_2$,*

$$\left(\varphi^{\rho_1}_{x_1 y_1} \mid \varphi^{\rho_2}_{x_2 y_2}\right) = \begin{cases} 0 & \text{if } \rho_1 \nsim \rho_2, \\ \frac{1}{\dim E}(x_2 \mid x_1)(y_1 \mid y_2) & \text{if } E_1 = E_2 = E \text{ and } \rho_1 = \rho_2 = \rho. \end{cases}$$

Proof. Generalizing the process used in the proof of Proposition 2.9 of Chapter 2, we define, for each continuous linear mapping $u : E_1 \to E_2$, the following operator intertwining ρ_1 and ρ_2,

$$T_u = \int_G \rho_2(g)\, u\, \rho_1(g^{-1})\, \mathrm{d}g.$$

For $y_1 \in E_1$, $y_2 \in E_2$, we consider the linear map $u_{y_1 y_2} : E_1 \to E_2$ defined by $u_{y_1 y_2}(x) = (y_1 \mid x)y_2$, for x in E_1. Using the fact that ρ_1 is unitary, we then obtain the equation $(\varphi^{\rho_1}_{x_1 y_1} \mid \varphi^{\rho_2}_{x_2 y_2}) = (x_2 \mid T_{u_{y_1 y_2}} x_1)$.

Then we apply Schur's lemma. The quantity above is zero if ρ_1 is not equivalent to ρ_2. If $E_1 = E_2 = E$ and $\rho_1 = \rho_2 = \rho$, then $T_{u_{y_1 y_2}} = \tau(y_1, y_2)\mathrm{Id}_E$, where $\tau(y_1, y_2)$ is antilinear in y_1 and linear in y_2. We calculate $\tau(y_1, y_2)$ by calculating the trace of $T_{u_{y_1 y_2}}$. This is equal to the trace of $u_{y_1 y_2}$ because, for each linear mapping u, $\operatorname{Tr} T_u = \int_G \operatorname{Tr}(\rho(g) \circ u \circ \rho(g^{-1}))\mathrm{d}g = \int_G \operatorname{Tr} u \, \mathrm{d}g = \operatorname{Tr} u$. Because $\operatorname{Tr}(u_{y_1 y_2}) = (y_1 \mid y_2)$, we obtain the desired result. □

In particular, if ρ_1 and ρ_2 are not equivalent, in every orthonormal basis,

$$\left(\varphi^{\rho_1}_{ij} \mid \varphi^{\rho_2}_{k\ell}\right) = 0 \tag{4.1}$$

and, if $\rho_1 = \rho_2 = \rho$, then

$$\left(\varphi^{\rho}_{ij} \mid \varphi^{\rho}_{k\ell}\right) = \frac{1}{\dim E}\delta_{ik}\delta_{j\ell}. \tag{4.2}$$

We denote by $\widehat{G}$ the set of equivalence classes of irreducible representations of a compact group G. When $L^2(G)$ is separable—which is usually true in the cases one meets in practice—the orthogonality relations above imply that $\widehat{G}$ is countable.

By (4.1) and (4.2) the matrix coefficients in orthonormal bases of inequivalent irreducible unitary representations of G form an orthogonal set in $L^2(G)$. We now state without proof the Peter–Weyl theorem, which states that these matrix coefficients form an orthogonal basis of $L^2(G)$ in the Hilbert space sense. For a proof, see, *e.g.*, Simon (1996), Knapp (2002), Bump (2013), Hall (2016).

Theorem 4.5 (Peter–Weyl Theorem for Compact Groups). *Every function $f \in L^2(G)$ has a "Fourier series," converging in the L^2 sense,*

$$f = \sum_{\alpha \in \widehat{G}} \sum_{i,j=1}^{\dim \rho^\alpha} c^\alpha_{ij} \rho^\alpha_{ij}, \tag{4.3}$$

where the ρ^α are unitary representatives of the classes of inequivalent irreducible representations of G, the ρ^α_{ij} are their matrix coefficients in orthonormal bases, and

$$c^\alpha_{ij} = (\dim \rho^\alpha)(\rho^\alpha_{ij} \mid f) = (\dim \rho^\alpha) \int_G f(g)\overline{\rho^\alpha_{ij}(g)}\mathrm{d}g. \tag{4.4}$$

Example. The irreducible representations of the Abelian compact group $G = \mathbb{R}/2\pi\mathbb{Z}$ are the exponentials, $e_k : x \mapsto e^{ikx}$, for $k \in \mathbb{Z}$. Thus in this case $\widehat{G} = \mathbb{Z}$.

We thus see that the expansion of functions of $L^2(G)$ as series, where G is a compact group, given by the Peter–Weyl theorem is a generalization of the

Fourier series expansion of 2π-periodic functions square-integrable on $[0, 2\pi]$: if $G = \mathbb{R}/2\pi\mathbb{Z}$, then for every $f \in L^2(G)$,

$$f = \sum_{k\in\mathbb{Z}} c_k e_k, \tag{4.5}$$

where

$$c_k = (e_k \mid f) = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-ik\theta}\mathrm{d}\theta. \tag{4.6}$$

In this case, each representation is one-dimensional and its unique matrix coefficient is the character of the representation. Formulas (4.3) and (4.4) are thus seen to be generalizations of the classic formulas (4.5) and (4.6) of Fourier analysis.

If (E, ρ) is a finite-dimensional representation of G, we define the *character* χ_ρ of ρ by

$$\forall g \in G,\ \chi_\rho(g) = \mathrm{Tr}(\rho(g)).$$

The complex-valued function thus defined on G is continuous. It is a class function which depends only on the equivalence class of the representation ρ.

Theorem 4.6 (Orthogonality Relations). *Let ρ_1 and ρ_2 be irreducible representations of G. Then*

$$(\chi_{\rho_1} \mid \chi_{\rho_2}) = \begin{cases} 0 & \text{if } \rho_1 \nsim \rho_2, \\ 1 & \text{if } \rho_1 \sim \rho_2. \end{cases}$$

Proof. Applying Theorem 4.1, we see that these relations are a consequence of the preceding formulas (4.1) and (4.2). □

A representation ρ is irreducible if and only if $(\chi_\rho \mid \chi_\rho) = 1$.

If ρ is a representation of G, we can decompose it as a Hilbert sum of irreducible representations, ρ_i, where $i \in \widehat{G}$. We shall write

$$\rho = \widehat{\bigoplus_{i\in\widehat{G}}} m_i\rho_i,$$

where

$$m_i = (\chi_{\rho_i} \mid \chi_\rho).$$

We may have $m_i = \infty$.

Theorem 4.1 of Chapter 2 concerning the projection operators can be generalized as follows. A projection on the isotypic component $m_i\rho_i$ is

$$P_i = \dim\rho_i \int_G \overline{\chi_i(g)}\rho(g)\mathrm{d}g.$$

In general, many results from the representation theory of finite groups extend to the case of compact groups, replacing averaging over the group with integration with respect to the Haar measure.

We shall determine the irreducible representations of the compact groups SU(2) and SO(3) in Chapter 6.

Below the reader will find a chart summarizing the principal properties of the representations of compact groups, compared to those of finite groups.

5 Summary of Chapter 3

We summarize the properties of finite groups that are true also for *compact* groups. The representations are assumed continuous on separable *complex* Hilbert spaces.

Representations of Finite G	**Representations of Compact G**
Every representation is unitarizable	*true*
Every finite-dimensional representation is completely reducible	*true*
Every irreducible representation is finite-dimensional	*true*
There is a finite number of equivalence classes of irreducible representations = number of conjugacy classes of the group = dimension of the vector space of class functions	*false*
The irreducible characters form an orthonormal basis of the vector space of class functions	*true* *as a Hilbert space basis*
The matrix coefficients in orthonormal bases of unitary irreducible representations form an orthogonal basis of the vector space $L^2(G)$	*true* *as an orthogonal set*
Every representation has a decomposition into a direct sum $\rho = \bigoplus_{i=1}^{N} m_i\rho_i$ $m_i = (\chi_i \mid \chi_\rho)$	*into a direct Hilbert sum* $\rho = \hat{\oplus}_{i\in\widehat{G}} m_i\rho_i$ $m_i = (\chi_i \mid \chi_\rho)$
Decomposition of the regular representation $\sum_{i=1}^{N}(\dim\rho_i)^2 = \lvert G\rvert$	*false*
Each irreducible representation is contained in the regular	*true*

representation a number of times equal to its dimension

Projection onto an isotypic component $P_i = \frac{\dim \rho_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g)$ $P_i = \dim \rho_i \int_G \overline{\chi_i(g)} \rho(g) \mathrm{d}g$

References

One will find a study of representations of compact groups in Barut and Raczka (1986), Naĭmark and Štern (1982), Sternberg (2003), Simon (1996), Bump (2013) and many other textbooks, such as Berndt (2007).

For a proof of the bicontinuity of continuous bijective operators, as well as many other results in analysis, see Rudin (1991) or Brezis (2011).

The existence of a left-invariant measure on a locally compact group is proved in the book by Weil (1979), in Loomis (1953), and in Hewitt–Ross (1979).

See Gurarie (1992) for a thorough introductory text, including a proof of Schur's lemma, and Naĭmark and Štern (1982) for a comprehensive study of irreducible representations on Hilbert spaces, including a proof of Theorem 4.3. For a generalization to locally convex topological vector spaces, see A. A. Kirillov (1976).

The Peter-Weyl theorem is proved in most of the books mentioned.

Topics in History

Hilbert

The name of Hilbert is ubiquitous in mathematics as well as in theoretical physics, and it has lent mathematics the adjective, "hilbertian". David Hilbert was born in 1862 in Königsberg—now, Kaliningrad—, where he studied at the university and where he defended his thesis when he was 22, "On invariant properties of special binary forms, in particular spherical functions", and spent the first part of his career. It was Felix Klein who succeeded in securing his appointment to a chair of mathematics in 1895 in Göttingen, a larger and more renowned university. There he remained for the rest of his career. He had sixty-nine doctoral students, among whom was Hermann Weyl (1885–1955)[1]. Three volumes of his collected works were published in his lifetime, from 1932 to 1935. He died in 1943 after immense contributions to mathematics and physics.

The concept of a Hilbert space arose in his work on integral equations which he presented in a forty-page paper, "Foundations of a general theory of linear integral equations", published in 1904 in the *Göttinger Nachrichten*,

[1] See more on Weyl below, especially on p. 113.

and later, in 1912, in a book with nearly the same title, re-issued in 1924. It was also present in his lecture, "Essence and goals of an analysis of infinitely many independent variables", delivered at the IVth International Congress of Mathematicians held in Rome in 1908. First held in Zurich in 1897, then in 1900 in Paris, this international congress was scheduled to take place every four years. It did throughout the twentieth century, with notable exceptions due to the wars, 1916, 1940, 1944. It resumed only in 1950 and has been held regularly since then, except for 1982 when it was scheduled to take place in Warsaw and had to be postponed for a year as a protest against the imposition of martial law in Poland that followed anti-government demonstrations in the country. It had been during the 1900 congress that Hilbert addressed the participants with a lecture listing twenty-three unsolved problems. Some of "Hilbert's problems" were solved soon thereafter but some have remained unsolved to this day.

The expression "Hilbert space" was coined by John von Neumann (1903–1957) in 1927 when he was establishing quantum mechanics on a firm mathematical basis in a 50-page article published in the *Göttingen Nachrichten*, and this term soon became current in the mathematics and mathematical physics literature.

Hilbert, Einstein and the equations of General Relativity

In early 1915, as Einstein was working on a preliminary version of his general theory of relativity, Hilbert was working with enthusiasm in this new field, seeking to formulate the laws of physics from well-chosen axioms. He corresponded with Einstein and invited him to come and deliver lectures in Göttingen. Einstein traveled from Berlin for a week's stay in early July 2015, lectured and had lively scientific exchanges with Hilbert, and returned to Berlin, immensely pleased with the visit which he described in a letter[2] to a friend written immediately upon his return, pointing out that Hilbert and his colleagues knew and understood his work and were "completely convinced". He added that scientific life in Göttingen was much superior to that of Berlin! As both Einstein and Hilbert kept working, Hilbert came very near being credited with the discovery of the equations of general relativity. He actually published the equations before Einstein's article with those same equations appeared. Hilbert's article, "The foundations of physics, Part I", was dated 20 Novenber 1915, while Einstein's publication, "The field equations of gravitation", carried the date 25 November 1915. So, has Hilbert priority for "Einstein's equations of general relativity"? In fact, no, because it was found in 1997 by historians of science who examined the printer's proofs of Hilbert's article—the final draft that the author was able to correct before the article went to press—that Hilbert had added the equations to his article on the proofs just before he returned the article to the printer but not before he had received a letter from Einstein giving him the latest news, relating his breakthrough and stating the correct form of the equations that he had very recently found!

[2]Einstein's correspondence has been edited and commented in *The Collected Papers of Albert Einstein*, published by the Princeton University Press.

Whence "Hermitian"?

There was indeed a mathematician, Charles Hermite (1822–1901), who was born in eastern France and died in Paris, who not only lent his name to "Hermitian matrices" and other "Hermitian" and "antihermitian" concepts but was an active mathematician. Elected a member of the Académie des Sciences in 1856, he was named in 1869 a professor of analysis both at the École Polytechnique and the Sorbonne, the seat of the Faculté des Sciences of the university of Paris. An important mathematician who worked in many areas of analysis, elliptic functions and Abelian integrals, among many other topics, he exchanged scientific letters with many of his well-known contemporaries, including C. G. J. Jacobi (1804–1851).

We find the name of Hermite three times and the adjective "Hermitesche" many times in Weyl's *Gruppentheorie and Quantenmechanik* of 1928, as well as in the second edition of 1931. This important book was translated into English in 1931 as *The Theory of Groups and Quantum Mechanics*. In the title "Unitary geometry and Hermitian forms" of paragraph 4 of the first chapter, we read "[this bilinear form] bears the name of the French mathematician Hermite", and we find many occurrences of the adjective "Hermitian". As to "Hilbert spaces", their definition is also in Weyl's book. In German, von Neumann had written about Hermitian operators and Hilbert spaces in his article entitled "General theory of the eigenvalues of Hermitian functional operators" that was published in the *Mathematische Annalen* in 1929, and these terms soon appeared again in his joint work with Wigner.

The "Hermite polynomials" have applications in numerical analyis, combinatorics, signal processing, and many other fields.

The Haar measure

It was von Neumann who proved the uniqueness of the Haar measure in 1934. The proof of the existence of a left-invariant measure on a locally compact topological group had first been given in 1932 in a communication to the Hungarian Academy by Alfréd Haar (1885–1933), following work of Adolf Hurwitz (1859–1919) at the end of the nineteenth century. The complete works of Haar that were published in Budapest in 1959 included the original article of 1932 in Hungarian followed by a translation into German by the author which had appeared in the *Annals of Mathematics* of 1933, "Der Massbegriff in der Theorie der kontinuierlichen Gruppen" (The notion of mass in the theory of continuous groups).

Hausdorff, philosophy, poetry, then topology

Felix Hausdorff (1868–1942) obtained his habilitation at the university of Leipzig in 1895 with a doctoral thesis on astronomy but his later work was mainly in topology and set theory, and he was also between 1897 and 1904—under the pseudonym Paul Mongré—the author of philosophical essays, poems and a play, "Der Arzt seiner Ehre" (The Physician of his honesty), first published in

1904, that was a great success, performed more than 300 times in more than 30 towns.

Hausdorff's work on the "Baker–Campbell–Hausdorff" formula was published in Leipzig in 1906. There had been preliminary results which had been published in the *Proceedings of the London Mathematical Society*, in 1897 by John Campbell (1862–1924), who was educated and taught in Oxford, and then, in three articles from 1902 to 1905, by Henry Baker (1866–1956), a Cambridge professor whose work was mainly in algebraic geometry and analysis.

Hausdorff's book *Mengenlehre* is a classic. First published in Berlin in 1914 as *Grundzüge der Mengenlehre* (Basic features of set theory) (reprint Chelsea, New York, 1949, 1965 and 1978), it had a second, revised and very different edition in 1927, and in 1935, a third edition, *Mengenlehre*, which was translated into English as *Set theory* in 1957, republished by Chelsea in 1962, 1978, 1991, by the American Mathematical Society in 2005, and many other editions: a veritable maze of editions and a bestseller. Hausdorff presented an axiomatic approach to topology, beginning with abstract set theory, then topological spaces and continuous mappings, properties of continuous functions of a real variable: a course suitable for college seniors or beginning graduate students, but hard to follow!

He had been appointed associate professor at the University of Bonn in 1910, had been a full professor since 1913, and was again teaching in Bonn when the Nazi regime came to power. As a Jew, although he was not observant, he was dismissed from his position and was forbidden to publish in Germany. A tragic ending: he committed suicide with his wife and her sister in January 1942, the night he was informed that they were to be deported to a concentration camp the next day.

His collected works, mathematical as well as literary and philosophical, comprising 9 volumes and a volume of biography, were published between 2001 and 2021.

Exercises

Exercise 3.1 *Continuity condition for a representation*

Let E be a Hilbert space and let ρ be a morphism of the locally compact group G into $\mathrm{GL}(E)$, the group of bijective continuous linear operators from E to E. (One knows that such an operator is bicontinuous.) We consider the following conditions:

- (*i*) ρ is a continuous mapping of G into $\mathrm{GL}(E)$ with the usual norm.
- (*ii*) $\lim_{g\to e} \|\rho(g) - \mathrm{Id}_E\| = 0$.
- (*iii*) The mapping $(g, x) \in G \times E \mapsto \rho(g)(x) \in E$ is continuous.
- (*iv*) For each $x \in E$, the mapping $g \in G \mapsto \rho(g)(x) \in E$ is continuous. (In this case ρ is called a *representation* of G on E.)

(a) Show that conditions (*i*) and (*ii*) are equivalent.
(b) Show that (*i*) implies (*iii*).
(c) Show that the conditions (*iii*) and (*iv*) are equivalent. [We know, by the Banach-Steinhaus theorem (uniform boundedness theorem), that if K is a compact subset of G and if $\sup_{g\in K}\|\rho(g)(x)\| < \infty$, for every $x \in E$, then $\sup_{g\in K}\|\rho(g)\| < \infty$.]
(d) Show that, if E is finite-dimensional, conditions (*i*)–(*iv*) are equivalent.

Exercise 3.2 *Dual of a unitary representation.*

Show that the dual representation of a unitary representation of a compact group is equivalent to the conjugate representation.

Exercise 3.3 *Regular representation of a compact group.*

Let G be a compact group, and let $E = L^2(G)$ be the Hilbert space of square-integrable functions on G with respect to the Haar measure. We set, for every $f \in L^2(G)$, $g, h \in G$,

$$(R(g)(f))(h) = f(g^{-1}h).$$

Show that R is a unitary representation of G on $L^2(G)$.

Exercise 3.4 *Equivalence of representations.*

Show that two equivalent unitary representations of a topological group are unitarily equivalent. [Hint: Introduce the adjoint T^* of the intertwining operator T and the positive square root of the positive self-adjoint operator TT^*.]

Exercise 3.5 *An application of Schur's lemma.*

Let (E, ρ) be an irreducible representation of a compact group G. Show that if two scalar products on E are invariant under the representation ρ, they are proportional.

Exercise 3.6 *A not completely reducible representation.*

Let G be the group of complex matrices $\left(\begin{smallmatrix} a & b \\ 0 & 1 \end{smallmatrix}\right)$ such that $|a| = 1$, with the usual topology.

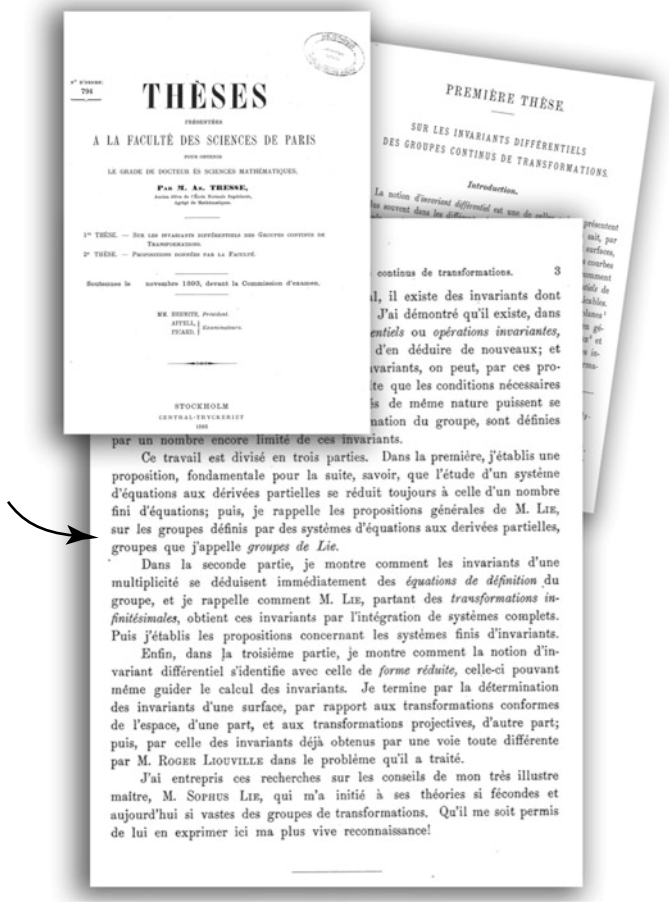
(a) Is this group compact?
(b) Show that the fundamental representation of G on $\mathbb{C}^2$ is reducible, but not completely reducible.
(c) Find the endomorphisms of $\mathbb{C}^2$ that commute with the fundamental representation of G.

Exercise 3.7 *Haar measure on the group* $\mathrm{O}(2)$.

(a) Find the Haar measure on $\mathrm{O}(2)$.
(b) Find the left-invariant and the right-invariant measures on the group of affine transformations of $\mathbb{R}$.

Exercise 3.8 *Representations of the direct product of two compact groups.*

Let (E_1, ρ_1) and (E_2, ρ_2) be finite-dimensional representations of compact Lie groups G_1 and G_2, respectively. Show that we can define a representation $\rho = \rho_1 \times \rho_2$ of $G_1 \times G_2$ on $E_1 \otimes E_2$ by $\rho(g_1, g_2)(v_1 \otimes v_2) = \rho_1(g_1)v_1 \otimes \rho_2(g_2)v_2$, for $g_1 \in G_1,\ g_2 \in G_2,\ v_1 \in E_1,\ v_2 \in E_2$. Show that each irreducible representation of $G_1 \times G_2$ is of the form $(E_1 \otimes E_2, \rho_1 \times \rho_2)$, where (E_i, ρ_i) is an irreducible representation of G_i, $i = 1, 2$.



THÈSES

PRÉSENTÉES

A LA FACULTÉ DES SCIENCES DE PARIS

POUR OBTENIR

LE GRADE DE DOCTEUR ÈS SCIENCES MATHÉMATIQUES.

PAR M. AD. TRESSE,

Ancien élève de l'École Normale Supérieure, Agrégé de Mathématiques.

1re THÈSE. — SUR LES INVARIANTS DIFFÉRENTIELS DES GROUPES CONTINUS DE TRANSFORMATIONS.

2e THÈSE. — PROPOSITIONS DONNÉES PAR LA FACULTÉ.

Soutenues le novembre 1893, devant la Commission d'examen.

MM. HERMITE, Président.
APPELL, }
PICARD, } Examinateurs.

STOCKHOLM
CENTRAL-TRYCKERIET
1893

PREMIÈRE THÈSE

SUR LES INVARIANTS DIFFÉRENTIELS DES GROUPES CONTINUS DE TRANSFORMATIONS.

Introduction.

costinus de transformations. 3

il, il existe des invariants dont
J'ai démontré qu'il existe, dans
entiels ou *opérations invariantes*,
d'en déduire de nouveaux; et
ivariants, on peut, par ces pro-
te que les conditions nécessaires
is de même nature puissent se
nation du groupe, sont définies

par un nombre encore limité de ces invariants.

Ce travail est divisé en trois parties. Dans la première, j'établis une proposition, fondamentale pour la suite, savoir, que l'étude d'un système d'équations aux dérivées partielles se réduit toujours à celle d'un nombre fini d'équations; puis, je rappelle les propositions générales de M. Lie, sur les groupes définis par des systèmes d'équations aux derivées partielles, groupes que j'appelle *groupes de Lie*.

Dans la seconde partie, je montre comment les invariants d'une multiplicité se déduisent immédiatement des *équations de définition* du groupe, et je rappelle comment M. Lie, partant des *transformations infinitésimales*, obtient ces invariants par l'intégration de systèmes complets. Puis j'établis les propositions concernant les systèmes finis d'invariants.

Enfin, dans la troisième partie, je montre comment la notion d'invariant différentiel s'identifie avec celle de *forme réduite*, celle-ci pouvant même guider le calcul des invariants. Je termine par la détermination des invariants d'une surface, par rapport aux transformations conformes de l'espace, d'une part, et aux transformations projectives, d'autre part; puis, par celle des invariants déjà obtenus par une voie toute différente par M. Roger Liouville dans le problème qu'il a traité.

J'ai entrepris ces recherches sur les conseils de mon très illustre maître, M. Sophus Lie, qui m'a initié à ses théories si fécondes et aujourd'hui si vastes des groupes de transformations. Qu'il me soit permis de lui en exprimer ici ma plus vive reconnaissance!

Arthur Tresse (1868–1958), a student of Lie, submitted his thesis in 1893 in Paris. It was published in Acta Mathematica, *18 (1893), p. 1–88. In the introduction, page 3, he gives the groups studied by Lie the name "groupes de Lie",* "then I recall the general propositions of Mr. Lie, concerning the groups that are defined by systems of partial differential equations, groups which I shall call *Lie groups*".

Chapter 4

Lie Groups and Lie Algebras



We restrict ourselves to the study of *linear Lie groups*, that is, to closed subgroups of $\mathrm{GL}(n,\mathbb{R})$, for a positive integer n, in other words, to groups of real matrices. We adopt the convention, introduced in Chapter 1, of calling such a group simply a *Lie group*. We shall show that to each Lie group there corresponds a Lie algebra. For ease of exposition, we start by defining Lie algebras, and return later to the study of groups.

1 Lie Algebras

1.1 Definition and Examples

We can define Lie algebras over any field. We restrict ourselves to the real or complex case. So let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Definition 1.1. *A* Lie algebra *over* $\mathbb{K}$ *is a* $\mathbb{K}$*-vector space* $\mathfrak{a}$ *with a* $\mathbb{K}$*-bilinear, antisymmetric composition law,* $[\ ,\ ]$*, satisfying the Jacobi identity,*

$$\forall X, Y, Z \in \mathfrak{a},\ \ [X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0. \tag{1.1}$$

The bilinear, antisymmetric composition law $[\ ,\ ]$ *is called the* Lie bracket *or simply the* bracket *of* $\mathfrak{a}$.

Examples.

- The vector space $\mathbb{R}^3$ with the cross-product is a real Lie algebra of dimension 3.
- Let E be a $\mathbb{K}$-vector space. We define the *commutator* of two $\mathbb{K}$-linear operators on E, u and v, by

$$[u,v] = u \circ v - v \circ u, \tag{1.2}$$

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Y. Kosmann-Schwarzbach, *Groups and Symmetries*, Universitext,
https://doi.org/10.1007/978-3-030-94360-8_4

where $\circ$ is the composition of operators. The verification of the Jacobi identity using the associativity of $\circ$ is immediate:

$$(u \circ (v \circ w - w \circ v) - (v \circ w - w \circ v) \circ u) + (v \circ (w \circ u - u \circ w) - (w \circ u - u \circ w) \circ v) + (w \circ (u \circ v - v \circ u) - (u \circ v - v \circ u) \circ w) = 0.$$

Thus the commutator defines a Lie algebra structure on the vector space of $\mathbb{K}$-linear operators on E.

In particular, the vector space of $n \times n$ matrices with coefficients in the field $\mathbb{K}$, with the commutator

$$[X, Y] = XY - YX, \tag{1.3}$$

is a Lie algebra over $\mathbb{K}$ of dimension n^2, denoted by $\mathfrak{gl}(n, \mathbb{K})$.

Definition 1.2. *A* Lie subalgebra *of a Lie algebra is a vector subspace closed under the bracket.*

Thus each $\mathbb{K}$-vector subspace of $\mathfrak{gl}(n, \mathbb{K})$ closed under the commutator is a Lie algebra over $\mathbb{K}$.

Examples.

- The vector space of real (resp., complex) matrices of trace zero is a real (resp., complex) Lie algebra.
- The vector space of complex antihermitian matrices is a real Lie algebra.
- The vector space of real antisymmetric matrices is a real Lie algebra.
- More generally, if $p + q = n$, we denote by J_{pq} the diagonal matrix with p diagonal elements 1 followed by q diagonal elements -1, and we set

$$\mathfrak{so}(p, q) = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid X J_{pq} \ + J_{pq} \ {}^t X = 0\}.$$

This is a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$. In fact, for X and $Y \in \mathfrak{so}(p, q)$,

$$J_{pq}({}^t Y \ {}^t X - \ {}^t X \ {}^t Y) = -Y J_{pq} \ {}^t X + X J_{pq} \ {}^t Y = -(XY - YX) J_{pq}.$$

Remark. Neither the vector space of symmetric matrices nor that of Hermitian matrices is stable under the commutator.

A Lie algebra is called *Abelian* if the bracket is identically zero.

An *ideal* of a Lie algebra $\mathfrak{a}$ is a Lie subalgebra $\mathfrak{b}$ such that $[\mathfrak{a}, \mathfrak{b}] \subset \mathfrak{b}$, *i.e.*, for every element X in $\mathfrak{a}$ and for every element Y in $\mathfrak{b}$, $[X, Y]$ is in $\mathfrak{b}$. In any Lie algebra $\mathfrak{a}$, the subspaces $\{0\}$ and $\mathfrak{a}$ itself are ideals, called the *trivial ideals*.

The *center* of a Lie algebra $\mathfrak{a}$ is the Abelian ideal of $\mathfrak{a}$,

$$\{X \in \mathfrak{a} \mid \forall Y \in \mathfrak{a}, \ [X, Y] = 0\}.$$

Definition 1.3. *A Lie algebra is called* simple *if it has no non-trivial ideals and is not of dimension* 0 *or* 1. *It is called* semi-simple *if it has no non-zero Abelian ideals.*

If E is a normed vector space over $\mathbb{K}$, we consider the vector space of continuous $\mathbb{K}$-linear maps of E into E, with the standard norm of linear maps which satisfies the inequality

$$\|u \circ v\| \leq \|u\|\|v\|, \tag{1.4}$$

for all linear maps u and v. The vector space of continuous linear maps with the commutator (1.2) is a Lie subalgebra of the Lie algebra of all linear operators on E. We denote it by $\mathfrak{gl}(E)$.

If E is finite-dimensional, every linear map on E is continuous, therefore $\mathfrak{gl}(E)$ is the Lie algebra of all linear operators on E. In particular, $\mathfrak{gl}(\mathbb{K}^n)$ can be identified with the Lie algebra $\mathfrak{gl}(n, \mathbb{K})$ of $n \times n$ matrices with coefficients in $\mathbb{K}$.

For all of the following, we shall consider *finite-dimensional* Lie algebras.

1.2 Morphisms

Let $\mathfrak{a}$ and $\mathfrak{b}$ be Lie algebras over $\mathbb{K}$. A *morphism* of Lie algebras of $\mathfrak{a}$ into $\mathfrak{b}$ is a $\mathbb{K}$-linear map φ of $\mathfrak{a}$ into $\mathfrak{b}$ such that

$$\forall X, Y \in \mathfrak{a},\ [\varphi X, \varphi Y]_{\mathfrak{b}} = \varphi([X, Y]_{\mathfrak{a}}).$$

Note that most authors use the term *homomorphism*.

An *isomorphism* of $\mathfrak{a}$ onto $\mathfrak{b}$ is a bijective morphism. Its inverse is also an isomorphism. An *automorphism* of $\mathfrak{a}$ is an isomorphism of $\mathfrak{a}$ onto $\mathfrak{a}$. We write $\mathrm{Aut}(\mathfrak{a})$ for the group of automorphisms of $\mathfrak{a}$.

1.3 Commutation Relations and Structure Constants

Let $\mathfrak{a}$ be a Lie algebra of finite dimension, d, and let (e_i), $i = 1, \ldots, d$, be a basis of the vector space $\mathfrak{a}$. The *commutation relations* of $\mathfrak{a}$ in the basis (e_i) are the Lie brackets, $[e_i, e_j], 1 \leq i < j \leq d$. The constants, real or complex numbers C^i_{jk} such that

$$[e_i, e_j] = \sum_{k=1}^{d} C^k_{ij} e_k,$$

are called *structure constants*. If two finite-dimensional Lie algebras have the same commutation relations, they are isomorphic. More precisely, if there exist in the Lie algebras $\mathfrak{a}$ and $\mathfrak{b}$ of the same finite dimension bases (e_i) and (f_i) such that the structure constants are the same, then the linear isomorphism of $\mathfrak{a}$ onto $\mathfrak{b}$ defined by $e_i \mapsto f_i$ is an isomorphism of Lie algebras.

1.4 Real Forms

On the one hand, each complex vector space, E, can be considered as a vector space over $\mathbb{R}$, denoted by $E^{\mathbb{R}}$, of twice the dimension. On the other hand, each real vector space E can be complexified by setting $E^{\mathbb{C}} = E \oplus iE = E \otimes \mathbb{C}$ and by setting $i(X + iY) = -Y + iX$, for every X and $Y \in E$. Note that if E is a complex (resp., real) vector space and $E \neq \{0\}$, then $(E^{\mathbb{R}})^{\mathbb{C}} \neq E$ (resp., $(E^{\mathbb{C}})^{\mathbb{R}} \neq E$), as one can easily see by calculating the dimensions.

If $\mathfrak{a}$ is a complex Lie algebra, we denote by $\mathfrak{a}^{\mathbb{R}}$ the Lie algebra over $\mathbb{R}$ (of twice the dimension) obtained by "realification". On the other hand, if $\mathfrak{a}$ is a real Lie algebra, the corresponding complexified vector space can be given the structure of a complex Lie algebra by setting $[X, iY] = i[X, Y]$ and $[iX, iY] = -[X, Y]$, for every X and $Y \in \mathfrak{a}$. We denote by $\mathfrak{a}^{\mathbb{C}}$ the complex Lie algebra thus obtained, called the *complexification* of the real Lie algebra $\mathfrak{a}$.

A *real form* of a complex Lie algebra $\mathfrak{g}$ is a real Lie algebra $\mathfrak{a}$ such that $\mathfrak{a}^{\mathbb{C}} = \mathfrak{g}$. For example $(\mathfrak{gl}(n, \mathbb{R}))^{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$. In general, a complex Lie algebra has several non-isomorphic real forms.

1.5 Representations of Lie Algebras

Here we define Lie algebra representations, restricting ourselves to the case of finite-dimensional representations. We shall see in Section 7.2 the relationship between representations of groups and representations of Lie algebras.

Definition 1.4. *Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{K}$. A* representation *of $\mathfrak{g}$ is a $\mathbb{K}$-linear mapping, R, of $\mathfrak{g}$ into $\mathfrak{gl}(E)$, where E is a finite-dimensional complex vector space, satisfying*

$$\forall X, Y \in \mathfrak{g},\ [R(X), R(Y)] = R([X, Y]).$$

A representation of a complex Lie algebra $\mathfrak{g}$ on a complex vector space E is thus a morphism of complex Lie algebras of $\mathfrak{g}$ into $\mathfrak{gl}(E)$. The dimension of E is called the *dimension* of the representation.

A representation of a real Lie algebra $\mathfrak{g}$ on a complex vector space E is a morphism of real Lie algebras of $\mathfrak{g}$ into $\mathfrak{gl}(E)^{\mathbb{R}}$. We show below (Proposition 1.7) that such a morphism has a unique extension to a morphism of complex Lie algebras of $\mathfrak{g}^{\mathbb{C}}$ into $\mathfrak{gl}(E)$. Later, we study numerous examples. Here is a very simple one. The matrices

$$\eta_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},\ \eta_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix},\ \eta_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis of a real Lie algebra $\mathfrak{g} = \mathfrak{so}(3)$, of dimension 3, whose commutation relations are

$$[\eta_k, \eta_\ell] = \eta_m,$$

for each cyclic permutation k, ℓ, m of $1, 2, 3$. To the elements η_1, η_2, η_3 of $\mathfrak{g}$, we associate the complex 2×2 matrices,

$$\xi_1 = \frac{i}{2}\sigma_1,\ \xi_2 = -\frac{i}{2}\sigma_2,\ \xi_3 = \frac{i}{2}\sigma_3,$$

where

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\ \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

are the Pauli matrices. Because

$$[\xi_k, \xi_\ell] = \xi_m,$$

we see immediately that the $\mathbb{R}$-linear map of $\mathfrak{so}(3)$ into $\mathfrak{gl}(2, \mathbb{C})^{\mathbb{R}}$ defined by $\eta_k \mapsto \xi_k$ is a representation of $\mathfrak{g}$ on $\mathbb{C}^2$. Furthermore, extending this map by $\mathbb{C}$-linearity defines a morphism of complex Lie algebras of $\mathfrak{so}(3)^{\mathbb{C}}$ into $\mathfrak{gl}(2, \mathbb{C})$.

Remark. We can also define the *real representations* of Lie algebras over $\mathbb{K}$, which are $\mathbb{R}$-linear maps into $\mathfrak{gl}(E)$, where E is a real vector space.

Definition 1.5. *A representation (E, R) of a Lie algebra $\mathfrak{g}$ is called* irreducible *if $E \neq \{0\}$ and if there are no non-trivial subspaces of E invariant under R, i.e., the only vector subspaces of E invariant under R are $\{0\}$ and E itself.*

Definition 1.6. *Representations (E_1, R_1) and (E_2, R_2) of a Lie algebra $\mathfrak{g}$ are* equivalent *if there is an isomorphism T of E_1 onto E_2 such that*

$$\forall X \in \mathfrak{g},\ R_2(X) \circ T = T \circ R_1(X).$$

Proposition 1.7. *Let $\mathfrak{g}^{\mathbb{C}}$ be the complexification of a real Lie algebra $\mathfrak{g}$. Every representation of $\mathfrak{g}$ can be extended uniquely to a representation of $\mathfrak{g}^{\mathbb{C}}$. There is a bijective correspondence between irreducible representations of $\mathfrak{g}$ and of $\mathfrak{g}^{\mathbb{C}}$.*

Proof. Let (E, R) be a representation of the Lie algebra $\mathfrak{g}$ on a complex vector space E. We set

$$\forall X, Y \in \mathfrak{g},\ R(X + iY) = R(X) + iR(Y) \in \mathfrak{gl}(E).$$

A calculation using the equation $R([X, X']) = R(X) \circ R(X') - R(X') \circ R(X)$, for X and $X' \in \mathfrak{g}$, shows that

$$R([X + iY, X' + iY']) = R(X + iY) \circ R(X' + iY') - R(X' + iY') \circ R(X + iY).$$

Thus the extension of R is a representation of $\mathfrak{g}^{\mathbb{C}}$. The uniqueness is obvious. On the other hand, from a representation of $\mathfrak{g}^{\mathbb{C}}$ one can define a representation of $\mathfrak{g}$ by restriction. Thus it is clear that there is a bijective correspondence between representations of $\mathfrak{g}$ and representations of $\mathfrak{g}^{\mathbb{C}}$. Furthermore, if a representation of $\mathfrak{g}$ is irreducible, the corresponding representation of $\mathfrak{g}^{\mathbb{C}}$ is irreducible as well. □

In summary, the equivalence relation on the representations of $\mathfrak{g}$ corresponds to the equivalence relation on the representations of $\mathfrak{g}^{\mathbb{C}}$. Thus the equivalence classes of irreducible representations of $\mathfrak{g}$ and of $\mathfrak{g}^{\mathbb{C}}$ are in bijective correspondence.

2 Review of the Exponential Map

Let $X \in \mathfrak{gl}(n, \mathbb{K})$. We set

$$\exp X = e^X = \sum_{p=0}^{\infty} \frac{X^p}{p!}. \tag{2.1}$$

For each compact set $K \subset \mathfrak{gl}(n, \mathbb{K})$ there exists a positive real number M such that $\|X\| \leq M$ for all $X \in K$. Therefore when we equip $\mathfrak{gl}(n, \mathbb{K})$ with the standard norm of linear maps, the series (2.1) is convergent and uniformly convergent on each compact set due to the inequality (1.4).

We have

$$\exp 0 = I, \tag{2.2}$$

$$\forall s, t \in \mathbb{R},\ \exp(sX) \exp(tX) = \exp(s+t)X. \tag{2.3}$$

Furthermore, if $XY = YX$ (but in this case only!), then $\exp(X + Y) = \exp X \exp Y$.

We deduce from (2.3) that, for every X, $\exp X$ is invertible with inverse $\exp(-X)$,

$$(\exp X)^{-1} = \exp(-X).$$

Thus the exponential map, exp, is a map of $\mathfrak{gl}(n, \mathbb{K})$ into the group of invertible matrices, $\mathrm{GL}(n, \mathbb{K}) \subset \mathfrak{gl}(n, \mathbb{K})$.

Lemma 2.1. *The map $t \in \mathbb{R} \mapsto \exp(tX) \in \mathrm{GL}(n, \mathbb{K})$ is differentiable and*

$$\frac{\mathrm{d}}{\mathrm{d}t} \exp(tX) = X \exp(tX) = \exp(tX)\ X. \tag{2.4}$$

Proof. The term-by-term differentiation of the power series for $\exp(tX)$ is legitimate and gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{p=0}^{\infty} \frac{X^p}{p!} t^p = \sum_{p=1}^{\infty} X^p \frac{t^{p-1}}{(p-1)!} = X \sum_{p=0}^{\infty} X^p \frac{t^p}{p!} = X \exp(tX) = \exp(tX)\ X,$$

for every $t \in \mathbb{R}$. □

In particular,

$$\frac{\mathrm{d}}{\mathrm{d}t} \exp(tX)\big|_{t=0} = X. \tag{2.5}$$

Proposition 2.2. *The exponential map of $\mathfrak{gl}(n, \mathbb{K})$ into $\mathrm{GL}(n, \mathbb{K}) \subset \mathfrak{gl}(n, \mathbb{K})$ is smooth and its differential at the origin is the identity map of $\mathfrak{gl}(n, \mathbb{K})$.*

Proof. We have

$$\exp X = I + X + X\sum_{p=2}^{\infty}\frac{X^{p-1}}{p!},$$

whence $\|\exp X - \exp 0 - X\| \leq \|X\|\,\varepsilon(X)$, where

$$\varepsilon(X) = \|\sum_{p=2}^{\infty}\frac{X^{p-1}}{p!}\| \leq \|X\|\|\sum_{q=0}^{\infty}\frac{X^q}{(q+2)!}\|.$$

This last series converges, so $\lim_{\|X\|\to 0}\varepsilon(X) = 0$, which shows that exp is differentiable at 0, with differential $\mathrm{Id}_{\mathfrak{gl}(n,\mathbb{K})}$.

One can show that the map exp is differentiable at each point and is of class C^∞ (see Exercise 4.6) and we shall use this result. □

By the inverse function theorem, we obtain the following result.

Corollary 2.3. *There exists an open neighborhood U_0 of 0 in $\mathfrak{g}l(n,\mathbb{K})$ and an open neighborhood V_0 of I in $\mathrm{GL}(n,\mathbb{K})$ such that* exp *is a diffeomorphism of U_0 onto V_0.*

We call the inverse diffeomorphism, defined on a neighborhood of $I \in \mathrm{GL}(n,\mathbb{K})$, the *logarithm*. We denote it by log (or ln). Then,

$$\forall X \in \mathfrak{g}l(n,\mathbb{K}) \text{ such that } \|X\| < 1,\ \log(I+X) = \sum_{p=0}^{\infty}(-1)^p\frac{X^{p+1}}{p+1}.$$

Proposition 2.4. *For every $X, Y \in \mathfrak{gl}(n,\mathbb{K})$,*

$$\exp(tX)\,\exp(tY) = \exp\left(t(X+Y) + \frac{t^2}{2}[X,Y] + \mathrm{O}(t^3)\right). \tag{2.6}$$

Proof. For t near 0, on the one hand,

$$\exp(tX)\,\exp(tY) = (I + tX + \frac{t^2}{2}X^2 + \mathrm{O}(t^3))(I + tY + \frac{t^2}{2}Y^2 + \mathrm{O}(t^3))$$

$$= I + t(X+Y) + \frac{t^2}{2}(X^2 + 2XY + Y^2) + \mathrm{O}(t^3).$$

On the other hand,

$$\exp\left(t(X+Y) + \frac{t^2}{2}(XY - YX) + \mathrm{O}(t^3)\right)$$

$$= I + t(X+Y) + \frac{t^2}{2}(XY - YX) + \frac{t^2}{2}(X^2 + XY + YX + Y^2) + \mathrm{O}(t^3).$$

The two expressions are thus equal. □

One could expand further and consider the coefficients of the terms of order t^3, t^4, etc. The result is known as the Baker-Campbell-Hausdorff formula.

Corollary 2.5. *The exponential of a sum can be expressed as*

$$\exp(X+Y) = \lim_{k\to+\infty} (\exp\frac{X}{k}\ \exp\frac{Y}{k})^k. \tag{2.7}$$

Proof. We have, for every positive integer k,

$$\left(\exp\frac{X}{k}\ \exp\frac{Y}{k}\right)^k = \left(\exp\left(\frac{1}{k}(X+Y)+\mathrm{O}(\frac{1}{k^2})\right)\right)^k$$

$$= \exp\left(X+Y+\mathrm{O}(\frac{1}{k})\right),$$

so the result follows by the continuity of the exponential map. □

Proposition 2.6. *For each* $X \in \mathfrak{gl}(n,\mathbb{K})$,

$$\exp(\mathrm{Tr}X) = \det(\exp X). \tag{2.8}$$

In particular, the exponential of a traceless matrix is a matrix of determinant 1.

Proof. Over $\mathbb{C}$, one can triangularize the matrix X. If X is similar to a triangular matrix T, then $\exp X$ is similar to $\exp T$. If the diagonal coefficients of T are $a_1 \ldots, a_n$, then the diagonal coefficients of the matrix $\exp T$ are $e^{a_1}, \ldots, e^{a_n}$. Hence

$$\det(\exp X) = \prod_{i=1}^{n} e^{a_i} = e^{\sum_{i=1}^{n} a_i} = e^{\mathrm{Tr}\,X}.$$

Over $\mathbb{R}$, the equation is valid for real numbers. □

Corollary 2.7. *We have*

$$\frac{\mathrm{d}}{\mathrm{d}t}\det(\exp(tX))\big|_{t=0} = \mathrm{Tr}\,X.$$

Proof. In Formula (2.8), we replace X by tX and differentiate. □

3 One-Parameter Subgroups of GL($n, \mathbb{K}$)

We shall show that in the case of one-parameter subgroups of the group of invertible matrices, the hypothesis of continuity implies differentiability.

Definition 3.1. *A* one-parameter subgroup *of a topological group* G *is a* continuous *map,* $f : \mathbb{R} \to G$*, such that*

$$\forall\ t, s \in \mathbb{R},\ \ f(t+s) = f(t)f(s).$$

We may also simply write *one-parameter group*. The definition implies that $f(0) = e$.

Lemma 3.2. *Let $f:\mathbb{R}\to \mathrm{GL}(n,\mathbb{K})$ be a one-parameter subgroup of $\mathrm{GL}(n,\mathbb{K})$. Then f is differentiable.*

Proof. Let $a\in\mathbb{R}$, $a>0$. The map f with values in $\mathrm{GL}(n,\mathbb{K})\subset \mathfrak{gl}(n,\mathbb{K})$ is continuous, and thus the integral $\int_0^a f(t)\,\mathrm{d}t$ exists and $\int_0^a f(t)\,\mathrm{d}t\in\mathfrak{gl}(n,\mathbb{K})$. For each $s\in\mathbb{R}$,

$$\int_0^a f(t+s)\,\mathrm{d}t = f(s)\int_0^a f(t)\,\mathrm{d}t = \int_s^{s+a} f(t)\,\mathrm{d}t.$$

This quantity is differentiable with respect to s. To prove that f is differentiable, it is sufficient to show that there is a real number $a>0$ such that $\int_0^a f(t)\,\mathrm{d}t$ is invertible. Now, because f is continuous at 0 and $f(0)=I$,

$$\exists a>0\ ,|t|\le a,\ \|f(t)-I\|\le\frac{1}{2},$$

whence we deduce that

$$\|\frac{1}{a}(\int_0^a f(t)\,\mathrm{d}t)-I\| = \|\frac{1}{a}\int_0^a (f(t)-I)\,\mathrm{d}t\|\le\frac{1}{2}<1,$$

and thus, for $|t|\le a$, $\int_0^a f(t)\,\mathrm{d}t$ is invertible. □

Proposition 3.3. *Let $f:\mathbb{R}\to\mathrm{GL}(n,\mathbb{K})$ be a one-parameter subgroup of $\mathrm{GL}(n,\mathbb{K})$. There exists a unique $X\in\mathfrak{gl}(n,\mathbb{K})$ such that, for each $t\in\mathbb{R}$,*

$$f(t)=\exp(tX). \tag{3.1}$$

Proof. The uniqueness is clear, since f is differentiable by the preceding lemma, and hence the desired matrix X must satisfy the equation

$$\frac{\mathrm{d}}{\mathrm{d}t}f(t)\big|_{t=0}=X. \tag{3.2}$$

We have

$$\frac{\mathrm{d}}{\mathrm{d}t}f(t)=\lim_{s\to0}\frac{f(t+s)-f(t)}{s}=\lim_{s\to0}f(t)\frac{f(s)-f(0)}{s}=f(t)\ X.$$

By Lemma 2.1, we see that $f(t)$ and $\exp(tX)$ are solutions of the same differential equation with the same initial condition, $f(0)=I$. The proposition follows. □

Definition 3.4. *The element X of $\mathfrak{g}l(n,\mathbb{K})$ satisfying (3.1) is called the* infinitesimal generator *of the one-parameter group $t\mapsto f(t)$.*

We have

$$X=f'(0)$$

and, more generally,

$$\forall t\in\mathbb{R},\ (f(t))^{-1}f'(t)=X. \tag{3.3}$$

Example. The rotations around the axes $0x$, $0y$, $0z$, respectively, constitute the one-parameter groups

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos t & -\sin t \\ 0 & \sin t & \cos t \end{pmatrix}, \quad \begin{pmatrix} \cos t & 0 & \sin t \\ 0 & 1 & 0 \\ -\sin t & 0 & \cos t \end{pmatrix}, \quad \begin{pmatrix} \cos t & -\sin t & 0 \\ \sin t & \cos t & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

of the form $\exp(t\eta_i), i = 1, 2, 3$, where the infinitesimal generators are the matrices η_1, η_2, η_3 defined in Section 1.5.

Important Remark. In physics, the name "infinitesimal generator" of a one-parameter group $f(t)$ is given to the matrix A satisfying

$$f(t) = \exp(-itA).$$

Thus $A = iX$. In particular if the one-parameter group consists of unitary transformations of a finite-dimensional Hilbert space, then the operator A is Hermitian (self-adjoint), and the operator X is antihermitian. In infinite dimensions, the problem is that the operator A is not necessarily bounded.

4 Lie Groups

Definition 4.1. *A linear Lie group is a closed subgroup of a group* $\mathrm{GL}(n, \mathbb{K})$.

We can identify $\mathrm{GL}(m, \mathbb{C})$ with a closed subgroup of $\mathrm{GL}(2m, \mathbb{R})$. In fact each complex $m \times m$ matrix A can be written $A = B + iC$, where B and C are real $m \times m$ matrices and the $\mathbb{R}$-linear map $A \mapsto \begin{pmatrix} B & -C \\ C & B \end{pmatrix}$ of $\mathrm{GL}(m, \mathbb{C})$ onto a closed subgroup of $\mathrm{GL}(2m, \mathbb{R})$ is a morphism of groups. *A linear Lie group is thus a closed subgroup of* $\mathrm{GL}(n, \mathbb{R})$, for a certain n. In Chapter 1 we have given an abstract definition of real Lie groups as smooth manifolds equipped with a group structure compatible with their differentiable structure, and we have stated that each linear Lie group is a real Lie group. In the following, we study only linear Lie groups and, *by convention*, we call them simply *Lie groups*. There exist Lie groups that are not linear Lie groups, such as the universal cover of $\mathrm{SL}(2, \mathbb{R})$. Most of the properties proved below are valid in the general case.

In Chapter 1, we saw several examples of Lie groups, the real and complex special linear groups in dimension n, the orthogonal groups and the unitary groups:

$$\begin{array}{ccccc} \mathrm{SO}(n) \subset \mathrm{O}(n) & \subset \mathrm{GL}(n, \mathbb{R}) & & \overbrace{\mathrm{SU}(n) \subset \mathrm{U}(n)}^{\text{real Lie groups}} & \subset \mathrm{GL}(n, \mathbb{C}) \\ \cap & & & \cap & \\ \mathrm{SL}(n, \mathbb{R}) & & & \mathrm{SL}(n, \mathbb{C}) & \end{array}$$

For each positive integer n, the additive group $\mathbb{R}^n$ is isomorphic to the group of $(n+1)\times(n+1)$ matrices of the form $\begin{pmatrix} I & x \\ 0 & 1 \end{pmatrix}$, where $x \in \mathbb{R}^n$, or to the group of matrices $\begin{pmatrix} e^{x_1} & & \\ & \ddots & \\ & & e^{x_n} \end{pmatrix}$, and is thus a Lie group.

The group of affine transformations of $\mathbb{R}^n$ is a Lie group.

The symplectic groups $\mathrm{Sp}(n,\mathbb{C})$ and $\mathrm{Sp}(n)$ are defined by

$$\mathrm{Sp}(n,\mathbb{C}) = \{A \in \mathrm{SL}(2n,\mathbb{C}) \mid AJ\,{}^tA = J\},$$

$$\mathrm{Sp}(n) = \mathrm{Sp}(n,\mathbb{C}) \cap \mathrm{U}(2n),$$

where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$, and are Lie groups.

5 The Lie Algebra of a Lie Group

Let G be a Lie group, *i.e.*, a closed subgroup of $\mathrm{GL}(n,\mathbb{R})$. We consider

$$\mathfrak{g} = \{X = \gamma'(0) \mid \gamma \,:\, \mathcal{I} \to G, \ \text{ of class } C^1,\ \gamma(0) = I\},$$

where $\mathcal{I}$ is an open interval of $\mathbb{R}$ containing 0. This is the set of vectors tangent to parametrized curves of class C^1 in G passing through I for the value 0 of the parameter. We shall show that $\mathfrak{g}$ is not only a real vector space, but also a Lie subalgebra of $\mathfrak{gl}(n,\mathbb{R})$.

Theorem 5.1. *Let G be a Lie group and let $\mathfrak{g}$ be defined as above.*

(i) $\mathfrak{g}$ is a vector subspace of $\mathfrak{gl}(n,\mathbb{R})$.

(ii) $X \in \mathfrak{g}$ if and only if, for every $t \in \mathbb{R}$, $\exp(tX) \in G$.

(iii) If $X \in \mathfrak{g}$ and if $g \in G$, then $gXg^{-1} \in \mathfrak{g}$.

(iv) $\mathfrak{g}$ is closed under the matrix commutator.

Proof. (i) Let $\gamma : \mathcal{I} \to G$ be of class C^1 and such that $\gamma(0) = I$. Set $\gamma'(0) = X$. For each $\lambda \in \mathbb{R}$, the parametrized curve $t \mapsto \gamma(\lambda t)$ has tangent vector λX at 0. Thus $\mathfrak{g}$ is closed under multiplication by real scalars. Let $\gamma_i : \mathcal{I} \to G$, $i = 1, 2$, be of class C^1 and such that $\gamma_i(0) = I$. Let $X_1 = \gamma_1'(0), X_2 = \gamma_2'(0)$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}(\gamma_1(t)\gamma_2(t))\big|_{t=0} = X_1 + X_2.$$

Thus $\mathfrak{g}$ is closed under addition.

(ii) It is clear that if $\exp(tX) \in G$, then $X = \frac{\mathrm{d}}{\mathrm{d}t}\exp(tX)\big|_{t=0} \in \mathfrak{g}$. Conversely, if $X \in \mathfrak{g}$, by hypothesis, $X = \frac{\mathrm{d}}{\mathrm{d}t}\gamma(t)\big|_{t=0}$, with $\gamma(t) \in G$. By the Taylor expansion, for any positive integer k,

$$\gamma(\frac{t}{k}) = I + \frac{t}{k}X + \mathrm{O}(\frac{1}{k^2}) = \exp\left(\frac{t}{k}X + \mathrm{O}(\frac{1}{k^2})\right).$$

We can deduce that

$$\left(\gamma(\frac{t}{k})\right)^k = \exp(tX + \mathrm{O}(\frac{1}{k})),$$

and hence

$$\lim_{k\to+\infty}\left(\gamma(\frac{t}{k})\right)^k = \exp(tX).$$

This equation implies that $\exp(tX) \in G$ because $\gamma(\frac{t}{k}) \in G$, and G is a closed subgroup.

(iii) If $X \in \mathfrak{g}$ and $g \in G$, then $gXg^{-1} = \frac{\mathrm{d}}{\mathrm{d}t}\exp(t(gXg^{-1}))\big|_{t=0}$, and for every t, the matrix $\exp(t(gXg^{-1})) = g(\exp(tX))g^{-1}$ belongs to G, so $gXg^{-1} \in \mathfrak{g}$.

(iv) Let X and $Y \in \mathfrak{g}$. Then, $\exp(tX) \in G$ and $\exp(tX)Y\exp(-tX) \in \mathfrak{g}$, by (iii). In the vector space $\mathfrak{g}$, we consider

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\exp(tX)Y\exp(-tX)\right)\big|_{t=0}.$$

This element of $\mathfrak{g}$ is $XY - YX = [X, Y]$. □

Definition 5.2. *The Lie algebra* $\mathfrak{g}$*, the tangent space to* G *at* I*, is called* the Lie algebra of the Lie group G.

Property (ii) of Theorem 5.1 is the fundamental characterization of the Lie algebra of a Lie group.

The dimension of the real vector space $\mathfrak{g}$ is called the *dimension* of the Lie group G. This is the number of independent parameters necessary to parametrize the points of the group. (This is also the dimension of the Lie group as a real manifold.)

If the group G is a discrete subgroup of $\mathrm{GL}(n, \mathbb{R})$, a continuous curve in G is necessarily constant, and the Lie algebra of the group is thus reduced to $\{0\}$ and the dimension of the group is 0.

Examples. The Lie algebras of $\mathrm{GL}(n, \mathbb{K}), \mathrm{SL}(n, \mathbb{K}), \mathrm{O}(n), \mathrm{SO}(n), \mathrm{U}(n), \mathrm{SU}(n)$ are denoted respectively by $\mathfrak{gl}(n, \mathbb{K}), \mathfrak{sl}(n, \mathbb{K}), \mathfrak{o}(n), \mathfrak{so}(n), \mathfrak{u}(n), \mathfrak{su}(n)$. The notation $\mathfrak{gl}(n, \mathbb{K})$ is consistent with our previous notation: it is the vector space of all the $n \times n$ matrices with coefficients in $\mathbb{K}$ equipped with the commutator.

Proposition 5.3. • $\mathfrak{sl}(n, \mathbb{K}) = \{X \in \mathfrak{gl}(n, \mathbb{K}) \mid \mathrm{Tr}\, X = 0\}$ *is the Lie algebra of traceless* $n \times n$ *matrices with coefficients in* $\mathbb{K}$*, and*

$$\dim_{\mathbb{R}} \mathfrak{sl}(n, \mathbb{R}) = n^2 - 1, \quad \dim_{\mathbb{C}} \mathfrak{sl}(n, \mathbb{C}) = n^2 - 1, \quad \dim_{\mathbb{R}} \mathfrak{sl}(n, \mathbb{C}) = 2(n^2 - 1).$$

- $\mathfrak{o}(n) = \mathfrak{so}(n) = \{X \in \mathfrak{gl}(n,\mathbb{R}) \mid X + {}^tX = 0\}$ *is the Lie algebra of antisymmetric real* $n \times n$ *matrices and* $\dim_{\mathbb{R}} \mathfrak{so}(n) = \frac{n(n-1)}{2}$.
- $\mathfrak{u}(n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) \mid X + {}^t\overline{X} = 0\}$ *is the Lie algebra of antihermitian complex* $n \times n$ *matrices and* $\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2$.
- $\mathfrak{su}(n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) \mid X + {}^t\overline{X} = 0, \operatorname{Tr} X = 0\}$ *is the Lie algebra of traceless antihermitian complex* $n \times n$ *matrices and* $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$.

Proof. In each case we express the condition for $\exp(tX)$ to belong to the indicated Lie group, for every $t \in \mathbb{R}$. □

In certain, but not all, cases, where the Lie group G is a subgroup of $\mathrm{GL}(n,\mathbb{C})$, its Lie algebra has the structure of a complex vector space. This is the case for $\mathfrak{sl}(n,\mathbb{C})$, but not for $\mathfrak{u}(n)$.

Theorem 5.4. *The exponential map* exp *is a diffeomorphism of a neighborhood of* 0 *in* $\mathfrak{g}$ *onto a neighborhood of* I *in* G.

Proof. Let U_0 be a neighborhood of 0 in $\mathfrak{gl}(n,\mathbb{R})$ and let V_0 be a neighborhood of I in $\mathrm{GL}(n,\mathbb{R})$ such that exp is a diffeomorphism of U_0 onto V_0, as in Corollary 2.3. We consider a supplementary vector subspace $\mathfrak{g}'$ of $\mathfrak{g}$ in $\mathfrak{gl}(n,\mathbb{R})$.

We shall first show by contradiction that there exists a neighborhood $U_0' \subset U_0$ of 0 in $\mathfrak{g}'$ such that the two conditions, $X \in U_0'$ and $\exp X \in G$, imply $X = 0$. Suppose then that in each neighborhood of 0 in $\mathfrak{g}'$ there is an $X \neq 0$ such that $\exp X \in G$. In particular, for every positive integer n, there is an $X_n \in \frac{1}{n}(U_0 \cap \mathfrak{g}'), X_n \neq 0$, such that $\exp X_n \in G$. Thus we construct a sequence (X_n) in $\mathfrak{g}'$ that tends to 0 as n tends to infinity. We set $Y_n = \frac{X_n}{\|X_n\|}$, which implies $\|Y_n\| = 1$. Because the unit sphere is compact, there is a subsequence of (Y_n) which converges in the vector space $\mathfrak{g}'$ to a limit Y of norm 1. We shall prove that $\exp(tY) \in G$ for all t and therefore $Y \in \mathfrak{g}$ which is the contradiction we sought since $\mathfrak{g} \cap \mathfrak{g}' = \{0\}$. For $t \in \mathbb{R}$, we consider the integral part, $p_n(t)$, of $\frac{t}{\|X_n\|}$ defined by

$$\frac{t}{\|X_n\|} = p_n(t) + u_n(t),\ p_n(t) \in \mathbb{Z},\ u_n(t) \in [0,1).$$

Then

$$\exp\left(\frac{t}{\|X_n\|} X_n\right) = (\exp X_n)^{p_n(t)} \exp(u_n(t) X_n).$$

The sequence $\exp(u_n(t)X_n)$ tends to I since the sequence $(u_n(t))$ is bounded and the sequence (X_n) tends to 0, and $(\exp X_n)^{p_n(t)}$ is in G since G is a group. Therefore, since G is a closed subgroup of $\mathrm{GL}(n,\mathbb{R})$, the sequence $\exp(\frac{t}{\|X_n\|} X_n)$ tends to an element in G. Because the exponential map is continuous, the limit of $\exp(\frac{t}{\|X_n\|} X_n)$ is $\exp(tY)$, which is therefore in G. By Theorem 5.1 (ii), this in turn implies that Y is in $\mathfrak{g}$, which contradicts the hypothesis $\mathfrak{g} \cap \mathfrak{g}' = \{0\}$.

Now let $\alpha : \mathfrak{g} \oplus \mathfrak{g}' \to \mathrm{GL}(n,\mathbb{R})$ be the map defined by

$$\alpha(X, X') = \exp X \exp X'.$$

The differential at 0 of α is the identity, and thus α is a local diffeomorphism of a product of neighborhoods of 0 in $\mathfrak{g}$ and $\mathfrak{g}'$, respectively, $U \times U'$, onto a neighborhood W of I in $\mathrm{GL}(n, \mathbb{R})$. We can assume that $U' \subset U'_0$. We show that $W \cap G = \exp U$. It is clear that $\exp U \subset W \cap G$. On the other hand, if $g \in W \cap G$, then $g = \exp X \exp X'$, where $X \in U \subset \mathfrak{g}$, $X' \in U'_0 \subset \mathfrak{g}'$. Then we have $X' = 0$ since $X' \in U'_0$ and $\exp X' = (\exp X)^{-1} g \in G$. Thus $W \cap G \subset \exp U$. □

6 The Connected Component of the Identity

Definition 6.1. *The* connected component *of a point $g \in G$ is the largest connected set contained in G and containing g. The* connected component of the identity *is often called* the connected component *of G for short. We shall denote it by G_0.*

The connected component of a point $g \in G$ is the union of the connected sets contained in G and containing g. Clearly, if G is connected, $G = G_0$ and G has only one connected component.

Proposition 6.2. *Each connected component of G is open and closed in G.*

Proof. Each connected component is closed because the closure of a connected set is connected. It is also open. In fact, each point of G has a connected neighborhood: for I, the image of a ball centered at the origin under the exponential map; for an arbitrary element of the group, the translate of such a neighborhood. For h in the connected component of $g \in G$, let V be a connected neighborhood of h. Then V is contained in the connected component of g, thus the connected component of g is a neighborhood of h. □

Recall that the only non-empty open subset of a connected set is the set itself. It follows that G is the disjoint union of its connected components.

Proposition 6.3. *The connected component of the identity of G is a subgroup of G. Any open and closed subgroup of G contains the connected component of the identity of G.*

Proof. If $g \in G_0$, the left translate of G_0 by g^{-1} is diffeomorphic to G_0, and thus is connected, and it contains I, so it is contained in G_0. Thus, G_0 is a subset containing I, and such that, for every g and $g' \in G_0$, $g^{-1}g'$ is in G_0, *i.e.*, G_0 is a subgroup.

Let H be an open and closed subgroup of G. Then $H \cap G_0$ is open and closed. Since it contains I, it is non-empty. Therefore $H \cap G_0 = G_0$. □

Example. The connected component of $\mathrm{O}(n)$ is $\mathrm{SO}(n)$. For $n = 3$, this fact will be proved in Section 2.1 of Chapter 5.

Proposition 6.4. *Every neighborhood of the identity in G generates a subgroup of G containing the connected component of the identity. In particular, if the Lie group G is connected, every neighborhood of the identity generates G.*

Proof. Let U be an open neighborhood of I in G, and let H be the subgroup generated by U. Because the translations are diffeomorphisms, for every $g \in H$, the subset Ug of H is open, and it is contained in H because H is a subgroup of G. Thus H is open. On the other hand, each open subgroup H of any topological group is also closed. In fact, the entire group is the union of left (or right) cosets modulo H and each class is open. Since H is the complement of the union of cosets other than that of the identity, and since this union of open sets is open, we deduce that H is closed. Thus the subgroup H is open and closed. We deduce that it contains the connected component of the identity. □

Theorem 6.5. *Let G be a Lie group and let $\mathfrak{g}$ be its Lie algebra. The subgroup of G generated by $\exp \mathfrak{g}$ is the connected component of the identity. In particular, if G is connected, each element of G is the product of a finite number of exponentials.*

Proof. By Theorem 5.4, the image of the exponential map is a neighborhood of the identity of G. Thus the subgroup that it generates contains the connected component of the identity, G_0, by Proposition 6.4. Because $\exp \mathfrak{g}$ is connected, it is contained in G_0 and so is the subgroup that it generates since G_0 is a subgroup of G, by Proposition 6.3. Therefore $\exp \mathfrak{g}$ generates G_0. □

Every $g \in G_0$ is thus a finite product, $\exp X_1 \dots \exp X_i \dots \exp X_k$, where each X_i, $1 \le i \le k$, is in $\mathfrak{g}$.

7 Morphisms of Lie Groups and of Lie Algebras

We defined in Section 1.2 the algebraic notion of a morphism of Lie algebras. Now we define the morphisms of Lie groups and consider the close relationship between these two concepts.

7.1 Differential of a Lie Group Morphism

Definition 7.1. *Let G and G' be Lie groups. A* morphism *of Lie groups, or Lie group morphism, $f : G \to G'$, is a continuous morphism of groups of G into G'. Most authors use the term* homomorphism.

Theorem 7.2. *Let f be a Lie group morphism of G into G', and let $\mathfrak{g}$ and $\mathfrak{g}'$ be the Lie algebras of G and G', respectively.*

(i) For each $X \in \mathfrak{g}$, the map $t \in \mathbb{R} \mapsto f(\exp(tX)) \in G'$ is a one-parameter subgroup of G'.

(ii) Let

$$\varphi(X) = \frac{\mathrm{d}}{\mathrm{d}t} f(\exp(tX))\big|_{t=0}. \tag{7.1}$$

Then, for every $t \in \mathbb{R}$,

$$f(\exp(tX)) = \exp\left(t\varphi(X)\right), \tag{7.2}$$

and, in particular,

$$f(\exp X) = \exp(\varphi(X)). \tag{7.3}$$

(iii) The map $\varphi : X \mapsto \varphi(X)$ is a morphism of Lie algebras from $\mathfrak{g}$ into $\mathfrak{g}'$.

Proof. (i) The map $t \mapsto f(\exp(tX))$ is continuous and also satisfies the group property

$$f\left(\exp((t+s)X)\right) = f(\exp(tX)\ \exp(sX)) = f(\exp(tX))\ f(\exp(sX)).$$

It is therefore a one-parameter subgroup of G', and by Lemma 3.2, we can differentiate the map $t \mapsto f(\exp(tX))$.

(ii) Because $\varphi(X) = \frac{\mathrm{d}}{\mathrm{d}t} f(\exp(tX))\big|_{t=0}$ is the infinitesimal generator of the one-parameter group, $t \mapsto f(\exp(tX))$, we obtain (7.2) and, in particular, (7.3).

(iii) Let us show that φ is $\mathbb{R}$-linear. Let s be a real number. Then $\varphi(sX) = s\varphi(X)$. In fact, $t \mapsto f(\exp(t(sX)))$ is a one-parameter subgroup with infinitesimal generator,

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\exp(tsX))\big|_{t=0} = s\frac{\mathrm{d}}{\mathrm{d}u} f(\exp(uX))\big|_{u=0} = s\varphi(X).$$

Now this infinitesimal generator is $\varphi(sX)$ by definition. Thus $\varphi(sX) = s\varphi(X)$.

Let us show that φ is additive. Let X and Y be elements of $\mathfrak{g}$. On the one hand, the map $t \mapsto f(\exp(t(X+Y)))$ is a one-parameter subgroup of G', with infinitesimal generator $\varphi(X+Y)$. On the other hand, using Corollary 2.5 and the continuity of f, we obtain

$$\exp(t\varphi(X+Y)) = f\left(\exp(t(X+Y))\right) = \lim_{k\to+\infty} f\left((\exp(\frac{t}{k}X)\ \exp(\frac{t}{k}Y))^k\right)$$

$$= \lim_{k\to+\infty} \left(f(\exp(\frac{t}{k}X))\ f(\exp(\frac{t}{k}Y))\right)^k$$

$$= \lim_{k\to+\infty} \left(\exp(\frac{t}{k}\varphi(X))\ \exp(\frac{t}{k}\varphi(Y))\right)^k,$$

because $\frac{1}{k}\varphi(X)$ is the infinitesimal generator of $t \mapsto f(\exp(\frac{t}{k}X))$. Thus

$$\exp(t\varphi(X+Y)) = \exp\left(t(\varphi(X)+\varphi(Y))\right).$$

Hence

$$\varphi(X+Y) = \varphi(X) + \varphi(Y).$$

It remains to show that φ preserves the commutator. Let us show first that

$$\forall g \in G, \forall X \in \mathfrak{g},\ \varphi(gXg^{-1}) = f(g)\varphi(X)(f(g))^{-1}.$$

In fact,

$$\exp(t\varphi(gXg^{-1})) = f(\exp(g(tX)g^{-1})) = f(g\exp(tX)g^{-1})$$

$$= f(g)f(\exp(tX))f(g^{-1}) = f(g)\ \exp(t\varphi(X))\ (f(g))^{-1}$$

$$= \exp(tf(g)\varphi(X)f(g^{-1}))$$

whence, for every $Y \in \mathfrak{g}$ and for every $t \in \mathbb{R}$,

$$\varphi(\exp(tY)\, X\, \exp(-tY)) = f(\exp(tY))\varphi(X)f(\exp(-tY)).$$

Differentiating at 0, as the map φ is linear, we obtain

$$\varphi(YX - XY) = \varphi(Y)\varphi(X) - \varphi(X)\varphi(Y),$$

which is the desired equation. □

Definition 7.3. *The morphism of Lie algebras $\varphi : \mathfrak{g} \to \mathfrak{g}'$ defined by (7.1) is called the* differential *of the Lie group morphism $f : G \to G'$, and is denoted by Df.*

Thus, by definition,

$$(Df)(X) = \frac{\mathrm{d}}{\mathrm{d}t} f(\exp(tX))\big|_{t=0}. \tag{7.4}$$

Definition 7.4. *A Lie group, more generally, a topological space is called* simply connected *if each continuous loop contained in it can be deformed continuously to a point.*

There is an imporatant result on the integration of Lie-algebra morphisms: if a Lie group G is connected and simply connected, then, for every morphism of real Lie algebras φ of $\mathfrak{g}$ into $\mathfrak{g}'$, there is a unique Lie group morphism of G into G' whose differential is φ.

7.2 Differential of a Lie Group Representation

One may define and study the infinite-dimensional representations of Lie groups. In particular, if the Lie group is compact, the results of Chapter 3 concerning the representations on separable complex Hilbert spaces apply. But we sha ll nonetheless limit ourselves here to the case of finite-dimensional representations. If E is a finite-dimensional complex vector space, then $\mathrm{GL}(E)$ is a (real) Lie group, with Lie algebra $\mathfrak{gl}(E)^{\mathbb{R}}$.

Definition 7.5. *A* representation *of a Lie group G is a Lie group morphism from G into $\mathrm{GL}(E)$, where E is a finite-dimensional complex vector space.*

Let ρ be a representation of a Lie group G on a finite-dimensional complex vector space E. We denote by $D\rho$ the differential of the Lie group morphism $\rho : G \to \mathrm{GL}(E)$. This is a morphism of Lie algebras of $\mathfrak{g}$ into $\mathfrak{gl}(E)^{\mathbb{R}}$ (and in particular into $\mathfrak{gl}(n,\mathbb{C})^{\mathbb{R}}$ if $E = \mathbb{C}^n$). Thus $D\rho$ is a representation of the Lie algebra $\mathfrak{g}$ on E in the sense of Definition 1.4. We have, by definition, for each $X \in \mathfrak{g}$,

$$(D\rho)(X) = \frac{\mathrm{d}}{\mathrm{d}t}\rho(\exp(tX))\big|_{t=0}. \tag{7.5}$$

In other words, $\rho(\exp(tX))$ is a one-parameter subgroup of $\mathrm{GL}(E)$, with infinitesimal generator $(D\rho)(X)$ and we then have, for every $t \in \mathbb{R}$,

$$\rho(\exp(tX)) = \exp(t(D\rho)(X)). \tag{7.6}$$

The following definition is thus justified.

Definition 7.6. *Let (E,ρ) be a representation of a Lie group G. The* differential *of the representation ρ is the Lie algebra representation, $D\rho : \mathfrak{g} \to \mathfrak{gl}(E)$, defined by (7.5).*

Theorem 7.7. *Let (E,ρ) be a representation of a Lie group G.*

(i) If $F \subset E$ is invariant under ρ, then F is invariant under $D\rho$.

(ii) If $D\rho$ is irreducible, then ρ is irreducible.

(iii) If $(E,(\ \mid\ ))$ is a finite-dimensional complex Hilbert space and if (E,ρ) is unitary*, then the Lie algebra representation $(E,D\rho)$ is* antihermitian*.*

(iv) Let (E_1,ρ_1) and (E_2,ρ_2) be representaions of G. If $\rho_1 \sim \rho_2$, then $D\rho_1 \sim D\rho_2$.

The converses are true if G is connected*.*

Proof. (i) If, for every $X \in \mathfrak{g}$ and for every $t \in \mathbb{R}$, F is invariant under $\rho(\exp(tX))$, then F is invariant under $(D\rho)(X)$.

(ii) is an immediate consequence of (i).

(iii) Suppose ρ is unitary. Then, for every $X \in \mathfrak{g}$ and every $t \in \mathbb{R}$,

$$(\rho(\exp(tX))u \mid \rho(\exp(tX))v) = (u \mid v),$$

for every u and $v \in E$. Differentiating with respect to t at $t = 0$, we obtain

$$(D\rho(X)u \mid v) + (u \mid D\rho(X)v) = 0, \tag{7.7}$$

which says that $D\rho$ is antihermitian.

(iv) If there exists a bijective linear map $T : E_1 \to E_2$ satisfying

$$\forall g \in G,\ \rho_2(g) \circ T = T \circ \rho_1(g)\ ,$$

then, by setting $g = \exp(tX)$ and differentiating with respect to t, we obtain the desired equation,

$$\forall X \in \mathfrak{g},\ (D\rho_2)(X) \circ T = T \circ (D\rho_1)(X).$$

The converses are true if G is connected because, in this case, every element of G is the product of a finite number of exponentials, by Theorem 6.5. We prove, for example, the converse of (iii). Suppose $D\rho$ is antihermitian. Using the equation $\frac{\mathrm{d}}{\mathrm{d}t}\rho(\exp tX) = D\rho(X) \circ \rho(\exp tX)$, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}(\rho(\exp tX)u \mid \rho(\exp tX)v)$$

$$= (D\rho(X)\rho(\exp tX)u \mid \rho(\exp tX)v) + (\rho(\exp tX)u \mid D\rho(X)\rho(\exp tX)v).$$

Now this quantity is zero for every $t \in \mathbb{R}$ by hypothesis. Thus the value of the scalar product $(\rho(\exp tX)u \mid \rho(\exp tX)v)$ remains constant and equal to the value $(u \mid v)$ it takes at $t = 0$. Thus ρ is unitary. □

If (E_1, ρ_1) and (E_2, ρ_2) are representations of G, then

$$D(\rho_1 \oplus \rho_2) = D\rho_1 \oplus D\rho_2,$$

and

$$D(\rho_1 \otimes \rho_2) = D\rho_1 \otimes \mathrm{Id}_{E_2} + \mathrm{Id}_{E_1} \otimes D\rho_2, \tag{7.8}$$

that is, explicitly, for every $X \in \mathfrak{g}$ and for every $v_1 \in E_1$, $v_2 \in E_2$,

$$D(\rho_1 \otimes \rho_2)(X)(v_1 \otimes v_2) = (D\rho_1)(X)v_1 \otimes v_2 + v_1 \otimes (D\rho_2)(X)v_2. \tag{7.9}$$

This equation is proved by taking the derivative at $t = 0$ of the equation defining $\rho_1 \otimes \rho_2$,

$$(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) = \rho_1(g)v_1 \otimes \rho_2(g)v_2,$$

for $g \in G$ and $v_1 \in E_1$, $v_2 \in E_2$, where $g = \exp(tX)$.

7.3 The Adjoint Representation

Let G be a Lie group with Lie algebra $\mathfrak{g}$. We consider the conjugation action of G on itself. For each $g \in G$,

$$\mathcal{C}_g\ : h \in G \mapsto ghg^{-1} \in G$$

is an automorphism of the Lie group G which leaves the identity element of G invariant. The differential of $\mathcal{C}_g$ at the identity is a linear mapping of $\mathfrak{g}$

into $\mathfrak{g}$, called the *adjoint action* of g and denoted by Ad_g. From the equation $\mathcal{C}_{gg'} = \mathcal{C}_g \circ \mathcal{C}_{g'}$, for g and $g' \in G$, we see that

$$\mathrm{Ad}_{gg'} = \mathrm{Ad}_g \circ \mathrm{Ad}_{g'}.$$

Since furthermore Ad_g is the identity of $\mathfrak{g}$ when $g = I$, this relation implies that $\mathrm{Ad} \ : g \in G \mapsto \mathrm{Ad}_g \in \mathrm{GL}(\mathfrak{g})$ is a representation of G on $\mathfrak{g}$.

Definition 7.8. *The representation* Ad *of* G *on* $\mathfrak{g}$ *is called the* adjoint representation of the Lie group G.

To the representation Ad of the Lie group G on $\mathfrak{g}$ there corresponds a representation $D(\mathrm{Ad})$ of the Lie algebra $\mathfrak{g}$ on itself, called the *adjoint representation of the Lie algebra* $\mathfrak{g}$, and denoted by ad. The following equation is thus true by definition, for every $t \in \mathbb{R}$ and each $X \in \mathfrak{g}$,

$$\mathrm{Ad}_{\exp(tX)} = \exp(t\mathrm{ad}_X). \tag{7.10}$$

Proposition 7.9. *(i) Let* A *be an invertible matrix belonging to the Lie group* G *and let* X *be a matrix belonging to the Lie algebra* $\mathfrak{g}$ *of* G*. Then,*

$$\mathrm{Ad}_A(X) = AXA^{-1}.$$

(ii) Let X *and* $Y \in \mathfrak{g}$*. Then,*

$$\mathrm{ad}_X(Y) = [X, Y].$$

(iii) Let X *and* $Y \in \mathfrak{g}$*. Then,*

$$\mathrm{ad}_{[X,Y]} = [\mathrm{ad}_X, \mathrm{ad}_Y]. \tag{7.11}$$

Proof. (i) By definition, for $B \in G, \mathcal{C}_A(B) = ABA^{-1}$, and thus

$$\mathrm{Ad}_A(X) = \frac{\mathrm{d}}{\mathrm{d}t} A \exp(tX) A^{-1}\big|_{t=0} = AXA^{-1}.$$

(ii) Furthermore,

$$\mathrm{ad}_X(Y) = \frac{\mathrm{d}}{\mathrm{d}t} \mathrm{Ad}_{\exp(tX)}(Y)\big|_{t=0} = \frac{\mathrm{d}}{\mathrm{d}t} \exp(tX) Y \exp(-tX)\big|_{t=0}$$

$$= XY - YX = [X, Y].$$

(iii) Equation (7.11) expresses the fact that ad is a Lie algebra representation. It is a direct consequence of the Jacobi identity. □

References

Most of the results of this chapter extend beyond the case of linear Lie groups to the case of Lie groups in general. Rossmann (2011) first treats the linear Lie groups and then introduces the general notion of Lie groups. In this spirit, one could also consult the book of Hall (2016), while Sagle and Walde (1973), Postnikov (1994), Marsden and Ratiu (2011), Sattinger and Weaver (1993), A. A. Kirillov (2008) treat the general case.

There are numerous more advanced textbooks, such as Bump (2013) and Godement (2017). The most classical book still available is Chevalley (1946). For the study of the differential geometry of Lie groups as manifolds, see for example Warner (1983) or Duistermaat–Kolk (2000); Warner's book also contains manifold cohomology theory, while Duistermaat and Kolk treat the representation theory of compact Lie groups. There one will find a proof of the Baker-Campbell-Hausdorff formula. The work of Naĭmark and Štern (1982) provides an exposition of the theory of Lie groups and of their representations. For a very complete treatment, see the 3 volumes of Bourbaki in the original French or in English translation.

For a proof of the theorem on the integration of a morphism of Lie algebras to a morphism of Lie groups in the simply connected case mentioned in Section 7.1, see, *e.g.*, Hall (2016), Naĭmark and Štern (1982), or Rossmann (2011).

Topics in History

Lie groups, "Liesche Gruppen" in German, "groupes de Lie" in French are named after Sophus Lie.

How old is "Lie theory"?

Lie created the theory of continuous groups in the winter of 1873–74, and the events of those few months were "the culminating tip of a historical iceberg"[1] that included Lie's previous research activity from 1869 on. Thus, in 2020, Lie theory turned approximately 150 years old.

Lie's earliest work was in collaboration with the young Felix Klein, seven years his junior, who was barely twenty years old when they met in Berlin in October 1869. They were together in Paris for 3 months in 1870 and, two years later, for several weeks in Göttingen, then in Erlangen. Their work dealt with a problem in geometry concerning the set, called a tetrahedral line complex, of the straight lines that meet a tetrahedron in 4 points whose anharmonic ratio is fixed. Lie studied the transformations leaving the vertices of a tetrahedron invariant. It was actually Klein who prepared the first results of Lie's research in 1869 for publication. When Lie and Klein later stayed in Berlin and Paris

[1] Thomas Hawkins, *Emergence of the Theory of Lie Groups*, Springer, 2000, p. 1. This book is a treasure trove of information on the history of Lie theory and I have largely borrowed from it, here and elsewhere.

they continued to collaborate on linear line complexes and the geometry of the curves in 3-space callled "W-curves", leading this time to joint articles in 1870 and 1871. In fact, the theory of Lie and Klein involved what was essentially a commutative Lie group of transformations but there was not yet a hint of the concept of a group in their joint publications. However, these publications mark the "emergence" of the theory of continuous groups of transformations, a forerunner of present-day theory of abstract Lie groups which would take shape as an analogue to the theory of permutation groups.

Lie is mistaken for a spy

The affairs of the world did not allow the fruitful cooperation of Lie with Klein to last undisturbed for a very long time. The beginning of the Franco-Prussian war on 17 July 1870 obliged Klein, a Prussian citizen, to leave Paris and return to Germany where he was soon appointed a lecturer in Göttingen. Meanwhile, Lie also, after some time, felt that he had to leave Paris. He decided to hike to Italy, and walked south until he was arrested in Fontainebleau, a distance of 65 kilometers (some 50 miles) from Paris. He was accused of being a German spy, his mathematical notes having been mistakenly identified by the non-mathematician police and military as a secret code! and imprisoned. When the French mathematician Gaston Darboux (1842–1917), whom Lie had visited in Paris, succeeded in obtaining his liberation, he fled to Italy, then returned to Germany and from there to Norway. In 1872 a chair of mathematics was created for him at the University of Christiania.

Lie groups

Weyl wrote in *The Theory of Groups and Quantum Mechanics*, the English translation of the book he first published in 1928: "The concept of a *group*, one of the oldest and most profound of mathematical concepts, was obtained by abstraction from that of a *group of transformations*." The word "group" was used by Lie and Klein only after 1871. When the concept first emerged in their work, the fact that the product of two elements of the group was an element of the group was its only defining property. It was later that they realized that it was necessary to require the inverse of each element to be an element of the group. Did they then write up a formal definition of a group, with an identity element and a composition law satisfying axioms? No, in fact Lie and Klein collaborated first on the geometry of surfaces, then on line complexes and later on the symplectic group of 4-dimensional complex space, always on groups of transformations – groups acting on spaces, the plane, or 3-dimensional space, or on a curve in the plane, or on a surface in 3-space – and it would be a long time before a formal definition of an abstract Lie group, as opposed to a Lie group of transformations, would be formulated.

After some time, what Lie had called a "continuous group of transformations" became known as a "Lie group of transformations". In fact, the term "Lie group" appeared in French while Lie was still alive, in Arthur Tresse's thesis of 1893, but Weyl in 1928 still wrote of "Lie's theory of continous groups

of transformations". Although Lie came close at that time to the modern definition of an abstract Lie group, a formal definition was introduced only much later in Oswald Veblen and J. H. C. Whitehead's *Foundations of Differential Geometry* in 1931, followed by Hassler Whitney's article "Differentiable manifolds" of 1936, and eventually the theory took shape at the hands of Claude Chevalley (1909–1984) in his book, *Lie Groups*, published in 1946.

Infinitesimal transformations

Nowadays, the Jacobi identity is an axiom, part of the definition of a Lie algebra, but in Lie's time it was a property that was satisfied by the "infinitesimal transformations" of a "continuous group of transformations".

The idea of an "infinitesimally small transformation" appeared in the mid-19th century, in the study of invariants and the theory of differential operators. James Joseph Sylvester (1814–1897) in 1852 and Cayley in 1854, then working as lawyers in London, introduced what they called "infinitely small transformations" which appear as differential operators, acting on real-valued functions of real variables, of the form

$$A(f) = \sum_{i=1}^{n} A_i(x_1, ..., x_n) \frac{\partial f}{\partial x_i}.$$

It was to Sylvester that the prestigious Copley medal, founded in 1731, was awarded in 1880 "for his long continued investigations and discoveries in mathematics" and in 1882 it was awarded to Arthur Cayley, "for his numerous profound and comprehensive researches in pure mathematics". When the De Morgan medal was awarded for the first time in 1887, the recipient was Sylvester.

The "Jacobi identity" is contained in a 180-page article (in Latin) by Jacobi on the integrability conditions for systems of first-order partial differential equations, which predated the work of Lie by more than 30 years. (Written in or before 1840, it was only published in 1862.) The identity that Jacobi proved does not concern the infinitesimal transformations of geometry but functions of variables $(q^1, ..., q^n, p_1, ..., p_n)$ and the bracket he considered was not the Lie bracket of vector fields – Lie would invent it in the mid-1870's –, but the "Poisson bracket" of such functions,

$$\{f, g\} = \frac{\partial f}{\partial q^1} \frac{\partial g}{\partial p_1} + ... + \frac{\partial f}{\partial q^n} \frac{\partial g}{\partial p_n}.$$

Jacobi proved the identity

$$\{\{f, g\}, h\} + \{\{g, h\}, f\} + \{\{h, f\}, g\} = 0,$$

and stated this property of the Poisson brackets as a theorem. Clearly Jacobi considered the identity that is now named after him to be a very important result since, after the statement of his theorem, he wrote: "Quod est gravissimum theorema".

In Lie's work, from 1874 on, the terminology can be confusing. We read of "groups of infinitesimal transformations" which are not groups and are what we now call "Lie algebras". We find brackets of functions which he simply called "expressions" but are "Poisson brackets" in modern terminology. We read about "Poisson brackets" of infinitesimal transformations which are nowadays generally called "Lie brackets". Some publications translated from Russian still use the term "Poisson brackets" to designate the Lie brackets of vector fields.

The following quotation from the English translation of Weyl's 1931 book summarizes the situation: "S. Lie was the first to undertake a systematic study of the construction of transformation groups from their infinitesimal elements. In fact, once they are known all the substitutions of the continuous group can be generated by integration, *i.e.*, by successive application of such infinitesimal elements – at least, all those which belong to the same connected "sheet" [connected component] as the identity" and he gives the following "Example: the proper orthogonal transformations can be obtained from the infinitesimal ones, but not the improper transformations with determinant -1".

Lie and the mathematicians of Paris

In 1882, Lie visited Paris a second time. He met the young mathematicians Henri Poincaré (1854–1912) and Émile Picard (1856–1941) who were genuinely interested in his theory of groups. In a letter to Klein, Lie wrote that he had met Poincaré who "would say occasionnally that all of mathematics is a story about groups". In 1888, Poincaré proposed to apply group-theoretic ideas to the theory of matrices (then called hypercomplex numbers). Later, when Poincaré and Picard had joined Darboux as members of the Académie des Sciences, they proposed the election of Lie as a corresponding member and he was elected by 30 votes against 3, and those 3 votes were split in favor of two other very distinguished candidates! The students at the École normale supérieure, Tresse among them, were encouraged to visit Lie in Leipzig to study his theory of groups in depth and, upon their return, Picard wrote to Lie: "Paris is actually becoming a center of groups; all this ferments in these young brains, and we shall have an excellent wine once the liquor has settled a little."

Lie and Engel

Friedrich Engel (1861–1941) received his doctorate in mathematics in Leipzig in 1883. On the suggestion of Felix Klein and Adolph Mayer, he went to work with Lie in Christiania in 1884. In less than a year they created a sizable manuscript of Lie's ideas that had been assembled by Engel. Their continued cooperation eventually produced the *Theorie der Transformationsgruppen* published in three volumes between 1888 and 1893. It was Engel who, much later, in the 20th century, edited Lie's papers, adding enlightening comments to the articles contained in the seven volumes of his collected works. In 1932, he was the co-author of a book in German entitled *The Lie theory of first-order partial differential equations*, a review of which actually tells us how difficult, by contrast, Lie's original publications were thought to be: "Only one who knows

the manner of writing in the difficult original papers of Lie can appreciate how much has been done in these lectures by the uniformisation and simplification of the proofs."

Exercises

Exercise 4.1 *The exponential mapping and Lie brackets.*

Let X and Y be real or complex $n \times n$ matrices. Show that for $t, s \in \mathbb{R}$, $k \in \mathbb{N}^*$,

(a)
$$\exp(tX)\exp(tY)\exp(-tX) = \exp\left(tY + t^2[X,Y] + \mathrm{O}(t^3)\right).$$

(b)
$$\exp(tX)\exp(tY)\exp(-tX)\exp(-tY) = \exp\left(t^2[X,Y] + \mathrm{O}(t^3)\right).$$

(c)
$$\exp t[X,Y] = \lim_{k\to+\infty}\left(\exp(t\frac{X}{k})\exp(\frac{Y}{k})\exp(-t\frac{X}{k})\exp(-\frac{Y}{k})\right)^{k^2}.$$

(d)
$$[X,Y] = \frac{\partial^2}{\partial t \partial s}\left(\exp(tX)\exp(sY)\exp(-tX)\exp(-sY)\right)\big|_{t=s=0}.$$

Exercise 4.2 *Infinite-dimensional Lie algebra: vector fields on $\mathbb{R}^n$.*

Consider the vector space of first-order linear differential operators on $\mathbb{R}^n$, of the form $D = \sum_{i=1}^n X^i \frac{\partial}{\partial x^i}$, where each X^i is a smooth function on $\mathbb{R}^n$. Show that the commutator gives this vector space the structure of an infinite-dimensional Lie algebra over $\mathbb{R}$.

Exercise 4.3 *Infinite-dimensional Lie algebra: vector fields on the circle (the Virasoro algebra).*

Consider the vector space E of smooth maps from the circle $S^1 = \{\mathrm{e}^{i\theta} \mid 0 \leq \theta < 2\pi\}$ into $\mathbb{C}$ and set, for $f \in E$, $X_f = f(\theta)\frac{\mathrm{d}}{\mathrm{d}\theta}$.

(a) Let $f, g \in E$. Show that there exists $h \in E$ such that $[X_f, X_g] = X_h$. Conclude that E is a Lie algebra, whose Lie bracket we denote as usual by $[\ ,\ ]$. Calculate $[\mathrm{e}^{ni\theta}, \mathrm{e}^{mi\theta}]$, for $n, m \in \mathbb{Z}$. This Lie algebra is called the *Witt algebra*.

(b) Set

$$c(f,g) = \int_0^{2\pi} f(\theta)\frac{\mathrm{d}^3 g}{\mathrm{d}\theta^3}(\theta)\,\mathrm{d}\theta.$$

Let $\widetilde{E} = E \oplus \mathbb{C}$, and let k be a complex constant. For every (f,λ) and $(g,\mu) \in \widetilde{E}$, set

$$[(f,\lambda),(g,\mu)]_k = ([f,g], k\,c(f,g)).$$

Show that $(\widetilde{E}, [\ ,\ ]_k)$ is a Lie algebra. Calculate $[(\mathrm{e}^{ni\theta},\lambda),(\mathrm{e}^{mi\theta},\mu)]_k$, for $n,m \in \mathbb{Z}$. This Lie algebra is called the *Virasoro algebra*.

Exercise 4.4 *Example of a non-surjective exponential map.*

(a) Show that the exponential map of $\mathfrak{sl}(2,\mathbb{C})$ into $\mathrm{SL}(2,\mathbb{C})$ is not surjective.

(b) Is the exponential map always injective?

Exercise 4.5 *The Heisenberg group.*

(a) Let $X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $H = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Calculate $[H,X]$, $[H,Y]$ and $[X,Y]$.

Calculate $\exp(tX)$, $\exp(tY)$ and $\exp(tH)$, for $t \in \mathbb{R}$.

The real Lie algebra $\mathfrak{g}$ generated by X,Y,H is called the *Heisenberg algebra* and the group G of real matrices of the form $\begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$ is called the *Heisenberg group*.

Show that the exponential map is a bijection of the Heisenberg algebra onto the Heisenberg group.

(b) What is the center of $\mathfrak{g}$? of G?

(c) More generally, we consider, for every positive integer n, the Lie algebra $\mathfrak{g}$ generated by Q_k, P_k, H, $k = 1,\dots,n$, with the commutation relations

$$[Q_k,H] = [P_k,H] = [Q_k,Q_\ell] = [P_k,P_\ell] = 0,$$

$$[Q_k,P_\ell] = \alpha\delta_{k\ell}H,$$

where α is a non-zero scalar. (The equations $[Q_k,P_\ell] = i\hbar\delta_{k\ell}$ are the *Heisenberg relations*.) Show that this Lie algebra has no finite-dimensional irreducible representation such that $\rho(H) \neq 0$, and that the irreducible finite-dimensional representations such that $\rho(H) = 0$ are one-dimensional.

Exercise 4.6 *Differential of the exponential map.*

Let G be a Lie group and $\mathfrak{g}$ its Lie algebra. Show that $\mathrm{d}_X(\exp)$, the differential at the point $X \in \mathfrak{g}$ of the map $\exp : \mathfrak{g} \to G$, satisfies

$$\exp(-X)\mathrm{d}_X(\exp) = \frac{1 - \exp(-\mathrm{ad}_X)}{\mathrm{ad}_X},$$

where the right-hand side denotes the sum of a series. [Hint: Consider, for X and Y in $\mathfrak{g}$, $A_s(X,Y) = \exp(-sX)\frac{\partial}{\partial t}\exp(s(X+tY))\big|_{t=0}$ and calculate $\frac{\partial}{\partial s}A_s(X,Y)$.]

Exercise 4.7 *Killing form.*

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra on $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The *Killing form* of $\mathfrak{g}$ is the $\mathbb{K}$-bilinear form on $\mathfrak{g}$ defined by

$$\forall X, Y \in \mathfrak{g}, \quad K(X,Y) = \mathrm{Tr}(\mathrm{ad}_X \circ \mathrm{ad}_Y).$$

(a) Show that for every $X, Y, Z \in \mathfrak{g}$,

$$K([X,Y],Z) + K(Y,[X,Z]) = 0.$$

(b) Let G be a Lie group with Lie algebra $\mathfrak{g}$. Show that K is an *invariant* bilinear form, that is, for every $g \in G$,

$$K(\mathrm{Ad}_g X, \mathrm{Ad}_g Y) = K(X,Y).$$

(c) Find the Killing form and determine its signature for each of the Lie algebras $\mathfrak{sl}(2,\mathbb{C}), \mathfrak{sl}(2,\mathbb{R})$ and $\mathfrak{so}(3)$.

Exercise 4.8 *Dimension of the symplectic groups.*

Let $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. Calculate the dimension of the Lie algebra

$$\mathfrak{sp}(n,\mathbb{C}) = \{X \in \mathfrak{sl}(2n,\mathbb{C}) \mid XJ + J\,{}^tX = 0\}.$$

What are the dimensions of $\mathrm{Sp}(n,\mathbb{C})$ and of $\mathrm{Sp}(n) = \mathrm{Sp}(n,\mathbb{C}) \cap \mathrm{U}(2n)$?

Exercise 4.9 *Derivations and the adjoint representation.*

A *derivation* of a Lie algebra $\mathfrak{g}$ is a linear mapping of $\mathfrak{g}$ into $\mathfrak{g}$ satisfying, for every $X, Y \in \mathfrak{g}$,

$$D[X,Y] = [DX,Y] + [X,DY].$$

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Let $\mathfrak{h}$ be a Lie algebra. Show that if ρ is a representation of G on $\mathfrak{h}$ by automorphisms, $D\rho$ is a representation of $\mathfrak{g}$ on $\mathfrak{h}$ by derivations. What can one say about the representations Ad and ad?

Exercise 4.10 *Coadjoint representation.*

Let G be a Lie group with Lie algebra $\mathfrak{g}$ and let $\mathfrak{g}^*$ be the dual vector space of $\mathfrak{g}$. By definition, the *coadjoint representation* of G on $\mathfrak{g}^*$ is the dual of the

adjoint representation. The *coadjoint orbit* of an element ξ of $\mathfrak{g}^*$ is its orbit under the coadjoint action of G.

Find the coadjoint orbits in the case where $G = \mathrm{SO}(3)$.

Exercise 4.11 *Real forms.*

Show that $\mathfrak{so}(3), \mathfrak{su}(2), \mathfrak{so}(2,1), \mathfrak{sl}(2,\mathbb{R})$ are real forms of $\mathfrak{sl}(2,\mathbb{C})$. Are they isomorphic as real Lie algebras?

Exercise 4.12 *The connected component of the identity of a Lie group.*

Let G be a Lie group and let G_0 be the connected component of the identity of G.

(a) Show that G_0 is a normal subgroup of G.

(b) Show that the Lie groups G and G_0 have the same Lie algebra.

Exercise 4.13 *Connected components.*

(a) Show that $\mathrm{GL}(n,\mathbb{C})$, for $n \geq 2$, and $\mathrm{U}(n)$, for $n \geq 1$, are connected. Find the connected component of the identity of $\mathrm{GL}(n,\mathbb{R})$.

(b) Study the connected component of the identity of the Lorentz group. How many connected components does the Lorentz group have?

Exercise 4.14 *Ideals of Lie algebras and normal subgroups.*

Let G be a connected Lie group with Lie algebra $\mathfrak{g}$, and let H be a closed connected subgroup of G. Show that H is normal if and only if the Lie algebra $\mathfrak{h}$ of H is an ideal of $\mathfrak{g}$.

Exercise 4.15 *Invariant measure on Lie groups.*

Suppose that the elements of a Lie group G of dimension p are parametrized by $x = (x_1, x_2, \ldots, x_p)$. We denote by $g^{-1}\frac{\mathrm{d}g}{\mathrm{d}x}$ the matrix depending on x whose columns are the components of the vectors $g^{-1}(x)\frac{\partial g}{\partial x_k}(x)$, $k = 1, 2, \ldots, p$, in a basis of $\mathfrak{g}$. Show that $|\det(g^{-1}\frac{\mathrm{d}g}{\mathrm{d}x})|$ is a left-invariant volume element on G.



Wilhelm Killing (1847-1923) around 1890, shortly after the publication of his innovative work on the properties of infinitesimal isometries of "space forms", generalizations of Euclidean space. This study brought him to the idea that much later was called "Lie algebra". Eventually he classified semi-simple Lie algebras.

(Portrait collection, Universitäts- und Landesbibliothek Münster)



Élie Cartan, who was born in Dolomieu (Isère) in 1869 and died in Paris in 1951, is famous for his work on Lie groups and algebras and his fundamental contributions to differential geometry. Here he is on vacation in 1931, with two of his sons, his daughter Hélène and a nephew. On the far left is his son Henri (1904-2008), one of the founders of the Bourbaki group and a great mathematician of the twentieth century.

(Private collection)

Chapter 5

Lie Groups SU(2) and SO(3)



We study the 3-dimensional Lie groups SU(2) and SO(3) whose Lie algebras are isomorphic, and we prove that SU(2) is the universal cover of SO(3).

1 The Lie Algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$

We know that $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are real Lie algebras of dimension 3. We shall show that they are isomorphic by showing that there are bases of each in which the commutation relations are the same.

1.1 Bases of $\mathfrak{su}(2)$

The Lie algebra $\mathfrak{su}(2) = \left\{X \in \mathfrak{gl}(2,\mathbb{C}) \mid {}^t\overline{X} + X = 0, \operatorname{Tr} X = 0\right\}$ is the real vector space of dimension 3 of traceless antihermitian 2×2 complex matrices.

(a) The three linearly independent matrices

$$\xi_1 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},\ \xi_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},\ \xi_3 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},$$

form a basis over $\mathbb{R}$ of $\mathfrak{su}(2)$, and satisfy the commutation relations

$$[\xi_k, \xi_\ell] = \xi_m,$$

where k, ℓ, m is a cyclic permutation of $1, 2, 3$.

(b) In physics, one considers the Pauli matrices, which are the traceless Hermitian matrices

$$\sigma_1 = -2i\xi_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \sigma_2 = 2i\xi_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\ \sigma_3 = -2i\xi_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy the commutation relations

$$[\sigma_k, \sigma_\ell] = 2i\sigma_m.$$

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One also sometimes introduces the matrices,

$$\widetilde{\sigma}_1 = \frac{1}{2}\sigma_1 = -i\xi_1,\ \widetilde{\sigma}_2 = \frac{1}{2}\sigma_2 = i\xi_2,\ \widetilde{\sigma}_3 = \frac{1}{2}\sigma_3 = -i\xi_3,$$

which satisfy the commutation relations

$$[\widetilde{\sigma}_k, \widetilde{\sigma}_\ell] = i\widetilde{\sigma}_m.$$

(c) We also introduce the matrices,

$$J_k = i\xi_k$$

which satisfy the commutation relations,

$$[J_k, J_\ell] = iJ_m.$$

We also consider the matrices,

$$J_3 = -\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},\ J_+ = J_1 + iJ_2 = \begin{pmatrix}0 & 0\\ -1 & 0\end{pmatrix},\ J_- = J_1 - iJ_2 = \begin{pmatrix}0 & -1\\ 0 & 0\end{pmatrix}.$$

We have

$$J_3 = i\xi_3 = -\widetilde{\sigma}_3 = -\frac{1}{2}\sigma_3,$$
$$J_\pm = J_1 \pm iJ_2 = i\xi_1 \mp \xi_2 = -\widetilde{\sigma}_1 \pm i\widetilde{\sigma}_2 = \frac{1}{2}(-\sigma_1 \pm i\sigma_2),$$

and so

$$\xi_3 = -iJ_3,\ \ \xi_1 = -\frac{i}{2}(J_+ + J_-),\ \ \xi_2 = -\frac{1}{2}(J_+ - J_-).$$

The matrices J_3, J_+, J_- satisfy the commutation relations,

$$[J_+, J_-] = 2J_3,\ [J_3, J_\pm] = \pm J_\pm.$$

(d) Finally, we introduce the basis of $\mathfrak{su}(2)$,

$$\mathcal{I} = \begin{pmatrix}0 & i\\ i & 0\end{pmatrix},\ \mathcal{J} = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix},\ \mathcal{K} = \begin{pmatrix}i & 0\\ 0 & -i\end{pmatrix}.$$

We see immediately that

$$\mathcal{I}^2 = \mathcal{J}^2 = \mathcal{K}^2 = -I,$$

$$\mathcal{I}\mathcal{J} = \mathcal{K} = -\mathcal{J}\mathcal{I},\ \mathcal{J}\mathcal{K} = \mathcal{I} = -\mathcal{K}\mathcal{J},\ \mathcal{K}\mathcal{I} = \mathcal{J} = -\mathcal{I}\mathcal{K},$$

and the commutation relations of $\mathfrak{su}(2)$ can thus be written

$$[\mathcal{I}, \mathcal{J}] = 2\mathcal{K},\ [\mathcal{J}, \mathcal{K}] = 2\mathcal{I},\ [\mathcal{K}, \mathcal{I}] = 2\mathcal{J}.$$

We have
$$\mathcal{I} = 2\xi_1 = 2i\widetilde{\sigma}_1 = i\sigma_1 = -2iJ_1,$$
$$\mathcal{J} = 2\xi_2 = -2i\widetilde{\sigma}_2 = -i\sigma_2 = -2iJ_2,$$
$$\mathcal{K} = 2\xi_3 = 2i\widetilde{\sigma}_3 = i\sigma_3 = -2iJ_3.$$

A matrix X of $\mathfrak{su}(2)$ can be written $X = x_1\mathcal{I} + x_2\mathcal{J} + x_3\mathcal{K}$, *i.e.*,

$$X = \begin{pmatrix} ix_3 & -x_2 + ix_1 \\ x_2 + ix_1 & -ix_3 \end{pmatrix},$$

where x_1, x_2, x_3 are real numbers. We have then

$$\det X = \|\overrightarrow{X}\|^2, \tag{1.1}$$

where $\|\overrightarrow{X}\|$ is the Euclidean norm of the vector $\overrightarrow{X}$ with components (x_1, x_2, x_3).

Interpretation in terms of quaternions. Let $\mathbb{H}$ be the associative algebra of the quaternions, the real vector space of dimension 4, with basis $(1, i, j, k)$, with the multiplication defined by

$$i^2 = j^2 = k^2 = -1\,,\; ij = -ji = k\,,\; jk = -kj = i\,,\; ki = -ik = j.$$

The map, $i \mapsto \mathcal{I}\,,\; j \mapsto \mathcal{J}\,,\; k \mapsto \mathcal{K}$, defines a bijection of the vector space $\mathfrak{su}(2)$ onto the vector subspace of $\mathbb{H}$ generated by i, j and k, called the vector space of *pure quaternions*, and denoted by $\mathbb{H}_0$. The vector subspace $\mathbb{H}_0$ is not a subalgebra of the associative algebra $\mathbb{H}$, but $\mathbb{H}_0$ is a Lie algebra under the commutator and the bijection above is an isomorphism of Lie algebras.

Remark. All the triples of matrices considered above form bases over $\mathbb{C}$ of $(\mathfrak{su}(2))^{\mathbb{C}} = \mathfrak{sl}(2, \mathbb{C})$. While the ξ_k and the matrices $\mathcal{I}, \mathcal{J}, \mathcal{K}$ belong to $\mathfrak{su}(2)$, the Pauli matrices σ_k and the matrices J_k belong to $i\mathfrak{su}(2)$.

1.2 Bases of $\mathfrak{so}(3)$

The Lie algebra $\mathfrak{so}(3)$ is the vector space of antisymmetric real 3×3 matrices. We have already seen that the matrices

$$\eta_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},\; \eta_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix},\; \eta_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

form a basis of this Lie algebra, with the commutation relations,

$$[\eta_k, \eta_\ell] = \eta_m.$$

It is clear that $\eta_k \mapsto e_k$, where (e_1, e_2, e_3) is the canonical basis of $\mathbb{R}^3$, defines an isomorphism of the Lie algebra $\mathfrak{so}(3)$ onto the Lie algebra $(\mathbb{R}^3, \wedge)$, the vector space $\mathbb{R}^3$ equipped with the cross product.

On the other hand,

$$\mathfrak{su}(2) \approx \mathfrak{so}(3).$$

More precisely, an isomorphism of Lie algebras is realized by the $\mathbb{R}$-linear map defined by $\xi_k \mapsto \eta_k$. From the fact that $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are isomorphic, we deduce that $(\mathfrak{so}(3))^{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$.

We may also consider the matrices

$$\widehat{J}_k = i\eta_k.$$

In the preceding isomorphism, $\widetilde{\sigma}_1 \mapsto -\widehat{J}_1,\ \widetilde{\sigma}_2 \mapsto \widehat{J}_2, \widetilde{\sigma}_3 \mapsto -\widehat{J}_3$.

1.3 Bases of $\mathfrak{sl}(2,\mathbb{C})$

Other than the bases already described above, the most frequently used basis is

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},\ X_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},\ X_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

which satisfies the commutation relations

$$[H, X_\pm] = \pm 2X_\pm,\ [X_+, X_-] = H.$$

We have

$$H = -2J_3 = -2i\xi_3 = 2\widetilde{\sigma}_3 = \sigma_3,$$

$$X_\pm = -J_\mp = -J_1 \pm iJ_2 = -(i\xi_1 \pm \xi_2) = \widetilde{\sigma}_1 \pm i\widetilde{\sigma}_2 = \frac{1}{2}(\sigma_1 \pm i\sigma_2),$$

and thus

$$J_3 = -\frac{1}{2}H, \quad , J_\pm = -X_\mp,$$

and

$$\mathcal{I} = i(X_+ + X_-), \quad \mathcal{J} = -(X_+ - X_-), \quad \mathcal{K} = iH.$$

2 The Covering Morphism of SU(2) onto SO(3)

We shall see that, although the Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ of the Lie groups SU(2) and SO(3) are isomorphic, the groups themselves are not. The one, SU(2), is connected and simply connected, but the other, SO(3), is connected but not simply connected, and there is a surjective group morphism of the first onto the second, whose kernel consists of the elements I and $-I$ of SU(2).

2.1 The Lie Group SO(3)

Every orthogonal transformation of $\mathbb{R}^3$ of determinant $+1$ leaves a unit vector a of $\mathbb{R}^3$ invariant. This is then a rotation through an angle $t \in \mathbb{R}/2\pi\mathbb{Z}$, denoted by $\mathrm{Rot}(a,t)$. Thus an element of SO(3) is determined by $a \in \mathbb{R}^3$, $\|a\| = 1$, and $t \in [0, 2\pi)$. The image of an element $x \in \mathbb{R}^3$ by $\mathrm{Rot}(a,t)$ is

$$\mathrm{Rot}(a,t)(x) = x + (1 - \cos t)a \wedge (a \wedge x) + \sin t \; a \wedge x. \tag{2.1}$$

For the proof, we observe first that if x is orthogonal to a, then,

$$\mathrm{Rot}(a,t)(x) = \cos t \; x + \sin t \; a \wedge x.$$

Next we decompose any x as $\lambda a + \mu b$, where b is orthogonal to a and we exploit the linearity of $\mathrm{Rot}(a,t)$. Hence

$$\mathrm{Rot}(a,t)(x) = \cos t \; x + (1 - \cos t)(x \mid a)a + \sin t \; a \wedge x.$$

Finally we use the triple product formula, $u \wedge (v \wedge w) = (u \mid w)v - (u \mid v)w$, which gives $(x \mid a)a = x + a \wedge (a \wedge x)$, which proves (2.1).

Proposition 2.1. *Two rotations are conjugate in* SO(3) *if and only if they have equal or opposite angles.*

Proof. It is clear that, for $t \neq 0$, the pair (a', t') defines the same rotation as the pair (a,t) if and only if $(a',t') = (a,t)$ or $(a',t') = (-a,-t)$. We shall now prove that rotations $\mathrm{Rot}(a,t)$ and $\mathrm{Rot}(a',t')$ are conjugate in SO(3) if and only if $t' = \pm t$. In fact, if a rotation R leaves a invariant, then, for each $g \in \mathrm{SO}(3)$, the rotation gRg^{-1} leaves ga invariant. More precisely,

$$\forall\, g \in \mathrm{SO}(3),\;\; g \; \mathrm{Rot}(a,t) \; g^{-1} = \mathrm{Rot}(ga,t). \tag{2.2}$$

Therefore, if $R' = \mathrm{Rot}(a',t')$ is a rotation conjugate to $R = \mathrm{Rot}(a,t)$ by an element $g \in \mathrm{SO}(3)$, then $\mathrm{Rot}(a',t') = \mathrm{Rot}(ga,t)$, and therefore R and R' have equal or opposite angles. Conversely, each rotation $\mathrm{Rot}(a',t)$ is conjugate to $\mathrm{Rot}(a,t)$ by an element $g \in \mathrm{SO}(3)$ such that $a' = ga$, and each rotation $\mathrm{Rot}(a',-t)$ is conjugate to $\mathrm{Rot}(a,t)$ by an element $g \in \mathrm{SO}(3)$ such that $-a' = ga$. □

Surjectivity of the exponential map of $\mathfrak{so}(3)$ *onto* SO(3). Let η_k be the infinitesimal generators of the one-parameter groups of rotations around the axes e_k, $k = 1, 2, 3$. By definition

$$\exp(t\eta_k) = \mathrm{Rot}(e_k, t).$$

Let a be any unit vector in $\mathbb{R}^3$. We choose $k = 1, 2$ or 3, and let g be an element of SO(3) such that $a = g(e_k)$. Then,

$$\mathrm{Rot}(a,t) = g \; \mathrm{Rot}(e_k,t) \; g^{-1} = g \;\; \exp(t\eta_k) \; g^{-1} = \exp(t \; g \; \eta_k \; g^{-1}).$$

By Theorem 5.1 (iii) of Chapter 4, the element $g\eta_k g^{-1}$ belongs to $\mathfrak{so}(3)$. We have thus shown that the exponential map is surjective from $\mathfrak{so}(3)$ onto SO(3). We deduce that SO(3) is arc-connected, hence connected, and that it is the connected component of the identity of O(3). We remark that the exponential map is clearly not injective from $\mathfrak{so}(3)$ into SO(3) because, for $X = \eta_k$ ($k = 1, 2$ or 3) and ℓ integer, $\exp((t + 2\ell\pi)X) = \exp(tX)$.

Identification of $\mathfrak{so}(3)$ *with* $\mathbb{R}^3$. Let $a \mapsto M_a$ be the $\mathbb{R}$-linear map from $\mathbb{R}^3$ onto $\mathfrak{so}(3)$ defined by $e_k \mapsto \eta_k$. The equation $M_{e_k}(x) = \eta_k(x) = e_k \wedge x$ implies by linearity that $M_a(x) = a \wedge x$, for every $x \in \mathbb{R}^3$ and, for $g \in \mathrm{SO}(3)$, $M_{ga}(x) = ga \wedge x = g(a \wedge g^{-1}x) = gM_a g^{-1}(x)$. Thus

$$\forall g \in \mathrm{SO}(3),\ \forall a \in \mathbb{R}^3,\ M_{ga} = \mathrm{Ad}_g\, M_a.$$

Proposition 2.2. *For each* $a \in \mathbb{R}^3$ *of unit norm and for every* $t \in \mathbb{R}$,

$$\exp(tM_a) = \mathrm{Rot}(a, t). \tag{2.3}$$

Proof. The two sides are one-parameter subgroups of SO(3). By (2.1),

$$\forall x \in \mathbb{R}^3,\ \frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Rot}(a,t)(x)\big|_{t=0} = a \wedge x,$$

while

$$\forall x \in \mathbb{R}^3,\ \frac{\mathrm{d}}{\mathrm{d}t}\exp(tM_a)(x)\big|_{t=0} = M_a(x).$$

which proves (2.3). □

Remark. Despite the differences of notation, formula (2.3) is the same as a formula to be found in physics books,

$$\mathcal{D}(\overrightarrow{u}, \alpha) = \mathrm{e}^{-i\alpha\,\overrightarrow{u}\cdot\overrightarrow{J}},$$

where $\overrightarrow{J}$ denotes the triple of matrices $i\eta_1, i\eta_2, i\eta_3$, which we have denoted by $\widehat{J}_1, \widehat{J}_2, \widehat{J}_3$ (but which are often denoted by J_1, J_2, J_3). This equation says that the rotation determined by the unit vector $\overrightarrow{u} = (u_1, u_2, u_3)$ and angle α is the exponential of the matrix $-i\alpha(u_1\widehat{J}_1 + u_2\widehat{J}_2 + u_3\widehat{J}_3)$. As this matrix is equal to $\alpha(u_1\eta_1 + u_2\eta_2 + u_3\eta_3)$, this equation is in fact the same as (2.3). In Tung (1985), this equation is (7.2-8), written

$$R_n(\psi) = \mathrm{e}^{-i\psi J_k n^k},$$

where n is a unit vector with components n^1, n^2, n^3 and ψ is the angle of the rotation.

2.2 The Lie Group SU(2)

The Lie group $\mathrm{SU}(2) = \{A \in \mathrm{GL}(2,\mathbb{C}) \mid A\,{}^t\overline{A} = I,\ \det A = 1\}$ is diffeomorphic to the sphere $\mathrm{S}^3 \subset \mathbb{R}^4$ because

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \mid a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\}.$$

In fact, for a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the equations $A\,{}^t\overline{A} = I$ and $\det A = 1$ imply that $|a|^2 + |b|^2 = 1$, $|c|^2 + |d|^2 = 1$, $ad - bc = 1$, $a\bar{c} + b\bar{d} = 0$, whence $ad\bar{c} + bd\bar{d} = 0$, whence $c = -\bar{b}$ and, similarly, $d = \bar{a}$. The elements of the Lie group SU(2) thus depend on three independent real parameters.

Like the sphere S^3, the group SU(2) is compact, connected and simply connected

Every unitary matrix is diagonalizable by means of a unitary matrix. Because the eigenvalues of a unitary matrix are of modulus 1, for every $A \in \mathrm{SU}(2)$, there is a real number t and a special unitary matrix g such that

$$A = g \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} g^{-1}.$$

Furthermore $\begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix}$ and $\begin{pmatrix} e^{-it} & 0 \\ 0 & e^{it} \end{pmatrix}$ are conjugate via $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ in SU(2).

Surjectivity of the exponential map of $\mathfrak{su}(2)$ *on* SU(2). Let $A = g\begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} g^{-1}$ be a matrix of SU(2). From the equation

$$\begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} = \exp(t\mathcal{K}),$$

we deduce that

$$A = \exp(tg\mathcal{K}g^{-1}),$$

with $g\mathcal{K}g^{-1} \in \mathfrak{su}(2)$. Thus the exponential map is surjective from $\mathfrak{su}(2)$ onto SU(2).

Identification of $\mathfrak{su}(2)$ *with* $\mathbb{R}^3$. Let $u \mapsto X_u$ be the $\mathbb{R}$-linear map from $\mathbb{R}^3$ onto $\mathfrak{su}(2)$ defined by

$$e_1 \mapsto \mathcal{I}, e_2 \mapsto \mathcal{J}, e_3 \mapsto \mathcal{K}.$$

Then (see Formula (1.1)),

$$\det X_u = \|u\|^2,$$

and, in addition,

$$(X_u)^2 = -(\det X_u) I.$$

We deduce that

Lemma 2.3. *For each* $X \in \mathfrak{su}(2)$, $X^2 = -(\det X)I$.

Proposition 2.4. *For each* $X \in \mathfrak{su}(2)$ *such that* $\det X = 1$*, and for every* $t \in \mathbb{R}$*,*

$$\exp(tX) = \cos t\ I + \sin t\ X.$$

Proof. It is clear that the two sides have X for derivative at $t = 0$. It thus suffices to show that, if $\det X = 1$, the map $t \mapsto \cos t\ I + \sin t\ X$ is a one-parameter subgroup of $\mathrm{GL}(2, \mathbb{C})$. For t and $s \in \mathbb{R}$,

$$(\cos t\ I + \sin t\ X)(\cos s\ I + \sin s\ X)$$

$$= \cos t \cos s\ I + \sin t \sin s\ X^2 + (\sin t \cos s + \cos t \sin s)X.$$

Thus, by the lemma, if $\det X = 1$,

$$(\cos t\ I + \sin t\ X)(\cos s\ I + \sin s\ X) = \cos(t+s)I + \sin(t+s)X.$$

One can also use the definition of the exponential as a series of matrices and the preceding lemma. □

As a consequence of Proposition 2.4, we see that each matrix of SU(2) can be written $\alpha_0 I + \alpha_1 \mathcal{I} + \alpha_2 \mathcal{J} + \alpha_3 \mathcal{K}$, where the vector $(\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ of $\mathbb{R}^4$ has norm 1. Thus SU(2) is identified with the group of quaternions of unit norm.

2.3 Projection of SU(2) onto SO(3)

We consider the adjoint representation of SU(2) on $\mathfrak{su}(2)$,

$$\mathrm{Ad} : \mathrm{SU}(2) \to \mathrm{GL}(\mathfrak{su}(2)).$$

When we identify $\mathfrak{su}(2)$ with $\mathbb{R}^3$ by means of the isomorphism $u \mapsto X_u$, the map Ad is identified with a map

$$\varphi : \mathrm{SU}(2) \to \mathrm{GL}(3, \mathbb{R}).$$

This is the map that, to $g \in \mathrm{SU}(2)$, associates the matrix of Ad_g in the basis $(\mathcal{I}, \mathcal{J}, \mathcal{K})$ of $\mathfrak{su}(2)$.

For each $g \in \mathrm{SU}(2)$, $\mathrm{Ad}_g : \mathfrak{su}(2) \to \mathfrak{su}(2)$ is the map $X \mapsto gXg^{-1}$. This map preserves determinants. By (1.1), in the identification of $\mathfrak{su}(2)$ with $\mathbb{R}^3$, the determinant of a matrix corresponds to the square of the Euclidean norm of the vector, hence for every $g \in \mathrm{SU}(2)$, $\varphi(g)$ preserves the norm, thus $\varphi(\mathrm{SU}(2)) \subset \mathrm{O}(3)$. In fact, since φ is continuous and SU(2) is connected, $\varphi(\mathrm{SU}(2)) \subset \mathrm{SO}(3)$. We know that the map $\varphi : \mathrm{SU}(2) \to \mathrm{SO}(3)$ is a morphism of groups. We shall show that the morphism φ is surjective from SU(2) onto SO(3), and we shall determine its kernel.

Proposition 2.5. (i) *In the basis* $(\mathcal{I}, \mathcal{J}, \mathcal{K})$ *of* $\mathfrak{su}(2)$*, the matrix of* ad_{ξ_k} *is* η_k*, for* $k = 1, 2, 3$*.*

(ii) *For each* $t \in \mathbb{R}$*,*

$$\varphi(\exp(t\xi_k)) = \exp(t\eta_k).$$

(*iii*) *For each* $u \in \mathbb{R}^3$ *of unit norm, and for every* $t \in \mathbb{R}$,

$$\varphi(\exp(tX_u)) = \mathrm{Rot}(u, 2t).$$

(*iv*) *The map* $\varphi : \mathrm{SU}(2) \to \mathrm{SO}(3)$ *is surjective.*

Proof. (i) It suffices to write $\mathrm{ad}_{\xi_k}(\xi_\ell) = [\xi_k, \xi_\ell] = \xi_m$ and we immediately obtain the matrix of ad_{ξ_k} in the basis (ξ_1, ξ_2, ξ_3), which is also its matrix in the basis $(\mathcal{I}, \mathcal{J}, \mathcal{K})$.

(ii) This equality is the consequence of (i).

(iii) Let the components of $u \in \mathbb{R}^3$ be u_1, u_2, u_3. Then

$$X_u = u_1\mathcal{I} + u_2\mathcal{J} + u_3\mathcal{K} = 2(u_1\xi_1 + u_2\xi_2 + u_3\xi_3).$$

Thus the matrix of ad_{X_u} in the basis $(\mathcal{I}, \mathcal{J}, \mathcal{K})$ is $2(u_1\eta_1 + u_2\eta_2 + u_3\eta_3) = 2M_u$. Therefore the matrix of $\mathrm{Ad}_{\exp(tX_u)}$ in this basis is $\exp(2tM_u)$. If u is of unit norm, then it is $\mathrm{Rot}(u, 2t)$.

(iv) The surjectivity of φ is a consequence of (iii). □

The kernel of Ad is $\{g \in \mathrm{SU}(2) \mid \mathrm{Ad}_g = \mathrm{Id}_{\mathfrak{su}(2)}\}$. Let $g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$ be such that $g\mathcal{K} = \mathcal{K}g$, $g\mathcal{I} = \mathcal{I}g$, $g\mathcal{J} = \mathcal{J}g$. One sees that $b = 0$ and a is real. Because $\det g = 1$, $g = I$ or $g = -I$. Thus $\mathrm{Ker}\,\varphi = \{I, -I\}$. The morphism φ is thus not injective, as its kernel contains two elements.

Because φ is continuous and surjective, and since SU(2) is simply connected, we can state:

Theorem 2.6. *The Lie group* SU(2) *is the double-sheeted universal cover of the Lie group* SO(3).

References

The books of Rossmann (2011) and of Hall (2016), among a great many others, study the group of rotations SO(3) and its universal cover SU(2). See Chapter 9 for more on local homeomorphisms and universal covers.

In this chapter and the next we introduced some notations used in physics for which see Basdevant–Dalibard (2005), Ludwig–Falter (2012), Tung (1985), or Edmonds (1996).

Topics in History

"At the foundation of much of what we do today"

It is difficult to over-estimate the importance of the work of Élie Cartan (1869–1951) in the representation theory of Lie groups. It began in 1893 with the

publication of two 3-page notes in the *Comptes rendus de l'Académie des sciences* and then an article in German which contained a detailed exposition of the results of those short notes, all three on the structure of simple groups of transformations. They were followed in 1894 by his 150 page "The structure of finite and continuous groups of transformations", the doctoral thesis which he had defended that same year at the University of Paris, in the venerable Sorbonne, whose new buildings had been inaugurated five years earlier (and were only completed in 1901).

In fact, Cartan's works covered a great many fields. Shlomo Sternberg places three portraits in chronological order on the first pages of his book *Curvature in Mathematics and Physics* (2012): "The first is Riemann who is the founder of our entire subject", then Einstein who needed no comment other than his dates, "Born: 14 March 1879 in Ulm, Württemberg, Germany. Died: 18 April 1955 in Princeton, New Jersey, USA", and then "Élie Cartan – Born: 9 April 1869 in Dolomieu, Savoie, Rhône, France. Died: 6 May 1951 in Paris, France – whose exterior differential calculus, method of moving frames, 'philosophy' of Riemannian geometry, symmetric spaces, etc. are key ingredients in [this] book. His spirit inhabits [its] every nook and cranny." The same must be said of any (good) book on Lie algebras and Lie groups.

"Cartan was one of the most profound mathematicians of the last hundred years, and his influence is still one of the most decisive in the development of modern mathematics" in the opinion of the American mathematician, Robert Hermann (1931–2020), writing in the *American Mathematical Monthly* in 1983. He concludes :"Cartan is certainly one of the greatest and most original minds of mathematics, whose work on Lie groups, differential geometry, and the geometric theory of differential equations is at the foundation of much of what we do today", a judgment that can be extended to present-day mathematics.

Élie Cartan in Dolomieu and after

Dolomieu is a small town, a village really, in the Isère department of the French Alps region, between Lyon and Grenoble, with a population which was close to 2,500 when Cartan was born. (It decreased later to about 1,300 and has since grown again to over 3,000.) Élie's father was the town's blacksmith and farrier[1]. "It is the sound of the anvil resounding [under the blows of the blacksmith's hammer] every morning from daybreak that was my childhood's rhyme", and, while his father melted and fashioned iron for utensils or shod horses, his mother would work on "le métier du canut [the loom for weaving silk], whenever she was free from the care of her children and the duty of cleaning and managing the house". Thus Cartan reminisced, at the beginning of his formal address to his peers, presumably assembled in the "grand amphithéâtre" of the Sorbonne on May 18, 1939, to celebrate his jubilee, two years before he retired at the age of seventy-two.

[1]The farrier (in French, maréchal-ferrant) is the smith who nails iron horseshoes onto horses' hoofs.

The young Élie studied in the local primary school until he was ten and was then encouraged by discerning local teachers and an "inspecteur" to go on to study with a scholarship, first in a "collège" in the nearby town of Vienne, where he was a boarder, then to a high school, in Grenoble, and then on to Paris, first at the newly built Lycée Janson-de-Sailly and then at the École normale supérieure which had a cast of eminent mathematicians as professors, among them Henri Poincaré, about whom Cartan wrote years later : "the giant of Mathematics whose teaching passed well above our heads".

There is nothing picturesque to be found in the career of Cartan as an assistant professor first in Montpellier, then in Lyon. He became a professor in Nancy in 1903 and was eventually named to a chair in the University of Paris in 1912. He spent vacations in the village of Dolomieu in a new house which was built for him in 1909. During the First World War, he was indeed drafted into the army but served in a military hospital which had been set up in the building of the École normale supérieure on the rue d'Ulm in Paris, where he was able to pursue his research and to continue his mathematical work all through the war. He kept publishing until his last article appeared in 1949!

After his retirement from the Sorbonne, Cartan kept teaching, but now he lectured in the all-girls school, the École normale supérieure de jeunes filles, otherwise called École de Sèvres, because it was located in Sèvres, west of Paris (in buildings dating to the mid-eighteenth century which had been the site of the factory making Sèvres porcelain). The war broke out in 1939 and German troops occupied the buildings. Temporary premises had to be found to house the girls and their school. They moved to Paris, to a building on the left bank where Cartan would lecture once a week. Soon, Vichy's racial laws would prevent three girls from taking the regular examinations. One of these young women, Paulette Libermann, had a taste for mathematics. Cartan proposed a research topic to her and directed her research so that she was able to complete a "diploma of higher studies". Later during the war, Cartan acted to provide her with a small scholarship at the time when, together with her parents, she was in hiding in Lyon. She would be the first alumna of the École de Sèvres (and the third woman) to become a full professor of mathematics in France, and, a quarter of a century after Cartan had guided her first steps in research, she was appointed to a chair at the University of Paris.

Posterity

Today the elementary school in the village of Dolomieu is called "École élémentaire Élie Cartan" and a street is named after him.

Cartan's mathematical works comprise nearly 200 publications, over 200 if one counts his published correspondence with Einstein, various texts on mathematics or mathematicians, and translations of his books into Russian. In the list of references of an advanced book on representation theory, one often finds more than a dozen references to works of Cartan, in addition to a reference to his complete works that were published in three volumes between 1952 and 1955 and reprinted in 1984, while, in the index, one could find more than a dozen items that bear Cartan's name: Cartan matrices, Cartan subalgebras, Cartan subgroups, Cartan involution, Cartan decomposition, etc.

For the notice on Cartan that he wrote for the *Dictionary of Scientific Biography*, Jean Dieudonné begins with "Cartan was one of the most profound mathematicians of the last hundred years, and his influence is still one of the most decisive in the development of modern mathematics", and he concludes "Cartan's recognition as a first-rate mathematician came to him only in his old age; before 1930 Poincaré and Weyl were probably the only prominent mathematicians who correctly assessed his uncommon powers and depth. This was due partly to his extreme modesty and partly to the fact that in France the main trend of mathematical research after 1900 was in the field of function theory, but chiefly to his extraordinary originality. It was only after 1930 that a younger generation started to explore the rich treasure of ideas and results that lay buried in his papers. Since then his influence has been steadily increasing, and with the exception of Poincaré and Hilbert, probably no one else has done so much to give the mathematics of our day its present shape and viewpoints." Dieudonné (1906–1992) is the author of a treatise on analysis and geometry in 9 volumes, *Éléments d'analyse*, and of innumerable preliminary versions of parts of the Bourbaki treatises, many of which, after discussion by the group, eventually became the published version.

The obituary of Cartan by Shiing-Shen Chern and Chevalley in the *Bulletin of the American Mathematical Society*, published in 1952, opens with "Undoubtedly one of the greatest mathematicians of this century, his career was characterized by a rare harmony of genius and modesty".

Élie Cartan's son: Henri Cartan

Of the children of Élie Cartan and his wife, one son was a gifted musician who died very young, another son was a physicist who published important books and articles, and was a professor at the University of Poitiers at the time he entered the Résistance against the Nazi forces occupying France, was arrested and taken to Germany where he was detained and executed in 1943, and two studied at the École normale supérieure, which was and remains the French equivalent of graduate school, Hélène, the youngest child would teach in high schools, and Henri, the eldest child, became a very important mathematician: "one of the world's most distinguished mathematicians" is how Ian Stewart, himself a very well known mathematician, described Henri Cartan after his death at the age of 104 in 2008. The obituary in *The Times* of London described him as "one of the most influential, versatile and distinguished pure mathematicians of the past century". He was one of the very small group of mathematicians who founded the Bourbaki group in 1935 and who went on to collectively write numerous treatises destined to eventually cover all of pure mathematics, an aim not achieved, but the existing treatises have had an enormous influence, especially, but not only in France.

Exercises

Exercise 5.1 *Conjugate rotations.*

We use the notation of Section 2.1. Let x be a vector in $\mathbb{R}^3$ and let $g \mapsto g(t)$ be a one-parameter subgroup of SO(3), with infinitesimal generator M_x.

(a) Show that the rotations $g(t)$ leave the vector x of $\mathbb{R}^3$ invariant.
(b) For $g_0 \in \mathrm{SO}(3)$, show that $t \mapsto g_0 g(t) g_0^{-1}$ is a one-parameter subgroup of SO(3) with infinitesimal generator $M_{g_0 x}$.

Exercise 5.2 *Eigenvalues* -1 *and* 1 *of* Ad_g.

(a) Show that the endomorphism $\mathrm{Ad}_g + \mathrm{Id}_{\mathfrak{su}(2)}$ of $\mathfrak{su}(2)$ is invertible for every $g \in \mathrm{SU}(2)$ except for a set of measure zero. Find this set of measure zero.
(b) Find the dimension of the kernel of $\mathrm{Ad}_g - \mathrm{Id}_{\mathfrak{su}(2)}$ as a function of $g \in \mathrm{SU}(2)$.

Exercise 5.3 *An isomorphism of Lie algebras.*

Show that the Lie algebra $\mathfrak{su}(2) \times \mathfrak{su}(2)$ is isomorphic to the Lie algebra $\mathfrak{so}(4)$.

Exercise 5.4 *Haar measure on the group* SU(2).

We parametrize SU(2) off a set of measure zero by setting, for an element $g \in \mathrm{SU}(2)$, $g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$, $a = a_1 + i a_2$ and $b = b_1 + i b_2$, and

$$a_1 = \cos\theta_3,\ a_2 = \sin\theta_3 \cos\theta_2,\ b_1 = \sin\theta_3 \sin\theta_2 \cos\theta_1,\ b_2 = \sin\theta_3 \sin\theta_2 \sin\theta_1,$$

for $\theta_1 \in (0, 2\pi)$, $\theta_2 \in (0, \pi)$, $\theta_3 \in (0, \pi)$.

(a) Show that the Haar integral on SU(2) can be written

$$I(f) = \frac{1}{2\pi^2} \int_{\mathcal{O}} f(\theta_1, \theta_2, \theta_3) \sin^2\theta_3 \sin\theta_2 \, \mathrm{d}\theta_1 \, \mathrm{d}\theta_2 \, \mathrm{d}\theta_3,$$

where $\mathcal{O} = (0, 2\pi) \times (0, \pi) \times (0, \pi)$.

(b) Set $\theta_3 = t$. Show that the Haar integral of a class function f is

$$\frac{2}{\pi} \int_0^{\pi} f(t) \sin^2 t \, \mathrm{d}t.$$

Exercise 5.5 *Euler angles and Haar measure on the group* SO(3).

(a) Show that each rotation $g \in \mathrm{SO}(3)$ can be written

$$g = \exp(\alpha \eta_3) \exp(\beta \eta_1) \exp(\gamma \eta_3)$$

with $0 \le \alpha < 2\pi$, $0 \le \beta \le \pi$, $0 \le \gamma < 2\pi$, uniquely if $\beta \neq 0$ and $\beta \neq \pi$.

The angles α, β, γ are called the *Euler angles* of the rotation.

(b) Show that the Haar integral of SO(3) can be written

$$I(f) = \frac{1}{8\pi^2} \int_{\Omega_0} f(\alpha, \beta, \gamma) \, \sin\beta \, \mathrm{d}\alpha \, \mathrm{d}\beta \, \mathrm{d}\gamma,$$

where $\Omega_0 = (0, 2\pi) \times (0, \pi) \times (0, 2\pi)$.

Exercise 5.6 *Euler angles and Haar measure on the group* SU(2).

Let $\Omega = (0, 2\pi) \times (0, \pi) \times (-2\pi, 2\pi)$.

(a) Show that, for every pair of complex numbers a and b such that $|a|^2 + |b|^2 = 1$, $\mathrm{Im}(a) \neq 0$ and $\mathrm{Re}(b) \neq 0$, there is a unique triple $(\phi, \theta, \psi) \in \Omega$ such that $a = \cos\frac{\theta}{2} e^{i\frac{\phi+\psi}{2}}$, $b = i \sin\frac{\theta}{2} e^{i\frac{\phi-\psi}{2}}$. Conclude that, off a set of measure zero of SU(2), each matrix $g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \in \mathrm{SU}(2)$ can be written in one and only way as

$$g = \exp(\phi\xi_3) \exp(\theta\xi_1) \exp(\psi\xi_3),$$

with $(\phi, \theta, \psi) \in \Omega$. Show that the image of $g(\phi, \theta, \psi)$ under the morphism φ of SO(3) onto SU(2) is the rotation with Euler angles (ϕ, θ, ψ).

(b) Show that the Haar integral on SU(2) can be written

$$I(f) = \frac{1}{16\pi^2} \int_{\Omega} f(\phi, \theta, \psi) \, \sin\theta \, \mathrm{d}\phi \, \mathrm{d}\theta \, \mathrm{d}\psi.$$

Chapter 6

Representations of SU(2) and SO(3)



To study the irreducible representations of the Lie groups SU(2) and SO(3), we first study the irreducible representations of their Lie algebra, $\mathfrak{su}(2)$, which coincide with those of the complexification, $\mathfrak{sl}(2,\mathbb{C})$. We recall that by our convention a *representation* is a representation on a finite-dimensional complex vector space.

1 Irreducible Representations of $\mathfrak{sl}(2,\mathbb{C})$

1.1 The Representations D^j

We consider $\mathfrak{sl}(2,\mathbb{C})$ equipped with the basis

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad X_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

in which the commutation relations can be written

$$\begin{cases} [H, X_+] = 2X_+, \\ [H, X_-] = -2X_-, \\ [X_+, X_-] = H. \end{cases} \tag{1.1}$$

Let (E, R) be an irreducible finite-dimensional representation of $\mathfrak{sl}(2,\mathbb{C})$. The operator $R(H)$ has at least one eigenvalue λ and an eigenvector v for this eigenvalue, satisfying $v \neq 0$ and

$$R(H)v = \lambda v.$$

By the commutation relations (1.1),

$$R(H)R(X_+)v = (R(X_+)R(H) + 2R(X_+))v = (\lambda + 2)R(X_+)v, \tag{1.2}$$

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Y. Kosmann-Schwarzbach, *Groups and Symmetries*, Universitext,
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and

$$R(H)R(X_-)v = (R(X_-)R(H) - 2R(X_-))v = (\lambda - 2)R(X_-)v. \tag{1.3}$$

Because there can be no more than a finite number of distinct eigenvalues of $R(H)$, there is an eigenvalue λ_0 and an associated eigenvector v_0 such that $R(H)v_0 = \lambda_0 v_0$ and, furthermore,

$$R(X_+)v_0 = 0. \tag{1.4}$$

Then we set, for $k \in \mathbb{N}$,

$$v_k = R(X_-)^k v_0.$$

By iterating equation (1.3), we find

$$R(H)v_k = (\lambda_0 - 2k)v_k. \tag{1.5}$$

The non-zero vectors v_k are thus eigenvectors of $R(H)$ with distinct eigenvalues.

We shall show by induction on k that, for any positive integer k,

$$R(X_+)v_k = k(\lambda_0 - k + 1)v_{k-1}.$$

The equation is satisfied for $k = 1$ because, by (1.1) and (1.4),

$$R(X_+)v_1 = R(X_+)R(X_-)v_0 = R(H)v_0 = \lambda_0 v_0.$$

Assume the equation holds for k. Using the hypothesis, $v_{k+1} = R(X_-)v_k$, and the third equation of (1.1), we obtain

$$\begin{aligned} R(X_+)v_{k+1} &= R(X_+)R(X_-)v_k = (R(X_-)R(X_+) + R(H))v_k \\ &= R(X_-)k(\lambda_0 - k + 1)v_{k-1} + (\lambda_0 - 2k)v_k \\ &= (k(\lambda_0 - k + 1) + \lambda_0 - 2k)v_k = (k+1)(\lambda_0 - k)v_k, \end{aligned}$$

which proves the equation.

As the non-zero vectors v_k are linearly independent and the vector space E was assumed finite-dimensional, there is an integer n such that

$$v_0 \neq 0, v_1 \neq 0, \ldots, v_n \neq 0, v_{n+1} = 0.$$

Hence, writing $R(X_+)v_{n+1} = 0$, we find

$$\lambda_0 = n.$$

One can then verify that, for every $k \in \mathbb{N}$,

$$[R(X_+), R(X_-)]v_k = R(H)v_k,$$

$$[R(H), R(X_\pm)]v_k = \pm 2R(X_\pm)v_k.$$

The vectors, $v_0, v_1, \ldots, v_n$ generate a subspace of E invariant under R, and since the desired representation is irreducible, they generate E, which is thus of dimension $n+1$. In summary, we have found a representation $(E^{(n)}, R^{(n)})$ of dimension $n+1$ of $\mathfrak{sl}(2,\mathbb{C})$ and a basis $v_0, v_1, \ldots, v_n$ of $E^{(n)}$ such that

$$\begin{cases} R^{(n)}(H)v_k &= (n-2k)v_k, \\ R^{(n)}(X_-)v_k &= v_{k+1}, \\ R^{(n)}(X_+)v_k &= k(n-k+1)v_{k-1}, \end{cases} \tag{1.6}$$

for $0 \leq k \leq n$, and $v_{-1} = 0$, $v_{n+1} = 0$. Let us show that the representation $(E^{(n)}, R^{(n)})$ is irreducible. We suppose that u is a non-zero vector in a vector subspace F of $E^{(n)}$, invariant under $R^{(n)}$. Then,

$$u = \sum_{k=0}^{n} u_k v_k, \ u_k \in \mathbb{C}.$$

Suppose $u_{k_0} \neq 0, u_{k_0+1} = \ldots = u_n = 0$, where $0 < k_0 \leq n$. Then $(R^{(n)}(X_+))^{k_0} u$ is proportional to v_0 with a non-zero coefficient. Thus v_0 belongs to F and consequently so do the v_k, for $k = 0, \ldots, n$. Thus each non-trivial invariant subspace coincides with the entire space. On the other hand, the proof above showed that every irreducible finite-dimensional representation of $\mathfrak{sl}(2,\mathbb{C})$ is of the form $(E^{(n)}, R^{(n)})$, for a certain $n \in \mathbb{N}$.

Proposition 1.1. *The irreducible finite-dimensional representations of* $\mathfrak{sl}(2,\mathbb{C})$ *are the* $(E^{(n)}, R^{(n)})$, $n \in \mathbb{N}$.

Often, particularly in physics texts, one writes simply Hv_k, X_-v_k, X_+v_k, where the name of the representation is understood. Physicists set $n = 2j$, with j integer or half-integer, and they prefer the basis J_3, J_+, J_- of $\mathfrak{sl}(2,\mathbb{C})$. They use (E^j, D^j) to denote the representation $(E^{(2j)}, R^{(2j)})$ and they use the basis of $E^j = E^{(2j)}$ indexed by the number m, $-j \leq m \leq j$ (m is integer if j is integer, half-integer if j is half-integer), denoted by $|j,m\rangle$ and called the *standard basis of the representation* D^j, defined by

$$|j,m\rangle = (-1)^{j+m} \sqrt{\frac{(j-m)\,!}{(j+m)\,!}}\, v_{j+m}. \tag{1.7}$$

The notation comes from the "bras" and "kets" formalism of Dirac in quantum mechanics. In the basis $|j,m\rangle$,

$$D^j \begin{cases} J_3|j,m\rangle &= m|j,m\rangle \\ J_+|j,m\rangle &= \sqrt{(j-m)(j+m+1)}|j,m+1\rangle \\ J_-|j,m\rangle &= \sqrt{(j+m)(j-m+1)}|j,m-1\rangle \end{cases} \tag{1.8}$$

Hence

$$J_\pm|j,m\rangle = \sqrt{j(j+1) - m(m\pm 1)}|j,m\pm 1\rangle. \tag{1.9}$$

1.2 The Casimir Operator

In each representation space (E, R) of $\mathfrak{sl}(2,\mathbb{C})$, we can consider the operator $(R(J_1))^2 + (R(J_2))^2 + (R(J_3))^2$ which we write in shorthand:

$$J^2 = J_1^2 + J_2^2 + J_3^2. \tag{1.10}$$

Because $J_+J_- = (J_1 + iJ_2)(J_1 - iJ_2) = J_1^2 + J_2^2 - i[J_1, J_2] = J_1^2 + J_2^2 + J_3$, we have

$$J^2 = J_+J_- + J_3(J_3 - I) = J_-J_+ + J_3(J_3 + I).$$

It is clear that, if $R = D^j$, each vector $|j, m\rangle$ is an eigenvector of J^2. More precisely, we find from (1.8) and (1.9) that

$$J^2|j, m\rangle = j(j + 1)|j, m\rangle.$$

Thus we see that, in the representation D^j, not only is each vector $|j, m\rangle$ an eigenvector of J^2, but in fact J^2 acts as the multiple of the identity $j(j + 1)I$. The operator J^2 is called the *Casimir operator* of the representation D^j. It commutes with each operator of the representation D^j.

1.3 Hermitian Nature of the Operators J_3 and J^2

We shall show that the operators J_3 and J^2 in the representation (E^j, D^j) of $\mathfrak{sl}(2,\mathbb{C})$ are Hermitian with respect to a well-chosen scalar product on E^j. We define the scalar product on E^j by requiring that $|j, m\rangle$ be an orthonormal basis. We denote by X^* the adjoint of a matrix X, *i.e.*, $X^* = {}^t\overline{X}$. Then

Proposition 1.2.

$$(J_+)^* = J_-, (J_-)^* = J_+,\ (J_3)^* = J_3,\ (J^2)^* = J^2.$$

Proof. We use the basis

$$(|j, -j\rangle,\ |j, -j + 1\rangle, \ldots, |j, j - 1\rangle,\ |j, j\rangle),$$

in the given order.

- The matrix of J_3 in the basis above is

$$\begin{pmatrix} -j & 0 & & 0 \\ 0 & -j+1 & & 0 \\ & & \ddots & \\ 0 & 0 & & j \end{pmatrix}.$$

Thus J_3 is real and symmetric, thus Hermitian (self-adjoint).

- The matrix of J_+ is

$$\begin{pmatrix} 0 & 0 & 0 & & & & \\ \sqrt{2j} & 0 & 0 & & & & \\ 0 & \sqrt{2(2j-1)} & 0 & & & & \\ & & \ddots & \ddots & & & \\ & & & \ddots & 0 & 0 & 0 \\ & & & & \sqrt{2(2j-1)} & 0 & 0 \\ & & & & 0 & \sqrt{2j} & 0 \end{pmatrix}.$$

- The matrix of J_- is

$$\begin{pmatrix} 0 & \sqrt{2j} & 0 & & & & \\ 0 & 0 & \sqrt{2(2j-1)} & & & & \\ 0 & 0 & 0 & & & & \\ & & \ddots & \ddots & & & \\ & & & \ddots & 0 & \sqrt{2(2j-1)} & 0 \\ & & & & 0 & 0 & \sqrt{2j} \\ & & & & 0 & 0 & 0 \end{pmatrix}.$$

We have $(J_-)_{m,m+1} = \sqrt{j(j+1) - m(m+1)} = (J_+)_{m+1,m}$. Thus the matrices of J_+ and J_- are real and are transposes of one another. Furthermore,

$$(J^2)^* = J_-^* J_+^* + J_3(J_3 - I) = J_+ J_- + J_3(J_3 - I) = J^2.$$

which proves the proposition. □

In quantum physics, the Hermitian operators J_3 and J^2 are observables that correspond, respectively, to the component along the z-axis of the *angular momentum* and the square of its norm.

By restricting the representation D^j to the real form $\mathfrak{su}(2)$ of $\mathfrak{sl}(2,\mathbb{C})$, we obtain a representation of $\mathfrak{su}(2)$ which we shall denote by the same symbol, D^j.

Because $J_3 = i\xi_3$, we deduce from Proposition 1.2 that

$$D^j(\xi_3)^* = -D^j(\xi_3).$$

Because $J_\pm = i\xi_1 \mp \xi_2$, we have

$$\xi_1 = -\frac{i}{2}(J_+ + J_-), \quad \xi_2 = \frac{1}{2}(J_- - J_+),$$

whence

$$D^j(\xi_1)^* = -D^j(\xi_1), \quad D^j(\xi_2)^* = -D^j(\xi_2).$$

As a consequence, for every $X \in \mathfrak{su}(2)$, $D^j(X)^* = -D^j(X)$. We thus have proved the following proposition.

Proposition 1.3. *The operators of the representation D^j of $\mathfrak{su}(2)$, $j \in \frac{1}{2}\mathbb{N}$, are antihermitian for the scalar product on E^j defined by the condition that $|j,m\rangle$ be an orthonormal basis.*

Thus, denoting this scalar product by $(\ \mid\ )$,

$$(D^j(X)x_1 \mid x_2) = -(x_1 \mid D^j(X)x_2),$$

for every $X \in \mathfrak{su}(2)$ and for every $x_1, x_2 \in E^j$. The matrix of $D^j(X)$ in the basis $|j,m\rangle$ is antihermitian.

2 Representations of SU(2)

2.1 The Representations $\mathcal{D}^j$

We shall study representations $\mathcal{D}^j$ of SU(2), $j \in \frac{1}{2}\mathbb{N}$, show that they have as differentials $D\mathcal{D}^j$, the representations D^j of $\mathfrak{su}(2)$ studied above, and show that they are the only irreducible representations of SU(2).

The group $\mathrm{SL}(2,\mathbb{C})$ acts on $\mathbb{C}^2$ by the fundamental representation, so that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} az_1 + bz_2 \\ cz_1 + dz_2 \end{pmatrix}.$$

As we have seen in Section 3.1 of Chapter 2, it is natural to make $\mathrm{SL}(2,\mathbb{C})$ act on the space of functions from $\mathbb{C}^2$ to the complex numbers by

$$\rho(g)f = f \circ g^{-1},$$

for each function f on $\mathbb{C}^2$. Thus one defines a representation of $\mathrm{SL}(2,\mathbb{C})$. (Here 'representation' only means that $\rho(gg') = \rho(g) \circ \rho(g')$ if g and $g' \in \mathrm{SL}(2,\mathbb{C})$.)

If $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has determinant 1, then $g^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, thus explicitly

$$(\rho(g)f)(z_1, z_2) = f(dz_1 - bz_2, -cz_1 + az_2). \tag{2.1}$$

We shall study the representation ρ restricted to SU(2), but it is certainly not irreducible, and we shall find vector subspaces on which SU(2) acts irreducibly.

For each $g \in \mathrm{SU}(2)$, $g^{-1} = {}^t\bar{g}$, we then have $\rho(g)f = f \circ {}^t\bar{g}$ and if $g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$,

$$(\rho(g)f)(z_1, z_2) = f(\bar{a}z_1 - bz_2, \bar{b}z_1 + az_2). \tag{2.2}$$

Let V^j be the vector space of homogeneous polynomials with complex coefficients in two variables, (z_1, z_2), of degree $2j$, where $j \in \frac{1}{2}\mathbb{N}$. This complex vector space is of dimension $2j+1$. A basis is

$$z_2^{2j},\ z_1 z_2^{2j-1},\ \ldots,\ z_1^{j+m} z_2^{j-m},\ \ldots,\ z_1^{2j}, \quad -j \le m \le j.$$

For example, these bases are

$$\begin{array}{llll} \text{if } j=\frac{1}{2}, & z_2, & z_1 & \\ \text{if } j=1, & z_2^2, & z_1z_2, & z_1^2 \\ \text{if } j=\frac{3}{2}, & z_2^3, & z_1z_2^2, & z_1^2z_2, \quad z_1^3. \end{array}$$

Clearly, V^j is invariant under the representation ρ of SU(2). Now we shall study the representation $\mathcal{D}^j$ of SU(2) on V^j thus obtained by restriction.

We introduce the notation

$$f_m^j(z_1,z_2) = z_1^{j+m}z_2^{j-m}.$$

Action of diagonal matrices in SU(2). In the representation $\mathcal{D}^j$, the diagonal matrices of SU(2) act in a simple way on V^j. We set

$$g_t = \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} = \exp(t\mathcal{K}) = \exp(2t\xi_3), \quad \text{where} \quad \xi_3 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

Then,

$$(\mathcal{D}^j(g_t)f_m^j)(z_1,z_2) = f_m^j(e^{-it}z_1, e^{it}z_2) = e^{-2imt}f_m^j(z_1,z_2),$$

and so

$$\mathcal{D}^j(g_t)f_m^j = e^{-2imt}f_m^j.$$

Each f_m^j is thus an eigenvector of $\mathcal{D}^j(g_t)$ for the eigenvalue e^{-2imt}.

Calculation of the differential of $\mathcal{D}^j$. Let $X = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathfrak{sl}(2,\mathbb{C})$ and $g(t) = \exp(tX) = \begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix}$. Then $g(0) = I$ and $g'(0) = X$, and thus $a(0) = d(0) = 1$, $b(0) = c(0) = 0$, and

$$a'(0) = \alpha,\ b'(0) = \beta,\ c'(0) = \gamma,\ d'(0) = \delta.$$

Thus, for every polynomial f in two variables,

$$\begin{aligned} ((D\rho)(-X)f)(z_1,z_2) &= \tfrac{\mathrm{d}}{\mathrm{d}t}(\rho(g(t)^{-1})f)(z_1,z_2)\big|_{t=0} = \tfrac{\mathrm{d}}{\mathrm{d}t}(f\circ g(t))(z_1,z_2)\big|_{t=0} \\ &= (\alpha z_1+\beta z_2)\partial_1 f(z_1,z_2) + (\gamma z_1+\delta z_2)\partial_2 f(z_1,z_2). \end{aligned}$$

Thus

$$((D\rho)(\xi_3)f)(z_1,z_2) = -\frac{i}{2}(z_1\partial_1 f - z_2\partial_2 f),$$

$$((D\rho)(\xi_1)f)(z_1,z_2) = -\frac{i}{2}(z_2\partial_1 f + z_1\partial_2 f),$$

$$((D\rho)(\xi_2)f)(z_1,z_2) = \frac{1}{2}(z_2\partial_1 f - z_1\partial_2 f).$$

Therefore we find

$$\begin{aligned}(D\rho)(J_3) &= i(D\rho)(\xi_3) = \tfrac{1}{2}(z_1\partial_1 - z_2\partial_2),\\ (D\rho)(J_+) &= i(D\rho)(\xi_1) - (D\rho)(\xi_2) = z_1\partial_2,\\ (D\rho)(J_-) &= i(D\rho)(\xi_1) + (D\rho)(\xi_2) = z_2\partial_1.\end{aligned}$$

We shall study the action $D\rho$ restricted to V^j. It suffices to determine the value of the operators $(D\mathcal{D}^j)(\xi_k)$, for $k = 1, 2, 3$, or $(D\mathcal{D}^j)(J_3), (D\mathcal{D}^j)(J_+)$, $(D\mathcal{D}^j)(J_-)$, on the basis vectors f^j_m, $-j \leq m \leq j$, of V^j.

It is clear that $(D\mathcal{D}^j)(\xi_3)f^j_m = -imf^j_m$. Using

$$\partial_1 f^j_m = (j+m)z_1^{j+m-1}z_2^{j-m},$$

$$\partial_2 f^j_m = (j-m)z_1^{j+m}z_2^{j-m-1},$$

we obtain

$$(D\mathcal{D}^j)(\xi_1)f^j_m = -\frac{i}{2}((j+m)f^j_{m-1} + (j-m)f^j_{m+1}),$$

$$(D\mathcal{D}^j)(\xi_2)f^j_m = \frac{1}{2}((j+m)f^j_{m-1} - (j-m)f^j_{m+1})\ ,$$

and the simple formulas

$$\begin{aligned}(D\mathcal{D}^j)(J_3)f^j_m &= mf^j_m,\\ (D\mathcal{D}^j)(J_+)f^j_m &= (j-m)f^j_{m+1},\\ (D\mathcal{D}^j)(J_-)f^j_m &= (j+m)f^j_{m-1}.\end{aligned}$$

We set

$$|j,m\rangle = \frac{1}{\sqrt{(j-m)!(j+m)!}}f^j_m.$$

We have then

$$\begin{cases}(D\mathcal{D}^j)(J_3)|j,m\rangle &= m|j,m\rangle\\ (D\mathcal{D}^j)(J_+)|j,m\rangle &= \sqrt{(j-m)(j+m+1)}|j,m+1\rangle\\ (D\mathcal{D}^j)(J_-)|j,m\rangle &= \sqrt{(j+m)(j-m+1)}|j,m-1\rangle\end{cases}$$

We deduce immediately from these formulas that the differential, $D\mathcal{D}^j$, of the representation $(V^j, \mathcal{D}^j)$ of $\mathrm{SU}(2)$ is equivalent to the representation D^j of the Lie algebra $\mathfrak{su}(2)$, because it is the restriction of the same representation of $\mathfrak{sl}(2,\mathbb{C})$. We thus have proved the following proposition.

Proposition 2.1. *For each $j \in \frac{1}{2}\mathbb{N}$, the differential of the representation $\mathcal{D}^j$ of* $\mathrm{SU}(2)$ *can be identified with the representation D^j of* $\mathfrak{su}(2)$.

From now on, we shall identify the support V^j of $D\mathcal{D}^j$ with the support E^j of D^j.

Proposition 2.2. *For each $j \in \frac{1}{2}\mathbb{N}$, the representation $(V^j, \mathcal{D}^j)$ of* $\mathrm{SU}(2)$ *is* unitary *for the scalar product on V^j defined by requiring that $|j,m\rangle$ be an orthonormal basis.*

Proof. In light of Theorem 7.7 of Chapter 4, this result is the consequence of Propositions 1.3 and 2.1, and of the fact that SU(2) is connected. □

For each $g \in \mathrm{SU}(2)$, the matrix of $\mathcal{D}^j(g)$ in the basis $|j,m\rangle$ is thus unitary.

Proposition 2.3. *For each j, integer or half-integer, $(V^j, \mathcal{D}^j)$ is an irreducible representation of* SU(2).

Proof. By Theorem 7.7 of Chapter 4, the proposition follows from the fact that the differential of $\mathcal{D}^j$ is an irreducible representation. (If V is a vector subspace of V^j invariant under $\mathcal{D}^j$, it is invariant under $D\mathcal{D}^j$, and thus V is trivial.) □

Theorem 2.4. *Every irreducible representation of* SU(2) *is equivalent to one of the representations $(V^j, \mathcal{D}^j)$ of dimension $2j+1$, for $j \in \frac{1}{2}\mathbb{N}$.*

Proof. Because the group SU(2) is compact, we know, by Theorem 4.3 of Chapter 3, that every irreducible representation of SU(2) is finite-dimensional. We have seen that each finite-dimensional irreducible representation of $\mathfrak{su}(2)$ is one of the $D^j, j \in \frac{1}{2}\mathbb{N}$. Because SU(2) is connected, the result follows. □

2.2 Characters of the Representations $\mathcal{D}^j$

By definition, for $j \in \frac{1}{2}\mathbb{N}$, $\chi_j(t) = \chi_{\mathcal{D}^j}(t) = \sum_{m=-j}^{j} e^{2imt}$. Therefore, for every $t \in (0,\pi)$,

$$\chi_j(t) = \frac{\sin(2j+1)t}{\sin t} .$$

For $t = 0$, we have $\chi_j(0) = 2j+1$, the dimension of the representation, and for $t = \pi$, $\chi_j(\pi) = (-1)^{2j}(2j+1)$.

Character of $\mathcal{D}^{j_1} \otimes \mathcal{D}^{j_2}$. One can express $\chi_j(t)$ as a function of $X = e^{2it}$, for $j \in \frac{1}{2}\mathbb{N}$. If $t \neq 0$ and $t \neq \pi$,

$$\chi_j(t) = \sum_{m=-j}^{j} X^m = \frac{X^{j+1} - X^{-j}}{X-1}.$$

Let j_1 and $j_2 \in \frac{1}{2}\mathbb{N}$. We have

$$\begin{aligned} \chi_{j_1}(t)\chi_{j_2}(t) &= \left(\sum_{m=-j_1}^{j_1} X^m\right)\left(\frac{X^{j_2+1} - X^{-j_2}}{X-1}\right) \\ &= \frac{1}{X-1}\left(\sum_{m=-j_1}^{j_1} X^{m+j_2+1} - \sum_{m=-j_1}^{j_1} X^{m-j_2}\right). \end{aligned}$$

We set $m = j_1 - k$ in the first sum and $m = k - j_1$ in the second. We obtain

$$\chi_{j_1}(t)\chi_{j_2}(t) = \frac{1}{X-1}\left(\sum_{k=0}^{2j_1} X^{j_1+j_2+1-k} - \sum_{k=0}^{2j_1} X^{-j_1-j_2+k}\right).$$

Suppose $j_1 \leq j_2$. Then $0 \leq k \leq 2j_1$ implies $j_1 + j_2 - k \geq 0$, and thus

$$\chi_{j_1}(t)\chi_{j_2}(t) = \sum_{k=0}^{2j_1} \chi_{j_1+j_2-k}(t) = \chi_{j_1+j_2}(t) + \chi_{j_1+j_2-1}(t) + \cdots + \chi_{j_2-j_1}(t).$$

For $t = 0$, the left-hand side is $\chi_{j_1}(0)\chi_{j_2}(0) = (2j_1 + 1)(2j_2 + 1)$. On the right-hand side, there are $2j_1 + 1$ terms whose sum is

$$\begin{aligned} \textstyle\sum_{k=0}^{2j_1} \chi_{j_2-j_1+k}(0) &= \textstyle\sum_{k=0}^{2j_1} (2(j_1 + j_2) + 1 - 2k) \\ &= (2j_1 + 1)(2(j_1 + j_2) + 1) - 2j_1(2j_1 + 1) \\ &= (2j_1 + 1)(2j_2 + 1). \end{aligned}$$

The formula thus holds in this case. The proof for $t = \pi$ is analogous. We have thus proved the formula

$$\chi_{j_1}\chi_{j_2} = \chi_{|j_2-j_1|} + \chi_{|j_2-j_1|+1} + \cdots + \chi_{j_1+j_2}. \tag{2.3}$$

From (2.3) and from equation (2.2) of Chapter 2, we deduce the decomposition of the tensor product $\mathcal{D}^{j_1} \otimes \mathcal{D}^{j_2}$ into a direct sum of irreducible representations, called the *Clebsch–Gordan formula*:

$$\mathcal{D}^{j_1} \otimes \mathcal{D}^{j_2} = \mathcal{D}^{|j_2-j_1|} \oplus \mathcal{D}^{|j_2-j_1|+1} \oplus \cdots \oplus \mathcal{D}^{j_1+j_2}. \tag{2.4}$$

3 Representations of SO(3)

We recall the existence of the surjective morphism φ from SU(2) onto SO(3), with kernel $\{I, -I\}$, studied in Section 2.3 of Chapter 5. If (E, ρ) is a representation of SU(2), then ρ factors through the projection φ if and only if

$$\rho(-I) = \rho(I) = \mathrm{Id}_E\,. \tag{3.1}$$

If ρ factors through φ as $\sigma \circ \varphi$, where $\sigma : \mathrm{SO}(3) \to \mathrm{GL}(E)$, then σ is a representation of SO(3), and ρ is irreducible if and only if σ is irreducible. The representation $\mathcal{D}^j$ of SU(2) satisfies the condition (3.1) if and only if j is an integer. In fact $-I = g_\pi$ and $\mathcal{D}^j(g_\pi)f_m^j = e^{-2im\pi} f_m^j$; the scalar factor is 1 if and only if m is an integer, thus if and only if j is an integer. Thus we see that the representations obtained by factorization of $\mathcal{D}^0, \mathcal{D}^1, \mathcal{D}^2, \ldots, \mathcal{D}^j, \ldots$, where j is an integer, are irreducible representations of SO(3). We denote these representations of SO(3) by $\underline{\mathcal{D}}^j$ or even $\mathcal{D}^j$, by abuse of notation. Because the Lie group SO(3) is connected, we can conclude:

Theorem 3.1. *Every irreducible representation of* SO(3) *is equivalent to one of the representations* $(V^j, \mathcal{D}^j)$, $j \in \mathbb{N}$.

References

One can find the results concerning the representations of SU(2) and of SO(3) in many sources including Fulton and Harris (2004), Sternberg (2003), and Talman (1968), since they are fundamental in mathematics and in physics. Also see the beginning of the comprehensive book, Knapp (2002). These results are used in the theory of angular momentum in quantum mechanics. See Basdevant and Dalibard (2005), Tung (1985), or Edmonds (1996).

Topics in History

The history of the theory of representations of finite-dimensional Lie groups began with Wilhelm Killing (1847–1923), who studied first in Münster, then in Berlin where he completed his doctorate in 1872, and went on to teach at the high school level for six years before publishing his major work on the structure of what we now call Lie algebras, "The representations of the finite continuous groups of transformations", in four parts in the *Mathematische Annalen* from 1888 to 1890[1]. He had a very distinguished career at the University of Münster, where he published, taught and accomplished heavy administrative tasks, becomimg "Rector magnificus" in 1897-1898, while performing charitable work in accordance with his deep catholic faith all his life.

Lie group theory was developed further on sound theoretical ground by Élie Cartan before acquiring its modern form in the work of Hermann Weyl.

Weyl, from Zurich to Princeton

In 1918, two years after Einstein gave its final form to the general theory of relativity and published it in *Annalen der Physik*, Weyl published *Space, Time, Matter* whose German original went through several revisions and many editions before it was translated into English and into French, both in 1922. He had given lectures on Einstein's theory of relativity in 1917 in Zurich and, in this book, he undertook a systematic presentation of that theory. His preface opens with a grandiose assessment of Einstein's achievement of which he is about to give an account: "It is as if a wall which separated us from Truth has collapsed". With this lofty aim, the book offers a careful presentation of the theory for the uninitiated, with all the necessary mathematical preliminaries, including "tensor algebra", "tensor analysis" and "Riemann's geometry".

Weyl went from relativity to representations, writing much later in 1949 that he was led to the study of group representations by his desire to understand "the mathematical substance behind the formal apparatus of relativity theory". In 1925 and 1926, he published the "Theory of representations of semi-simple continuous groups by linear transformations", a series of three articles, one hundred pages in all, which constituted a significant advance in the theory of Lie groups and their infinitesimals. While the terminology in use at the time

[1] On Killing, see Hawkins's book, part II.

has changed since then, the style of these articles is still very clear and they remain very readable today, that is, if you can read German. However, if you can read French, a French translation with a very thorough commentary is now available[2]. Weyl's account of the representation theory of semi-simple Lie groups, beginning with the detailed study of the special linear groups – a very pedagogical presentation – is a classic.

The Peter–Weyl theorem for compact topological group was published in 1927 by Weyl and one of his students, Fritz Peter (1899–1949), who became headmaster of a secondary school. Weyl had given an account of the results of their article in his lecture on May 7 at the Spring meeting of the Swiss mathematical society that was published in French translation in the journal *L'Enseignement mathématique*.

Shortly after his three articles on Lie group representations, Weyl brought out *Gruppentheorie und Quantenmechanik*, with a second edition in 1931 that was soon translated into English as *The Theory of Groups and Quantum Mechanics*. In it, he emphasized "the 'reciprocity' between the representations of the symmetric permutation group and those of the complete linear group [which] follows most naturally from the conceptual structure of quantum mechanics".

Weyl had been in danger in Germany because he was opposed to the Nazis and because his wife was Jewish. He resigned in 1933 from the directorship of the Göttingen mathematics department – where he was replaced by Hasse. He emigrated to the United States and was soon appointed as a member of the Institute in Princeton. He published still another book in 1939, *The Classical Groups, their Invariants and Representations*, whose "fundamentally algebraic character" he mentioned in the preface, adding that "neither the infinitesimal nor the topological methods have been omitted", while "the relations to other parts of mathematics are emphasized". At that time, he had already been in the United States for several years and he wrote this book, which became another classic, in English, deploring that "the gods have imposed upon my writing the yoke of a foreign tongue".

"Infinitesimal groups" are called "Lie algebras"

It was Weyl who, in his lectures at the Institute in Princeton in 1934, introduced the name "Lie algebra" for what until then had been called an "infinitesimal group". Nathan Jacobson, then a 24-year old algebraist with a very recent doctorate from Princeton University, was Weyl's assistant. According to some accounts, he had suggested this name to Weyl and the new terminolgy is to be found a few months later in the title of his article, "Rational methods in the theory of Lie algebras", published in 1935. Later, Jacobson was the author of an important body of work in algebra, including many books, and received many honors, testifying to "his leadership in the algebraic theory of Lie algebras" – as was written in one of several obituaries published after his death at age 88.

[2] Christophe Eckes, *Les groupes de Lie dans l'œuvre de Hermann Weyl* (2011).

Wigner in Berlin and Princeton

Eugene Wigner, born in 1902 in Budapest in Hungary which was part of the Austrian empire, was much younger than Weyl but their publications on group representations were roughly contemporary. He studied in Berlin where he received his doctorate in 1925 in chemical engineering and then he spent a year as assistant to Hilbert in Göttingen in 1927-28. Weyl, in the 1931 translation of his book on groups and quantum mechanics, refers briefly to Wigner's two papers published in the *Zeitschrift für Physik* of 1926 and 1927, where the "Wigner principle" was first formulated, while Wigner, in his own 1931 book on group theory that dealt with the application of group theory to quantum mechanics, cited Weyl. In an interview in 1984, cited in Eckes's book, Wigner complained that Weyl had ignored his work "even though practically all that is in [Weyl's] book was contained in publications by me and in joint publications by Johnny von Neumann and me". Many years after it first appeared in German, Wigner's book was translated into English, "with additions and corrections", under the title *Group Theory and its Application to the Quantum Mechanics of Atomic Spectra* (1931). It was hailed as "one of the three pioneering works on the application of group theory to quantum mechanics", the other two being Weyl's book of 1928 and van der Waerden's book of 1932, translated into English much later, in 1974, as *Group Theory and Quantum Mechanics*.

In a celebrated article of 1939, published in the *Annals of Mathematics*, Wigner achieved the determination of all the irreducible unitary representations of the semi-direct product of the 6-dimensional Lorentz group and the 4-dimensional Abelian group of translations of the Minkowski space-time, naming this group "the Poincaré group".

There is an often cited article by Wigner with an engaging title, "The unreasonable effectiveness of mathematics in the natural sciences", published in 1960 in which he argued that "mathematical concepts turn up in entirely unexpected connections [and] they often permit an unexpectedly close and accurate description of the phenomena in these connections [but] we cannot know whether a theory formulated in terms of mathematical concepts is uniquely appropriate".

In Wigner's list of publications, the shift from German to English occurred in 1933. While he had held dual positions in Berlin and at Princeton University since 1930, he settled in the United States after that date and pursued his career in Princeton. During the war he worked on the Manhattan project, then participated in design studies for nuclear reactors.

He was awarded the Nobel prize for Physics in 1963 and died in 1995, as his collected works were being edited. "In his later years, he found himself in the unusual position of being highly esteemed by physicists, mathematicians, chemists, engineers, and philosophers" wrote the physicist A. S. Wightman in an obituary in the *Notices of the American Mathematical Society*.

The Pauli matrices

Did Wolfgang Pauli (1900–1958), another of the great theoretical physicists of the 20th century, who was a contemporary of Wigner but had a much shorter life, introduce the 2×2 matrices that are now called the Pauli matrices? Indeed, he did and when Weyl discussed the spinning electron in chapter IV.4 of *The Theory of Groups and Quantum Mechanics* of 1931, as a reference for the matrices, S_x, S_y, S_z, he cited Pauli's article, "On the quantum mechanics of the magnetic electron", in the *Zeitschrift für Physik* of 1927.

Hendrick Casimir, the spinning top, and a career in industry

A great admirer of Weyl was the Dutch theoretical physicist, Hendrik Casimir (1909–2000). After a year's stay at the University of Copenhagen in 1929, then a center of research on the new, emerging theory of quantum mechanics, he completed his thesis, at the University of Leiden in 1931. He dealt with "[what] is usually called quantum mechanics of the *spinning top*", and it is there, in the study of the representations of the group of rotations, that he defined the operator that now bears his name. The following year, he became assistant to Pauli in Zurich and he went on from there to a career in research in theoretical physics, publishing in 1936 "On the Interaction between atomic nuclei and electrons", already with an interest in instrumentation, and then to a very successful career in industry. His work in electrotechnics was recognized with many awards.

Exercises

Exercise 6.1 *Equivalent representations.*

We set, for $g \in \mathrm{SU}(2)$, and for each complex-valued function f on $\mathbb{C}^2$,

$$\tilde{\rho}(g)f = f \circ \bar{g}^{-1}.$$

Show that $\tilde{\rho} : \mathrm{SU}(2) \to \mathrm{GL}(V^j)$ is a representation of $\mathrm{SU}(2)$ equivalent to $\mathcal{D}^j$.

Exercise 6.2 *The representations $\mathcal{D}^{\frac{1}{2}}$ and $\mathcal{D}^1$.*

(a) Write the matrices of J_3, J_+ and J_- in the basis f_m^j for the representations $D^{\frac{1}{2}}$ and D^1. Same question for ξ_1, ξ_2, ξ_3.

(b) Is the representation $\mathcal{D}^{\frac{1}{2}}$ equivalent to the fundamental representation of $\mathrm{SU}(2)$ on $\mathbb{C}^2$?
Is the representation $\underline{\mathcal{D}}^1$ equivalent to the complexification of the fundamental representation of $\mathrm{SO}(3)$ on $\mathbb{R}^3$?

(c) Decompose $\mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^{\frac{1}{2}}$, $\mathcal{D}^1 \otimes \mathcal{D}^1$, $\mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^1$, and $\mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^{\frac{1}{2}}$ into direct sums of irreducible representations.

Exercise 6.3 *Differential operators on polynomials.*

(a) Using the Euler identity for the homogeneous polynomials of degree $2j$, show that
$$(D\mathcal{D}^j)(J_3)f = (j - z_1\partial_1)f = (z_2\partial_2 + j)f.$$

(b) Express $J_+J_- + J_-J_+$, $(J_3)^2$ and J^2 as differential operators of order less than or equal to 2 on the homogeneous polynomials of two variables of degree $2j$.

Exercise 6.4 *Clebsch–Gordan coefficients.*

In the tensor product $V^{j_1} \otimes V^{j_2}$, we consider the two orthonormal bases
$$|j_1, m_1\rangle \otimes |j_2, m_2\rangle,\ -j_1 \leq m_1 \leq j_1,\ -j_2 \leq m_2 \leq j_2,$$
and
$$|J, M\rangle,\ |j_1 - j_2| \leq J \leq j_1 + j_2,\ -J \leq M \leq J.$$
The *Clebsch–Gordan coefficients* are the coefficients $C(J, M, j_1, m_1, j_2, m_2)$ of the change-of-basis matrix
$$|J, M\rangle = \sum_{m_1, m_2} C(J, M, j_1, m_1, j_2, m_2)\ |j_1, m_1\rangle \otimes |j_2, m_2\rangle.$$

Calculate the Clebsch–Gordan coefficients in the case of $V^{\frac{1}{2}} \otimes V^{\frac{1}{2}}$ and of $V^{\frac{1}{2}} \otimes V^1$.

Exercise 6.5 *Restriction to* SO(2) *of the representations* $\underline{\mathcal{D}}^j$.

(a) Find the irreducible representations of the group SO(2).
(b) Find the character of the representation $\underline{\mathcal{D}}^j$ of SO(3), for $j \in \mathbb{N}$.
(c) Decompose the restriction to SO(2) of the representation $\underline{\mathcal{D}}^j$ of SO(3), for $j \in \mathbb{N}$, into a direct sum of irreducible representations.
(d) Identify $\underline{\mathcal{D}}^1$ with the complexification of the fundamental representation of SO(3), and find a basis of $\mathbb{C}^3$ adapted to the decomposition of the restriction of $\underline{\mathcal{D}}^1$ to SO(2). Is the restriction to SO(2) of the fundamental representation of SO(3) reducible?

Exercise 6.6 *Matrix coefficients of* $\mathcal{D}^j$.

We set, for $\theta \in [0, \pi]$,
$$d^j(\theta) = \mathcal{D}^j(\exp(\theta\xi_1)).$$
Calculate the matrix coefficients d^1_{00}, d^1_{10} and $d^1_{-1,0}$ in the orthonormal basis $|j, m\rangle$.

Exercise 6.7 *Orthogonality of characters.*

(a) Find the value of the integral

$$\int_0^{2\pi} \frac{\sin(j_1+\frac{1}{2})t\ \sin(j_2+\frac{1}{2})t\ \sin(j+\frac{1}{2})t}{\sin\frac{t}{2}}\,\mathrm{d}t$$

in terms of the values of j_1, j_2, $j \in \mathbb{N}$.

(b) Same question for j_1, j_2, $j \in \frac{1}{2}\mathbb{N}$.

Chapter 7

Spherical Harmonics



Spherical harmonics play an important role in electrodynamics and in quantum mechanics. We shall show how they appear in the representation theory of the rotation group SO(3). Each irreducible representation of SO(3) can be realized in a finite-dimensional Hilbert space of functions on the sphere, the restrictions of harmonic homogeneous polynomials of a given degree, and this representation is unitary. We shall determine an orthonormal basis of this space that is transformed in a simple way under the action of the group of rotations: the elements of such a basis are simultaneous eigenfunctions for the operators J_3 and J^2, defined below. The functions thus defined on the sphere are called spherical harmonics.

1 Review of $L^2(S^2)$

We denote by S^2 the unit sphere in $\mathbb{R}^3$,

$$S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid (x_1)^2 + (x_2)^2 + (x_3)^2 = 1\}.$$

We introduce spherical coordinates (r, θ, ϕ) on $\mathbb{R}^3$, $r \geq 0$, $\theta \in [0, \pi]$, $\phi \in [0, 2\pi)$, such that

$$x_1 = r \sin\theta \cos\phi, \;\; x_2 = r \sin\theta \sin\phi, \;\; x_3 = r \cos\theta.$$

The angle ϕ is the longitude and θ is the co-latitude. On $\mathbb{R}^3$ minus the axis Ox_3 ($r > 0$ and $0 < \theta < \pi$), the passage from Cartesian coordinates to spherical coordinates is smooth. The volume element of $\mathbb{R}^3$ is then $r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\phi$, and the surface area element on the unit sphere is

$$\mathrm{d}\mu = \frac{1}{4\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi.$$

We denote by $L^2(S^2)$ the separable Hilbert space of (equivalence classes of) complex functions on S^2 that are square-integrable for the scalar product

$$(f_1 \mid f_2) = \frac{1}{4\pi} \int_{S^2} \overline{f_1(\theta, \phi)} f_2(\theta, \phi) \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi.$$

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Y. Kosmann-Schwarzbach, *Groups and Symmetries*, Universitext,
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In spherical coordinates, the *Laplacian*, $\Delta = (\frac{\partial}{\partial x_1})^2 + (\frac{\partial}{\partial x_2})^2 + (\frac{\partial}{\partial x_3})^2$, can be written

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\Delta_{S^2},$$

where Δ_{S^2} is the *spherical Laplacian*,

$$\Delta_{S^2} = \frac{\partial^2}{\partial \theta^2} + \cot\theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial \phi^2}.$$

2 Harmonic Polynomials

2.1 Representations of Groups on Function Spaces

We shall make use of the important fact, stated in Section 3.1 of Chapter 2, that each group of transformations of a space can be represented on the vector space of complex-valued functions on that space. If f is a function on $\mathbb{R}^3$, and if $g \in \mathrm{SO}(3)$, we set, for $x \in \mathbb{R}^3$,

$$(g \cdot f)(x) = f(g^{-1}x),$$

and this defines a representation of SO(3) on the vector space of functions on $\mathbb{R}^3$. (This turns out to be an infinite-dimensional representation. Here we talk of representation from the algebraic point of view only, without insisting on continuity.) We shall denote this representation by σ, defined by

$$\sigma(g)f = g \cdot f.$$

By abuse of language, we denote its restriction to certain subspaces of functions also by σ.

We introduce the harmonic polynomials and show that, by restricting σ to spaces of harmonic polynomials, we obtain all the irreducible representations of SO(3), which were determined in Chapter 6.

2.2 Spaces of Harmonic Polynomials

Definition 2.1. *We call a function f of class C^2 harmonic on $\mathbb{R}^3$ if*

$$\Delta f = 0.$$

For any non-negative integer ℓ we denote by $P^{(\ell)}$ the vector space of homogeneous polynomials of degree ℓ with complex coefficients on $\mathbb{R}^3$. We then consider the vector subspace of $P^{(\ell)}$ consisting of harmonic polynomials, *i.e.*, polynomials with vanishing Laplacian by $H^{(\ell)}$.

Lemma 2.2. *The vector space $H^{(\ell)}$ is of dimension $2\ell + 1$.*

Proof. First we calculate the dimension of the vector space $P^{(\ell)}$. A homogeneous polynomial of degree ℓ on $\mathbb{R}^3$ is determined by $\ell+1$ homogeneous polynomials in two variables, of respective degrees $0, 1, \ldots, \ell$. Because the vector space of homogeneous polynomials in two variables of degree k is of dimension $k+1$, we obtain

$$\dim P^{(\ell)} = 1 + 2 + \ldots + (\ell+1) = \frac{(\ell+1)(\ell+2)}{2}.$$

We observe next that the operator Δ sends $P^{(\ell)}$ into $P^{(\ell-2)}$, for $\ell \geq 2$. Let us show that the linear map $\Delta : P^{(\ell)} \to P^{(\ell-2)}$ is surjective. First we remark that, for every $q_3 \in \mathbb{N}$, $x_3^{q_3}$ belongs to the image of Δ because

$$\Delta(x_3^{q_3+2}) = (q_3+2)(q_3+1)x_3^{q_3}.$$

Similarly, one can easily see that $x_1 x_3^{q_3}$ and $x_2 x_3^{q_3}$ are in $\operatorname{Im}\Delta$. The following formula, which holds for every $q_1, q_2, q_3 \in \mathbb{N}$,

$$\begin{aligned}\Delta(x_1^{q_1} x_2^{q_2} x_3^{q_3}) = {} & q_1(q_1-1)x_1^{q_1-2}x_2^{q_2}x_3^{q_3} + q_2(q_2-1)x_1^{q_1}x_2^{q_2-2}x_3^{q_3} \\ & + q_3(q_3-1)x_1^{q_1}x_2^{q_2}x_3^{q_3-2},\end{aligned}$$

shows that if the property $x_1^{q_1}x_2^{q_2}x_3^{q_3} \in \operatorname{Im}\Delta$ is true for $q_1+q_2 = q-2$, it is true for $q_1+q_2 = q$. As this property is true for $q=0$ and for $q=1$, the surjectivity of the linear map $\Delta : P^{(\ell)} \to P^{(\ell-2)}$ is thus proved by induction on q. Consequently, $\dim H^{(\ell)} = \dim P^{(\ell)} - \dim P^{(\ell-2)} = 2\ell+1$. □

2.3 Representations of SO(3) on Spaces of Harmonic Polynomials

Let us prove that SO(3) acts on $H^{(\ell)}$. It is clear that, if $P \in P^{(\ell)}$, and if $g \in \mathrm{SO}(3)$, then $\sigma(g)P = g \cdot P$ is still a homogeneous polynomial of degree ℓ.

Proposition 2.3. *The subspace $H^{(\ell)}$ of $P^{(\ell)}$ is invariant under σ.*

Proof. Let f be a function of three variables of class C^2 and let g be in SO(3). We denote by (A_{ij}), $i,j = 1,2,3$, the matrix of g and by y_i the components of $y = g(x)$, for $x \in \mathbb{R}^3$. We have

$$\frac{\partial}{\partial x_i}(f \circ g)(x) = \sum_{j=1}^{3} A_{ji} \frac{\partial f}{\partial y_j}(y).$$

Hence

$$(\Delta(f \circ g))(x) = \sum_{i,j,k=1}^{3} A_{ji}A_{ki} \frac{\partial^2 f}{\partial y_j \partial y_k}(y).$$

Because (A_{ij}) is an orthogonal matrix, we obtain

$$(\Delta(f \circ g))(x) = \sum_{j=1}^{3} \frac{\partial^2 f}{\partial y_j^2}(y) = (\Delta f)(g(x)),$$

that is, $\Delta(f \circ g) = (\Delta f) \circ g$. Consequently, if P is a harmonic polynomial, for every $g \in \mathrm{SO}(3)$, the polynomial $\sigma(g)P = g \cdot P = P \circ g^{-1}$ is also harmonic. □

On the other hand, for a given homogeneous polynomial, P, the coefficients of the polynomials $g \cdot P$ depend continuously on the coefficients of the matrix $g \in \mathrm{SO}(3)$. Thus we can state the following.

Proposition 2.4. *By restricting σ to $H^{(\ell)}$, we obtain, for each $\ell \in \mathbb{N}$, a representation $(H^{(\ell)}, \sigma^\ell)$ of $\mathrm{SO}(3)$ of dimension $2\ell + 1$.*

We shall show that the representation σ^ℓ of $\mathrm{SO}(3)$ on $H^{(\ell)}$ is equivalent to the representation $\mathcal{D}^\ell$, defined in Section 3 of Chapter 6. For this, we shall use the following lemma.

Lemma 2.5. *For $\ell \in \frac{1}{2}\mathbb{N}$, let (V, ρ) be a representation of dimension $2\ell + 1$ of $\mathrm{SU}(2)$. If $e^{2i\ell\theta}$ or $e^{-2i\ell\theta}$ is an eigenvalue of $\rho(g_\theta)$, where $g_\theta = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$, then ρ is equivalent to $\mathcal{D}^\ell$.*

Proof. We can decompose the given representation into a direct sum of irreducible representations,

$$\rho = \bigoplus_{j=0}^{j_0} m_j \mathcal{D}^j,$$

and, since $\dim V = 2\ell + 1$, it follows that $j_0 \leq \ell$. If we had $j_0 < \ell$, all the eigenvalues of $\rho(g_\theta)$ would be of the form $e^{2im\theta}$, with $|m| < \ell$. Thus, one of the $\mathcal{D}^j$ in the direct sum is equal to $\mathcal{D}^\ell$. The condition on the dimension of the representation then implies that $\rho = \mathcal{D}^\ell$. □

We immediately deduce from this lemma that a $(2\ell + 1)$ -dimensional representation (V, σ) of $\mathrm{SO}(3)$ is equivalent to $\mathcal{D}^\ell$ if and only if $e^{2i\ell\theta}$ or $e^{-2i\ell\theta}$ is an eigenvalue of $\sigma(\varphi(g_\theta))$. Here φ is the morphism of $\mathrm{SU}(2)$ into $\mathrm{SO}(3)$ defined in Section 2.3 of Chapter 5.

Proposition 2.6. *The representation σ^ℓ of $\mathrm{SO}(3)$ on $H^{(\ell)}$ is equivalent to the representation $\mathcal{D}^\ell$.*

Proof. The polynomial $p_\ell = (x_1 + ix_2)^\ell$ belongs to $P^{(\ell)}$. We can easily see that it is harmonic. Furthermore, in the representation σ^ℓ of $\mathrm{SO}(3)$ on $H^{(\ell)}$, it is an eigenfunction of $\varphi(g_\theta)$ for the eigenvalue $e^{-2i\ell\theta}$, because

$$\varphi(g_\theta) = \varphi(\exp 2\theta\xi_3) = \mathrm{Rot}(e_3, 2\theta) = \begin{pmatrix} \cos 2\theta & -\sin 2\theta & 0 \\ \sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and, consequently,

$$(\varphi(g_\theta)) \cdot p_\ell = e^{-2i\ell\theta} p_\ell.$$

Lemma 2.5 then gives the desired result.

Proposition 2.7. *For each $\ell \geq 2$,*

$$P^{(\ell)} = H^{(\ell)} \oplus r^2 P^{(\ell-2)}.$$

Proof. The sum of the dimensions of the subspaces $H^{(\ell)}$ and $r^2P^{(\ell-2)}$ of $P^{(\ell)}$ is equal to the dimension of $P^{(\ell)}$. Let us show that their intersection is trivial. If $P \in P^{(\ell)}$, with the help of the Euler identity, $x_1\frac{\partial P}{\partial x_1} + x_2\frac{\partial P}{\partial x_2} + x_3\frac{\partial P}{\partial x_3} = \ell P$, we establish, for every integer $k \geq 0$, the formula

$$\Delta(r^{2k}P) = 2k(2\ell + 2k + 1)r^{2k-2}P + r^{2k}\Delta P.$$

For $P \in H^{(\ell)}$, let k be the largest integer such that there exists a polynomial $Q \in P^{(\ell-2k)}$ satisfying $P = r^{2k}Q$. We thus have $0 = 2k(2\ell - 2k + 1)r^{2k-2}Q + r^{2k}\Delta Q$. We must have $k = 0$, since if not Q would be divisible by r^2, which contradicts the hypothesis on k. □

We deduce from this proposition that

$$P^{(\ell)} = H^{(\ell)} \oplus r^2H^{(\ell-2)} \oplus \dots, \tag{2.1}$$

where the last term is $r^\ell H^{(0)}$ if ℓ is even, and $r^{\ell-1}H^{(1)}$ if ℓ is odd.

3 Definition of Spherical Harmonics

A homogeneous polynomial on $\mathbb{R}^3$ is entirely determined by its restriction to the unit sphere S^2.

Definition 3.1. *The functions on the sphere obtained by restriction of harmonic homogeneous polynomials are called* spherical harmonics.

For each non-negative integer ℓ, the spherical harmonics of degree ℓ form a vector space $\widetilde{H}^{(\ell)}$ of dimension $2\ell+1$ which is isomorphic to $H^{(\ell)}$ and contained in the space of smooth functions on the sphere, itself contained in $L^2(S^2)$.

We remark first that, by equation (2.1), the space of restrictions to the sphere of homogeneous polynomials of degree ℓ can be written

$$\widetilde{P}^{(\ell)} = \widetilde{H}^{(\ell)} \oplus \widetilde{H}^{(\ell-2)} \oplus \dots, \tag{3.1}$$

where the last term is $\widetilde{H}^{(0)}$ if ℓ is even, and $\widetilde{H}^{(1)}$ if ℓ is odd.

3.1 Representations of SO(3) on Spaces of Spherical Harmonics

For each $\ell \in \mathbb{N}$, by the identification of $\widetilde{H}^{(\ell)}$ with $H^{(\ell)}$, we obtain a representation, also denoted by σ^ℓ, of SO(3) on the space of spherical harmonics of degree ℓ. These representations are *unitary*, as one can easily see using the rotational

invariance of the measure on S^2. In fact, for all functions f_1 and $f_2 \in L^2(S^2)$, and each $g \in \mathrm{SO}(3)$, setting $\sigma(g)f = f \circ g^{-1}$,

$$(\sigma(g)f_1|\sigma(g)f_2) = \int_{S^2} \overline{f_1(g^{-1}x)} f_2(g^{-1}x) \mathrm{d}\mu(x) = \int_{S^2} \overline{f_1(x)} f_2(x) \mathrm{d}\mu(x) = (f_1|f_2).$$

We have already seen (see Chapter 6, Section 2.1) that, when a Lie group is represented on a function space, the differential of the representation associated to each element of the Lie algebra of the group is a differential operator on the functions (when these are differentiable). By Theorem 7.7 of Chapter 4, if the representation of the group is *unitary*, the representation of the Lie algebra is *antihermitian*.

Let η_1, η_2, η_3 be the basis of the Lie algebra $\mathfrak{so}(3)$ of $\mathrm{SO}(3)$ introduced in Section 1.2 of Chapter 5. We see easily, by calculating $\frac{\mathrm{d}}{\mathrm{d}t} f(\exp(-t\eta_k)x)\big|_{t=0}$, for each $f \in C^\infty(\mathbb{R}^3)$, $x \in \mathbb{R}^3$ with components x_1, x_2, x_3 and $k = 1, 2, 3$, that the differential, $D\sigma$, of the representation σ of $\mathrm{SO}(3)$ on $C^\infty(\mathbb{R}^3)$ is such that

$$(D\sigma)\eta_1 = x_3 \frac{\partial}{\partial x_2} - x_2 \frac{\partial}{\partial x_3},$$

$$(D\sigma)\eta_2 = x_1 \frac{\partial}{\partial x_3} - x_3 \frac{\partial}{\partial x_1},$$

$$(D\sigma)\eta_3 = x_2 \frac{\partial}{\partial x_1} - x_1 \frac{\partial}{\partial x_2}.$$

By restriction, these formulas define the action of $\mathfrak{so}(3)$ on $P^{(\ell)}$ and $H^{(\ell)}$, and thus on $\widetilde{H}^{(\ell)}$. In spherical coordinates,

$$(D\sigma)\eta_1 = \sin\phi \frac{\partial}{\partial\theta} + \cos\phi \cot\theta \frac{\partial}{\partial\phi},$$

$$(D\sigma)\eta_2 = -\cos\phi \frac{\partial}{\partial\theta} + \sin\phi \cot\theta \frac{\partial}{\partial\phi},$$

$$(D\sigma)\eta_3 = -\frac{\partial}{\partial\phi}.$$

Each of these operators is *antihermitian* for the scalar product of $L^2(S^2)$, since the operators associated to elements of the group $\mathrm{SO}(3)$ by σ are unitary, and this fact can also be proved by a direct calculation. We introduce the following operators,

$$J_1 = i(\sin\phi \frac{\partial}{\partial\theta} + \cos\phi \cot\theta \frac{\partial}{\partial\phi}),$$

$$J_2 = i(-\cos\phi \frac{\partial}{\partial\theta} + \sin\phi \cot\theta \frac{\partial}{\partial\phi}),$$

$$J_3 = -i \frac{\partial}{\partial\phi},$$

which are thus *Hermitian*. We introduce also $J_\pm = J_1 \pm iJ_2$,

$$J_\pm = e^{\pm i\phi}(\pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}).$$

Then the operator J_3 is Hermitian and the operators J_+ and J_- are adjoint to one another.

Remark. The notation is consistent with that of Chapter 5. The Hermitian operators J_k are associated to the matrices $J_k = i\xi_k$ or $\widehat{J}_k = i\eta_k$, while the operators $J_\pm$ are associated to $J_\pm = J_1 \pm iJ_2 = i\xi_1 \mp \xi_2$ or $\widehat{J}_\pm = \widehat{J}_1 \pm i\widehat{J}_2 = i\eta_1 \mp \eta_2$. But to be prefectly rigorous, the operators we denote by J_k are by definition $i(D\sigma)\eta_k$ rather than $(D\sigma)(i\eta_k)$, since the $i\eta_k$ do not belong to $\mathfrak{so}(3)$.

3.2 The Casimir Operator

Now we introduce the Casimir operator of the representation in question (see Section 1.2 of Chapter 6),

$$J^2 = J_1^2 + J_2^2 + J_3^2 = J_+J_- + J_3^2 - J_3,$$

with the notation as above. The operator J^2 is *Hermitian*, since J_3 as well as J_+J_- are Hermitian.

A simple calculation shows that J^2 is equal to the negative of the spherical Laplacian,

$$J^2 = -\Delta_{S^2}, \tag{3.2}$$

that is,

$$\sum_{k=1}^{3}((D\sigma)\eta_k)^2 = \Delta_{S^2}.$$

3.3 Eigenfunctions of the Casimir Operator

We can write for $P \in P^{(\ell)}$,

$$P(r,\theta,\phi) = r^\ell Y(\theta,\phi).$$

The condition $\Delta P = 0$ is equivalent to

$$\Delta_{S^2}Y = -\ell(\ell+1)Y.$$

Therefore, the numbers $-\ell(\ell+1)$, for non-negative integer ℓ, are eigenvalues of the operator Δ_{S^2}, of multiplicity $2\ell+1$ since the corresponding eigenspace is $\widetilde{H}^{(\ell)}$. Thus,

$$\Delta_{S^2}\big|_{\widetilde{H}^{(\ell)}} = -\ell(\ell+1)\mathrm{Id}_{\widetilde{H}^{(\ell)}}.$$

Theorem 3.2. *The space $L^2(S^2)$ is the Hilbert direct sum of the $\widetilde{H}^{(\ell)}$, for $\ell \in \mathbb{N}$.*

Proof. We show first that the $\widetilde{H}^{(\ell)}$ are pairwise orthogonal in $L^2(S^2)$. This is the consequence of the fact that these are the eigenspaces of the operator Δ_{S^2}, which is Hermitian, as we have just shown.

Because the space of continuous functions on S^2 is dense in $L^2(S^2)$, it suffices to show that each continuous function on S^2 is a uniform limit of sums of elements of the $\widetilde{H}^{(\ell)}$. By the Weierstrass theorem, we can approximate any continuous function on S^2 by the restriction of polynomials on $\mathbb{R}^3$. We then decompose the polynomials into sums of homogeneous polynomials and, in view of the decomposition (3.1), the proof is then complete. □

We see that the space of square-integrable functions on the sphere decomposes into a Hilbert sum of eigenspaces of the spherical Laplacian, which are the spaces of spherical harmonics. Thus we can conclude as follows.

Corollary 3.3. *The spectrum of the Casimir operator acting on the functions of class C^2 on the sphere is discrete, its eigenvalues are the integers $\ell(\ell+1)$, $\ell \in \mathbb{N}$, and the associated eigenfunctions are the spherical harmonics of degree ℓ.*

Remark. The observables of quantum mechanics are Hermitian operators on the Hilbert space of states. As we observed in Chapter 6, the Hermitian operators J_k, $k = 1, 2, 3$, and J^2 have a physical interpretation: up to a factor of $\hbar$ ($\hbar = \frac{h}{2\pi}$, where h is Planck's constant), J_1, J_2, J_3 are the observables associated to the components of the *angular momentum* of a particle and, up to a factor of $\hbar^2$, J^2 is the observable associated to the square of the norm of the angular momentum. Thus the numbers $\hbar^2\ell(\ell+1)$, for non-negative integers ℓ, are the eigenvalues of the square of the norm of the angular momentum.

3.4 Bases of the Spaces of Spherical Harmonics

More particularly, we use the name *spherical harmonics* for the elements of an orthonormal basis, Y_m^ℓ, $-\ell \le m \le \ell$, of $\widetilde{H}^{(\ell)} \subset L^2(S^2)$, for each non-negative integer ℓ. Some authors seek to realize the orthonormalization condition according to the normalized scalar product on $L^2(S^2)$ introduced in Section 1, while others use the unnormalized scalar product defined by

$$\langle f_1 \mid f_2 \rangle = \int_0^\pi \int_0^{2\pi} \overline{f_1(\theta,\phi)} f_2(\theta,\phi) \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi.$$

We define the following functions, for $\ell \ge 0$. If $0 \le m \le \ell$, we set

$$Y_m^\ell(\theta,\phi) = \gamma_m^\ell Z_m^\ell(\theta) e^{im\phi},$$

where

$$Z_m^\ell(\theta) = \sin^m\theta \, Q_m^\ell(\cos\theta), \quad Q_m^\ell(x) = \frac{\mathrm{d}^{\ell+m}}{\mathrm{d}x^{\ell+m}}(1-x^2)^\ell,$$

and γ_m^ℓ is the real number

$$\gamma_m^\ell = \frac{(-1)^{\ell+m}}{2^\ell \ell!}\sqrt{\frac{2\ell+1}{4\pi}}\sqrt{\frac{(\ell-m)!}{(\ell+m)!}}.$$

If $-\ell \le m < 0$, we set

$$Y_m^\ell = (-1)^m \overline{Y_{-m}^\ell}.$$

Thus we define, for each $\ell \ge 0$, a family Y_m^ℓ, $-\ell \le m \le \ell$, of $2\ell+1$ functions on the sphere. We shall show that these functions are spherical harmonics in the preceding sense, and that they form an orthonormal basis of the space $\widetilde{H}^{(\ell)}$ for the unnormalized scalar product $\langle \ | \ \rangle$.

On the one hand, the functions Y_m^ℓ are eigenvectors of the operator $J_3 = -i\frac{\partial}{\partial\phi}$ with eigenvalue m,

$$J_3 Y_m^\ell = m Y_m^\ell. \tag{3.3}$$

On the other hand, the functions Y_m^ℓ satisfy the equations

$$J_+ Y_m^\ell = \sqrt{(\ell-m)(\ell+m+1)}\, Y_{m+1}^\ell, \tag{3.4}$$

and

$$J_- Y_m^\ell = \sqrt{(\ell+m)(\ell-m+1)}\, Y_{m-1}^\ell, \tag{3.5}$$

which we can easily verify by distinguishing the cases $m \ge 0$ and $m < 0$. Thus we see that the functions Y_m^ℓ are eigenvectors of $J^2 = -\Delta_{S^2}$ with eigenvalue $\ell(\ell+1)$,

$$J^2 Y_m^\ell = \ell(\ell+1) Y_m^\ell, \tag{3.6}$$

and we remark that the eigenvalue does not depend on m. Each function Y_m^ℓ is thus seen to be a spherical harmonic. Because the operator J_3 is Hermitian, the functions Y_m^ℓ are pairwise orthogonal. These $2\ell+1$ functions of $\widetilde{H}^{(\ell)}$ thus form an orthogonal basis.

The functions Y_m^ℓ correspond to the "kets" $|\ell, m\rangle$ introduced in Section 1.1 of Chapter 6, equation (1.7). More precisely, Formulas (3.3), (3.4) and (3.5) show that one defines an intertwining operator from $(\widetilde{H}^{(\ell)}, \sigma^\ell)$ to $(V^\ell, \mathcal{D}^\ell)$ by mapping the basis Y_m^ℓ to the basis $|\ell, m\rangle$.

Remark. We know that J^2 commutes with the operators J_k, $k = 1, 2, 3$. Because the operators J^2 and J_3 commute, we can diagonalize them simultaneously. This is what we have done by our choice of the basis (Y_m^ℓ).

We would like to show that, for the scalar product $\langle \ | \ \rangle$, this basis is orthonormal. We know already that it is orthogonal. We show next that, for

fixed ℓ, all the basis vectors Y_m^ℓ of $\widetilde{H}^{(\ell)}$ have the same norm, using the equalities (3.4) and (3.5) and the fact that J_+ and J_- are adjoints of one another. For $-\ell+1 \le m \le \ell$,

$$\langle J_+ Y_{m-1}^\ell \mid Y_m^\ell \rangle = \sqrt{(\ell+m)(\ell-m+1)}\langle Y_m^\ell \mid Y_m^\ell \rangle,$$

while

$$\langle Y_{m-1}^\ell \mid J_- Y_m^\ell \rangle = \sqrt{(\ell+m)(\ell-m+1)}\langle Y_{m-1}^\ell \mid Y_{m-1}^\ell \rangle.$$

Because these two quantites are equal we conclude, by induction on m, that all the vectors Y_m^ℓ have the same norm. It suffices then to calculate $\langle Y_\ell^\ell \mid Y_\ell^\ell \rangle$. The polynomial Q_ℓ^ℓ is the constant $(-1)^\ell(2\ell)!$ and, consequently,

$$Y_\ell^\ell(\theta,\phi) = (-1)^\ell \, \gamma_\ell^\ell \, (2\ell)! \, \sin^\ell \theta \, e^{i\ell\phi}.$$

We then have

$$\langle Y_\ell^\ell \mid Y_\ell^\ell \rangle = \gamma_\ell \, \mathcal{I}_\ell,$$

where

$$\gamma_\ell = 2\pi \, (\gamma_\ell^\ell (2\ell)!)^2$$

and

$$\mathcal{I}_\ell = \int_0^\pi \sin^{2\ell+1}\theta \, \mathrm{d}\theta.$$

We have $\mathcal{I}_0 = 2$ and $\gamma_0 = \frac{1}{2}$, and therefore $\langle Y_0^0 \mid Y_0^0 \rangle = 1$. An integration by parts shows that $\mathcal{I}_{\ell+1} = \mathcal{I}_\ell - \frac{1}{2\ell+2}\mathcal{I}_{\ell+1}$, that is, $(2\ell+3)\mathcal{I}_{\ell+1} = (2\ell+2)\mathcal{I}_\ell$. Because $(2\ell+2)\gamma_{\ell+1} = (2\ell+3)\gamma_\ell$, we can deduce that

$$\langle Y_\ell^\ell \mid Y_\ell^\ell \rangle = 1,$$

for every $\ell \in \mathbb{N}$. By Theorem 3.2, we can thus state the following theorem.

Theorem 3.4. *The spherical harmonics Y_m^ℓ, $\ell \in \mathbb{N}$, $-\ell \le m \le \ell$, form a Hilbert space basis of $L^2(S^2)$ equipped with the unnormalized scalar product.*

In other words, each function belonging to $L^2(S^2)$ has a series expansion in spherical harmonics, convergent in the sense of the norm of $L^2(S^2)$,

$$f = \sum_{\ell=0}^{\infty} \sum_{-\ell \le m \le \ell} f_m^\ell Y_m^\ell = f_0^0 Y_0^0 + f_1^1 Y_1^1 + f_0^1 Y_0^1 + f_{-1}^1 Y_{-1}^1 + \dots,$$

where the coefficients of the expansion are given by the unnormalized scalar products of functions on the sphere,

$$f_m^\ell = \langle Y_m^\ell \mid f \rangle.$$

Remark. The *Legendre polynomials* are defined, for $\ell \in \mathbb{N}$, by

$$P_\ell(x) = \frac{(-1)^\ell}{2^\ell \ell!} \frac{\mathrm{d}^\ell}{\mathrm{d}x^\ell}(1-x^2)^\ell,$$

and the *Legendre functions* are defined, for $m \in \mathbb{N}$ and for $x \in [-1, 1]$, by

$$P_{\ell,m}(x) = (-1)^m (1-x^2)^{\frac{m}{2}} \frac{\mathrm{d}^m}{\mathrm{d}x^m} P_\ell(x) = \frac{(-1)^{\ell+m}}{2^\ell \ell!} (1-x^2)^{\frac{m}{2}} \frac{\mathrm{d}^{\ell+m}}{\mathrm{d}x^{\ell+m}} (1-x^2)^\ell.$$

In general one expresses the spherical harmonics, Y_m^ℓ, by means of the Legendre functions, $P_{\ell,m}$. We see that, for $m \geq 0$,

$$Y_m^\ell(\theta, \phi) = \gamma_{\ell,m} P_{\ell,m}(\cos\theta) e^{im\phi},$$

where

$$\gamma_{\ell,m} = \sqrt{\frac{2\ell+1}{4\pi}} \sqrt{\frac{(\ell-m)!}{(\ell+m)!}}.$$

The functions denoted by $Z_m^\ell(\theta)$ above differ from the Legendre functions $P_{\ell,m}(\cos\theta)$ only by a numerical factor,

$$Z_m^\ell(\theta) = (-1)^{\ell+m} 2^\ell \ell! P_{\ell,m}(\cos\theta),$$

and the constants γ_m^ℓ and $\gamma_{\ell,m}$ are related by

$$\gamma_m^\ell = \frac{(-1)^{\ell+m}}{2^\ell \ell!} \gamma_{\ell,m}.$$

This close link between spherical harmonics and Legendre functions allows us to deduce the properties of Legendre functions from those of spherical harmonics which were proved using the representation theory of groups. This method generalizes to other groups to which there correspond other *special functions*.

We can obtain the orthonormality property of the basis functions (Y_m^ℓ) by expressing them as functions of the matrix coefficients of the unitary representations σ^ℓ and using the orthogonality relations proved in Theorem 4.4 of Chapter 3 (see Problem 9).

3.5 Explicit Formulas

In spectroscopy, the levels $\ell = 0, 1, 2, 3, 4, 5$ are denoted by the letters s, p, d, f, g, h. Here are the explicit formulas for the levels s, p and d.

s	$Y_0^0 = \sqrt{\frac{1}{4\pi}}$
p_0	$Y_0^1 = \sqrt{\frac{3}{4\pi}} \cos\theta$
$p_\pm$	$Y_{\pm1}^1 = \mp\sqrt{\frac{3}{8\pi}} \sin\theta \; e^{\pm i\phi}$
d_0	$Y_0^2 = \sqrt{\frac{5}{16\pi}} (3\cos^2\theta - 1)$
$d_{\pm1}$	$Y_{\pm1}^2 = \mp\sqrt{\frac{15}{8\pi}} \sin\theta\cos\theta \; e^{\pm i\phi}$
$d_{\pm2}$	$Y_{\pm2}^2 = \sqrt{\frac{15}{32\pi}} \sin^2\theta \; e^{\pm 2i\phi}$

We also calculate for the level f,

$$Y_0^3 = \sqrt{\frac{7}{16\pi}} \cos\theta(5\cos^2\theta - 3),$$

and so forth.

We also introduce, for $m > 0$, real linear combinations of spherical harmonics,

$$Y_{m,c}^{\ell} = (-1)^m \sqrt{2}\gamma_m^{\ell} Z_m^{\ell}(\theta) \cos m\phi$$

and

$$Y_{m,s}^{\ell} = (-1)^m \sqrt{2}\gamma_m^{\ell} Z_m^{\ell}(\theta) \sin m\phi.$$

When one restricts a representation $\mathcal{D}^{\ell}$ to a finite subgroup of SO(3), such as a dihedral group, one obtains a representation, in general not irreducible, of this subgroup. With linear combinations of spherical harmonics, Y_m^{ℓ}, one can construct bases for the spaces of irreducible representations of finite subgroups of SO(3).

The theory of spherical harmonics has vast generalizations, where the groups considered are compact Lie groups more general than SO(3). What generalizes the spheres are the homogeneous spaces, which are manifolds that are quotients of Lie groups by closed subgroups.

References

Sternberg (2003) treats the theory of spherical harmonics, and one can find a survey of some of their properties in Basdevant-Dalibard (2005) and Edmonds (1996). See also Singer (2005). For a more complete theory, one might consult Vilenkin (1968), Vilenkin-Klimyk (1991), Talman (1968), or Dieudonné (2012). The book by Dym and McKean (2010) deals with spherical harmonics in terms of Fourier analysis, and so does that of Axler, Bourdon and Ramey (2001) on a more advanced level. The comprehensive textbook of Gurarie (1992) relates representation theory and harmonic analysis. See Faraut (2008) for an excellent self-contained text and Hall (2016).

Topics in History

The study of spherical harmonics is a chapter in the immense domain of special functions, with links to the theory of group representations, and this domain of pure mathematics has had an enormous number of applications to fields as diverse as, for example, neutron transport problems or optical imaging or "acoustic direction of arrival analysis", as well as in 3D computer graphics. The fields of applications of the special functions are as numerous as they are diverse.

The diversity of special functions

Some special functions are called hypergeometric, others bear the names of mathematicians, Bessel (1784–1846), Jacobi, Hankel (1839–1873), Macdonald (1865–1935), Whittaker (1905–1984) and of many others. Some can be studied in terms of representations of Lie groups of 2×2 matrices, others play a role in the representation theory of the linear, orthogonal, unitary or symplectic groups of higher order.

From a historical point of view, already at the end of the 20th century there were books on 'special functions and operator calculus', 'special functions and probability theory', 'special functions and computer science'. A search for "special functions" in the titles of books on MathSciNet in 2021 offers over 150 answers, many of them conference proceedings, and a large number of these books deal with applications and numerical methods. One book published in 2010 contains the proceedings of the annual conference of the "Society for Special Functions and their Applications", based in India but with an international advisory committee. A review of a book published in 2013 opens with "Special functions play a fundamental role in mathematical, physical, bio-medical and engineering science advanced study and research", and adds that "it is almost impossible to obtain analytical and numerical solutions of numerous problems in the above-mentioned fields without detailed knowledge of special functions".

Special functions and group representations

The 1965 book in Russian by N. Ya. Vilenkin, *Representations of Lie Groups and Special Functions*, was "the first attempt in the literature to give a unified systematization of basic special functions of mathematical physics from the point of view of the theory of group representations". It was translated into English three years later and, in 1969, into French. This book and the three volumes in Russian subsequently published by Vilenkin and A. U. Klimyk, which appeared in English translation in 1991–1993, treat the properties of special functions, in particular the differential equations, recurrence formulas and addition formulas which they satisfy. They constitute 'the' reference for the theory of special functions and its close connection with representation theory, up to the latest developments concerning the q-special functions, that had expanded considerably after quantum groups had been defined by Vladimir Drinfel'd in his communication at the International Congress of Mathematicians in 1986, and continued to progress ever since. Much additional theory was covered in a volume of "Recent advances" by the same authors that appeared in 1995, after Vilenkin's death in 1991. This volume, unlike the preceding ones, was published in English, "translated from the Russian manuscript", marking a trend in the realm of publications which has since become prevalent: scientists from Russia and, more recently from China and practically all countries, including Germany and France, publish in English, not in their native language.

Whence spherical harmonics?

The differential operator "Laplacian" is named after Pierre Simon de Laplace (1749–1827), mathematician, physicist, author of immensely important books and papers. The "Laplacian" operator, harmonic functions and spherical harmonics can be found in his "Theory of the attraction of spheroids and of the configuration of the planets" of 1782 and in his *Treatise on Celestial Mechanics* whose 3 volumes appeared between 1799 and 1825. Laplace lived and worked through the ancien régime, the Revolution, Napoléon's reign and the return of the monarchy in France. He published on calculus, celestial mechanics, differential equations, the theory of probability, the theory of heat, in sum he made advances in nearly all fields of the mathematics and mathematical physics that had been developed before Gauss, Jacobi and Riemann. How many scientists have a railway stop in the immediate proximity of Paris *intra muros* named after them? (There is also a street named after him in the heart of the student quarter, it is short and narrow, but very close to where the École Polytechnique – which Laplace had co-founded in 1796 – stood until it moved to its vast new campus in Palaiseau, south-west of Paris.)

Legendre polynomials and Legendre functions were named after Adrien-Marie Legendre (1752–1833), another French mathematician with very diverse and important publications, in number theory as well as on elliptic functions, who had been associated with the Académie des Sciences since 1783. He is known for, among other results, his proof of the irrationality of π. He introduced the polynomials now called Legendre polynomials in his work on celestial mechanics of 1784, "An inquiry on the configuration of planets". And he too has a street named after him in Paris.

A disclaimer

I shall not attempt even a fragmentary sketch of the history of special functions[1]. It would have to recall the Babylonians and trigonometry, Copernicus, the logarithms of Napier in 1614, the calculus of Newton, Leibniz and Euler, and all the greats of the 18th and 19th centuries, Laplace, Legendre, Hermite, Laguerre, as well as Gauss, Jacobi, and Maxwell. It would have to record the progress of almost all branches of physics and the advent of quantum mechanics, the creation of tables of values of more and more special functions, the initial ideas of Élie Cartan, the important contributions from Russian mathematicians, the classic handbook of Milton Abramowitz and Irene A. Stegun published in 1964 by the U.S. Government Printing Office in Washington, D.C., and evidently much, much more.

[1] I found such a sketch by Stephen Wolfram on his web page and I recommend it.

Exercises

Exercise 7.1 *Representations of* SO(2).

Let (r, θ) be polar coordinates on $\mathbb{R}^2$ minus the origin. We recall that the Laplacian in polar coordinates can be written

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}.$$

(a) Show that $(\sigma(g)f)(x) = f(g^{-1}(x))$, $x \in \mathbb{R}^2$, defines a unitary representation σ of SO(2) on $L^2(\mathbb{R}^2)$.

Denote by $\mathcal{H}^{(\ell)}$ the vector space of harmonic homogeneous polynomials on $\mathbb{R}^2$, of degree ℓ, where $\ell \in \mathbb{N}$.

(b) Find a basis of $\mathcal{H}^{(\ell)}$.
(c) Show that $\mathcal{H}^{(\ell)}$ is invariant under σ. Is the restriction of σ to $\mathcal{H}^{(\ell)}$ irreducible for every $\ell \geq 0$? If not, find its decomposition into a direct sum of irreducible representations of SO(2).

Exercise 7.2 *Orthogonality relations for the Legendre functions.*

Deduce from the properties of spherical harmonics the orthogonality relations for the Legendre functions,

$$\int_{-1}^{1} P_{\ell,m}(x) P_{\ell',m}(x) \mathrm{d}x = 0 \qquad \text{if} \qquad \ell \neq \ell',$$

and calculate the value of the integral for $\ell = \ell'$.

Exercise 7.3 *Addition theorem for spherical harmonics.*

Let $d^{\ell}_{mn}(\theta)$, $-\ell \leq m \leq \ell$, $-\ell \leq n \leq \ell$, $\theta \in [0, \pi]$, be the matrix coefficients of the rotation $\exp(\theta \eta_1)$ in the representation $\mathcal{D}^{\ell}$ in the orthonormal basis $|\ell, m\rangle$. Then,

$$d^{\ell}_{00}(\theta) = P_{\ell}(\cos\theta),$$

where P_ℓ is the ℓ-th Legendre polynomial, and the coefficients $d^{\ell}_{m0}(\theta)$ are related to the spherical harmonics $Y^{\ell}_{m}(\theta, \phi)$ by

$$Y^{\ell}_{m}(\theta, \phi) = \sqrt{\frac{2\ell + 1}{4\pi}} (-i)^m d^{\ell}_{m0}(\theta) e^{im\phi}.$$

[These formulas are proved in Problem 9. The factor $(-i)^m$ appears here instead of the factor i^m because $\varphi^{\ell}_{m} = (-1)^{\ell - m} |\ell, m\rangle$ implies $\rho^{\ell}_{m,0} = (-1)^m d^{\ell}_{m0}$.]

(a) Verify these formulas when $\ell = 1$ using the result of Exercise 6.6.
(b) Prove the equation

$$\frac{4\pi}{2\ell + 1} \sum_{m=-\ell}^{\ell} \overline{Y^{\ell}_{m}(\theta, \phi)} Y^{\ell}_{m}(\theta', \phi) = P_{\ell}(\cos(\theta - \theta')).$$

This equation, called the *addition theorem* for spherical harmonics, has important applications in quantum mechanics.

Exercise 7.4 *Recurrence relation for the spherical harmonics.*

Show that there are constants $\alpha(\ell, m)$ and $\beta(\ell, m)$ such that

$$\cos\theta\, Y_m^\ell = \alpha(\ell, m) Y_m^{\ell+1} + \beta(\ell, m) Y_m^{\ell-1}.$$

Chapter 8

Representations of SU(3) and Quarks



In the study of the representations of the Lie group SU(3) we introduce the notion of weight and we present the root diagrams of several low-dimensional representations, in particular of its fundamental and adjoint representations. We show how the classification of elementary particles appears in the representation theory of SU(3).

1 Representations of $\mathfrak{sl}(3,\mathbb{C})$ and SU(3)

1.1 Review of $\mathfrak{sl}(n,\mathbb{C})$

In $\mathfrak{sl}(n,\mathbb{C})$, the Lie algebra of traceless $n\times n$ matrices with complex coefficients, we denote by $\mathfrak{h}$ the Abelian Lie subalgebra of the traceless diagonal matrices. We note that $\mathfrak{h}$ is maximal among the Abelian Lie subalgebras of $\mathfrak{sl}(n,\mathbb{C})$. In fact, if a Lie subalgebra of $\mathfrak{sl}(n,\mathbb{C})$ contains $\mathfrak{h}$ and a non-diagonal matrix, it is non-Abelian since no non-diagonal matrix commutes with every traceless diagonal matrix. The Lie subalgebra $\mathfrak{h}$ of $\mathfrak{sl}(n,\mathbb{C})$ is called a *Cartan subalgebra* of $\mathfrak{sl}(n,\mathbb{C})$.

We denote by E_{ij}, $1\leq i\leq n$, $1\leq j\leq n$, the matrix having 1 on the intersection of the i-th line and the j-th column and having 0 everywhere else. For $1\leq i\leq n-1$, we set $H_i=E_{ii}-E_{i+1,i+1}$. Recall that a basis of $\mathfrak{sl}(n,\mathbb{C})$ is formed by the E_{ij}, $1\leq i\leq n$, $1\leq j\leq n$, $i\neq j$, and the H_i, $i=1,\dots,n-1$, where the matrices H_i, $i=1,\dots,n-1$, are a basis of $\mathfrak{h}$.

We have, $1\leq i\leq n$, $1\leq j\leq n$, $1\leq k\leq n$, $1\leq \ell\leq n$, $E_{ij}E_{kl}=\delta_{jk}E_{il}$, whence the commutation relations of these matrices,

$$[E_{ij},E_{kl}]=\delta_{jk}E_{il}-\delta_{il}E_{kj}, \tag{1.1}$$

$$[H,E_{ij}]=(\lambda_i-\lambda_j)(H)E_{ij}, \tag{1.2}$$

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where $\lambda_i : H \mapsto \lambda_i(H)$ denotes the linear map of $\mathfrak{h}$ into $\mathbb{C}$ that associates to each traceless diagonal matrix, H, the element of its diagonal on the i-th line.

1.2 The Case of $\mathfrak{sl}(3,\mathbb{C})$

In the case of $\mathfrak{sl}(3,\mathbb{C})$, the Cartan subalgebra $\mathfrak{h}$ is of dimension 2 and has as a basis

$$H_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ H_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

We can take as a basis of the vector space of strictly upper triangular matrices,

$$E_1 = E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ E_2 = E_{23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},\ E_3 = E_{13} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and, as a basis of the vector space of strictly lower triangular matrices,

$$F_1 = {}^tE_1 = E_{21},\ F_2 = {}^tE_2 = E_{32},\ F_3 = {}^tE_3 = E_{31}.$$

We set

$$H_3 = H_1 + H_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The commutation relations can be written:

$$[H_1, H_2] = 0,\ [E_i, F_i] = H_i,\ i = 1, 2, 3,$$

$$[E_1, E_2] = E_3,\ [E_1, E_3] = 0,\ [E_2, E_3] = 0,$$

$$[F_1, F_2] = -F_3,\ [F_1, F_3] = 0,\ [F_2, F_3] = 0,$$

and

$$[E_1, F_2] = 0,\ [E_2, F_1] = 0,$$

$$[E_1, F_3] = -F_2,\ [E_3, F_1] = -E_2,$$

$$[E_2, F_3] = F_1,\ [E_3, F_2] = E_1.$$

The remaining commutation relations will interest us more particularly in what follows. Some of them say that E_1, E_2 and E_3 are common eigenvectors of the endomorphisms ad_{H_1} and ad_{H_2} of $\mathfrak{g} = \mathfrak{sl}(3,\mathbb{C})$,

$$[H_1, E_1] = 2E_1,\ [H_2, E_1] = -E_1,$$

$$[H_1, E_2] = -E_2,\ [H_2, E_2] = 2E_2,$$

$$[H_1, E_3] = E_3,\ [H_2, E_3] = E_3.$$

The others say that, similarly, F_1, F_2 and F_3 are common eigenvectors of the endomorphisms ad_{H_1} and ad_{H_2},

$$[H_1, F_1] = -2F_1,\ [H_2, F_1] = F_1,$$

$$[H_1, F_2] = F_2,\ [H_2, F_2] = -2F_2,$$

$$[H_1, F_3] = -F_3,\ [H_2, F_3] = -F_3.$$

We let $\alpha_1(H_1)$ (resp., $\alpha_1(H_2)$) denote the eigenvalue of ad_{H_1} (resp., ad_{H_2}) corresponding to the eigenvector E_1. By the preceding formulas,

$$\alpha_1(H_1) = 2,\ \alpha_1(H_2) = -1.$$

In fact, each element of the Cartan subalgebra $\mathfrak{h}$ can be written as a linear combination of H_1 and H_2 and we thus define a linear form α_1 on $\mathfrak{h}$. The values of this linear form on the basis vectors (H_1, H_2) are $(2, -1)$. These are also the components of this linear form in the basis dual to the basis (H_1, H_2), by the definition of the notion of dual basis. We introduce similarly the notation $\alpha_2(H)$ for the eigenvalues of $H \in \mathfrak{h}$ corresponding to the eigenvector E_2,

$$\alpha_2(H_1) = -1,\ \alpha_2(H_2) = 2.$$

Finally, let $\alpha_3(H)$ be the eigenvalue of $H \in \mathfrak{h}$ corresponding to the eigenvector E_3,

$$\alpha_3(H_1) = 1,\ \alpha_3(H_2) = 1.$$

We clearly have

$$\alpha_3 = \alpha_1 + \alpha_2.$$

By definition,

$$[H_i, E_j] = \alpha_j(H_i)E_j,\quad 1 \le i \le 3,\quad 1 \le j \le 3,$$

and we also see that

$$[H_i, F_j] = -\alpha_j(H_i)F_j,\quad 1 \le i \le 3,\quad 1 \le j \le 3.$$

The linear forms $\alpha_1,\ \alpha_2,\ \alpha_3,\ -\alpha_1,\ -\alpha_2,\ -\alpha_3$ on $\mathfrak{h}$ play an important role in representation theory. They are called the *roots* of $\mathfrak{sl}(3,\mathbb{C})$.

Above we have introduced, for $i = 1, 2, 3$, the linear form on $\mathfrak{h}$, denoted by λ_i, that associates to each traceless diagonal matrix H in $\mathfrak{h}$ the diagonal element $\lambda_i(H)$ on its i-th line. We see immediately, directly or from (1.2), that

$$\alpha_1 = \lambda_1 - \lambda_2,\ \alpha_2 = \lambda_2 - \lambda_3,\ \alpha_3 = \lambda_1 - \lambda_3.$$

1.3 The Bases (I_3, Y) and (I_3, T_8) of $\mathfrak{h}$

In the Cartan subalgebra, $\mathfrak{h}$, of $\mathfrak{sl}(3,\mathbb{C})$ we introduce a new basis, (I_3, Y), defined by

$$I_3 = \frac{1}{2}H_1, \quad Y = \frac{1}{3}(H_1 + 2H_2) = \frac{1}{3}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

We introduce also

$$T_8 = \frac{\sqrt{3}}{2}Y = \frac{1}{2\sqrt{3}}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix},$$

and we consider the basis (I_3, T_8) of $\mathfrak{h} \subset \mathfrak{sl}(3,\mathbb{C})$ as well.

Remark. In many physics books, I_3 is denoted by T_3. The quantity $Q = \frac{1}{2}Y + I_3$ is introduced as well. The eigenvalues of the operators I_3, Y and Q of the symmetry algebra in a representation correspond to the third component of the *isospin*, to the *hypercharge* and to the *electric charge*, respectively.

1.4 Representations of $\mathfrak{sl}(3,\mathbb{C})$ and SU(3)

There is a bijection between the irreducible representations of $\mathfrak{sl}(3,\mathbb{C})$ and those of $\mathfrak{su}(3)$, by complexification and, conversely, by restriction. Furthermore, the irreducible representations of the Lie algebra $\mathfrak{su}(3)$ are in bijective correspondence with those of the Lie group SU(3) since this group is connected and simply connected.

2 The Adjoint Representation and Roots

In what follows we are interested in the representations of the Lie group SU(3). By the preceding remarks, it suffices to study the representations of the Lie algebra $\mathfrak{sl}(3,\mathbb{C})$, which is of complex dimension 8. We make use of the notations of Section 1.1. The commutation relations of $\mathfrak{sl}(3,\mathbb{C})$ imply: if $H = \begin{pmatrix} \lambda_1(H) & 0 & 0 \\ 0 & \lambda_2(H) & 0 \\ 0 & 0 & \lambda_3(H) \end{pmatrix}$, then $[H, E_{ij}] = (\lambda_i(H) - \lambda_j(H))E_{ij}$. The roots of $\mathfrak{sl}(3,\mathbb{C})$ are thus given by the following table:

root	relations		basis(I_3, Y)	basis(I_3, T_8)
α_1	$[I_3, E_1] = E_1$	$[Y, E_1] = 0$	$(1, 0)$	$(1, 0)$
α_2	$[I_3, E_2] = -\frac{1}{2}E_2$	$[Y, E_2] = E_2$	$(-\frac{1}{2}, 1)$	$(-\frac{1}{2}, \frac{\sqrt{3}}{2})$
α_3	$[I_3, E_3] = \frac{1}{2}E_3$	$[Y, E_3] = E_3$	$(\frac{1}{2}, 1)$	$(\frac{1}{2}, \frac{\sqrt{3}}{2})$
α_1'	$[I_3, F_1] = -F_1$	$[Y, F_1] = 0$	$(-1, 0)$	$(-1, 0)$
α_2'	$[I_3, F_2] = \frac{1}{2}F_2$	$[Y, F_2] = -F_2$	$(\frac{1}{2}, -1)$	$(\frac{1}{2}, -\frac{\sqrt{3}}{2})$
α_3'	$[I_3, F_3] = -\frac{1}{2}F_3$	$[Y, F_3] = -F_3$	$(-\frac{1}{2}, -1)$	$(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$

The following equalities are obvious:

$$\alpha_3 = \alpha_1 + \alpha_2,\ \alpha_1' = -\alpha_1,\ \alpha_2' = -\alpha_2,\ \alpha_3' = -\alpha_3.$$

The first can also be seen as a consequence of the Jacobi identity because

$$\begin{aligned}[H_i, E_3] &= [H_i, [E_1, E_2]] = [[H_i, E_1], E_2] + [E_1, [H_i, E_2]] \\ &= \alpha_1(H_i)[E_1, E_2] + \alpha_2(H_i)[E_1, E_2] = (\alpha_1 + \alpha_2)(H_i)E_3.\end{aligned}$$

We recall also that $\alpha_1 = \lambda_1 - \lambda_2$ and $\alpha_2 = \lambda_2 - \lambda_3$.

Now we draw the roots of $\mathfrak{sl}(3, \mathbb{C})$ in the plane relative to the axes I_3 and T_8. We see that these six roots form a regular hexagon, which would not have been the case had we used the basis (I_3, Y). Technically 0 is not called a root but, for every $H \in \mathfrak{h}$, 0 is a double eigenvalue of the endomorphism ad_H. This is why we draw the point 0 as a double point (a point surrounded by a little circle). The adjoint representation is of dimension 8 and we shall denote it in the rest of this chapter by **8**. In the following sections, we shall talk more generally of *weights* of representations of the algebra $\mathfrak{sl}(3, \mathbb{C})$ and we shall see that *the roots are the weights of the adjoint representation* (Fig. 1).

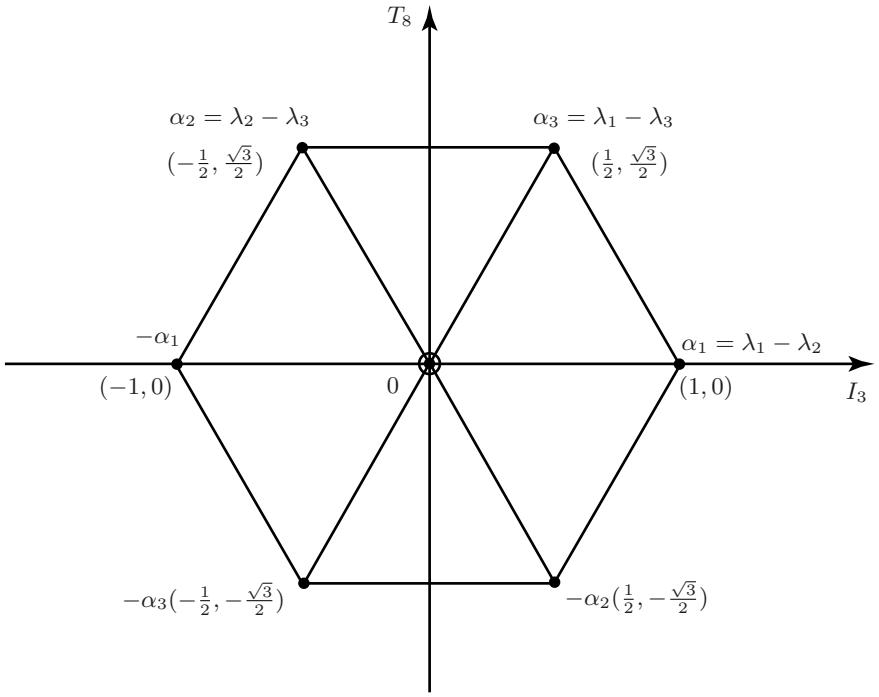


Figure 1. The representation **8**.

3 The Fundamental Representation and Its Dual

3.1 The Fundamental Representation

Now we consider the fundamental representation. Thus

$$\begin{aligned} H_1e_1 &= e_1, & H_2e_1 &= 0,\\ H_1e_2 &= -e_2, & H_2e_2 &= e_2,\\ H_1e_3 &= 0, & H_2e_3 &= -e_3. \end{aligned}$$

By definition the *weights* of this representation are the linear forms λ_1, λ_2 and λ_3 on $\mathfrak{h}$, satisfying, for $H \in \mathfrak{h}$ and $i = 1, 2, 3$,

$$He_i = \lambda_i(H)e_i.$$

Thus in the basis (H_1, H_2) the components of the three weights are

$$\lambda_1 = (1, 0),\ \lambda_2 = (-1, 1),\ \lambda_3 = (0, -1).$$

Hence we have the following table:

weights	relations		basis(I_3, Y)	basis(I_3, T_8)	quarks
λ_1	$I_3e_1 = \frac{1}{2}e_1$	$Ye_1 = \frac{1}{3}e_1$	$(\frac{1}{2}, \frac{1}{3})$	$(\frac{1}{2}, \frac{1}{2\sqrt{3}})$	u
λ_2	$I_3e_2 = -\frac{1}{2}e_2$	$Ye_2 = \frac{1}{3}e_2$	$(-\frac{1}{2}, \frac{1}{3})$	$(-\frac{1}{2}, \frac{1}{2\sqrt{3}})$	d
λ_3	$I_3e_3 = 0$	$Ye_3 = -\frac{2}{3}e_3$	$(0, -\frac{2}{3})$	$(0, -\frac{1}{\sqrt{3}})$	s

Thus there are three linearly independent vectors e_1, e_2, e_3 which are common eigenvectors of all elements H of $\mathfrak{h}$. We denote them by $u,\ d,\ s$, which symbols designate the *quarks*,

$$u = up,\ d = down,\ s = strange.$$

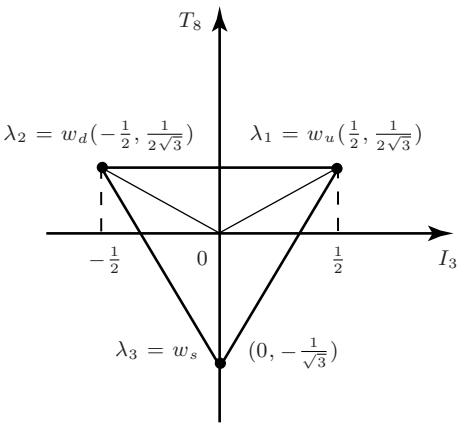


Figure 2. The representation **3**.

(See more information on the quarks in Sections 6 and 7 below.)

In the basis (I_3, T_8), the weights, $\lambda_1 = w_u, \lambda_2 = w_d, \lambda_3 = w_s$, of the fundamental representation form an equilateral triangle. Thus

$$\lambda_1 + \lambda_2 + \lambda_3 = 0.$$

Above, we drew the weight diagram of the fundamental representation of $\mathfrak{sl}(3, \mathbb{C})$, which we shall call **3** from now on (Fig. 2).

3.2 The Dual of the Fundamental Representation

We pass from the fundamental representation to its dual by replacing the matrices $X \in \mathfrak{sl}(3, \mathbb{C})$ by $X' = -{}^tX$, and so $H'_j e_i = -\lambda_i(H_j)e_i$. The equations

$$\begin{array}{ll} H'_1 e_1 = -e_1, & H'_2 e_1 = 0, \\ H'_1 e_2 = e_2, & H'_2 e_2 = -e_2, \\ H'_1 e_3 = 0, & H'_2 e_3 = e_3, \end{array}$$

show that the components of the three weights, $\lambda'_1, \lambda'_2, \lambda'_3$, of this representation in the basis (H_1, H_2) are

$$\lambda'_1 = (-1, 0),\ \lambda'_2 = (1, -1),\ \lambda'_3 = (0, 1).$$

One obtains thus the following table for the weights of the dual representation of the fundamental representation.

weights	relations		basis(I_3, Y)	basis(I_3, T_8)	antiquarks
λ'_1	$I_3 e_1 = -\frac{1}{2} e_1$	$Y e_1 = -\frac{1}{3} e_1$	$(-\frac{1}{2}, -\frac{1}{3})$	$(-\frac{1}{2}, -\frac{1}{2\sqrt{3}})$	$\overline{u}$
λ'_2	$I_3 e_2 = \frac{1}{2} e_2$	$Y e_2 = -\frac{1}{3} e_2$	$(\frac{1}{2}, -\frac{1}{3})$	$(\frac{1}{2}, -\frac{1}{2\sqrt{3}})$	$\overline{d}$
λ'_3	$I_3 e_3 = 0$	$Y e_3 = \frac{2}{3} e_3$	$(0, \frac{2}{3})$	$(0, \frac{1}{\sqrt{3}})$	$\overline{s}$

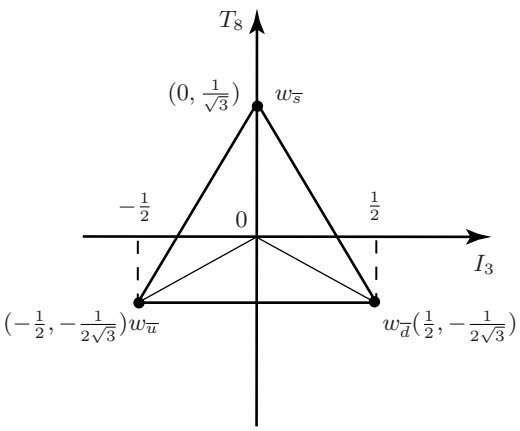


Figure 3. The representation $\overline{\mathbf{3}}$.

The weight vectors $\overline{u}, \overline{d}$ and $\overline{s}$ are called the *antiquarks*.

Above, we drew the weight diagram of the representation dual to the fundamental representation of $\mathfrak{sl}(3,\mathbb{C})$, which we shall call $\overline{\mathbf{3}}$ from now on (Fig. 3).

4 Highest Weight of a Finite-Dimensional Representation

4.1 Highest Weight

A *weight* of a representation (E,ρ) is a linear form ω on $\mathfrak{h}$ such that there is a non-zero vector $v \in E$ satisfying $\rho(H)v = \omega(H)v$ for every $H \in \mathfrak{h}$.

We shall see that in each of the three representations described above, one of the weights plays a particular role. It will be called the *highest weight* of the representation. An eigenvector corresponding to the highest weight, which is defined up to multiplication by a non-zero scalar, is called a *highest weight vector*.

Example 1. In $\mathbf{3}$, the highest weight is $\lambda_1 = w_u$, with highest weight vector $e_1 = u$. We verify the following properties:

$$\begin{array}{ll} (i) & H_1e_1 = e_1,\ H_2e_1 = 0 \quad (e_1 \text{ is an eigenvector for each } H \in \mathfrak{h}), \\ (ii) & E_ie_1 = 0, \qquad\qquad\quad i = 1,2,3. \end{array}$$

Furthermore, we obtain e_2 and e_3 by the action of F_1, F_2 and F_3 on e_1. More precisely,

$$F_1e_1 = e_2,\ F_2e_1 = 0,\ F_3e_1 = e_3.$$

Example 2. In $\overline{\mathbf{3}}$, the highest weight is $\lambda_3' = w_{\overline{s}}$, with highest weight vector $e_3 = \overline{s}$, that is,

$$\begin{array}{ll} (i) & H_1'e_3 = 0,\ H_2'e_3 = e_3 \quad (e_3 \text{ is an eigenvector for each } H \in \mathfrak{h}), \\ (ii) & E_i'e_3 = -F_ie_3 = 0, \qquad\quad i = 1,2,3. \end{array}$$

As in the preceding example, we obtain scalar multiples of e_1 and e_2 by the action of F_1', F_2' and F_3' on e_3. More precisely,

$$F_1'e_3 = -E_1e_3 = 0,\ F_2'e_3 = -E_2e_3 = -e_2,\ F_3'e_3 = -E_3e_3 = -e_1.$$

Example 3. In $\mathbf{8}$, the highest weight is α_3, with highest weight vector E_3. We have

$$\begin{array}{ll} (i) & [H_1,E_3] = E_3, [H_2,E_3] = E_3 \quad (E_3 \text{ is an eigenvector for each } H \in \mathfrak{h}), \\ (ii) & [E_i,E_3] = 0, \qquad\qquad\qquad\quad i = 1,2,3. \end{array}$$

Here we obtain scalar multiples of $E_1, E_2, F_1, F_2, F_3, H_1, H_2$ by the action of F_1, F_2 and F_3 on E_3. More precisely,

$$[F_1,E_3] = E_2,\ [F_2,E_3] = -E_1,\ [F_3,E_3] = -H_3,$$

$$[F_2,[F_1,E_3]] = -H_2,\ [F_3,[F_1,E_3]] = -F_1,$$
$$[F_1,[F_2,E_3]] = -H_1,\ [F_3,[F_2,E_3]] = F_2,$$
$$[F_3,[F_3,E_3]] = -2F_3.$$

In summary, we have seen that for the representations ρ studied so far, there is a *highest weight*, w_{max}, and a vector, e_{max}, called *highest weight vector*, defined up to multiplication by a non-zero scalar, satisfying the following properties:

(i) for every $H \in \mathfrak{h}$, the vector e_{max} is an eigenvector of $\rho(H)$ for the eigenvalue $w_{max}(H)$,
(ii) for $i = 1, 2, 3$,
$$\rho(E_i)(e_{max}) = 0,$$
(iii) the images of the vector e_{max} by repeated application of the operators $\rho(F_i)$, for $i = 1, 2, 3$, generate the entire space of the representation.

4.2 Weights as Linear Combinations of the λ_i

We have set
$$H = \begin{pmatrix} \lambda_1(H) & 0 & 0 \\ 0 & \lambda_2(H) & 0 \\ 0 & 0 & \lambda_3(H) \end{pmatrix},$$
and thus defined linear forms λ_1, λ_2 and λ_3 on $\mathfrak{h}$. We shall see that each weight of the representations above can be written

$$w = m_1\lambda_1 + m_2\lambda_2 + m_3\lambda_3$$

(which means that $w(H) = m_1\lambda_1(H) + m_2\lambda_2(H) + m_3\lambda_3(H)$, for every $H \in \mathfrak{h}$), where the m_i are integers. We can denote a weight by a triple (m_1, m_2, m_3). Such a triple is not unique. In fact, $\lambda_1 + \lambda_2 + \lambda_3 = 0$, and consequently we can always reduce to the case where one of the m_i is zero. For example, we can write

$$w = (m_1 - m_3)\lambda_1 + (m_2 - m_3)\lambda_2 + m_3(\lambda_1 + \lambda_2 + \lambda_3) = m_1'\lambda_1 + m_2'\lambda_2.$$

We can also reduce to the case where the m_i are all three positive or zero, as we shall show in the following examples.

Example 1. The representation **3**. In **3**, the weights are

$$w_u = \lambda_1,\ w_d = \lambda_2,\ w_s = \lambda_3.$$

We see that
$$w_u = \lambda_1 = \lambda_1 + 0\lambda_2 + 0\lambda_3 \to (1,0,0),$$
$$w_d = \lambda_2 = 0\lambda_1 + 1\lambda_2 + 0\lambda_3 \to (0,1,0),$$
$$w_s = \lambda_3 = 0\lambda_1 + 0\lambda_2 + 1\lambda_3 \to (0,0,1).$$

Example 2. The representation $\overline{\mathbf{3}}$. In $\overline{\mathbf{3}}$, the weights are

$$w_{\overline{u}} = -\lambda_1,\ w_{\overline{d}} = -\lambda_2,\ w_{\overline{s}} = -\lambda_3.$$

We then have

$$w_{\overline{u}} = -\lambda_1 = \lambda_2 + \lambda_3 \to (0,1,1),$$
$$w_{\overline{d}} = -\lambda_2 = \lambda_1 + \lambda_3 \to (1,0,1),$$
$$w_{\overline{s}} = -\lambda_3 = \lambda_1 + \lambda_2 \to (1,1,0).$$

Example 3. The representation **8**. In the representation **8**, the weights are the roots $\alpha_i, -\alpha_i$, $i = 1,2,3$. We see that

$$\alpha_1 = \lambda_1 - \lambda_2 \to (2,0,1), \alpha_1' = \lambda_2 - \lambda_1 \to (0,2,1),$$
$$\alpha_2 = \lambda_2 - \lambda_3 \to (1,2,0), \alpha_2' = \lambda_3 - \lambda_2 \to (1,0,2),$$
$$\alpha_3 = \lambda_1 - \lambda_3 \to (2,1,0), \alpha_3' = \lambda_3 - \lambda_1 \to (0,1,2).$$

In these three examples, we see that the highest weight is the one for which the triple (m_1, m_2, m_3) is the largest in lexicographic order.

4.3 Finite-Dimensional Representations and Weights

For each finite-dimensional irreducible representation ρ of $\mathfrak{sl}(3,\mathbb{C})$, there is a highest weight w and a highest weight vector v, defined up to multiplication by a non-zero scalar, satisfying, for every $H \in \mathfrak{h}$,

$$\rho(H)v = w(H)v,\ \ \rho(E_i)v = 0,\ \ i = 1,2,3.$$

Furthermore, the action of compositions of powers of $\rho(F_i)$, $i = 1,2,3$, on v generates the entire space of the irreducible representation of $\mathfrak{sl}(3,\mathbb{C})$. These facts are a special case of general theorems on the representations of semi-simple Lie algebras, for which see the references at the end of this chapter.

From the calculation

$$\rho(H)\rho(F_i)v = \rho(F_i)\rho(H)v + \rho([H,F_i])v = w(H)\rho(F_i)v - \alpha_i(H)\rho(F_i)v,$$

we obtain

$$\rho(H)\rho(F_i)v = (w - \alpha_i)(H)\rho(F_i)v, \tag{4.1}$$

and it follows that each vector $\rho(F_1)^{k_1}\rho(F_2)^{k_2}\rho(F_3)^{k_3}v$, for non-negative integers k_i, $i = 1,2,3$, is a weight vector corresponding to the weight $w - k_1\alpha_1 - k_2\alpha_2 - k_3\alpha_3$. The weights thus obtained are part of that translate of the root diagram which contains the point w. The following is a useful rule:

Rule. In the diagram, the weights of an irreducible representation of $\mathfrak{sl}(3,\mathbb{C})$ lie on concentric hexagons that may degenerate into triangles. The multiplicity

of each weight is 1 on the edge, it increases by 1 on each concentric hexagon approaching the center, and it is constant on the triangles.

For a proof of this rule, see Fulton and Harris (2004).

4.4 Another Example: the Representation 6

We consider the representation of highest weight $w = 2\lambda_1$. The other weights will be constructed by adding the negative integer multiples of $\lambda_1 - \lambda_2$, $\lambda_2 - \lambda_3$ and $\lambda_1 - \lambda_3$ such that the weights obtained are not higher than the highest weight. Let v be a highest weight vector. By definition,

$$H_1 v = 2v,\ H_2 v = 0,\ E_1 v = E_2 v = E_3 v = 0.$$

Knowing that $Hv = w(H)v$, we obtain, using equation (4.1),

$$H(F_i v) = (w - \alpha_i)(H)(F_i v),$$

whence

$$\begin{aligned} H(F_1 v) &= (w - (\lambda_1 - \lambda_2))(H)(F_1 v) = (\lambda_1 + \lambda_2)(H)(F_1 v) = -\lambda_3(H)(F_1 v), \\ H(F_3 v) &= (w - (\lambda_1 - \lambda_3))(H)(F_3 v) = (\lambda_1 + \lambda_3)(H)(F_3 v) = -\lambda_2(H)(F_3 v). \end{aligned}$$

By repeated action of F_1, F_2, F_3, we obtain the following diagram. This representation is of dimension 6 and we shall denote it by **6** (Fig. 4).

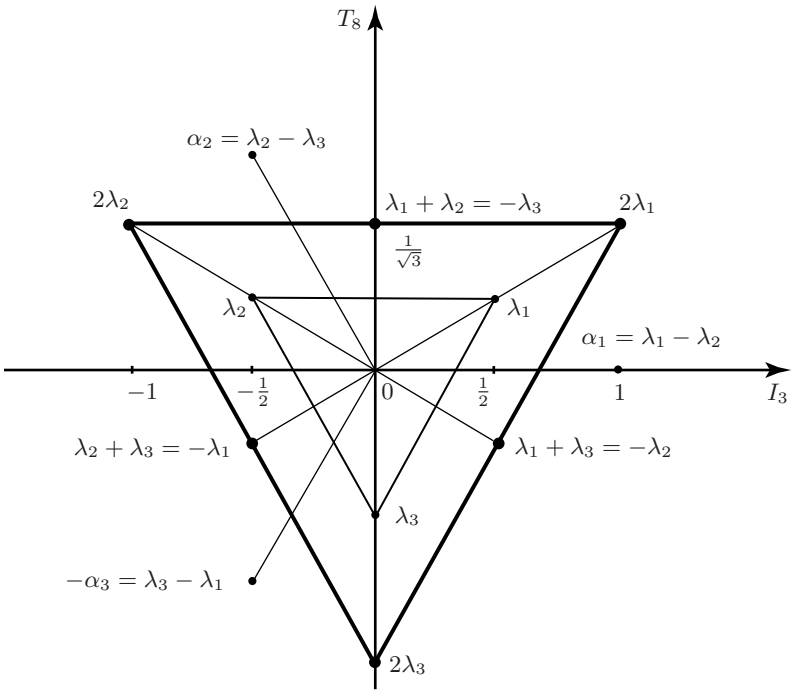


Figure 4. The representation **6**.

4.5 One More Example: the Representation 10

We consider the representation of highest weight $w = 3\lambda_1$. Below is the list of weights of this representation. Because of the applications to particle physics, which we shall discuss briefly in Section 6, we have given the names of the elementary particles associated to each weight, as well as the value of the component of each weight on the element $Q = \frac{1}{2}Y + I_3$ which corresponds to the charge of each particle.

weight w	$w(I_3)$	$w(Y)$	$w(T_8)$	$w(Q)$	
$3\lambda_1$	$\frac{3}{2}$	1	$\frac{\sqrt{3}}{2}$	2	Δ^{++}
$2\lambda_1 + \lambda_2$	$\frac{1}{2}$	1	$\frac{\sqrt{3}}{2}$	1	Δ^{+}
$\lambda_2 - \lambda_3 = \lambda_1 + 2\lambda_2$	$-\frac{1}{2}$	1	$\frac{\sqrt{3}}{2}$	0	Δ^{0}
$3\lambda_2$	$-\frac{3}{2}$	1	$\frac{\sqrt{3}}{2}$	-1	Δ^{-}
$\lambda_2 - \lambda_1 = 2\lambda_2 + \lambda_3$	-1	0	0	-1	Σ^{*-}
$\lambda_3 - \lambda_1 = \lambda_2 + 2\lambda_3$	$-\frac{1}{2}$	-1	$-\frac{\sqrt{3}}{2}$	-1	Ξ^{*-}
$3\lambda_3 = 2\lambda_3 - \lambda_1 - \lambda_2$	0	-2	$-\sqrt{3}$	-1	Ω^{-}
$\lambda_3 - \lambda_2 = 2\lambda_3 + \lambda_1$	$\frac{1}{2}$	-1	$-\frac{\sqrt{3}}{2}$	0	Ξ^{*0}
$\lambda_3 + 2\lambda_1 = \lambda_1 - \lambda_2$	1	0	0	1	Σ^{*+}
0	0	0	0	0	Σ^{*0}

Below we draw the weight diagram of this representation, which is of dimension 10, and which we denote by the symbol **10** (Fig. 5). To each weight corresponds a unique weight vector up to scalar multiplication. This representation is irreducible as one can show, by examining the weight diagram, that it is not the direct sum of representations of smaller dimension. In fact, we can give a complete list of these irreducible representations of small dimension:

- the trivial representation, denoted by **1**,
- the fundamental representation, **3**,
- the dual $\overline{\mathbf{3}}$ of the fundamental representation, **3**,
- the representation **6**,
- the dual $\overline{\mathbf{6}}$ of the representation **6**,
- the adjoint representation, **8**.

See Section 6 for other versions of the following diagram, with the notation used in physics.

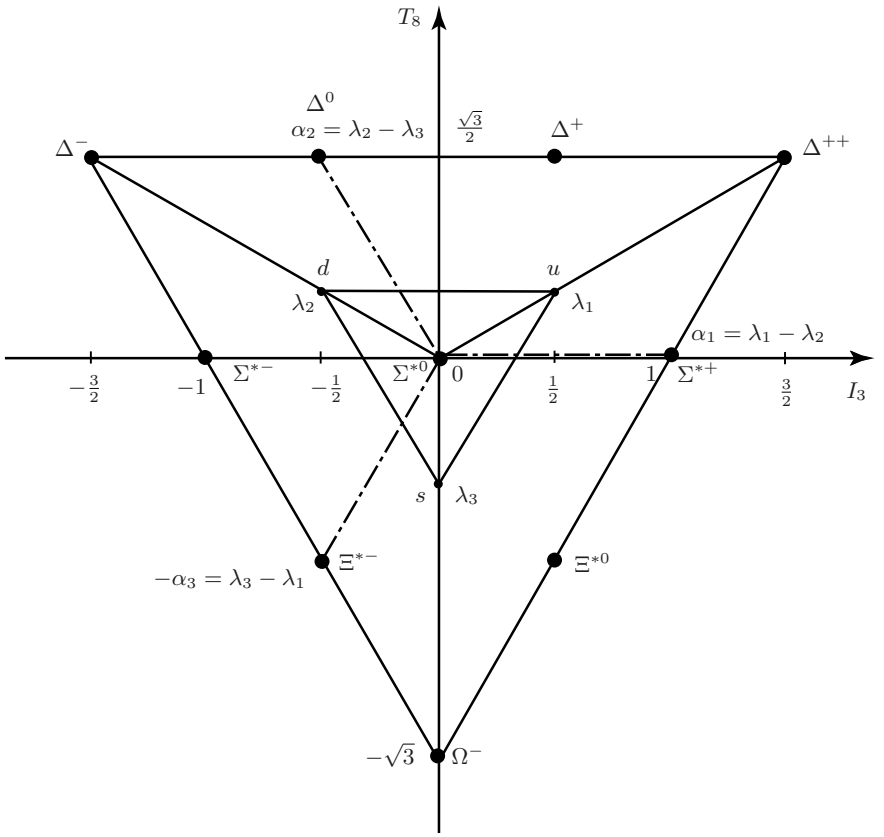


Figure 5. The representation **10**.

5 Tensor Products of Representations

Let (E_1, ρ_1) and (E_2, ρ_2) be representations of the same group G. We recall that, by definition, $\rho_1 \otimes \rho_2$ is the representation of G on $E_1 \otimes E_2$ defined by

$$(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) = \rho_1(g)v_1 \otimes \rho_2(g)v_2,$$

for $g \in G$ and $v_1 \in E_1$, $v_2 \in E_2$. Then the differential of this representation satisfies equation (7.9) of Chapter 4,

$$D(\rho_1 \otimes \rho_2)(X)(v_1 \otimes v_2) = (D\rho_1)(X)v_1 \otimes v_2 + v_1 \otimes (D\rho_2)(X)v_2,$$

for every $X \in \mathfrak{g}$. Abusing language, we denote

$$D(\rho_1 \otimes \rho_2) = D\rho_1 \otimes D\rho_2.$$

The following result is almost obvious given the formula above, but it is very important, and is in fact valid in general.

Proposition 5.1. *If v_1 is a weight vector of a representation R_1 of $\mathfrak{sl}(3,\mathbb{C})$ for a weight w_1, and if v_2 is a weight vector of a representation R_2 of $\mathfrak{sl}(3,\mathbb{C})$ for a weight w_2, then $v_1 \otimes v_2$ is a weight vector of the representation $R_1 \otimes R_2$ for the weight $w_1 + w_2$.*

Example 1: $\mathbf{3} \otimes \overline{\mathbf{3}}$. We shall treat the example of the representation $\mathbf{3} \otimes \overline{\mathbf{3}}$ graphically, where we see that

$$\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}.$$

The weight 0 is of multiplicity 3, and the other weights are of multiplicity 1 (Fig. 6).

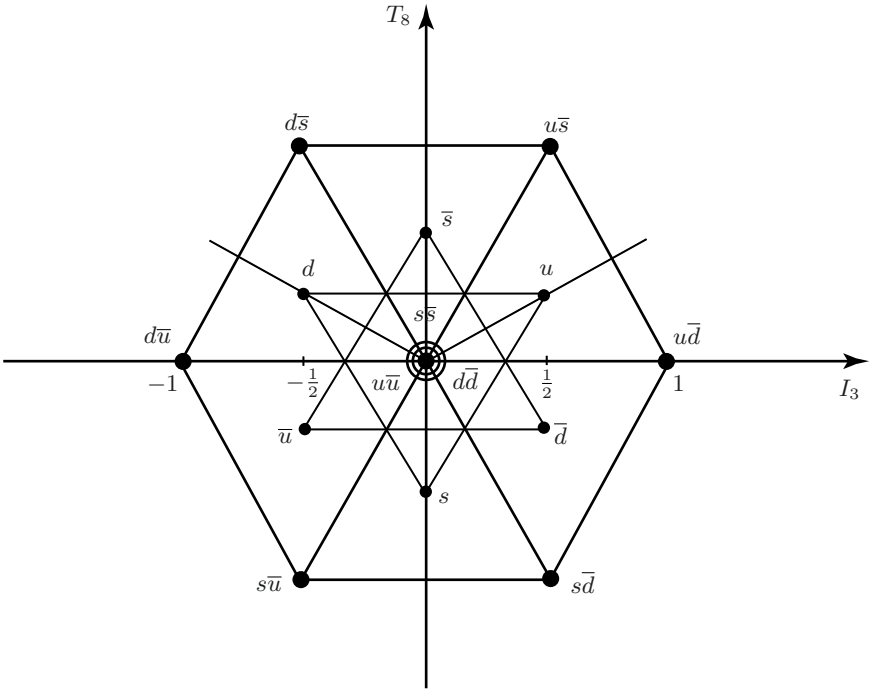


Figure 6. $\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$.

Example 2: $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$. In this tensor product, there are three weights of multiplicity 1, six weights of multiplicity 3, and the weight 0 is of multiplicity 6. We see that

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}.$$

The table and diagram below give the components of the weights with the corresponding weight vectors (Fig. 7).

weight vectors					I_3	Y	
					uuu	$\frac{3}{2}$	1
					ddd	$-\frac{3}{2}$	1
					sss	0	-2
			uud	udu	duu	$\frac{1}{2}$	1
			uus	usu	suu	1	0
			ddu	dud	udd	$-\frac{1}{2}$	1
			dds	dsd	sdd	-1	0
			ssu	sus	uss	$\frac{1}{2}$	-1
			ssd	sds	dss	$-\frac{1}{2}$	-1
uds	dus	sud	usd	dsu	sdu	0	0

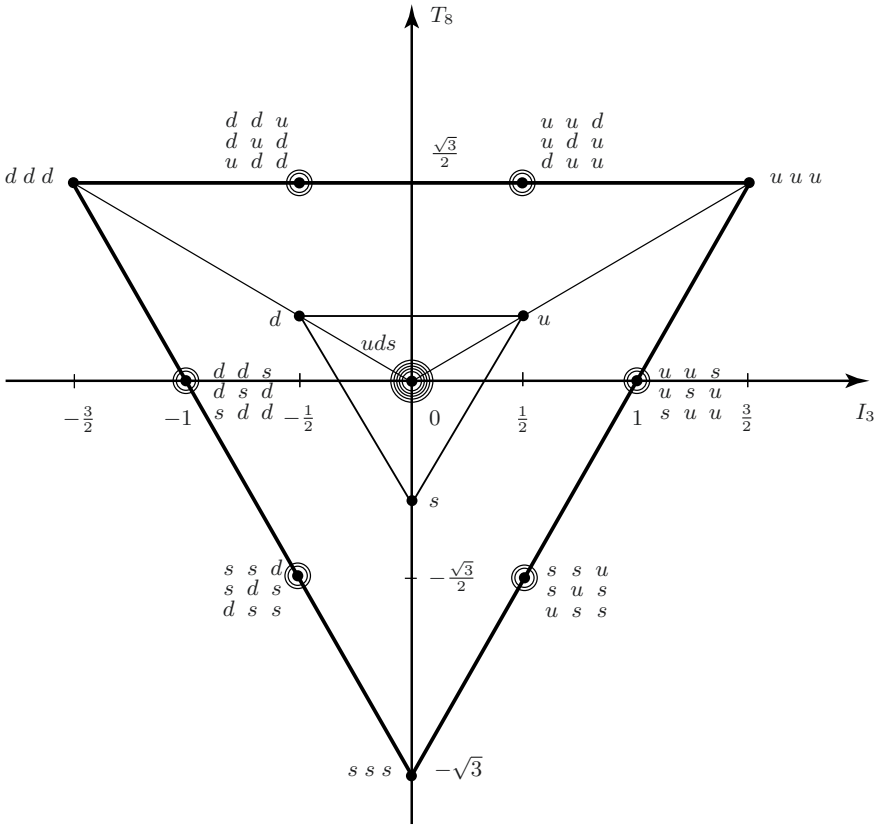


Figure 7. $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$.

6 The Eightfold Way

Up until 1955, scientists had observed seven baryons, and an eighth was observed in 1958, the Ξ^0. On the other hand there were seven known mesons. This set of observations led Murray Gell-Mann and Yuval Ne'eman independently, during the winter of 1960–1961, to propose the symmetry group SU(3) for the classification of the elementary particles. The eight baryons were grouped in the octet of the adjoint representation, and the seven known mesons formed part of such an octet. The publication of Ne'eman's results took place in 1961, while the internal report of the California Institute of Technology by Gell-Mann was not published as an article in *Physical Review* until 1962. The particle η completing the octet of mesons and whose existence was predicted by this theory was discovered in 1962.

The basic idea is that each elementary particle "is" a weight vector of the representation. To each weight vector corresponds a weight that is the collection of the eigenvalues of the operators associated to the chosen basis vectors in the Cartan subalgebra. Thus each weight is defined by two numbers, the eigenvalues of H_1 and H_2, or of I_3 and Y, or of I_3 and T_8, according to the basis chosen. These two numbers are the *quantum numbers*, which characterize the particle when the weight is simple:

- the quantum number corresponding to I_3 is the third component of the *isospin*,
- the quantum number corresponding to Y is the *hypercharge*.

Then the charge is $Q = \frac{1}{2}Y + I_3$, and furthermore, $Y = B + S$, where B is the baryon number and S is the *strangeness*, which is zero for the proton and the neutron, and for the three π mesons.

The *baryon* and *meson octets* were obtained as the representation **8** of SU(3), whence the name of *Eightfold Way* given to the theory.

Other extremely short-lived elementary particles called *resonances* were observed in 1961, the four Δ and the three Σ^*. These corresponded to seven of the ten points of the weight diagram of the representation **10** of SU(3). The theory predicted the existence of two particles of hypercharge -1, the Ξ^*, which were discovered in 1962, as well as the particle corresponding to the tenth point, which was called Ω^-, and was discovered later, in 1964. This discovery was a great success for the SU(3) theory.

We reproduce below tables of baryons and of mesons with the corresponding diagrams, and the diagram of baryon resonances, adapted from two different sources. We recognize immediately the weight diagram of the representation **8** for the baryons and the mesons, and that of the representation **10** for the baryon resonances. In these tables, J denotes the spin and I the isospin of the particle.

6.1 Baryons ($B = 1$)

	S	I	I_3	Q	J	Mass ($\mathrm{MeV/c^2}$)	Half-life (s)
Ξ^-	-2	$\frac{1}{2}$	$-\frac{1}{2}$	-1	$\frac{1}{2}$	1321	1.6×10^{-10}
Ξ^0	-2	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1315	2.9×10^{-10}
Σ^-	-1	1	-1	-1	$\frac{1}{2}$	1197	1.5×10^{-10}
Σ^0	-1	1	0	0	$\frac{1}{2}$	1193	7.4×10^{-20}
Σ^+	-1	1	1	1	$\frac{1}{2}$	1189	0.8×10^{-10}
Λ^0	-1	0	0	0	$\frac{1}{2}$	1116	2.6×10^{-10}
n	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	940	0.9×10^{3}
p	0	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	938	*stable*

The eight observed baryons belong to the weight diagram of the representation **8** as shown in the following figures.

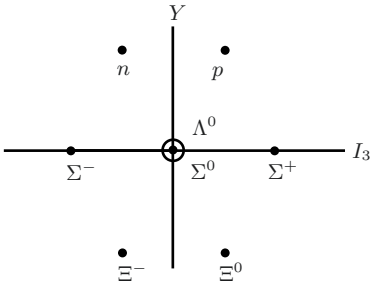


Figure 8a.
The baryon octet.

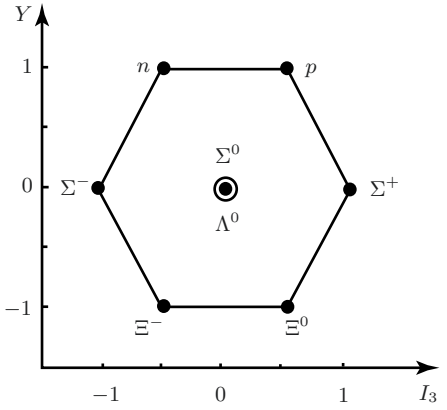


Figure 8b. The baryon octet.

6.2 Mesons ($B = 0$)

	S	I	I_3	Q	J	Mass ($\mathrm{MeV/c^2}$)	Half-life (s)	
K^+	1	$\frac{1}{2}$	$\frac{1}{2}$	1	0	494	1.2×10^{-8}	
K^0	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	498	$K_S = \frac{1}{\sqrt{2}}(K^0 + \overline{K}^0)$	0.9×10^{-10}
$\overline{K}^0$	-1	$\frac{1}{2}$	$\frac{1}{2}$	0	0	498	$K_L = \frac{1}{\sqrt{2}}(K^0 - \overline{K}^0)$	5.2×10^{-8}
K^-	-1	$\frac{1}{2}$	$-\frac{1}{2}$	-1	0	494	1.2×10^{-8}	
π^+	0	1	1	1	0	140	2.6×10^{-8}	
π^0	0	1	0	0	0	135	8.4×10^{-17}	
π^-	0	1	-1	-1	0	140	2.6×10^{-8}	

For the meson η, $S = 0$, $I = 0$, $I_3 = 0$, $Q = 0$, $J = 0$. Its mass is 548 MeV/c^2.

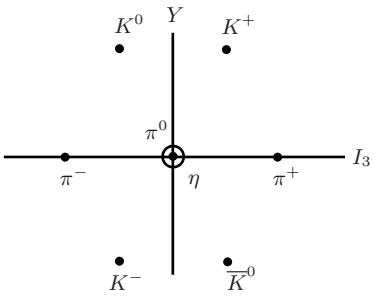


Figure 9a.
The meson octet.

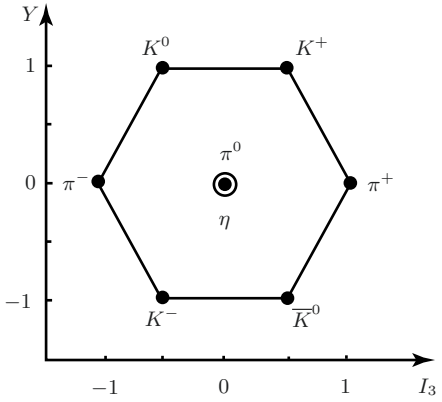


Figure 9b. The meson octet.

6.3 Baryon Resonances

The observed and predicted baryon resonances belong to the weight diagram of the representation **10** as shown in the following figures.

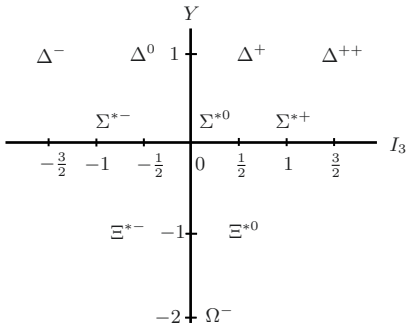


Figure 10a.
The baryon resonances.

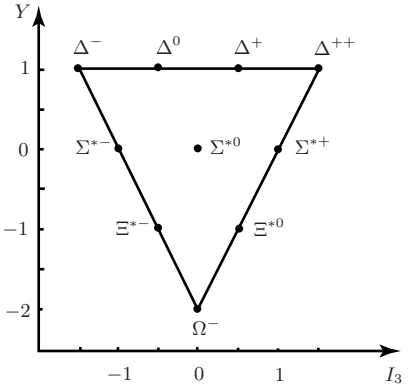


Figure 10b. The baryon resonances.

The baryon resonances have a very short half-life, of the order of 10^{-23} s for $\Delta^{++}, \Delta^+, \Delta^0$ and Δ^-.

7 Quarks and Antiquarks

However, the representations **3** and $\overline{\mathbf{3}}$ did not correspond to any observed particle. The quark model allows us to consider the hadrons (baryons and mesons) as "states composed of quarks and antiquarks." Mathematically, this means the following:

- The *quarks* are the particles corresponding to the representation **3** of SU(3): the three quarks, u (*up*), d (*down*) and s (*strange*), are the weight vectors of this representation.

3	w	$w(I_3)$	$w(Y)$	$w(T_8)$	$w(Q)$
u	$w_u = \lambda_1$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{2\sqrt{3}}$	$\frac{2}{3}$
d	$w_d = \lambda_2$	$-\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{2\sqrt{3}}$	$-\frac{1}{3}$
s	$w_s = \lambda_3 = -\lambda_1 - \lambda_2$	0	$-\frac{2}{3}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{3}$

- The *antiquarks* are the particles corresponding to the representation $\overline{\mathbf{3}}$ of SU(3): the three antiquarks, $\overline{u}, \overline{d}$, and $\overline{s}$, are the weight vectors of this representation.

$\overline{\mathbf{3}}$	w	$w(I_3)$	$w(Y)$	$w(T_8)$	$w(Q)$
$\overline{u}$	$w_{\overline{u}} = \lambda_2 + \lambda_3 = -\lambda_1$	$-\frac{1}{2}$	$-\frac{1}{3}$	$-\frac{1}{2\sqrt{3}}$	$-\frac{2}{3}$
$\overline{d}$	$w_{\overline{d}} = \lambda_1 + \lambda_3 = -\lambda_2$	$\frac{1}{2}$	$-\frac{1}{3}$	$-\frac{1}{2\sqrt{3}}$	$\frac{1}{3}$
$\overline{s}$	$w_{\overline{s}} = \lambda_1 + \lambda_2 = -\lambda_3$	0	$\frac{2}{3}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{3}$

- One obtains the meson octet in the decomposition of the tensor product $\mathbf{3} \otimes \overline{\mathbf{3}}$, more precisely,

$$\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1},$$

  "a meson is composed of a quark-antiquark pair."
- One obtains the baryon octet and decuplet in the decomposition of the tensor product

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1},$$

  "a baryon is composed of three quarks."

These decompositions were proved graphically in Section 5 above.

To avoid difficulties due to fractional charges and an incompatibility with the Pauli exclusion principle, an additional SU(3) symmetry was in fact introduced later, corresponding to a new quantum number the *color* and leading to the theory of *quantum chromodynamics*.

References

Roots, Dynkin diagrams, which are an "efficient way to convey the structure of a semi-simple algebra", and weights of the representations of Lie algebras can be studied in Fulton and Harris (2004) and they are the subject of the book by Humphreys (1997). There one can find the classification of semi-simple complex Lie algebras, established by Killing and Cartan at the end of the nineteenth century. See Serre (2001) and Knapp (2002), or Sattinger and Weaver (1993) who describe applications to particle physics. The case of $\mathfrak{sl}(3,\mathbb{C})$ is treated in detail in Hall (2016). The study of the representations of low-dimensional Lie algebras precedes the general theory in Fulton and Harris (2004), where the rule stated in Section 4.3 above is proved. For a treatise oriented towards theoretical physics, see, *e.g.*, Georgi (1999).

For the proof of the simple connectedness of SU(3), see for example Rossmann (2011) or Hall (2016).

The tables of baryons and mesons in Section 6 are taken from K. Hagiwara *et al.*, "Review of Particle Properties", *Physical Review* D 66, 010001 (2002). See <http://pdg.lbl.gov> for more recent and much more detailed experimental results. Weight diagrams of the representations of SU(3) can be found in Fulton and Harris (2004), and also in Ludwig and Falter (2012), Tung (1985), or Greiner and Müller (1994). Figures 8a–10a were reproduced from Sternberg (2003) and figures 8b–10b are adapted from a course in French on symmetries and microscopic physics by Blaizot and Tolédano, last published in 1997. For information on the theoretical and experimental discoveries in the years 1950–1962, see Sternberg (2003). The original articles are reproduced in the collection edited by Gell-Mann and Ne'eman in 1964, with a new edition in 2000. Gell-Mann first called the quarks "the leptons ν, e^- and μ^-". Subsequently, in 1963, he introduced the term *quark*, taken from *Finnegan's Wake* by James Joyce, "Three quarks for Muster Mark!...".

Topics in History

To illustrate this chapter, we need only turn to *Symmetries in Physics*, a collection of lectures and discussions which took place in 1983 and were published in book form in 1987[1]. The lectures of Murray Gell-Mann and Yuval Ne'eman are particularly relevant to the topics of this chapter: these two physicists relate their own discoveries.

In his "Particle theory from S-matrix to quarks", Gell-Mann wrote: "If I had understood Lie algebras in 1959 and early 1960, I would have realized that (without charm) the system of commutation relations could close on the chiral algebra of $\mathrm{SU}(3) \times \mathrm{SU}(3)$. But I did not. The notion that all the relevant algebraic systems were tabulated and that I could look them up in a book was still strange to me". Indeed, things have changed since then! He explains that,

[1] *Symmetries in Physics (1600–1980)*, ed. Doncel *et al.*

when he was working in Paris in late 1959, he was looking for the possible generalizations of both the theory of Yang and Mills concerning the isotopic spin, involving the Lie group SU(2), and quantum electrodynamics which "was, of course, an example of a gauge theory using U(1)". He added: "We could readily conceive, therefore, of a gauge theory involving what we would call today a product of SU(2) factors and U(1) factors". While a scheme with one factor of each type was in fact suggested by a colleague, Gell-Mann thought that "the question was whether any other generalization existed". So,

> I worked and worked in my office in the Collège de France[2] and finally I wrote down, as the necessary and sufficient conditions, the canonical relations:
>
> $$[F_i, F_j] = \mathrm{i}\ c_{ijk} F_k,$$
>
> where c_{ijk} is real and totally antisymmetric and the F_i are Hermitian charge operators. I had no idea what were the possible realizations of this formula.
>
> Every day I would have lunch with my French friends and drink wine, and afterwards I would come back and struggle with drowsiness in my office. I worked through the case of three operators, four operators, five operators, six operators, and seven operators, trying to find algebras that did not correspond to what we would now call products of SU(2) factors and U(1) factors. I got all the way to seven dimensions and found none, of course. At that point, I said: 'That's enough!' I did not have the strength after drinking all that wine to try eight dimensions. Unfortunately, I did not pay sufficient attention to the identity of one of my regular companions at lunch. It was Professor Serre, one of the world's greatest experts on Lie algebras. I knew, of course, that [Jean-Pierre] Serre was a famous pure mathematician, but I did not know what his specialty was. It never occurred to me to ask him about my equations, and I doubt whether he would have given the answer if I had asked him. Probably this canonical form for the commutation relations was far too explicit for Serre. He would have preferred something like:
>
> $$T : s \to u.$$
>
> At least that is my impression of the kind of formula that mathematicians like to write.

True, but this kind of formula is not the only kind of formulas that mathematicians like! Gell-Mann then related his attempts at explaining the baryons and other elementary particles: "Since at that time, I did not understand SU(3), I used the permutation group S_3 to analyze the symmetry properties of the model".

[2] He was then visiting the Collège de France in Paris where most offices were notoriously small and uncomfortable.

He wrote his Caltech Report on the *Eightfold Way* in January 1961, then converted his report into an article for the *Physical Review* while harboring many doubts "over the correctness of the octet assignment for baryons [which] led to a useful presentation of the subject" and he added: "A very natural groundwork was being laid for the quark model of 1963."

In early 1963, he lectured at the Massachusetts Institute of Technology (MIT) and in March he visited Columbia University where, at the Faculty Club, he was asked "why it was not possible to use [his] formula

$$3 \times 3 \times 3 = 1 + 8 + 8 + 10$$

to obtain the baryons". He replied that he had tried but that "the fundamental entities would then have fractional charges [...] $\frac{2}{3}$ and $-\frac{1}{3}$ [...]. But thinking about the matter afterwards, that day and the next morning, it occurred to me that, if [the present theory] were correct, any fundamental hadrons would have to be unobservable and [if so] they might as well have fractional charge." In the fall of 1963, Gell-Mann was back at UCLA in Pasadena:

> My only attempt to convey the notion of quarks to [the physicists at the] CERN[3] during that fall was in a telephone call to my old teacher Viki [Viktor] Weisskopf, who was the Director General at that time. After I had explained to him that I had been working for a few months on an exciting idea that each baryon was made of three fractionally charged particles, he said: 'Please, Murray, let us be serious, this is an international call'.

In other words, long-distance telephone calls are expensive, let us not waste the money of CERN, which is paying for the call, on extravagant ideas of no interest for genuine science! Gell-Mann concludes: "Painful as it is to describe, this story of confusion, ignorance, mistakes, difficulties, and vacillations may be the most interesting aspect of our particular corner of the history of science". Does the physicist's outlook apply to the progress of mathematics?

Yuval Ne'eman, in his "Hadron symmetry, classification and compositeness", published in the same volume, told his story of the *Eightfold Way* after general comments on symmetries. He wrote, speaking of the physicists in Princeton in the years 1956–1960, that "they had kept using the orthogonal groups only (rotations) going from 3 dimensions (isospin) to 4, then to 7, 8, 9, etc. Nobody had tried to generalize isospin as a unitary group, by going from SU(2) to SU(3). [...] What was needed was a compact, simple Lie group of rank 2." Ne'eman thus spent the summer of 1960 "learning about Lie groups and their classification".

> In September, I had drawn up the root diagrams of $A_2(= SU(3))$, $B_2(= SO(5))$, $C_2(= Sp(4))$, $D_2(= SU(4))$, G_2 (one of 5 "exceptional" [Lie] groups) [...]. A_2 appeared to fit perfectly. The mesons would fit in an octet, with a prediction for an eighth 0-particle (the η^0).

[3]CERN is the European Center for Nuclear Research with a particle accelerator located near Geneva.

> I tried representations for the baryons – and the octet assignment seemed so obvious!

Ne'eman then relates his hesitations going "against the current belief at the time" and his eventual submission of "Derivation of strong interactions from gauge invariance" to the journal *Nuclear Physics*, which was received on 13 February 1961. One day in March, he was suddenly informed of Gell-Mann's Caltech internal document, first issued 20 January 1961, and he realized that "indeed Gell-Mann was introducing precisely the same model, with similar arguments"! Ne'eman's position was then in the department of physics at Tel Aviv University. Before his return to Israel in August 1961, "favourable experimental evidence began to accumulate." Before the end of 1961, "a group of experimental physicists at Johns Hopkins University found the missing eighth meson, the η". The Ω^- was finally observed, but only in February 1964. Ne'eman tells the story as follows. He had attended a conference in Florida a week earlier and was then told by the director of the Brookhaven laboratory that "there was no Ω^- in 50 000 pictures" and he adds "they finally found it, after looking at 100 000 pictures..." When the Ω^- was finally detected, Ne'eman was at Caltech, the experimentalists neglected to inform him but he heard rumors. "I sent in a request for a reprint, and received a nice set of bubble chamber pictures, with the note 'Please excuse the over-sight, but you knew it existed before we did'!"

Exercises

Exercise 8.1 *Orthonormal basis of the Cartan subalgebra of* $\mathfrak{sl}(3,\mathbb{C})$.

1. Show that the Killing form of $\mathfrak{sl}(3,\mathbb{C})$ is $K(X,Y) = 6\ \mathrm{Tr}XY$.
2. Equip the Cartan subalgebra $\mathfrak{h}$ with the restriction of the Killing form of $\mathfrak{sl}(3,\mathbb{C})$. Show that (I_3, Y) and (I_3, T_8) are orthogonal bases of $\mathfrak{h}$. Find an orthonormal basis of $\mathfrak{h}$.

Exercise 8.2 *The representation* $\mathbf{3}\otimes\mathbf{3}$ *of* SU(3).

Decompose the representation $\mathbf{3}\otimes\mathbf{3}$ of SU(3) into a direct sum of irreducible representations with the help of the weight diagram.

Exercise 8.3 *The representation* **10** *of* SU(3).

Prove that the representation **10** of SU(3) is irreducible.

Exercise 8.4 *On the representation* $\mathbf{3}\otimes\overline{\mathbf{3}}$ of SU(3).

Prove that the adjoint representation of $\mathfrak{sl}(3,\mathbb{C})$ is equivalent to the restriction of $\mathbf{3}\otimes\overline{\mathbf{3}}$ to the smallest invariant subspace containing the vector $u\otimes\overline{u}$.

Exercise 8.5 *Restrictions to* $\mathfrak{sl}(2,\mathbb{C})$.

The subalgebra $\mathfrak{g}$ of $\mathfrak{sl}(3,\mathbb{C})$ generated by E_1, F_1, H_1 is isomorphic to $\mathfrak{sl}(2,\mathbb{C})$. Decompose the restriction to $\mathfrak{g}$ of the adjoint representation of $\mathfrak{sl}(3,\mathbb{C})$ into a direct sum of irreducible representations of $\mathfrak{g}$. Same question for the representations **6** and **10**.

Exercise 8.6 *Representations of dimension* 15 *of* $\mathfrak{sl}(3,\mathbb{C})$.

1. Show that there are four inequivalent irreducible representations of dimension 15 of $\mathfrak{sl}(3,\mathbb{C})$, which can be denoted by $\mathbf{15}$, $\overline{\mathbf{15}}$, $\mathbf{15}'$, $\overline{\mathbf{15}}'$, and draw their weight diagrams in the basis (I_3, T_8).
2. Show that $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = 3\,\mathbf{3} \oplus 2\,\overline{\mathbf{6}} \oplus 3\,\mathbf{15} \oplus \mathbf{15}'$.



Murray Gell-Mann was born in New York in 1929 and died in 2019. Professor at the California Institute of Technology since 1956, he won the Nobel prize in physics in 1969. He proposed the theory of the Eightfold Way and quarks in 1961, at about the same time as Ne'eman.

(American Institute of Physics, Emilio Segrè Visual Archives)

Yuval Ne'eman (1925–2006), founder and director of the physics and astronomy department of Tel Aviv University, discovered the SU(3) *symmetry of subatomic particles in 1961, at about the same time as Gell-Mann.*

(American Institute of Physics, Emilio Segrè Visual Archives)





Paul Adrien Maurice Dirac (1902–1984), in his celebrated work on the theory of the electron in 1928, introduced the equation on spinors now called the Dirac equation – which, when squared, yields the Schrödinger equation. He was a professor at Cambridge University in England when he was awarded the Nobel prize in physics in 1933, jointly with Erwin Schrödinger.

(American Institute of Physics, Emilio Segrè Visual Archives)

Chapter 9

Spin Groups and Spinors



We present some results on spin groups and their representations which are meant to be an introduction to the subject but which are far from being a comprehensive theory of spinors. For full proofs and a more in-depth study, we refer to the books and articles in the bibliography. We shall first introduce Clifford algebras because the representations of the spin groups are conveniently defined as restrictions of their representations.

1 Clifford Algebras

1.1 Definition

Let V be a finite-dimensional vector space over $\mathbb{K}$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Recall that the tensor powers of the vector space V are defined by recursion,

$$V^{\otimes 0} = \mathbb{K}, \quad V^{\otimes(k+1)} = V^{\otimes k} \otimes V, \ \forall k \in \mathbb{N}^*,$$

where $V^{\otimes k} = V \otimes V \otimes ... \otimes V$ with k factors, and that the tensor algebra of V is the direct sum of all tensor powers $V^{\otimes k}$, for $k \in \mathbb{N}$.

Recall that an ideal in an associative algebra is a vector subspace which is invariant under the left and the right multiplication by elements of the algebra. Denote by 1 the unit element of $\mathbb{K}$.

Definition 1.1. *Let Q be a non-degenerate quadratic form on V. The* Clifford algebra *of V equipped with the quadratic form Q is the quotient of the tensor algebra of V by the ideal generated by the elements $x \otimes x + Q(x)1$ for $x \in V$. It is denoted by $Cl(V, Q)$.*

Elements in $V^{\otimes 0} = \mathbb{K}$ and $V^{\otimes 1} = V$ are identified with their image in $Cl(V, Q)$. The Clifford algebra $Cl(V, Q)$ is an associative algebra with unit, 1. Denote by γ the projection of the tensor algebra of V onto the quotient

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Y. Kosmann-Schwarzbach, *Groups and Symmetries*, Universitext,
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$Cl(V,Q)$ and denote the product of elements u and v in $Cl(V,Q)$ by uv, and in particular $uu = u^2$. By definition, for all $x \in V$,

$$\gamma(x)^2 = -Q(x)1.$$

Let B denote the symmetric bilinear form associated to the quadratic form Q. By definition, for all x and $y \in V$,

$$B(x,y) = \frac{1}{2}\big(Q(x+y) - Q(x) - Q(y)\big),$$

and therefore $B(x,x) = Q(x)$. Then, for all $x, y \in V$,

$$\gamma(x)\gamma(y) + \gamma(y)\gamma(x) = -2\,B(x,y)1.$$

Let $n \geq 1$ be the dimension of V and let $(e_\alpha), \alpha = 1, ..., n$, be a basis of V. Elements, $\gamma(e_{\alpha_1}), ..., \gamma(e_{\alpha_r})$, for $0 \leq \alpha_1 < ... < \alpha_r \leq n$, $1 \leq r \leq n$, where, by convention, the element corresponding to $r = 0$ is the element 1 of $Cl(V,Q)$, constitute a basis of $Cl(V,Q)$. Therefore $Cl(V,Q)$ is of dimension 2^n.

The Clifford algebra $Cl(V,Q)$ is a $\mathbb{Z}_2$-graded associative algebra. The vector subspace generated by the products of an even number of elements $\gamma(e_\alpha)$, $1 \leq \alpha \leq n$, is a subalgebra of dimension 2^{n-1} of $Cl(V,Q)$ which is independent of the choice of a basis in V. It is called the *reduced Clifford algebra* and denoted by $Cl^+(V,Q)$.

1.2 Universal Property

Let A be a $\mathbb{K}$-algebra with unit 1 and let u be a $\mathbb{K}$-linear map from V to A satisfying $(u(x))^2 = -Q(x)1$, for all $x \in V$. There exists a unique morphism of $\mathbb{K}$-algebras, $U : Cl(V,Q) \to A$, such that $U \circ j = u$, where j is the injection of V into $Cl(V,Q)$.

$$\begin{array}{ccc} & Cl(V,Q) & \\ j & \nearrow \quad \searrow & U \\ V & \xrightarrow{\ u\ } & A \end{array}$$

This property, called the *universal mapping property* of Clifford algebras, characterizes the Clifford algebra of a vector space equipped with a quadratic form.

1.3 Complex and Real Clifford Algebras

Recall that, by definition, the direct sum $A \oplus B$ of algebras A and B is the direct sum of the vector spaces A and B equipped with the multiplication defined by $(a_1, b_1)(a_2, b_2) = (a_1a_2, b_1b_2)$, for $a_1, a_2 \in A,\ b_1, b_2 \in B$.

We shall first describe the Clifford algebras in the complex case. Let $\mathbb{K} = \mathbb{C}$, $V = \mathbb{C}^n$ and Q be the canonical quadratic form $Q(z) = z_1^2 + z_2^2 + ... + z_n^2$. Denote the complex Clifford algebra $C(V,Q)$ by $\mathbb{C}l(n)$.

- If n is even, $n = 2\nu$, the Clifford algebra $\mathbb{C}l(n)$ is isomorphic as a vector space to the algebra of endomorphisms of $\mathbb{C}^{2^\nu}$, thus it is isomorphic as

a vector space to the algebra of endomorphisms of the exterior algebra of $\mathbb{C}^\nu$. The reduced Clifford algebra $\mathbb{C}l^+(n)$ is isomorphic to the direct sum of two copies of End($\mathbb{C}^{2^{\nu-1}}$).

- If n is odd, $n = 2\nu + 1$, the Clifford algebra $\mathbb{C}l(n)$ is isomorphic to the direct sum of two copies of the algebra of endomorphisms of End($\mathbb{C}^{2^\nu}$). The reduced Clifford algebra $\mathbb{C}l^+(n)$ is isomorphic to End($\mathbb{C}^{2^{\nu-1}}$).

The Clifford algebras over $\mathbb{K} = \mathbb{R}$, for the quadratic form with signature (p, q) defined by $Q(x) = x_1^2 + x_2^2 + ... + x_p^2 - x_{p+1}^2 - ... - x_{p+q}^2$, are algebras over the reals denoted by $Cl(p, q)$. The complexification of $Cl(p, q)$ is the complex Clifford algebra $\mathbb{C}l(p + q)$. We list the real Clifford algebras $Cl(p, q)$ according to the value of $p - q$ modulo 8.

The real Clifford algebra $Cl(p, q)$ is isomorphic to the space of endomorphisms of a real (for $p - q \equiv 0$ or 6), or a complex (for $p - q \equiv 1$ or 5), or a quaternionic (for $p-q \equiv 2$ or 4) vector space, or to the direct sum of two copies of the space of endomorphisms of a real (for $p - q \equiv 7$) or a quaternionic (for $p - q \equiv 3$) vector space.

The reduced Clifford algebra $Cl^+(p, q)$ of $Cl(p, q)$ is isomorphic to the space of endomorphisms of a real (for $p-q \equiv 1$ or 7), or a complex (for $p-q \equiv 2$ or 6), or a quaternionic vector space (for $p - q \equiv 3$ or 5), or to the direct sum of two copies of the space of endomorphisms of a real (for $p-q \equiv 0$) or a quaternionic (for $p - q \equiv 4$) vector space.

2 The Groups Pin(n) and Spin(n)

In this section, $V = \mathbb{R}^n$, $n \geq 1$ and Q is the Euclidean (*i.e.*, positive definite) quadratic form on V. We denote by $Cl_n = Cl(n, 0)$ the corresponding Clifford algebra. The elements of Cl_n which are the images of the canonical basis (e_α), $\alpha = 1, ..., n$, of $\mathbb{R}^n$ under the canonical injection of $\mathbb{R}^n$ into Cl_n will be denoted by γ_α. Thus, for each $\alpha \in \{1, ..., n\}$,

$$(\gamma_\alpha)^2 = -1,$$

and, for all $\alpha, \beta \in \{1, ..., n\}$ such that $\alpha \neq \beta$,

$$\gamma_\alpha \gamma_\beta + \gamma_\beta \gamma_\alpha = 0.$$

2.1 The Group Pin(n)

Let n be an integer, $n \geq 1$. In the vector space Cl_n, the elements, γ_α, $\alpha = 1, ..., n$, generate an n-dimensional vector subspace, E, isomorphic to $\mathbb{R}^n$. We equip the vector space E with the Euclidean norm defined by requiring that the basis $\gamma_\alpha, \alpha = 1, ..., n$, be orthonormal. We consider the set

$$\Sigma = \{\sum_{\alpha=1}^{n} \lambda_\alpha \gamma_\alpha \big| \lambda_\alpha \in \mathbb{R}, \sum_{\alpha=1}^{n} (\lambda_\alpha)^2 = 1\} \subset E,$$

which is the unit sphere in the Euclidean vector space E. Each element g of Σ is invertible in the algebra Cl_n, with inverse $g^{-1} = -g$. In fact, if $g = \sum_{\alpha=1}^{n} \lambda_\alpha \gamma_\alpha \in \Sigma$, then $-g$ is in Σ and

$$g\,(-g) = (-g)\,g = -\sum_{\alpha=1}^{n} (\lambda_\alpha)^2 (\gamma_\alpha)^2 = \sum_{\alpha=1}^{n} (\lambda_\alpha)^2 = 1.$$

Definition 2.1. *The group* $\mathrm{Pin}(n)$ *is the subgroup of the group of invertible elements of* Cl_n *generated by* Σ.

In other words, an element of $\mathrm{Pin}(n)$ is a finite product of elements of Σ.

2.2 Adjunction and Conjugation

We shall now define two endomorphisms of Cl_n, the adjunction and the conjugation.

- The *adjunction* is the algebra automorphism, $\mathcal{A}$, of Cl_n defined by $\mathcal{A}(\gamma_\alpha) = -\gamma_\alpha$, $\alpha = 1, ..., n$. Thus

  $$\mathcal{A}(\gamma_{\alpha_1} ... \gamma_{\alpha_r}) = (-1)^r \gamma_{\alpha_1} ... \gamma_{\alpha_r}$$

  for $\alpha_1, ..., \alpha_r \in \{1, ..., 2^n\}$, $r = 1, ..., n$.
- The *conjugation* is the endomorphism, $\mathcal{C}$, of the vector space Cl_n defined by

  $$\mathcal{C}(\gamma_{\alpha_1} ... \gamma_{\alpha_r}) = (-1)^r \gamma_{\alpha_r} ... \gamma_{\alpha_1},$$

  for $\alpha_1, ..., \alpha_r \in \{1, ..., 2^n\}$, $r = 1, ..., n$. It is an algebra anti-automorphism.

2.3 Orthogonal Transformations are Products of Reflections

We shall need the following elementary results.

Reflections. In $\mathbb{R}^n$ with the Euclidean scalar product, $\langle\ ,\ \rangle$, the reflection with respect to the hyperplane orthogonal to a non-zero vector v is the map

$$u \mapsto u - 2\frac{\langle v, u\rangle}{\langle v, v\rangle} v.$$

In fact, the image of a vector $u = \lambda v + w$, where $\lambda \in \mathbb{R}$ and w is in the hyperplane orthogonal to v, is $\lambda v + w - 2\frac{\langle v, \lambda v + w\rangle}{\langle v, v\rangle} v = -\lambda v + w$. In particular, the image of v is $-v$ and the vectors in the hyperplane orthogonal to v are invariant.

A reflection with respect to a hyperplane is an orthogonal transformation with determinant -1. Therefore the product of an even (resp., odd) number of reflections is an orthogonal transformation with determinant 1 (resp., -1).

Change of hyperplane. If the reflection with respect to the hyperplane orthogonal to v has matrix $A \in \mathrm{O}(n)$, the transformation with matrix BAB^{-1}, for $B \in \mathrm{O}(n)$, is the reflection with respect to the hyperplane orthogonal to Bv. In fact, consider the map $w \mapsto BAB^{-1}w = B(B^{-1}w - 2\frac{\langle v, B^{-1}w\rangle}{\langle v,v\rangle}v) = w - 2\frac{\langle v, B^{-1}w\rangle}{\langle v,v\rangle}Bv$. Because B is orthogonal, $\langle v, B^{-1}w\rangle = \langle Bv, w\rangle$ and $\langle Bv, Bv\rangle = \langle v, v\rangle$. Hence $w \mapsto BAB^{-1}w$ is the reflection with respect to the hyperplane orthogonal to Bv.

Rotations in the plane. We shall prove that in the plane any rotation is the product of two reflections.

Lemma 2.2. *In $\mathbb{R}^2$, the rotation of angle θ is equal to the product $s \circ s_{\frac{\theta}{2}}$ where s is the reflection about the x-axis and $s_{\frac{\theta}{2}}$ is the reflection about the axis with polar angle $\frac{\theta}{2}$.*

Proof. In $\mathbb{R}^2$, the matrix of the rotation of angle θ is $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$. The reflection with respect to the x-axis maps a vector $\begin{pmatrix} u \\ v \end{pmatrix}$ to the vector $\begin{pmatrix} u \\ -v \end{pmatrix}$. The reflection $s_{\frac{\theta}{2}}$ about the axis with unit vector $\begin{pmatrix} \cos\frac{\theta}{2} \\ \\ \sin\frac{\theta}{2} \end{pmatrix}$ maps a vector $\begin{pmatrix} u \\ v \end{pmatrix}$ to the vector $\begin{pmatrix} u - 2(u\cos\frac{\theta}{2} + v\sin\frac{\theta}{2})\cos\frac{\theta}{2} \\ \\ v - 2(u\cos\frac{\theta}{2} + v\sin\frac{\theta}{2})\sin\frac{\theta}{2} \end{pmatrix}$. Replacing $\begin{pmatrix} u \\ v \end{pmatrix}$ by $\begin{pmatrix} u \\ -v \end{pmatrix}$ and using the identities $1 - 2\cos^2\frac{\theta}{2} = 2\sin^2\frac{\theta}{2} - 1 = \cos\theta$ and $2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \sin\theta$, yields the result. □

Orthogonal transformations are products of reflections. The matrix of any orthogonal transformation in $\mathbb{R}^n$ is written BAB^{-1} where B is orthogonal and A is block-diagonal with elementary blocks which are either (2×2)-matrices of the form $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ or the (1×1)-block 1 or -1. If n is even, $n = 2\nu$, the block-diagonal matrix, A, has ν blocks, each of which is 2×2 and A is the matrix of the product of ν rotations in the planes generated by (e_1, e_2), (e_3, e_4), ..., $(e_{2\nu-1}, e_{2\nu})$, respectively. If n is odd, $n = 2\nu + 1$, the matrix A has ν blocks, each of which is 2×2, followed by 1 or -1 on the diagonal. Thus A is the matrix of the product of a number of reflections.

We conclude from the statements in this section that any orthogonal transformation is the product of a number of reflections.

Theorem 2.3. *The group* $\mathrm{O}(n)$ *is generated by reflections. Any element of* $\mathrm{SO}(n)$ *is the product of an even number of reflections.*

2.4 The Group Morphism from Pin(n) to O(n)

For each element u of the Clifford algebra Cl_n, consider the linear map $\varphi(u)$ from $E \subset Cl_n$ to Cl_n defined by

$$\varphi(u)\, x = \mathcal{A}(u)\ x\ \mathcal{C}(u),$$

for $x \in E$.

Theorem 2.4. *The restriction of the map φ to* Pin(n) *is a group morphism from* Pin(n) *onto* O(n)*, with kernel* $\{1, -1\}$.

Proof. We shall show that, for each $u \in \Sigma$, $\varphi(u)$ leaves $E \simeq \mathbb{R}^n$ invariant and $\varphi(u)$ is in O(n). Then we shall show that $\varphi : \mathrm{Pin}(n) \to \mathrm{O}(n)$ is a surjective group morphism and we shall determine its kernel.

A priori, $\varphi(u)$, for $u \in E$, is a linear map from E to Cl_n. We show that, for each $\alpha = 1, ..., n$ and $x \in E$, $\varphi(\gamma_\alpha)x$ is in E. In fact, for every element γ_β, $\beta = 1, ..., n$, in the basis of E, $\varphi(\gamma_\alpha)\gamma_\beta = \gamma_\alpha\gamma_\beta\gamma_\alpha$ is equal to either γ_β or $-\gamma_\beta$. Therefore $\varphi(\gamma_\alpha)$ is an endomorphism of E.

If $u \in \Sigma$, then $\varphi(u)$ is the reflection in the hyperplane perpendicular to u. In fact, choose an orthonormal basis of E, $e_1, \ldots, e_n$, such that $e_1 = u$. Then $\varphi(u)(e_1) = -e_1$, while $\varphi(u)(e_\alpha) = e_\alpha$, if $\alpha \neq 1$. Reflections preserve the norm therefore $\varphi(u) \in \mathrm{O}(n)$.

Since, for u, $v \in \Sigma$, and $x \in E$,

$$\varphi(uv)(x) = \mathcal{A}(uv)\ x\ \mathcal{C}(uv) = \mathcal{A}(u)\mathcal{A}(v)\ x\ \mathcal{C}(v)\mathcal{C}(u) = \varphi(u)(\varphi(v)x),$$

and since any element of Pin(n) is a product of elements of Σ, φ is a group morphism from Pin(n) into O(n). We proved in Theorem 2.3 that O(n) is generated by reflections. Therefore we see that φ is a morphism of groups from Pin(n) onto O(n).

The kernel of φ contains 1 and -1. No element e_α is in the kernel of φ. Let us show that no element $e_{\alpha_1} \cdots e_{\alpha_r}$, $r \geq 2$, is in the kernel of φ. In fact, if $\varphi(e_{\alpha_1} \cdots e_{\alpha_r})$ is the identity, then r must be even since the product of an odd number of reflections cannot be the identity. But when r is even, $r \geq 2$, $\varphi(e_{\alpha_1} \cdots e_{\alpha_r})(e_{\alpha_1}) = -e_{\alpha_1}$ so that no product of an even number of reflections is in the kernel of φ. Therefore the kernel of φ is $\{1, -1\}$. □

2.5 Definition and Properties of the Group Spin(n)

We shall now define Spin(n) as a subgroup of Pin(n). Both groups are closed subgroups of GL(n).

Definition 2.5. *The group* Spin(n) *is the inverse image of the subgroup* SO(n) *of* O(n) *under the morphism* $\varphi : \mathrm{Pin}(n) \to \mathrm{O}(n)$.

The group Spin(n) is contained in the reduced Clifford algebra Cl_n^+.

We recall the definition of finite covers and properties of simply connected Lie groups (Chapter 4, Definition 7.4).

Definition 2.6. *Let G and H be Lie groups with units e_G and e_H, respectively, and let φ be a continuous morphism from G onto H. Let p be a positive integer.*

(i) (G, φ) is called a p-fold cover *of H if $\varphi : G \to H$ is a local homeomorphism of a neighborhood of e_G in G onto a neighborhood of e_H in H and $\varphi^{-1}(e_H)$ has p elements. A 2-fold cover is called a* double cover.

(ii) A p-fold cover (G, φ) of H is called a universal cover *of H if G is simply connected.*

The universal cover of a given Lie group is essentially unique: if two finite covers of H are simply connected, they are homeomorphic.

Lemma 2.7. *If a Lie group G is a finite cover of a Lie group H, the Lie algebras of G and H coïncide. If simply connected Lie groups G and H have isomorphic Lie algebras, the connected components of the identity of G and H are isomorphic.*

Proof. In fact, the Lie algebras of H and of a finite cover of H depend only on a neighborhood of the identity. Therefore their Lie algebras are the same.

If Lie groups G and H have the same tangent space at the identity, they are homeomorphic in the neighborhood of the identity. Because they are Lie groups (with continuous translations) this local diffeomorphism defines a local diffeomorphism of some neighborhood of any point g onto its image, provided the groups are simply connected.

It follows from Theorem 2.4 and Definition 2.5 that for each $n \in \mathbb{N}$, the Lie algebra of Spin(n) is isomorphic to the Lie algebra of SO(n) which is of dimension $\frac{n(n-1)}{2}$.

For $n = 1$, the Lie algebra of Spin(1) is $\{0\}$. For $n = 2$, the Lie algebra of Spin(2) is $\mathfrak{so}(2) \simeq \mathbb{R}$. We shall show that, if $n \geq 3$, Spin(n) is simply connected by exhibiting a continuous path from 1 to -1 contained in Spin(n).

Lemma 2.8. *The map $c : t \in [0, \pi] \mapsto c_t = \cos t\, 1 + \sin t\, \gamma_1\gamma_2 \in Cl_n$ is a continuous path from 1 to -1 lying in* Spin(n).

Proof. Let $\varphi : \mathrm{Pin}(n) \to \mathrm{O}(n)$ be the group morphism defined in Section 2.4. To determine the restriction to $\mathbb{R}^n$ of the endomorphisms $\varphi(c_t)$ of Cl_n, we compute $\varphi(c_t)(\gamma_\alpha) = (\cos t\, 1 + \sin t\, \gamma_1\gamma_2)\gamma_\alpha(\cos t\, 1 + \sin t\, \gamma_2\gamma_1)$ for each positive integer α. We find $\varphi(c_t)(\gamma_1) = \cos 2t\, \gamma_1 + \sin 2t\, \gamma_2$, $\varphi(c_t)(\gamma_2) = -\sin 2t\, \gamma_1 + \cos 2t\, \gamma_2$ and, for $\alpha \geq 3$, $\varphi(c_t)(\gamma_\alpha) = \gamma_\alpha$. Thus $\varphi(c_t)$ acts as the rotation of angle $2t$ in the plane generated by γ_1 and γ_2 and it is the identity on the supplementary $(n-2)$-dimensional space. Therefore the path $t \in [0, \pi] \mapsto \varphi(c_t)$ lies in SO(n). □

Therefore, for $n \geq 3$, Spin(n) is a simply connected Lie group. From the preceding lemmas we obtain immediately:

Theorem 2.9. *For $n \geq 2$, the restriction of the morphism φ to* Spin(n) *is a double cover of* SO(n). *For $n \geq 3$,* Spin(n) *is simply connected and is the universal cover of* SO(n).

2.6 The Groups Spin(1), Spin(2), and Spin(3)

Spin(1)
The Clifford algebra Cl_1, generated by $\gamma_1 = \gamma$ with $\gamma^2 = -1$, is isomorphic to the algebra of complex numbers, $\mathbb{C}$. The group Pin(1) is generated by $\{\gamma, -\gamma\}$, so Pin(1) has 4 elements, γ, $\gamma^2 = -1$, $\gamma^3 = -\gamma$, $\gamma^4 = 1$, and therefore $\mathrm{Pin}(1) = \mathbb{Z}_4$. Now, $\varphi(\pm 1) = I$ while $\varphi(\pm\gamma)$ is the reflection $\gamma \mapsto -\gamma$. Therefore Spin(1) is isomorphic to the group $\mathbb{Z}_2$.

Spin(2)
The Clifford algebra Cl_2 is generated by γ_1, γ_2 with relations ${\gamma_1}^2 = -1$, ${\gamma_2}^2 = -1$, $\gamma_1\gamma_2 + \gamma_2\gamma_1 = 0$. It is easy to verify that, by mapping γ_1 to i, γ_2 to j, and $\gamma_1\gamma_2$ to k, one defines an isomorphism of the algebra Cl_2 onto the algebra of quaternions, $\mathbb{H}$, generated by 1, i, j, k. Thus Σ is the circle $\{\lambda_1 i + \lambda_2 j \,|\, \lambda_1, \lambda_2 \in \mathbb{R}, \lambda_1^2 + \lambda_2^2 = 1\}$. The group generated by Σ in Cl_2 is $\{\lambda_0 1 + \lambda_1 i + \lambda_2 j + \lambda_3 k \,|\, \sum_{i=0}^{3} \lambda_i^2 = 1, \lambda_i \in \mathbb{R}\}$. Thus Pin(2) is the group of quaternions of unit norm which is isomorphic to SU(2), itself isomorphic to the group O(2). By definition, Spin(2) is the connected component of Pin(2). Therefore Spin(2) is isomorphic to the group SO(2).

Spin(3)
From Theorem 2.6 of Chapter 5, we know that the group SU(2) is a double cover of SO(3). Since the group Spin(3) is the universal double cover of SO(3), we conclude that Spin(3) is isomorphic to the group SU(2).

3 Spinor Representations of the Clifford Algebras

In this section, n is an integer, $n \geq 3$, and ν denotes the integer part of $\frac{n}{2}$, so that, by definition, if n is even, $n = 2\nu$ and, if n is odd, $n = 2\nu + 1$. We consider the representations of the Clifford algebras of even- or odd-dimensional vector spaces on finite-dimensional complex vector spaces.

3.1 Representations of Algebras

The representations of the spin groups will be defined below in Section 4 in terms of the restriction to $\mathrm{Spin}(V, Q)$ of representations of the Clifford algebra $Cl(V, Q)$. Here, we recall the definitions of representations and of ideals of associative algebras.

Definition 3.1. *An* algebra homomorphism *from an algebra, A, to an algebra, B, is a linear map from A to B which respects multiplication.*

A representation *of an algebra A in a vector space V is an algebra homomorphism from A to the algebra of endomorphisms of V. The representation is* irreducible *if the only invariant vector subspaces of V are $\{0\}$ and V.*

Definition 3.2. *A left (resp., right)* ideal *in an algebra is a vector subspace which is invariant under the left (resp., right) multiplication by elements of the algebra. It is called a* minimal *left (resp., right) ideal if it does not strictly contain a left (resp., right) non-zero ideal.*

3.2 Spinor Representations of the Complex Clifford Algebras

We outline the construction of the irreducible representations of the complex Clifford algebras, omitting several proofs. Recall that a vector subspace W of a complex vector space V equipped with a quadratic form Q is *isotropic* if the restriction of Q to W vanishes.

Spinors in even dimension
Let V be a vector space of even dimension, $n = 2\nu$, over the field $\mathbb{K} = \mathbb{C}$ of complex numbers, equipped with a non-degenerate quadratic form Q. The space V can be decomposed into a direct sum, $V = W \oplus W'$, of two isotropic subspaces, W and W', each of dimension ν. The exterior algebra of W is a minimal left ideal, S, of dimension 2^ν, of the Clifford algebra, $Cl(V, Q)$. A map, ρ, from $Cl(V, Q)$ into the algebra of endomorphisms of S is defined by

$$\rho(w)u = w\,u,$$

for $w \in Cl(V, Q)$ and $u \in S \subset Cl(V, Q)$. The map ρ is an isomorphism of algebras and an irreducible representation of $Cl(V, Q)$ in the vector space S. By definition, the map ρ is a *spinor representation* of the Clifford algebra $Cl(V, Q)$ and the space S, of dimension 2^ν, is a *space of spinors* of (V, Q). This representation is irreducible and it is the unique irreducible representation of $Cl(V, Q)$, up to equivalence.

Theorem 3.3. *If V is a complex vector space of even dimension $n = 2\nu$, the complex Clifford algebra $Cl(V, Q)$ has a* unique class of irreducible representations, of dimension 2^ν.

The spinor representation of the complex Clifford algebra $\mathbb{C}l(2\nu)$ is a representation in the vector space $\mathbb{C}^{2^\nu}$. In particular, if $n = 4$ the space of spinors is $\mathbb{C}^4$.

Spinors in odd dimension
If V is a complex vector space of odd dimension, $n = 2\nu + 1$, the reduced complex Clifford algebra $Cl^+(V, Q)$ has a unique class of irreducible representations, of dimension 2^ν. There exists a minimal left ideal S in the reduced Clifford algebra, $Cl^+(V, Q)$, of dimension 2^ν, and a map, ρ, from $Cl(V, Q)$ into the algebra of endomorphisms of S, defined by $\rho(w)u = w\,u$, for $w \in Cl(V, Q)$ and $u \in S \subset Cl^+(V, Q)$, which is a homomorphism and whose restriction to $Cl^+(V, Q)$ is an irreducible representation of $Cl^+(V, Q)$ in S. By definition, the map ρ is the *spinor representation* of the complex Clifford algebra $Cl(V, Q)$ and the space S is the *space of spinors* of (V, Q). This representation is irreducible, and it is the unique irreducible representation of $Cl(V, Q)$, up to equivalence.

Theorem 3.4. *If V is a complex vector space of odd dimension $n = 2\nu + 1$, the complex Clifford algebra $Cl(V, Q)$ has a* unique class of irreducible representations, of dimension 2^ν.

The spinor representation of the complex Clifford algebra $\mathbb{C}l(2\nu + 1)$ is a representation in the vector space $\mathbb{C}^{2^\nu}$. In particular, if $n = 3$ the space of spinors is $\mathbb{C}^2$.

Restriction of the spinor representation to the reduced Clifford algebra

The restriction of the spinor representation to the reduced Clifford algebra $Cl^+(V,Q)$ is irreducible or not according to whether n is odd or even.

- If n is odd, the restriction of the spinor representation to the reduced Clifford algebra $Cl^+(V,Q)$ is irreducible.
- If n is even, $n = 2\nu$, the restriction of the spinor representation to the reduced Clifford algebra $Cl^+(V,Q)$ is reducible. In fact, it is the sum of two inequivalent irreducible representations, each of dimension $2^{\nu-1}$, which are called the positive and the negative *Weyl representations*. The elements of the representation spaces $\mathbb{C}^{2^{\nu-1}}$ are called *half-spinors* or *semi-spinors*.

For example, if $n = 4$, the spaces of positive and negative Weyl spinors are complex vector spaces of dimension 2.

3.3 The Real Case

Next, we consider the case of a real vector space, V, with a quadratic form, Q. By definition, a *space of spinors* for (V,Q) is a space of spinors for the complexified vector space $V \otimes \mathbb{C}$ equipped with the complexification of the quadratic form, Q. It is a complex vector space of dimension 2^ν, whether V is of dimension 2ν or $2\nu+1$. The *spin representation* of V is the restriction to the Clifford algebra $Cl(V,Q)$ of the spin representation of the complexifed Clifford algebra. It is not irreducible in general.

In the case where $V = \mathbb{R}^n$ is equipped with the canonical quadratic form of signature (p,q), $p+q = n$, there exists a homomorphism from the complexified Clifford algebra to the algebra of $2^\nu \times 2^\nu$ complex matrices. If n is even, $n = 2\nu$, it is an isomorphism. If n is odd, $n = 2\nu + 1$, its restriction to the reduced Clifford algebra is an isomorphism. In all cases, the *space of spinors* of $\mathbb{R}^n$ is the complex vector space $\mathbb{C}^{2^\nu}$.

4 Representations of the Spin Groups

4.1 The Complex Spin Groups

For each positive integer n, the real spin group $\mathrm{Spin}(n)$ was defined as a subgroup of $\mathrm{Pin}(n)$, corresponding to the definite positive quadratic form on $\mathbb{R}^n$, itself a group of invertible elements in the real Clifford algebra Cl_n. Similarly, in the complex case, the complex Lie groups $\mathrm{Spin}^{\mathbb{C}}(n)$ are defined as groups of invertible elements in the complex Clifford algebra $\mathbb{C}l_n$, satisfying an additional condition which implies that, for $n \geq 3$, $\mathrm{Spin}^{\mathbb{C}}(n)$ is the universal cover of the Lie group $\mathrm{SO}(n,\mathbb{C})$, the special orthogonal group over $\mathbb{C}$.

The group $\mathrm{Spin}^{\mathbb{C}}(n)$ is contained in the reduced Clifford algebra $\mathbb{C}l_n^+$.

4.2 The Groups Spin(p, q)

For each positive integer n, the group $\mathrm{Spin}(n)$ was defined as a subset of the real Lie algebra Cl_n corresponding to the positive definite quadratic form on $\mathbb{R}^n$. In Section 1, the real Clifford algebras $Cl(p, q)$, corresponding to non-degenerate quadratic forms of arbitrary signature (p, q) on $\mathbb{R}^n$, $n = p+q$, were introduced. This, in turn, leads to a definition of the *real spin groups*, $\mathrm{Spin}(p, q)$, associated to a quadratic form of signature (p, q).

For all $n \geq 3$, as well as for signatures $(2, 0)$ and $(0, 2)$, the group $\mathrm{Spin}(p, q)$ is a *connected double cover* of $\mathrm{SO}_0(p, q)$, the connected component of the special orthogonal group $\mathrm{SO}(p, q)$. For $n \geq 3$, $\mathrm{Spin}(1, n-1)$ is the the *universal cover* of $\mathrm{SO}_0(1, n-1)$ and $\mathrm{Spin}(n-1, 1)$ is the *universal cover* of $\mathrm{SO}_0(n-1, 1)$.

In the case of the theory of relativity, the space (V, Q) is the 4-dimensional Minkowski space-time, M_4, which is $\mathbb{R}^4$ with the canonical quadratic form of signature $(1, 3)$ or $(3, 1)$, according to the conventions. The group $\mathrm{Spin}(1, 3)$ is the universal double cover of the connected component of the Lorentz group, $\mathrm{SO}_0(1, 3)$.

4.3 Representations of the Spin Groups and Spinors

The real spin groups, $\mathrm{Spin}(p, q)$, are subsets of the reduced real Clifford algebras, $Cl^+(p, q)$. In particular, $\mathrm{Spin}(n)$ is a subset of the reduced real Clifford algebra, Cl_n^+, while $\mathrm{Spin}^{\mathbb{C}}(n)$ is a subset of the reduced complex Clifford algebra $\mathbb{C}l_n^+$.

Definition 4.1. *The* spinor representations of the spin groups *Spin(p, q) (resp., Spin$^{\mathbb{C}}(n)$) are defined as the restrictions of the representations of the reduced Clifford algebras $Cl^+(p, q)$ (resp., $\mathbb{C}l_n^+$).*

In a representation ρ of $\mathrm{Spin}(p, q)$, whenever a basis (e_α), $\alpha = 1, ..., n$, of (V, Q) is chosen, $\rho(e_\alpha)$ is a $2^\nu \times 2^\nu$ complex matrix, where $n = 2\nu$ or $2\nu + 1$. Denote it by γ_α. Then

$$\gamma_\alpha \gamma_\beta + \gamma_\beta \gamma_\alpha = -2\, g_{\alpha\beta},$$

where $g_{\alpha\beta} = B(e_\alpha, e_\beta)$ and B denotes the symmetric bilinear form associated to Q. In particular, if the basis is orthogonal and $g_{\alpha\alpha} = \pm 1$, then

$$(\gamma_\alpha)^2 = \mp 1 \text{ and}$$

$$\gamma_\alpha \gamma_\beta + \gamma_\beta \gamma_\alpha = 0, \text{ for } \alpha \neq \beta.$$

Thus, for $\mathrm{Spin}(p, q)$, $(\gamma_\alpha)^2 = -1$ if $\alpha = 1, ..., p$, and $(\gamma_\alpha)^2 = 1$ if $\alpha = p+1, ..., p+q$.

If S is a space of spinors for $\mathrm{Spin}(n)$, then $S \otimes S^*$ is isomorphic to either $\mathbb{C}l_n$, if n is even, $n = 2\nu$, or to $\mathbb{C}l_n^+$, if n is odd, $n = 2\nu + 1$. In either case, the endomorphisms γ_α of S are represented, once a basis has been selected, by $2^\nu \times 2^\nu$ complex matrices.

4.4 Spinors in 3 Dimensions

We saw in Section 2.6 that the group Spin(3) is isomorphic to SU(2). Therefore the irreducible representations of the group Spin(3) coincide with those of SU(2). From the results and in the notations of Section 2 of Chapter 6, we obtain immediately:

Theorem 4.2. *The irreducible representations of* Spin(3) *are the representations* $(V^j, \mathcal{D}^j)$ *for* j *an* integer *or a* half-integer.

As we saw in Chapter 6, the kernel of the projection $q : \mathrm{Spin}(3) \to \mathrm{SO}(3)$ is $\{-I, I\}$, so we can re-phrase Theorem 3.1 of Chapter 6 as follows.

Theorem 4.3. *Those representations of* Spin(3) *which factor as representations of* SO(3) *are the representations* $(V^j, \mathcal{D}^j)$ *for* j *an* integer.

Definition 4.4. *The elements of the representation space,* V^j*, of* Spin(3)*, of complex dimension* $2j + 1$*, are called* spinors of spin j.

Since the representation $\mathcal{D}^{\frac{1}{2}}$ is 2-dimensional, after the choice of a basis of $V^{\frac{1}{2}}$, the spinors of spin $\frac{1}{2}$ are vectors in $\mathbb{C}^2$. The operators γ_α, $\alpha = 1, 2, 3$, are represented by 2×2 complex matrices, called the *Pauli matrices*. For an orthonormal basis, they are the matrices, σ_1, σ_2, σ_3, that were introduced in Chapter 4, Section 1.5 (see also Chapter 5, Section 1.1). When representing spinors of spin $\frac{1}{2}$ by vectors in $\mathbb{R}^4$, the operators γ_α are represented by real 4×4 matrices, called the *Majorana matrices*.

4.5 Spinors in 4 Dimensions and the Dirac Equation

The spinors that figure in the Dirac equation are the elements of the spin $\frac{1}{2}$ representation of the proper orthochronous Lorentz group, $\mathrm{SO}_0(1, 3)$, the group of special relativity, acting on 4-dimensional *Minkowski space-time*, M_4, with the indefinite scalar product g of signature $(1, 3)$. In an orthogonal basis of M_4, (e_0, e_1, e_2, e_3), such that $g(e_0, e_0) = 1, g(e_k, e_k) = -1$ for $k = 1, 2, 3$, the γ-matrices, $\gamma_0, \gamma_1, \gamma_2, \gamma_3$, of the representation of Spin$(1, 3)$ are 4×4 complex matrices such that

$$\gamma_0^2 = -1, \ \gamma_k^2 = 1 \text{ for } k = 1, 2, 3.$$

In this representation, spinors have 4 complex components and half-spinors have 2 complex components.

A *spinor field* of spin $\frac{1}{2}$ is a spinor-valued function ψ over M_4. The 4 components of the spinor field depend on the coordinates, x_0, x_1, x_2, x_3, of points in M_4. The *Dirac operator* is the differential operator on differentiable fields of spinors of spin $\frac{1}{2}$,

$$\psi \mapsto \gamma^\alpha \partial_\alpha \psi,$$

where $\partial_\alpha = \frac{\partial}{\partial x^\alpha}$, $\alpha = 0, 1, 2, 3$, and summation over repeated indices in different positions is understood (the Einstein convention). The *Dirac equation* is

$$\gamma^\alpha \partial_\alpha \psi = 0.$$

The square of the Dirac operator, $\gamma^\alpha \partial_\alpha$, is the Laplacian on M_4, also called the *d'Alembertian*,

$$\Box = -(\frac{\partial}{\partial x^0})^2 + (\frac{\partial}{\partial x^1})^2 + (\frac{\partial}{\partial x^2})^2 + (\frac{\partial}{\partial x^3})^2.$$

The Klein–Gordon equation,

$$\Box \psi = m^2 \psi,$$

where ψ is the wave function of an electron and m is its mass, is then a consequence of the Dirac equation for a spinor field of spin $\frac{1}{2}$,

$$\gamma^\alpha \partial_\alpha \psi = m\, \psi.$$

Finding a 'square root' of the Laplacian was the motivation for P. A. M. Dirac to introduce spinors into the theory of the electron.

4.6 Important Remark

The literature on spinors is very large, both in mathematics and in physics. The definitions of the spin groups and spinors vary from reference to reference and may differ from the ones above. In particular, with some definitions, the Dirac equation is written $\gamma^\alpha \nabla_\alpha \psi = i\, m\, \psi$ or $\gamma^\alpha \nabla_\alpha \psi = -\, i\, m\, \psi$, where $i^2 = -1$. In addition, when the Klein–Gordon and the Dirac equations are written in the usual physical variables, coefficients involving the speed of light and Planck's constant enter the equations.

References

For a very accessible text that treats Clifford algebras and the spin groups, see chapter 10 of the book on matrix groups by M. Curtis (1984). For a more advanced treatment of spin groups and spinor representations, see chapters 5 and 6 of Varadarajan's book on supersymmetry (2004), or chapter 6 of M. J. D. Hamilton's book (2017), a detailed theory of spinors with applications to theoretical physics, including the classification of the real Clifford algebras and reduced Clifford algebras in all signatures. For a treatise written for theoretical physicists, see Woit (2017). An advanced mathematical and highly recommended treatment is in Meinrenken's *Clifford Algebras and Lie Theory* (2014). Note that what we have denoted by $Cl(V, Q)$ is denoted by $Cl(V, -Q)$ in many publications.

Topics in History

William Clifford

Born in 1845, the English mathematician William Clifford studied at King's College in London, at Trinity College in Cambridge, and then became professor

of applied mathematics at the Uuniversity of London in 1871 and, three years later, was elected a member of the Royal Sociely. His search for generalizations of Hamilton's theory of quaternions led him to the discovery in 1878 of hypercomplex numbers that came to be known as elements of a "Clifford algebra". An important philosopher as well as a very gifted mathematician, he worked in many different areas of mathematics, including non-euclidean geometry. His "preliminary sketch of biquaternions" appeared in 1873 followed by more work on this generalization of the quaternions and his book on the dynamics of solid and fluid bodies was published in 1878, a year before his death, while his article on "The philosophy of the pure sciences" appeared posthumously in 1879 as did his collected works in 1882. His treatise "On the space-theory of matter" published in 1876 is considered a precursor of the general theory of relativity.

Dirac

The family of Paul Adrien Maurice Dirac (1902–1984) was English on his mother's side and Swiss on his father's. When he published his celebrated paper, "The quantum theory of the electron", in 1928, he was already a Research Fellow at St John's College in Cambridge where he would be awarded the prestigious Lucasian chair of mathematics in the university in 1931. He received the Nobel prize in physics in 1933 "for the discovery of new productive forms of atomic theory". The citation for the prize, which was awarded jointly to him and to Erwin Schrödinger, further tells us that "the importance of Dirac's work lies essentially in his famous wave equation, which introduced special relativity into Schrödinger's equation". His later career after 1972 was at Florida State University in the United States where a collection of his scientific papers is preserved.

In his work on the relativistic theory of the electron that had appeared in the Proceedings of the Royal Society in 1928, he derived the "Dirac equation" as a first-order equation such that, by squaring it, the Schödinger equation is obtained. To this effect he introduced 4 matrices which, after normalization, he denoted by α_μ and in formula (6), p. 613, he wrote the equations $\alpha_\mu^2 = 1,\ \alpha_\mu\alpha_\nu + \alpha_\nu\alpha_\mu = 1\ (\mu \neq \nu),\ \mu, \nu = 1, 2, 3, 4$. His book "The Principles of Quantum Mechanics", first published in 1930, appeared in many editions and continues to be regularly reprinted.

Spinor representations and quantum mechanics: an outline

The history of spinors begins – though they did not acquire this name until much later – in 1913, when Élie Cartan determined, among other results, the representations of the orthognal groups, including those that would eventually be called spin representations. But it was only with the development of the theory of quantum mechanics in the late 1920's that the name "spinors" was invented and adopted by both the mathematics and the theoretical physics communities. Who was the inventor? According to van der Waerden writing in 1929 it was the physicist Paul Ehrenfest (1880–1933) who lent them their name:

> Let us call spinors for short the new quantities which, in addition to the vectors and tensors, have appeared in the quantum mechanics of the spinning electron, and whose transformation laws under the Lorentz group are entirely different from those of tensors. 'Is there no spinor analysis that every physicist can learn as tensor analysis, and by means of which one can form all the possible spinors, as well as all the invariant equations in which spinors occur?' So Mr Ehrenfest asked me and the answer will be given below.[1]

Spinors have played an increasingly important role in quantum theories and in 1938 Cartan published a general study, *Leçons sur la théorie des spineurs*, eventually translated into English as *The Theory of Spinors*. In fact, "Spinors in n dimensions", an article by Brauer and Weyl which surveyed the representations of the spin groups, had already appeared in 1935.

The French mathematician Claude Chevalley was a professor of mathematics at Columbia University in New York after the Second World War when he wrote "The Algebraic Theory of Spinors" as part of a series of books on the humanities and sciences celebrating the bicentennial of the foundation of the university. Published by Columbia University Press in 1954, it remained a hard to find classic until it was re-published in 1997, with a foreword by Pierre Cartier and Catherine Chevalley and a postface by Jean-Pierre Bourguignon. In the introduction to his book, Chevalley explained:

> The simplest case of a spin representation is the one which presents itself for the orthogonal Lie algebra in 3 variables; this Lie algebra is well-known to be isomorphic to the special unitary Lie algebra on 2 variables, which shows that it has a faithful representation of degree 2: this is its spin representation.

This 2-dimensional representation of the Lie algebra $\mathfrak{so}(3)$, which is isomorphic to $\mathfrak{su}(2)$, is the differential of a representation, not of the Lie group SO(3), but of its 2-fold cover, SU(2). In fact, the group SU(2) is isomorphic to the spin group Spin(3) and this is indeed why this representation is called the spin representation or the spinor representation.

On publications and terminology

A 1935 article by Brauer and Weyl, Cartan's treatise of 1938, *Leçons sur la théorie des spineurs*, which was eventually translated into English in 1966, and the successive editions of the 1938 book of Dirac, *The Principles of Quantum Mechanics*, followed by Claude Chevalley's treatise, *The Algebraic Theory of Spinors* in 1954 were all influential publications on the mathematical theory of spinors.

[1] Translation of the first paragraph of B. L. van der Waerden, "Spinoranalyse", *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse* (1929), p. 100.

To explain the origin of the notation $\mathrm{Pin}(n)$ and the term "Pin group", we turn to another important article, "Clifford modules", of 1964, by Michael Atiyah, Raoul Bott and Arnold Shapiro, where they write that it was Jean-Pierre Serre who argued that, since $\mathrm{SO}(n)$ is the connected component of $\mathrm{O}(n)$ and $\mathrm{SU}(n)$ is the connected component of $\mathrm{U}(n)$, in the same way $\mathrm{Spin}(n)$ must be the connected component of a group called $\mathrm{Pin}(n)$!

The Majorana spinors are named after the extremely gifted Italian physicist, Ettore Majorana. Born in Sicily in 1906, he disappeared in 1938, and so far no amount of searching has given a clue to his disappearance.

Regarding "the Cartan–Dieudonné theorem" on the decomposition of orthogonal transformations into products of reflections, it is difficult to find its proof in the numerous publications of either Cartan or Dieudonné. Actually it was proved by Cartan, in 1938 or before, and was generalized by Dieudonné to the case of orthogonality with respect to quadratic forms over fields more general than the field of real numbers.

The Klein–Gordon equation

Klein–Gordon equation is named after Oskar Klein (1894–1977) – not to be confused with the German mathematician Felix Klein who figured so prominently in previous topics – and Walter Gordon (1893–1940) – not to be confused with the Erlangen algebraist, Paul Gordan, who was Emmy Noether's thesis adviser.

Oskar Klein, the Swedish theoretical physicist who worked with Niels Bohr in Copenhagen, was recruited to teach the new physics at the University of Michigan in the United States where he stayed for two years, and in 1930 was appointed to a chair in Stockholm. He is the Klein who gave his name to the Kaluza–Klein theory.

Walter Gordon was a German theoretical physicist who studied in Berlin with Max Planck and became an associate professor in Hamburg, before moving to Stockholm in 1933.

The "Klein–Gordon equation" was in fact formulated by Erwin Schrödinger (1887-1961) shortly before it appeared in print in articles by Walter Gordon in 1926 and, the following year, by Oskar Klein.

Exercises

Exercise 9.1. *Complex Clifford algebras in dimensions* 1*,* 2 *and* 3.
Study $\mathbb{C}l(1)$, $\mathbb{C}l(2)$, and $\mathbb{C}l(3)$.

Exercise 9.2. *The real Clifford algebras* $Cl(p,q)$ *for* $p+q \leq 2$.

(a) Show that $Cl(0,1) \simeq \mathbb{C}$, $Cl(1,0) \simeq \mathbb{R} \oplus \mathbb{R}$, $Cl(0,2) \simeq \mathbb{H}$. Show that $Cl(1,1) \simeq Cl(2,0) \simeq \mathrm{End}(\mathbb{R}^2)$.
(b) For each case, determine the adjunction.

Exercise 9.3. *The real Clifford algebra in dimension* 3 *is a direct sum.*

(a) Let A and B be algebras. Verify that the multiplication $(a_1, b_1)(a_2, b_2) = (a_1a_2, b_1b_2)$, for $a_1, b_1 \in A,\ a_2, b_2 \in B$ gives $A \oplus B$ the structure of an algebra.
(b) Show that $Cl_3 \simeq \mathbb{H} \oplus \mathbb{H}$.

Exercise 9.4. *The real Clifford algebra in dimension* 4.

Show that $Cl_4 \simeq \operatorname{End} \mathbb{H}^2$, the algebra of 2×2 matrices with coefficients in $\mathbb{H}$.

Exercise 9.5. *More real Clifford algebras in dimension* 4.

Show that $Cl(1,3) \simeq \operatorname{End} \mathbb{R}^4$, $Cl(3,1) \simeq \operatorname{End} \mathbb{H}^2$, $Cl^+(1,3) \simeq Cl^+(3,1) \simeq \operatorname{End} \mathbb{C}^2$.

Exercise 9.6. *The real Clifford algebra in dimension* 5.

Show that $Cl_5 \simeq \operatorname{End} \mathbb{C}^4$ (Hint: to $\gamma_1, \gamma_2, \gamma_3$ associate complex block diagonal matrices, $\begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix}$, where A and D are 2×2 matrices, diagonal in the case of γ_1 and anti-diagonal in the case of γ_2 and γ_3, and to γ_4 and γ_5 associate block anti-diagonal matrices, $\begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix}$, where B and C are anti-diagonal).

Exercise 9.7 *Divisors of zero.*

(a) In Cl_n, calculate $\gamma_1\gamma_2...\gamma_k$ for $1 \leq k \leq n$.
(b) A *divisor of zero* in an algebra A is an element $u \in A$ such that $u \neq 0$ and there exists an element $v \neq 0$ such that $uv = 0$. Show that, if $n \geq 3$, Cl_n has divisors of zero.

Exercise 9.8. *Complex spin groups of dimensions 1, 2, and 3.*

Study the groups $\operatorname{Spin}(1, \mathbb{C})$, $\operatorname{Spin}(2, \mathbb{C})$ and $\operatorname{Spin}(3, \mathbb{C})$.

Exercise 9.9. *An isomorphism.*

Show that the symplectic group $\operatorname{Sp}(1)$ is isomorphic to $\operatorname{SU}(2)$. Conclude that $\operatorname{Spin}(3)$ is isomorphic to $\operatorname{Sp}(1)$.

Exercise 9.10. *The center of* $\operatorname{Spin}(n)$.

For each $n \geq 1$, determine the center of the group $\operatorname{Spin}(n)$.

Exercise 9.11. *The Clifford algebra acts on the exterior algebra.*

Show that there is an action of the algebra $\operatorname{Cl}(V, Q)$ on $\wedge V = \oplus_{k \geq 0} \wedge^k V$.

Exercise 9.12. *The group* $\operatorname{Spin}(4)$.

Show that the group $\operatorname{Spin}(4)$ is isomorphic to $\operatorname{SU}(2) \times \operatorname{SU}(2)$.



Eugene Wigner was born in Budapest in 1902 and died in Princeton, NJ, in 1995. He introduced the theory of groups into quantum mechanics as early as 1927 and received the Nobel prize for physics in 1963 for his discovery of the fundamental symmetry principles in the theory of elementary particles.

(American Institute of Physics, Emilio Segrè Visual Archives)

Problems and Solutions

1 Restriction of a Representation to a Finite Group

Let $\mathrm{Rot}(a, \theta)$ be the rotation of the space $\mathbb{R}^3$ through the angle θ leaving invariant the unit vector a. Consider the rotations, for $p = 0, 1, 2$,

$$g_p = \mathrm{Rot}(e_3, \frac{2p\pi}{3}) \quad \text{and} \quad h_p = \mathrm{Rot}(g_p e_1, \pi),$$

where (e_1, e_2, e_3) is the canonical basis of $\mathbb{R}^3$. Set

$$\Gamma = \{g_0, g_1, g_2, h_0, h_1, h_2\}.$$

1. Prove the equalities

$$g_1 h_0 = h_0 g_2,\ g_2 h_0 = h_0 g_1,\ h_1 = h_0 g_1,\ h_2 = h_0 g_2.$$

Show that Γ is a subgroup of the group of rotations $\mathrm{SO}(3)$ isomorphic to the symmetric group on 3 elements, $\mathfrak{S}_3$.

2. Find the character table of the group Γ.

For $j \in \mathbb{N}$, we denote by $\underline{\mathcal{D}}^j$ the representation of $\mathrm{SO}(3)$ obtained by passing to the quotient in the representation $\mathcal{D}^j$ of $\mathrm{SU}(2)$.

3. Show that the value of the character $\underline{\chi}_j$ of $\underline{\mathcal{D}}^j$ on a rotation through an angle θ, $\theta \in (0, \pi]$, is

$$\underline{\chi}_j(\theta) = \frac{\sin(2j+1)\frac{\theta}{2}}{\sin\frac{\theta}{2}}.$$

What is $\underline{\chi}_j(0)$?

4. Find, for each $j \in \mathbb{N}$, the decomposition of the restriction to Γ of the representation $\underline{\mathcal{D}}^j$ into a direct sum of irreducible representations of Γ. [Hint: Consider the cases $j = 3k$, $j = 3k+1$, $j = 3k+2$, where $k \in \mathbb{N}$.]

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Solutions

1. It is clear that g_0 is the identity and that

$$g_2 = (g_1)^2,\ (g_p)^3 = g_0,\ (h_p)^2 = g_0,\ p = 0, 1, 2.$$

We have $h_0 = \exp\pi\eta_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$, $g_1 = \exp\frac{2\pi}{3}\eta_3 = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $g_2 = \exp\frac{4\pi}{3}\eta_3 = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$. By calculating the products of the matrices, we see that $g_1h_0 = h_0g_2 = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -1 \end{pmatrix}$. We deduce that $g_2h_0g_2 = g_2g_1h_0 = g_0h_0 = h_0$, whence $g_2h_0 = h_0g_1$.

The rotation h_1 leaves g_1e_1 invariant. The same is true of h_0g_1 because $h_0g_1(g_1e_1) = h_0(g_2e_1) = g_1(h_0e_1) = g_1e_1$. On the other hand, h_1 and h_0g_1 both square to the identity; the rotations h_1 and h_0g_1 are thus the same rotation around g_1e_1 and through an angle π. Similarly, the rotation h_0g_2 leaves g_2e_1 invariant, and squares to the identity, thus it is equal to h_2. The set Γ is thus

$$\Gamma = \{g_0, g_1, g_2, h_0, h_0g_1, h_0g_2\}.$$

The symmetric group on 3 elements can be written

$$\mathfrak{S}_3 = \{e, c, c^2, t, tc, tc^2\}.$$

It is generated by c and t with the relations $c^3 = e$, $t^2 = e$, $ct = tc^2$. We define a bijection α of Γ onto $\mathfrak{S}_3$ by setting $\alpha(g_p) = c^p$ and $\alpha(h_p) = tc^p$. The equations shown above imply that Γ is a group, and that α is an isomorphism of groups.

2. We denote by $\underline{1}$ the trivial one-dimensional representation, by ϵ the signature representation and by ρ_0 the irreducible representation of dimension 2 of $\mathfrak{S}_3$. The character table of Γ is then as follows.

	(1)	(2)	(3)
Γ	g_0	g_1	h_0
$\chi_{\underline{1}}$	1	1	1
χ_ϵ	1	1	−1
χ_{ρ_0}	2	−1	0

3. The conjugacy classes of SO(3) are parametrized by $\theta \in [0, \pi]$. By definition, $\underline{\mathcal{D}}^j(\mathrm{Rot}(e_3, \theta)) = \mathcal{D}^j(A)$, where $A \in \mathrm{SU}(2)$ is such that $\varphi(A) = \mathrm{Rot}(e_3, \theta)$,

where φ is the covering morphism of SU(2) onto SO(3). Because $\varphi(\pm\exp(\theta\xi_3)) = \exp(\theta\eta_3)$,

$$\underline{\mathcal{D}}^j(\mathrm{Rot}(e_3,\theta)) = \mathcal{D}^j\begin{pmatrix} e^{i\frac{\theta}{2}} & 0 \\ 0 & e^{-i\frac{\theta}{2}} \end{pmatrix}.$$

Thus $\underline{\chi}_j(\theta) = \chi_j(\frac{\theta}{2})$, where $\chi_j(t)$ is the value of the character of $\mathcal{D}^j$ on the matrix $\begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix}$, that is, $\frac{\sin(2j+1)t}{\sin t}$. Consequently

$$\underline{\chi}_j(\theta) = \frac{\sin(2j+1)\frac{\theta}{2}}{\sin\frac{\theta}{2}}, \quad 0 < \theta \leq \pi,$$

and $\underline{\chi}_j(0) = 2j+1$.

4. We thus have the following characters.

Γ	(1) g_0	(2) g_1	(3) h_0
$\underline{\chi}_{3k}$	$6k+1$	1	$(-1)^k$
$\underline{\chi}_{3k+1}$	$6k+3$	0	$(-1)^{k+1}$
$\underline{\chi}_{3k+2}$	$6k+5$	-1	$(-1)^k$

The restriction to the subgroup Γ of the representation $\underline{\mathcal{D}}^j$ can be written

$$\underline{\mathcal{D}}^j\big|_\Gamma = m_{\underline{1}}\underline{1} \oplus m_\epsilon\epsilon \oplus m_{\rho_0}\rho_0,$$

where, for $\rho = \underline{1}, \epsilon, \rho_0$,

$$m_\rho = (\underline{\chi}_j \mid \chi_\rho) = \frac{1}{6}\left((2j+1)\chi_\rho(g_0) + 2\underline{\chi}_j(g_1)\chi_\rho(g_1) + 3\underline{\chi}_j(h_0)\chi_\rho(h_0)\right).$$

Let $j = 3k+r$, where k is an integer and $r = 0, 1$ or 2. The calculation of the coefficients $m_{\underline{1}}$, m_ϵ and m_{ρ_0} gives us the following decomposition. If j is even (k and r have the same parity),

$$\underline{\mathcal{D}}^j\big|_\Gamma = (k+1)\underline{1} \oplus k\epsilon \oplus (2k+r)\rho_0.$$

If j is odd (k and r have opposite parity),

$$\underline{\mathcal{D}}^j\big|_\Gamma = k\underline{1} \oplus (k+1)\epsilon \oplus (2k+r)\rho_0.$$

In each case we verify that the direct sum is of dimension $2j+1$.

2 The Group O(2)

Let O(2) be the group of rotations and symmetries of the plane preserving the origin, generated by the matrices

$$r_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},\ 0 \le \theta < 2\pi,\ \text{and } s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

1. Prove that $r_\theta s = s r_{-\theta}$. Find the conjugacy classes of O(2).

2. (a) Show that each irreducible representation of O(2) is of dimension ≤ 2. [Hint: Consider the restriction of the representation to the subgroup SO(2).]

(b) Find the representations of dimension 1 of O(2).

(c) Prove that the two-dimensional irreducible representations of O(2) are indexed by positive integers. Denote them by π_n. [Hint: Consider the restriction of a two-dimensional representation to SO(2).]

(d) Find the character χ_{π_n} of π_n.

3. Let Ad be the adjoint representation of O(2) in its Lie algebra. Show that it is equivalent to one of the representations studied in Question 2.

4. Decompose $\pi_n \otimes \pi_m$, for $n \ge 1$ and $m \ge 1$, into a direct sum of irreducible representations.

5. (a) Show that $r_{\frac{2\pi}{3}}$ and s generate a subgroup G of O(2) isomorphic to the symmetric group on 3 elements, $\mathfrak{S}_3$.

(b) Find, for every integer $n \ge 1$, the decomposition of the restriction σ_n of π_n to the subgroup G as a direct sum of irreducible representations. Give a geometric interpretation of the result.

Solutions

1. The equations $s^2 = I$ and $s r_\theta s^{-1} = r_{-\theta}$ imply $r_\theta s = s r_{-\theta}$, whence $r_\theta s r_{-\theta} = s r_{-2\theta}$. Thus the conjugacy class of s is $\{s r_\theta \mid \theta \in [0, 2\pi)\}$. On the other hand r_θ and $r_{\theta'}$ are conjugate if and only if $\theta' = \pm\theta$. Thus the other conjugacy classes of O(2) are $\{I\}$, $\{-I\}$ and $\{r_\theta, r_{-\theta}\}$, for $0 < \theta < \pi$.

2. (a) Let ρ be an irreducible representation of O(2) on $\mathbb{C}^d$. Because SO(2) is Abelian, the restriction of ρ to SO(2) is a direct sum of representations of dimension 1. Let $x \in \mathbb{C}^d$ be a vector generating one of the lines invariant under the restriction of ρ. The vector space generated by x and $\rho(s)x$ is invariant under O(2), since $\rho(r_\theta)\rho(s)x = \rho(s)\rho(r_{-\theta})x$ is colinear with $\rho(s)x$. Because the representation is assumed irreducible, this is the whole space, and thus $d \le 2$. The representation is of dimension 1 if and only if x and $\rho(s)x$ are colinear.

(b) Let $\rho : \mathrm{O}(2) \to \mathbb{C}$ be a representation of dimension 1 of $\mathrm{O}(2)$. Then either $\rho(s) = 1$ or $\rho(s) = -1$. Because sr_θ is conjugate to s, for $\theta \in [0, 2\pi)$, $\rho(sr_\theta) = \rho(s)$, and we must have $\rho(r_\theta) = 1$, for every $\theta \in [0, 2\pi)$. Thus, either $\rho = \chi_1$ (the *trivial representation*) or $\rho = \chi_1'$ (the *determinant representation*).

(c) Let $\rho : \mathrm{O}(2) \to \mathrm{GL}(2, \mathbb{C})$ be a representation of dimension 2 of $\mathrm{O}(2)$. There exists $x \in \mathbb{C}^2$ such that x and $\rho(s)x$ are linearly independent (otherwise the representation is of dimension 1). In the basis $(x, \rho(s)x)$ of $\mathbb{C}^2$, the matrix of $\rho(s)$ is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Because the restriction of ρ to $\mathrm{SO}(2)$ is a direct sum of representations of dimension 1, there are integers n and m such that the matrix of $\rho(r_\theta)$ in the basis $(x, \rho(s)x)$ of $\mathbb{C}^2$ is $\begin{pmatrix} e^{in\theta} & 0 \\ 0 & e^{im\theta} \end{pmatrix}$. The equation $r_\theta s = sr_{-\theta}$ implies $m = -n$. We verify that for every integer $n \neq 0$, $\pi_n(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\pi_n(r_\theta) = \begin{pmatrix} e^{in\theta} & 0 \\ 0 & e^{-in\theta} \end{pmatrix}$ define an irreducible two-dimensional representation of $\mathrm{O}(2)$. The representations π_n and π_{-n} are equivalent.

(d) The only lines in $\mathbb{C}^2$ invariant under $\pi_n(s)$ are $\mathbb{C}(e_1 + e_2)$ and $\mathbb{C}(e_1 - e_2)$. These are invariant under $\pi_n(r_\theta)$ for every θ if and only if $n = 0$. We have $\pi_0 = \chi_1 \oplus \chi_1'$. For $n \geq 1$, π_n is irreducible. The character χ_{π_n} of π_n is such that

$$\chi_{\pi_n}(r_\theta) = 2\cos n\theta, \ \chi_{\pi_n}(s) = 0.$$

3. The Lie algebra of $\mathrm{O}(2)$ is $\mathfrak{o}(2) = \mathfrak{so}(2)$, a real one-dimensional Abelian Lie algebra. We can choose the basis $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ in $\mathfrak{o}(2)$ to identify $\mathfrak{o}(2)$ to $\mathbb{R}$. Thus $\mathrm{Ad} : \mathrm{O}(2) \to \mathrm{GL}(\mathfrak{o}(2))$ is a representation of $\mathrm{O}(2)$ of dimension 1. It must be equivalent to either χ_1 or χ_1'. We have $\mathrm{Ad}_s : X \in \mathfrak{o}(2) \mapsto sXs^{-1} \in \mathfrak{o}(2)$ and

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

thus $\mathrm{Ad}_s = -1$. Hence $\mathrm{Ad} \sim \chi_1'$, the determinant representation.

4. Let $n \geq 1$ and $m \geq 1$ be integers. We have

$$\chi_{\pi_n \otimes \pi_m} = \chi_{\pi_n} \chi_{\pi_m}.$$

This character is zero on s and satisfies

$$(\chi_{\pi_n} \chi_{\pi_m})(r_\theta) = 4\cos n\theta \cos m\theta = 2(\cos(n+m)\theta + \cos(|n-m|\theta)).$$

If $n \neq m$, we find that

$$\chi_{\pi_n} \otimes \chi_{\pi_m} = \chi_{\pi_{n+m}} + \chi_{\pi_{|n-m|}},$$

whence

$$\pi_n \otimes \pi_m = \pi_{n+m} \oplus \pi_{|n-m|}.$$

If $n = m, (\chi_{\pi_n}\chi_{\pi_n})(r_\theta) = \chi_{\pi_{2n}}(r_\theta) + 2$. The two-dimensional representation π_0 whose character takes the value 2 on r_θ and 0 on s is reducible, with $\pi_0 = \chi_1 \oplus \chi_1'$. We then have

$$\pi_n \otimes \pi_n = \pi_{2n} \oplus \chi_1 \oplus \chi_1'.$$

5. (a) We set $c = r_{\frac{2\pi}{3}}$. We verify that c and s satisfy the relations $c^3 = I, s^2 = I$ and $c^2 s = sc$. The subgroup G of O(2) generated by c and s is the group of six elements

$$\{e, c, c^2, s, sc, sc^2\}$$

isomorphic to $\mathfrak{S}_3$.

(b) The representation $\sigma_n = \pi_n\big|_G$ decomposes as a direct sum of representations of $G \approx \mathfrak{S}_3$. We have

$$\chi_{\sigma_n}(c) = 2\cos\frac{2n\pi}{3} = \begin{cases} 2 & \text{if } n \equiv 0 \pmod 3, \\ -1 & \text{if } n \not\equiv 0 \pmod 3. \end{cases}$$

On the other hand, the restriction of χ_1 to $\mathfrak{S}_3$ is the trivial representation $\underline{1}$, while the restriction of χ_1' is the signature representation ϵ. The restriction σ_1 of π_1 is the two-dimensional irreducible representation of $\mathfrak{S}_3$. One obtains the following character table.

	e	c	s
$\underline{1}$	1	1	1
ϵ	1	1	-1
σ_1	2	-1	0
$n \equiv 0 (\mathrm{mod}\ 3) \sigma_n$	2	2	0
$n \not\equiv 0 (\mathrm{mod}\ 3) \sigma_n$	2	-1	0

Hence, for $k \in \mathbb{N}$, $\begin{cases} \sigma_{3k} & = \underline{1} \oplus \epsilon, \\ \sigma_{3k+1} & = \sigma_{3k+2} = \sigma_1. \end{cases}$

Geometric interpretation. If $n \equiv 0 \pmod 3$, the rotations $c = r_{\frac{2\pi}{3}}$ and $c^2 = r_{\frac{4\pi}{3}}$ act by π_n as the identity on the plane $\mathbb{C}^2$. On the other hand, $\pi_n(s)$ always leaves invariant the line D_1 generated by the vector $e_1 + e_2$ and the line D_2 generated by the vector $e_1 - e_2$. More precisely $\pi_n(s)(e_1 + e_2) = e_1 + e_2$ and $\pi_n(s)(e_1 - e_2) = -(e_1 - e_2)$. Because these two lines are invariant under σ_n, we have

$$\sigma_n\,|_{D_1} = \underline{1} \quad \text{and} \quad \sigma_n\,|_{D_2} = \epsilon.$$

If $n \not\equiv 0 \pmod 3$, $\sigma_n(c) = \pi_n(r_{\frac{2\pi}{3}})$ leaves none of the lines in $\mathbb{C}^2$ invariant. Thus σ_n is a two-dimensional irreducible representation of $\mathfrak{S}_3$. This is the complexification of the representation of $\mathfrak{S}_3$ in $\mathbb{R}^2$ by the symmetries of an equilateral triangle centered at the origin: rotations through the angles 0, $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$ and symmetries with respect to the three altitudes.

3 Representations of the Dihedral and Quaternion Groups

Let n be an integer, $n \geq 2$. We denote by C_n the cyclic group of order n and by $D_{(n)}$ the dihedral group of order $2n$. We know that

$$C_n = \{I, r, r^2, \ldots, r^p, \ldots, r^{n-1}\}$$

and

$$D_{(n)} = \{I, r, r^2, \ldots, r^{n-1}, s, sr, sr^2, \ldots, sr^{n-1}\},$$

where $r^n = I, s^2 = I$ and $srs = r^{-1}$.

Part I

For $0 \leq k \leq n-1$, denote by χ_k the character of C_n such that

$$\chi_k(r) = e^{\frac{2i\pi k}{n}}.$$

1. What are the inequivalent irreducible representations of C_n?

2. Find the conjugacy classes of $D_{(n)}$. [Hint: Distinguish the case of n odd from the case of n even.]

3. Find the one-dimensional representations of $D_{(n)}$.

For each k, $0 \leq k \leq n-1$, consider the induced representation of a representation with character χ_k from C_n to $D_{(n)}$, and denote it by ρ_k.

4. (a) Show that the representation ρ_k is of dimension 2. Choose a basis of the vector space of this representation and write the matrix of $\rho_k(r)$ and the matrix of $\rho_k(s)$ in this basis.

(b) Find the character of ρ_k.

5. (a) Show that ρ_k and ρ_{n-k} are equivalent. Is the representation ρ_0 irreducible? Show that if n is even, with $n = 2m$, the representation ρ_m is not irreducible.

(b) Find the inequivalent irreducible representations of $D_{(n)}$.

Part II

In the group O(2), each element can be written r_θ or $s_0 r_\theta$, $0 \leq \theta < 2\pi$, with

$$r_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad \text{and} \quad s_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We set

$$\widehat{r_\theta} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \widehat{s_0} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

6. (a) What is the geometric interpretation of $\widehat{r_\theta}$ and of $\widehat{s_0}$?

(b) Show that there is a unique injective group morphism $\psi : \mathrm{O}(2) \to \mathrm{SO}(3)$ such that

$$\psi(r_\theta) = \widehat{r_\theta} \quad \text{and} \quad \psi(s_0) = \widehat{s_0}.$$

Let φ be the projection (covering morphism) from SU(2) onto SO(3). We introduce the matrices $\mathcal{I} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$, $\mathcal{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathcal{K} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$.

7. (a) What is the inverse image under φ of $\widehat{r_\theta}$? of $\widehat{s_0}$?

(b) Show that $D_{(n)}$ can be identified with a subgroup of O(2) generated by a rotation and s_0. What is the order of the group $\Gamma_n = \varphi^{-1}(\psi(D_{(n)}))$? What are the elements of this group?

8. Show that Γ_2 is the quaternion group,

$$\{\pm I, \pm\mathcal{I}, \pm\mathcal{J}, \pm\mathcal{K}\}.$$

9. Create the character table of the group Γ_2.

Let $\mathcal{R}^{(N)}$, where N is a non-negative integer, be the representation of Γ_2 obtained as the restriction to Γ_2 of the irreducible representation of dimension $N+1$ of SU(2).

10. (a) Find the character of $\mathcal{R}^{(N)}$.

(b) Find the decomposition of $\mathcal{R}^{(N)}$ as a direct sum of irreducible representations of Γ_2.

(c) Same question for each of the representations $\rho \otimes \mathcal{R}^{(1)}$, where ρ is an irreducible representation of Γ_2.

Solutions

Part I

1. Because C_n is an Abelian group, all the irreducible representations of C_n are one-dimensional. The group C_n thus has $|C_n| = n$ inequivalent irreducible representations: these are the characters χ_k, $0 \le k \le n-1$, such that, for $0 \le p \le n-1$, $\chi_k(r^p) = e^{\frac{2i\pi kp}{n}}$.

2. From $srs = r^{-1}$ we deduce the equation $sr = r^{-1}s$, whence, for every integer p,

$$sr^p s = r^{-p}.$$

(i) *Conjugacy classes of rotations.* We deduce from the preceding formula that r^p and r^{n-p} are conjugate in $D_{(n)}$. For $r^{p'}$ to be conjugate to r^p, it is necessary and sufficient that there be an integer q such that $r^{p'} = sr^q r^p sr^q$, which is equivalent to $r^{p'} = sr^p s$, or to $p + p' \equiv 0$, mod n.

- If n is odd, $n = 2m+1$, there are $m+1 = \frac{n+1}{2}$ conjugacy classes of rotations,
$$\{1\}, \{r, r^{n-1}\}, \ldots, \{r^m, r^{m+1}\}.$$
- If n is even, $n = 2m$, there are $m = \frac{n}{2}$ conjugacy classes of rotations,
$$\{1\}, \{r, r^{n-1}\}, \ldots, \{r^{m-1}, r^{m+1}\}, \{r^m\}.$$

(ii) *Conjugacy classes of symmetries.* In order that $sr^{p'}$ be conjugate to sr^p, it is necessary and sufficient that there be an integer q such that $sr^{p'} = r^q sr^p r^{-q}$ or $sr^{p'} = sr^q sr^p r^{-q} s$. The first condition gives $p' - p + 2q \equiv 0$, mod n, and the second condition gives $p' + p - 2q \equiv 0$, mod n.

- If n is odd, such a choice of q is always possible. Thus there is only one conjugacy class of symmetries, $\{s, sr, \ldots, sr^{n-1}\}$.
- If n is even, there are two conjugacy classes of symmetries, $\{s, sr^2, \ldots, sr^{n-2}\}$ and $\{sr, sr^3, \ldots, sr^{n-1}\}$, each having $\frac{n}{2}$ elements.

To summarize, if n is odd, there are $\frac{n+3}{2}$ conjugacy classes,

(1)	(2)		(2)	(n)
1	r	$\ldots$	$r^{\frac{n-1}{2}}$	s

and if n is even, there are $\frac{n}{2} + 3$ conjugacy classes,

(1)	(2)		(2)	(1)	$(\frac{n}{2})$	$(\frac{n}{2})$
1	r	$\ldots$	$r^{\frac{n}{2}-1}$	$r^{\frac{n}{2}}$	s	sr

In both cases, we verify that there are $2n$ elements in the group $D_{(n)}$.

3. Let $\pi : D_{(n)} \to \mathbb{C}^*$ be a one-dimensional representation of $D_{(n)}$. Let $a = \pi(r)$ and $b = \pi(s)$. It must be true that

$$a^n = 1, \quad b^2 = 1, \quad bab = a^{-1}.$$

Hence $a = \pm 1$ and $b = \pm 1$.

If n is odd, we must have $a = +1$; there are two one-dimensional representations, the trivial representation, π_1, and the representation π_1' defined by

$$\pi_1'(r) = 1, \quad \pi_1'(s) = -1.$$

If n is even, there are four one-dimensional representations, π_1, π_1', and π_2, π_2' defined by

$$\pi_2(r) = -1, \pi_2(s) = 1,$$

$$\pi_2'(r) = -1, \pi_2'(s) = -1,$$

whence $\pi_2(r^p) = \pi_2'(r^p) = (-1)^p$, $\pi_2(sr^p) = (-1)^p$, $\pi_2'(sr^p) = (-1)^{p+1}$.

4. (a) By definition, the space E_k of the representation ρ_k is the vector space of maps f of $D_{(n)}$ into $\mathbb{C}$ such that

$$\forall g \in D_{(n)}, \forall h \in C_n,\ f(gh) = \chi_k(h^{-1})f(g),$$

i.e., $f(gr^p) = e^{-\frac{2i\pi kp}{n}} f(g)$, for $0 \leq p \leq n-1$. Therefore

$$f(r^p) = e^{-\frac{2i\pi kp}{n}} f(I)$$

and

$$f(sr^p) = e^{-\frac{2i\pi kp}{n}} f(s),$$

and the function f is entirely determined by $f(I)$ and $f(s)$ which are arbitrary complex numbers. Thus the representation ρ_k is of dimension 2.

Let f_1 be the element of E_k such that $f_1(I) = 1, f_1(s) = 0$ and let f_2 be the element of E_k such that $f_2(I) = 0$, $f_2(s) = 1$. Then (f_1, f_2) is a basis of E_k.

By definition, for $f \in E_k$, and $g_0, g \in D_{(n)}$,

$$(\rho_k(g_0)f)(g) = f(g_0^{-1}g),$$

whence

$$(\rho_k(r)f)(I) = f(r^{-1}) = \chi_k(r)f(I) = e^{\frac{2i\pi k}{n}} f(I),$$

$$(\rho_k(r)f)(s) = f(r^{-1}s) = f(sr) = \chi_k(r^{-1})f(s) = e^{-\frac{2i\pi k}{n}} f(s).$$

Thus $\rho_k(r)f_1 = e^{\frac{2i\pi k}{n}} f_1$ and $\rho_k(r)f_2 = e^{-\frac{2i\pi k}{n}} f_2$, whence the matrix of $\rho_k(r)$ in the basis (f_1, f_2) of E_k,

$$\rho_k(r) = \begin{pmatrix} e^{\frac{2i\pi k}{n}} & 0 \\ 0 & e^{-\frac{2i\pi k}{n}} \end{pmatrix}.$$

Similarly, $(\rho_k(s)f)(I) = f(s)$ and $(\rho_k(s)f)(s) = f(I)$, whence $\rho_k(s)f_1 = f_2$ and $\rho_k(s)f_2 = f_1$. Hence we have

$$\rho_k(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(b) We have $\rho_k(r^p) = \begin{pmatrix} e^{\frac{2i\pi kp}{n}} & 0 \\ 0 & e^{-\frac{2i\pi kp}{n}} \end{pmatrix}$ and $\rho_k(sr^p) = \begin{pmatrix} 0 & e^{-\frac{2i\pi kp}{n}} \\ e^{\frac{2i\pi kp}{n}} & 0 \end{pmatrix}$.

The character of ρ_k is thus given by

$$\begin{cases} \operatorname{Tr}(\rho_k(r^p)) = 2\cos\frac{2kp\pi}{n}, & 0 \leq p \leq n-1, \\ \operatorname{Tr}(\rho_k(sr^p)) = 0. \end{cases}$$

We verify that it takes the same value on each conjugacy class of $D_{(n)}$.

5. (a) The representations ρ_k and ρ_{n-k} are equivalent since they have same character.

The representation ρ_0 is not irreducible. In fact, $\rho_0 = \pi_1 \oplus \pi'_1$.

If n is even, $n = 2m$, the character of ρ_m takes the value $2(-1)^p$ at r^p and the value 0 at sr^p. We have $\chi_{\rho_m} = \chi_{\pi_2} + \chi_{\pi'_2}$, whence $\rho_m = \pi_2 \oplus \pi'_2$.

(b) (i) Case of odd n, $n = 2m+1$. The representations $\rho_0, \rho_1, \ldots, \rho_{m-1}, \rho_m$ are not equivalent since the values of their characters at r are different.

Each of the representations ρ_k, $1 \leq k \leq m$, is irreducible. In fact, if $1 \leq k \leq m$, the only vector subspaces of E_k invariant under $\rho_k(r)$ are the lines $\mathbb{C}f_1$ and $\mathbb{C}f_2$, and these are not invariant under $\rho_k(s)$. [One can also show that the modulus of the character of each of these representations is equal to 1.]

The representations $\pi_1, \pi'_1, \rho_1, \ldots, \rho_m$ are $2 + m = \frac{n+3}{2}$ inequivalent irreducible representations. This is the number of conjugacy classes of $D_{(n)}$ for odd n. Thus we have determined all the irreducible representations of $D_{(n)}$. [Another argument is as follows. We compute the sum of the squares of the dimensions of these representations, $(\dim \pi_1)^2 + (\dim \pi'_1)^2 + \sum_{k=1}^{m} (\dim \rho_k)^2 = 2 + 4m = 2n$, which is the order of $D_{(n)}$.]

(ii) Case of even n, $n = 2m$. The representations $\rho_1, \ldots, \rho_{m-1}$ are inequivalent and irreducible (see the odd case).

The representations $\pi_1, \pi'_1, \pi_2, \pi'_2, \rho_1, \ldots, \rho_{m-1}$ are $4 + m - 1 = \frac{n}{2} + 3$ inequivalent irreducible representations and this number is the number of conjugacy classes of $D_{(n)}$ for even n. We then have the complete list. [Another argument: the sum of the squares of the dimensions of these representations is $4 + 4(m-1) = 2n$, which is the order of $D_{(n)}$.]

In both cases, the irreducible representations of $D_{(n)}$ are one- or two-dimensional.

Part II

6. (a) The transformation $\widehat{r_\theta}$ of $\mathbb{R}^3$ is the rotation around the axis Oz and through an angle θ, $\mathrm{Rot}(e_3, \theta)$, and $\widehat{s_0}$ is the symmetry with respect to the axis Ox, $\mathrm{Rot}(e_1, \pi)$.

(b) We see immediately that $\widehat{r_\theta}$ and $\widehat{s_0}$ belong to SO(3). We set

$$\psi(s_0 r_\theta) = \widehat{s_0}\widehat{r_\theta}.$$

Thus we define an injective map from O(2) into SO(3). It is clear that $\widehat{r_\theta}\widehat{r_{\theta'}} = \widehat{r_\theta r_{\theta'}} = \widehat{r_{\theta+\theta'}}$, $(\widehat{s_0})^2 = (\widehat{s_0^2}) = I$. The matrix product $\widehat{s_0}\widehat{r_\theta}\widehat{s_0}$ is equal to $\widehat{r_{-\theta}} = (\widehat{r_\theta})^{-1}$. We deduce that ψ is a group morphism. In fact,

$$\psi(r_\theta(s_0 r_{\theta'})) = \psi(s_0 r_{\theta'-\theta}) = \widehat{s_0}\widehat{r_{\theta'-\theta}},$$

while

$$\psi(r_\theta)\psi(s_0 r_{\theta'}) = \widehat{r_\theta}\widehat{s_0}\widehat{r_{\theta'}} = \widehat{s_0}\widehat{r_{-\theta}}\widehat{r_{\theta'}} = \widehat{s_0}\widehat{r_{\theta'-\theta}}.$$

Similarly,

$$\psi((s_0 r_\theta)(s_0 r_{\theta'})) = \psi(r_{\theta'-\theta}) = \widehat{r_{\theta'-\theta}},$$

while

$$\psi(s_0 r_\theta)\psi(s_0 r_{\theta'}) = \widehat{s}_0 \widehat{r}_\theta \widehat{s}_0 \widehat{r_{\theta'}} = \widehat{r_{\theta'-\theta}}.$$

Geometrically, we associate to the rotation through an angle θ in the plane $x\mathrm{O}y$ the rotation of $\mathbb{R}^3$ with the same angle and with axis $\mathrm{O}z$, and to the symmetry s_0 of the plane $x\mathrm{O}y$ that exchanges the axes, the symmetry of $\mathbb{R}^3$ with respect to the axis x. (While in $\mathbb{R}^2$ a symmetry with respect to a line has determinant -1, a symmetry with respect to a line in $\mathbb{R}^3$ is a rotation through an angle π, with determinant $+1$.)

7. (a) We know that, on the one hand, $\varphi(\exp \frac{\theta}{2}\mathcal{K}) = \mathrm{Rot}\ (e_3, \theta) = \widehat{r}_\theta$ and, on the other hand, $\varphi(\exp \frac{\pi}{2}\mathcal{I}) = \mathrm{Rot}(e_1, \pi) = \widehat{s}_0$. We then have

$$\begin{aligned} \varphi^{-1}(\widehat{r}_\theta) &= \{\exp \tfrac{\theta}{2}\mathcal{K}, -\exp \tfrac{\theta}{2}\mathcal{K}\}, \\ \varphi^{-1}(\widehat{s}_0) &= \{\exp \tfrac{\pi}{2}\mathcal{I}, -\exp \tfrac{\pi}{2}\mathcal{I}\}. \end{aligned}$$

We have $\exp t\mathcal{I} = \cos t I + \sin t\ \mathcal{I}$, whence $\exp \frac{\pi}{2}\mathcal{I} = \mathcal{I}$. Thus

$$\varphi^{-1}(\widehat{s}_0) = \{\mathcal{I}, -\mathcal{I}\},$$

Similarly $\exp t\mathcal{K} = \cos t I + \sin t\ \mathcal{K}$, whence

$$\varphi^{-1}(\widehat{r}_\theta) = \{\cos \frac{\theta}{2} I + \sin \frac{\theta}{2}\mathcal{K}, -\cos \frac{\theta}{2} I - \sin \frac{\theta}{2}\mathcal{K}\}.$$

(b) The group $D_{(n)}$ can be identified with the subgroup of $\mathrm{O}(2)$ generated by the rotation $r_{\frac{2\pi}{n}}$ and the symmetry s_0.

The group $\psi(D_{(n)})$ is of the same order as $D_{(n)}$, *i.e.*, $2n$. The group $\Gamma_n = \varphi^{-1}(\psi(D_{(n)}))$ is of order $4n$ since, for each element g of $\psi(D_{(n)})$, $\varphi^{-1}(g)$ contains two opposite elements of $\mathrm{SU}(2)$.

The group $\psi(D_{(n)})$ is generated by $\widehat{r_{\frac{2\pi}{n}}}$ and $\widehat{s}_0$. It consists of transformations of $\mathbb{R}^3$, $\widehat{r_{\frac{2p\pi}{n}}}$ and $\widehat{s}_0\widehat{r_{\frac{2p\pi}{n}}}$, for $0 \leq p \leq n-1$. The group $\Gamma_n = \varphi^{-1}(\psi(D_{(n)}))$ is thus generated by the matrices of $\mathrm{SU}(2)$, $\pm(\cos \frac{\pi}{n} I + \sin \frac{\pi}{n}\mathcal{K})$, $\mathcal{I}$ and $-\mathcal{I}$. It consists of the matrices

$$\cos \frac{p\pi}{n} I + \sin \frac{p\pi}{n}\mathcal{K}, \quad -\cos \frac{p\pi}{n} I - \sin \frac{p\pi}{n}\mathcal{K},$$

and the matrices

$$\cos \frac{p\pi}{n}\mathcal{I} - \sin \frac{p\pi}{n}\mathcal{J}, \quad -\cos \frac{p\pi}{n}\mathcal{I} + \sin \frac{p\pi}{n}\mathcal{J},$$

for $0 \leq p \leq n-1$.

8. If $n = 2$, the group $\Gamma_2 = \varphi^{-1}(\psi(D_{(2)}))$ consists of the matrices

$$I, -I, \mathcal{I}, -\mathcal{I}, \mathcal{J}, -\mathcal{J}, \mathcal{K}, -\mathcal{K}.$$

Therefore it is the quaternion group. We can study the case where $n = 2$ directly. The group $D_{(2)}$ is the group $\{I, r, s, sr\}$, with $r^2 = s^2 = I,\ rs = sr$. It can be realized in $\mathbb{R}^2$ as the group $\{I, r_\pi = -I,\ s_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ s_0 r_\pi = -s_0\}$, and in $\mathbb{R}^3$ as the group $\{I, \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}\}$.

We have $\varphi^{-1}(I) = \{I, -I\}$. (The letter I denotes the identity on $\mathbb{R}^2$ or on $\mathbb{C}^2$, as well as on $\mathbb{R}^3$.) On the other hand,

$$\begin{aligned} \varphi^{-1}(\widehat{r_\pi}) &= \{\mathcal{K}, -\mathcal{K}\}\ , \\ \varphi^{-1}(\widehat{s_0}) &= \{\mathcal{I}, -\mathcal{I}\}, \\ \varphi^{-1}(\widehat{s_0 r_\pi}) &= \{\mathcal{J}, -\mathcal{J}\}, \qquad \text{because } \mathcal{K}\mathcal{I} = \mathcal{J}. \end{aligned}$$

Thus we find the list of elements of the group Γ_2.

9. We note that $\mathcal{I}^{-1} = -\mathcal{I},\ \mathcal{J}^{-1} = -\mathcal{J},\ \mathcal{K}^{-1} = -\mathcal{K}$.

The conjugacy classes of Γ_2 are

$$\{I\},\ \{-I\},\ \{\mathcal{I}, \mathcal{I}^{-1}\},\ \{\mathcal{J}, \mathcal{J}^{-1}\},\ \{\mathcal{K}, \mathcal{K}^{-1}\}.$$

In fact, $\mathcal{J}\mathcal{I}\mathcal{J}^{-1} = -\mathcal{J}\mathcal{I}\mathcal{J} = -\mathcal{J}\mathcal{K} = -\mathcal{I}$, and $\mathcal{K}\mathcal{I}\mathcal{K}^{-1} = -\mathcal{K}\mathcal{I}\mathcal{K} = -\mathcal{J}\mathcal{K} = -\mathcal{I}$. These calculations show that $\mathcal{I}, \mathcal{J}, \mathcal{K}$ are pairwise not conjugate.

There are four one-dimensional representations of Γ_2. To show this, let a and b be the values of a one-dimensional representation χ of Γ_2 at $\mathcal{I}$ and $\mathcal{K}$ respectively. Then

$$a^2 b^2 = 1 \quad \text{and} \quad a^2 = 1,$$

since

$$\mathcal{I}^2\mathcal{K}^2 = I \text{ and } \mathcal{K}^{-1}\mathcal{I}\mathcal{K}\mathcal{I} = \mathcal{K}^{-1}\mathcal{I}\mathcal{J} = I.$$

Thus $a = \pm 1$ and $b = \pm 1$. These choices determine the value of χ at $\mathcal{J} = \mathcal{K}\mathcal{I}$ and at $-I = \mathcal{I}^2$.

Because this group, Γ_2, has 5 conjugacy classes, there is another irreducible representation, α, of dimension $d \geq 2$. We must have $4 + d^2 = |\Gamma_2|$. Because the order of Γ_2 is 8, necessarily $d = 2$. We complete the character table using the orthogonality of columns. The representation α is in fact the fundamental

representation of $\Gamma_2 \subset \mathrm{SU}(2)$. One obtains the table

	(1) I	(1) $-I$	(2) $\mathcal{I}$	(2) $\mathcal{J}$	(2) $\mathcal{K}$
α_0	1	1	1	1	1
α_1	1	1	1	-1	-1
α_2	1	1	-1	1	-1
α_3	1	1	-1	-1	1
α	2	-2	0	0	0

(We have used the same letter to denote the representation α and its character.)

10. (a) Let $\chi^{(N)}$ be the character of the irreducible representation of dimension $N+1$ of SU(2). We have $\chi^{(N)}(I) = N+1$.

We know that if $g \in \mathrm{SU}(2)$ is conjugate to $\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, 0 < \theta < \pi$,

$$\chi^{(N)}(g) = \sum_{m=-\frac{N}{2}}^{\frac{N}{2}} e^{2im\theta} = \frac{\sin(N+1)\theta}{\sin\theta}.$$

Since $\mathcal{I}, \mathcal{J}$ and $\mathcal{K}$ are conjugate in SU(2) to the matrix $\begin{pmatrix} e^{i\frac{\pi}{2}} & 0 \\ 0 & e^{-i\frac{\pi}{2}} \end{pmatrix}$,

$$\chi^{(N)}(\mathcal{I}) = \chi^{(N)}(\mathcal{J}) = \chi^{(N)}(\mathcal{K}) = \sin(N+1)\frac{\pi}{2} = \cos N\frac{\pi}{2}.$$

Thus we find

$$\chi^{(N)}(\mathcal{I}) = 0, \text{ if } N \text{ is odd},$$
$$\chi^{(N)}(\mathcal{I}) = 1, \text{ if } N \equiv 0 \mod 4,$$
$$\chi^{(N)}(\mathcal{I}) = -1, \text{ if } N \equiv 2 \mod 4.$$

We have $-I = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} e^{i\pi} & 0 \\ 0 & e^{-i\pi} \end{pmatrix}$. If N is even, $N = 2N'$,

$$\chi^{(N)}(-I) = \sum_{m=-N'}^{N'} e^{2im\pi} = 2N' + 1 = N + 1,$$

while, if N is odd, $N = 2N' + 1$, the index m takes half-integer values, and thus each of the $N+1$ terms of the sum is equal to $e^{-i\pi} = -1$ and we find

$$\chi^{(N)}(-I) = -(N+1).$$

Hence

$$\chi^{(N)}(-I) = (-1)^N(N+1).$$

(b) In order to find the decomposition of $\mathcal{R}^{(N)}$ as a direct sum of irreducible representations, we calculate $(\alpha_i \mid \chi^{(N)})$, $0 \leq i \leq 3$, and $(\alpha \mid \chi^{(N)})$. We have

$$(\alpha_0 \mid \chi^{(N)}) = \frac{1}{8}(\chi^{(N)}(I) + \chi^{(N)}(-I) + 6\chi^{(N)}(\mathcal{I})),$$

$$(\alpha_i \mid \chi^{(N)}) = \frac{1}{8}(\chi^{(N)}(I) + \chi^{(N)}(-I) - 2\chi^{(N)}(\mathcal{I})), \quad 1 \leq i \leq 3,$$

$$(\alpha \mid \chi^{(N)}) = \frac{1}{4}(\chi^{(N)}(I) - \chi^{(N)}(-I)).$$

If N is odd, $(\alpha_i \mid \chi^{(N)}) = 0$, for $0 \leq i \leq 3$ and $(\alpha \mid \chi^{(N)}) = \frac{1}{2}(N+1)$, whence

$$\mathcal{R}^{(N)} = \frac{N+1}{2}\alpha.$$

In particular, $\mathcal{R}^{(1)} = \alpha$.

If N is even and $N \equiv 0$, mod 4, $(\alpha_0 \mid \chi^{(N)}) = \frac{N}{4} + 1$, $(\alpha_i \mid \chi^{(N)}) = \frac{N}{4}$, for $1 \leq i \leq 3$, and $(\alpha \mid \chi^{(N)}) = 0$, whence

$$\mathcal{R}^{(N)} = (\frac{N}{4} + 1)\alpha_0 \oplus \frac{N}{4}(\alpha_1 \oplus \alpha_2 \oplus \alpha_3).$$

If N is even and $N \equiv 2$, mod 4, $(\alpha_0 \mid \chi^{(N)}) = \frac{1}{8}(2(N+1)-6) = \frac{N-2}{4}$, $(\alpha_i \mid \chi^{(N)}) = \frac{1}{8}(2(N+1)+2) = \frac{N+2}{4}$, for $1 \leq i \leq 3$, and $(\alpha \mid \chi^{(N)}) = 0$, whence

$$\mathcal{R}^{(N)} = \frac{N-2}{4}\alpha_0 \oplus \frac{N+2}{4}(\alpha_1 \oplus \alpha_2 \oplus \alpha_3).$$

We verify that, in each case, the dimension of $\mathcal{R}^{(N)}$ is $N+1$.

(c) We have $\mathcal{R}^{(1)} = \alpha$. It is clear that $\alpha_0 \otimes \alpha = \alpha$. By calculating the scalar products of the characters, we find $\alpha_i \otimes \alpha = \alpha$, for $0 \leq i \leq 3$, and

$$\alpha \otimes \alpha = \alpha_0 \oplus \alpha_1 \oplus \alpha_2 \oplus \alpha_3.$$

The preceding results can be generalized to the subgroups $\varphi^{-1}(\psi(D_{(n)}))$, $n > 2$, and, more generally, to the finite subgroups of SU(2). These are either of the form $\varphi^{-1}(G)$, for a finite subgroup G of SO(3), or are cyclic groups of odd order. For each finite subgroup of SU(2), one can decompose each tensor product of an irreducible representation and the restriction of the two-dimensional representation of SU(2) into a direct sum of irreducible representations. One can show that the matrix of coefficients appearing in these decompositions can be calculated from a matrix, called the Cartan matrix, associated to a simple Lie algebra. The correspondence between finite subgroups of SU(2) and simple Lie algebras is called the *McKay correspondence*. There are two papers on this subject by B. Kostant, "The McKay correspondence, the Coxeter element and representation theory", in *Élie Cartan et les mathématiques d'aujourd'hui*, Astérisque, Société Mathématique de France, 1985, p. 209–255, and "The Coxeter element and the branching law for the finite subgroups of SU(2)", in *The Coxeter Legacy*, C. Davis and E. W. Ellers, eds., American Mathematical Society, 2006, p. 63–70.

4 Representations of SU(2) and of $\mathfrak{S}_3$

We denote by σ the fundamental representation of SU(2) in $\mathbb{C}^2$, and by (u_0, u_1) the canonical basis of $\mathbb{C}^2$.

Let $u_{klm} = u_k \otimes u_l \otimes u_m$, where $k, l, m = 0$ or 1, be the vectors of the canonical basis of the vector space $E = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$.

1. Show that σ is equivalent to $\mathcal{D}^{\frac{1}{2}}$. Find an isomorphism of $\mathbb{C}^2$ onto the support $V^{\frac{1}{2}}$ of the representation $\mathcal{D}^{\frac{1}{2}}$ that intertwines σ and $\mathcal{D}^{\frac{1}{2}}$.

2. Decompose the representation $\rho = \sigma \otimes \sigma \otimes \sigma$ of SU(2) in E into a direct sum of irreducible representations.

3. Let $D\rho$ be the differential of the representation ρ. Set $\xi_3 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in \mathfrak{su}(2)$. Is each u_{klm} an eigenvector of $(D\rho)(\xi_3)$? Find the eigenvalues of $(D\rho)(\xi_3)$ and the multiplicity of each one.

4. Let $\mathfrak{S}_3$ be the symmetric group on three elements.

(a) Give the list of inequivalent irreducible representations and the character table of $\mathfrak{S}_3$.

(b) Define a representation π of $\mathfrak{S}_3$ in E by

$$\pi(g)(u_{k\ell m}) = u_{g(k)g(\ell)g(m)},$$

for $g \in \mathfrak{S}_3$. Find the character of π and decompose π as a direct sum of irreducible representations.

5. Find a basis of each of the isotypic components of (E, π).

6. Show that each isotypic component of (E, π) is the support of a representation of SU(2) and find the representation.

Find a basis of E adapted to the decomposition of the representation ρ of SU(2) into irreducible representations.

Solutions

1. Because σ is two-dimensional, in order to show that σ is equivalent to $\mathcal{D}^{\frac{1}{2}}$, it suffices to show that σ is irreducible. In fact, the eigenspaces of $g_t = \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix}$ are $\mathbb{C}u_0$ and $\mathbb{C}u_1$ and these lines are not invariant under $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in$ SU(2).

By definition, if $g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \in \mathrm{SU}(2)$,

$$\sigma(g)u_0 = au_0 - \bar{b}u_1,\ \sigma(g)u_1 = bu_0 + \bar{a}u_1,$$

while

$$(\mathcal{D}^{\frac{1}{2}}(g)f)(z_1, z_2) = f(\bar{a}z_1 - bz_2, \bar{b}z_1 + az_2)$$

and $f^{\frac{1}{2}}_{\frac{1}{2}}(z_1, z_2) = z_1,\ f^{\frac{1}{2}}_{-\frac{1}{2}}(z_1, z_2) = z_2$. Thus

$$\mathcal{D}^{\frac{1}{2}}(g)f^{\frac{1}{2}}_{\frac{1}{2}} = \bar{a}f^{\frac{1}{2}}_{\frac{1}{2}} - bf^{\frac{1}{2}}_{-\frac{1}{2}},\quad \mathcal{D}^{\frac{1}{2}}(g)f^{\frac{1}{2}}_{-\frac{1}{2}} = \bar{b}f^{\frac{1}{2}}_{\frac{1}{2}} + af^{\frac{1}{2}}_{-\frac{1}{2}}.$$

The isomorphism T of $\mathbb{C}^2$ onto $V^{\frac{1}{2}}$ defined by

$$T(u_0) = f^{\frac{1}{2}}_{-\frac{1}{2}},\ T(u_1) = -f^{\frac{1}{2}}_{\frac{1}{2}}$$

intertwines σ and $\mathcal{D}^{\frac{1}{2}}$. This can be easily verified by calculating $\mathcal{D}^{\frac{1}{2}}(g)T(u_k)$ and $T(\sigma(g)u_k)$, for $k = 0$ and 1.

2. Because the representation σ is equivalent to $\mathcal{D}^{\frac{1}{2}}$, $\rho = \sigma \otimes \sigma \otimes \sigma$ is equivalent to $\mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^{\frac{1}{2}}$. We can thus apply the Clebsch–Gordan formula which yields $\mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^{\frac{1}{2}} = \mathcal{D}^0 \oplus \mathcal{D}^1$, $\mathcal{D}^0$ being the trivial one-dimensional representation. Thus

$$\rho = (\mathcal{D}^0 \oplus \mathcal{D}^1) \otimes \mathcal{D}^{\frac{1}{2}} = \mathcal{D}^{\frac{1}{2}} \oplus (\mathcal{D}^1 \otimes \mathcal{D}^{\frac{1}{2}}).$$

Because $\mathcal{D}^1 \otimes \mathcal{D}^{\frac{1}{2}} = \mathcal{D}^{\frac{1}{2}} \oplus \mathcal{D}^{\frac{3}{2}}$, we obtain

$$\rho = 2\mathcal{D}^{\frac{1}{2}} \oplus \mathcal{D}^{\frac{3}{2}}.$$

3. Because $\rho = \sigma \otimes \sigma \otimes \sigma$, for $X \in \mathfrak{su}(2)$,

$$\begin{aligned}(D\rho)(X)u_{k\ell m} &= (D\sigma)(X)u_k \otimes u_\ell \otimes u_m \\ &\quad + u_k \otimes (D\sigma)(X)u_\ell \otimes u_m \\ &\quad + u_k \otimes u_\ell \otimes (D\sigma)(X)u_m.\end{aligned}$$

If $X = \xi_3$, we have $(D\sigma)(\xi_3)u_k = \xi_3 u_k = (-1)^k \frac{i}{2}u_k$, whence

$$(D\rho)(\xi_3)u_{k\ell m} = ((-1)^k + (-1)^\ell + (-1)^m)\frac{i}{2}u_{k\ell m}.$$

Thus each $u_{k\ell m}$ is an eigenvector of $(D\rho)(\xi_3)$. The eigenvalues are $\frac{3i}{2}, -\frac{3i}{2}, \frac{i}{2}, -\frac{i}{2}$.

- The eigenvalues $\frac{3i}{2}$ and $-\frac{3i}{2}$ are simple, with eigenvectors u_{000} and u_{111}, respectively.
- The eigenvalues $\frac{i}{2}$ and $-\frac{i}{2}$ have multiplicity 3, with eigenvectors u_{001}, u_{010}, u_{100} and $u_{011}, u_{101}, u_{110}$, respectively.

4. (a) We know that $\mathfrak{S}_3$ has two irreducible one-dimensional representations, the trivial representation $\underline{1}$ and the signature representation ϵ, and a two-dimensional irreducible representation α. The conjugacy class of a cyclic permutation c contains two elements (the two cyclic permutations different from the identity) and the conjugacy class of a transposition t contains three elements (the three transpositions). Thus the character table is

	(1) e	(2) c	(3) t
$\chi_{\underline{1}}$	1	1	1
χ_ϵ	1	1	-1
χ_α	2	-1	0

(b) The group $\mathfrak{S}_3$ acts on ordered triples (k,ℓ,m) by permutations. The trace $\chi_\pi(g)$ of $\pi(g)$ is the number of triples which are left invariant by g^{-1}. We have $\chi_\pi(e) = 8$. If g is a cyclic permutation, only u_{000} and u_{111} are invariant, and thus $\chi_\pi(c) = 2$. If t is a transposition, $\chi_\pi(t) = 4$. In fact, $t = (12)$ leaves $u_{000}, u_{111}, u_{001}, u_{110}$ invariant.

We find the decomposition of $\pi = m_{\underline{1}}\underline{1} \oplus m_\epsilon\epsilon \oplus m_\alpha\alpha$, by calculating, by means of the character table,

$$\begin{aligned} m_{\underline{1}} &= (\chi_{\underline{1}}|\chi_\pi) = 4, \\ m_\epsilon &= (\chi_\epsilon|\chi_\pi) = 0, \\ m_\alpha &= (\chi_\alpha|\chi_\pi) = 2. \end{aligned}$$

Thus

$$\pi = 4\ \underline{1} \oplus 2\ \alpha.$$

5. We have seen that $E = 4E^{\underline{1}} \oplus 2E^\alpha$. We denote by $V^{\underline{1}} = 4E^{\underline{1}}$ and $V^\alpha = 2E^\alpha$ the isotypic components of E. Thus $E = V^{\underline{1}} \oplus V^\alpha$.

The projection of E onto $V^{\underline{1}}$ is defined by

$$P_{\underline{1}}u_{k\ell m} = \frac{1}{6}\sum_{g\in\mathfrak{S}_3}\pi(g)u_{k\ell m} = \frac{1}{6}\sum_{g\in\mathfrak{S}_3}u_{g(k)g(\ell)g(m)}.$$

We find

$$P_{\underline{1}}u_{000} = u_{000}, \quad P_{\underline{1}}u_{111} = u_{111},$$

$$P_{\underline{1}}u_{001} = P_{\underline{1}}u_{010} = P_{\underline{1}}u_{100} = \frac{1}{3}(u_{001} + u_{010} + u_{100}),$$

$$P_{\underline{1}}u_{011} = P_{\underline{1}}u_{101} = P_{\underline{1}}u_{110} = \frac{1}{3}(u_{011} + u_{101} + u_{110}).$$

Set $\frac{1}{3}(u_{001} + u_{010} + u_{100}) = u$ and $\frac{1}{3}(u_{011} + u_{101} + u_{110}) = u'$. A basis of $V^{\underline{1}} = 4E^{\underline{1}}$ is thus $(u_{000}, u_{111}, u, u')$.

For the projection onto V^α, we find

$$P_\alpha u_{k\ell m} = \frac{1}{3}(2u_{k\ell m} - u_{\ell mk} - u_{mk\ell}).$$

Thus $P_\alpha u_{000} = P_\alpha u_{111} = 0$ and

$$P_\alpha u_{001} = \tfrac{1}{3}(2u_{001} - u_{100} - u_{010}) = v, \quad P_\alpha u_{100} = \tfrac{1}{3}(2u_{100} - u_{010} - u_{001}) = v',$$
$$P_\alpha u_{110} = \tfrac{1}{3}(2u_{110} - u_{011} - u_{101}) = w, \quad P_\alpha u_{011} = \tfrac{1}{3}(2u_{011} - u_{101} - u_{110}) = w',$$

and

$$P_\alpha u_{010} = -P_\alpha(u_{001} + u_{100}),$$
$$P_\alpha u_{101} = -P_\alpha(u_{110} + u_{011}).$$

A basis of $V^\alpha = E^\alpha \oplus E^\alpha$ is thus given by the four vectors v, v', w, w'.

We can also obtain this result by observing that V^α is the orthogonal complement of $V^{\underline{1}}$ for the scalar product on E such that the vectors u_{klm} form an orthonormal basis.

Set $j = e^{\frac{2i\pi}{3}}$, so that $j^2 + j + 1 = 0$. The vectors $x = (j+2)v + (2j+1)v' = u_{001} + ju_{100} + j^2 u_{010}$ and $y = (j-1)v - (j+2)v' = u_{010} + ju_{001} + j^2 u_{100}$ on the one hand, and the vectors $x' = (j+2)w + (2j+1)w' = u_{110} + ju_{011} + j^2 u_{101}$ and $y' = (j-1)w - (j+2)w' = u_{101} + ju_{110} + j^2 u_{011}$, on the other hand, form a basis of a vector space E^α, the support of the representation α of $\mathfrak{S}_3$.

6. Let us show that the subspace $V^{\underline{1}}$ is closed under the representation $D\rho$ of $\mathfrak{su}(2)$, and thus under the representation ρ of SU(2), which is connected.

For convenience we denote $(D\rho)(X)u_{klm}$ by $X \cdot u_{klm}$, for $X \in \mathfrak{su}(2)$. The four vectors of the basis given above are eigenvectors of $(D\rho)(\xi_3)$,

$$\xi_3 \cdot u_{000} = \frac{3i}{2}u_{000}, \ \xi_3 \cdot u_{111} = -\frac{3i}{2}u_{111}, \ \xi_3 \cdot u = \frac{i}{2}u, \ \xi_3 \cdot u' = -\frac{i}{2}u'.$$

On the other hand,

$$(D\sigma)(\xi_1)u_0 = \frac{i}{2}u_1, (D\sigma)(\xi_1)u_1 = \frac{i}{2}u_0,$$

$$(D\sigma)(\xi_2)u_0 = \frac{1}{2}u_1, \ (D\sigma)(\xi_2)u_1 = -\frac{1}{2}u_0.$$

Thus,

$$\xi_1 \cdot u_{000} = \frac{3i}{2}u, \ \xi_1 \cdot u_{111} = \frac{3i}{2}u', \ \xi_1 \cdot u = \frac{i}{2}(u_{000} + 2u') \in V^{\underline{1}},$$

and, similarly, $\xi_1 \cdot u' = \frac{i}{2}(u_{111} + 2u) \in V^{\underline{1}}$. Furthermore,

$$\xi_2 \cdot u_{000} = \frac{3}{2}u, \ \xi_2 \cdot u_{111} = -\frac{3}{2}u', \ \xi_2 \cdot u = -\frac{1}{2}(u_{000} - 2u') \in V^{\underline{1}},$$

and, similarly, $\xi_2 \cdot u' = \frac{1}{2}(u_{111} - 2u) \in V^{\underline{1}}$.

The restriction of $D\rho$ to $V^{\underline{1}}$ is a representation of dimension 4 of $\mathfrak{su}(2)$, in which ξ_3 has eigenvalue $\frac{3i}{2}$. We know that this property implies that $\rho|_{V^{\underline{1}}}$ is equivalent to $\mathcal{D}^{\frac{3}{2}}$.

We verify that $T : V^{\underline{1}} \to V^{\frac{3}{2}} = E^{\frac{3}{2}}$, defined by $T(u_{000}) = -f^{\frac{3}{2}}_{-\frac{3}{2}}$, $T(u_{111}) = f^{\frac{3}{2}}_{\frac{3}{2}}$, $Tu = f^{\frac{3}{2}}_{-\frac{1}{2}}$, $Tu' = -f^{\frac{3}{2}}_{\frac{1}{2}}$, intertwines $D\rho\big|_{V^{\underline{1}}}$ and $D^{\frac{3}{2}}$.

Next we consider the space $V^{\alpha} = 2E^{\alpha}$. We have

$$\xi_3 \cdot v = \frac{i}{2}v,\ \xi_3 \cdot v' = \frac{i}{2}v',\ \xi_3 \cdot w = -\frac{i}{2}w,\ \xi_3 \cdot w' = -\frac{i}{2}w'.$$

The vector subspace generated by v and w is the support of a representation $\mathcal{D}^{\frac{1}{2}}$. In fact, it is closed under $D\rho$. We have $\xi_1 \cdot v = -\frac{i}{2}w$, and $\xi_1 \cdot w = -\frac{i}{2}v$, because $\xi_1^2 = -\frac{1}{4}I$. Furthermore, $\xi_2 \cdot v = -\frac{1}{2}w$ and $\xi_2 \cdot w = \frac{1}{2}v$.

We verify that, if we set $Tv = f^{\frac{1}{2}}_{-\frac{1}{2}}$ and $Tw = f^{\frac{1}{2}}_{\frac{1}{2}}$, the isomorphism T intertwines the restriction of $D\rho$ to the vector space generated by v and w with $D^{\frac{1}{2}}$.

Analogous equations for v' and w' show that the vector subspace generated by v' and w' is also the support of a representation $\mathcal{D}^{\frac{1}{2}}$. In conclusion, the two vector subspaces generated by v and w on the one hand, and by v' and w' on the other hand, are each the support of a representation $\mathcal{D}^{\frac{1}{2}}$.

Using the decomposition of E under the action of $\mathfrak{S}_3$, we have obtained a basis of E adapted to the decomposition of $\rho = \sigma \otimes \sigma \otimes \sigma$.

We see that the isotypic components in the decompositions of the representation ρ of SU(2) and of the representation π of $\mathfrak{S}_3$ correspond to one another: $V^{\underline{1}} = 4E^{\underline{1}} \sim E^{\frac{3}{2}}$ and $V^{\alpha} = 2E^{\alpha} \sim 2E^{\frac{1}{2}}$. We have proved

$$E = (\dim E^{\alpha})E^{\frac{1}{2}} \oplus (\dim E^{\underline{1}})E^{\frac{3}{2}} = (\dim E^{\frac{1}{2}})E^{\alpha} \oplus (\dim E^{\frac{3}{2}})E^{\underline{1}}.$$

The fact that $\rho = \bigoplus_{k\in I} p_k\rho_k = \bigoplus_{k\in I} m_k\pi_k$, where the ρ_k are irreducible representations of SU(2) and the π_k irreducible representations of $\mathfrak{S}_3$, with $p_k = \dim \pi_k$ and $m_k = \dim \rho_k$, is a particular case of a general property, the *Schur–Weyl duality* between representations of $\mathrm{GL}(V)$ and of $\mathfrak{S}_r$ on $V^{\otimes r}$, for a given vector space V and a positive integer r (here $V = \mathbb{C}^2$ and $r = 3$). See, *e.g.*, Goodman and Wallach (2009).

5 Pseudo-Unitary and Pseudo-Orthogonal Groups

We consider the Lie group

$$\mathrm{SU}(1,1) = \{A \in \mathrm{SL}(2,\mathbb{C}) \mid AJ\,{}^t\overline{A} = J\}, \quad \text{where} \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

1. Show that $A \in \mathrm{SU}(1,1)$ if and only if $A = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix}$, where a and b are complex numbers such that $|a|^2 - |b|^2 = 1$.

Is the Lie group SU(1, 1) compact?

2. For $g \in \mathrm{SL}(2,\mathbb{R})$, we set $\mu(g) = \tau g \tau^{-1}$, where $\tau = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$. What is the image of μ? Show that the Lie groups $\mathrm{SU}(1,1)$ and $\mathrm{SL}(2,\mathbb{R})$ are isomorphic.

3. Let $\xi_1 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$, $\xi_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\xi_3 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$. Show that the matrices $M_1 = -i\xi_1, M_2 = -i\xi_2, M_3 = \xi_3$ form a basis of the Lie algebra $\mathfrak{su}(1,1)$ of $\mathrm{SU}(1,1)$. Write the commutation relations of $\mathfrak{su}(1,1)$ in this basis.

On $\mathbb{R}^3$ we consider the quadratic form defined by $Q(x) = x_1^2 + x_2^2 - x_3^2$, where x_1, x_2, x_3 denote the components of $x \in \mathbb{R}^3$ in the canonical basis (e_1, e_2, e_3).

4. Define an isomorphism α from $\mathbb{R}^3$ onto $\mathfrak{su}(1,1)$ by setting

$$\alpha(e_k) = M_k, \ k = 1, 2, 3.$$

(a) For each $x \in \mathbb{R}^3$, calculate the determinant of $\alpha(x)$ and compare it to $Q(x)$.

(b) Denote by K the Killing form of $\mathfrak{su}(1,1)$. Show that, for every $x \in \mathbb{R}^3$,

$$K(\alpha(x), \alpha(x)) = 2Q(x).$$

Consider the Lie group

$$\mathrm{O}(2,1) = \{A \in \mathrm{GL}(3,\mathbb{R}) \mid A J_{21} {}^t\!A = J_{21}\}, \quad \text{where} \quad J_{21} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Let $\mathrm{SO}(2,1) = \mathrm{O}(2,1) \cap \mathrm{SL}(3,\mathbb{R})$, and let $\mathrm{SO}_0(2,1)$ be the connected component of the identity of $\mathrm{SO}(2,1)$.

5. Let $A \in \mathrm{GL}(3,\mathbb{R})$. Show that $A \in \mathrm{O}(2,1)$ if and only if, for every $x \in \mathbb{R}^3$,

$$Q(Ax) = Q(x).$$

6. Find a basis of the Lie algebra $\mathfrak{so}(2,1)$ of $\mathrm{SO}(2,1)$. Write the commutation relations of $\mathfrak{so}(2,1)$ in this basis.

What can one say about the Lie algebra of $\mathrm{O}(2,1)$?

7. Are the Lie algebras $\mathfrak{su}(1,1)$ and $\mathfrak{so}(2,1)$ isomorphic? Are the Lie algebras $\mathfrak{sl}(2,\mathbb{R})$ and $\mathfrak{so}(2,1)$ isomorphic?

8. Consider the adjoint representation Ad of the group $\mathrm{SU}(1,1)$ on its Lie algebra. To $g \in \mathrm{SU}(1,1)$ we associate the matrix $\psi(g)$ of Ad_g in the basis (M_1, M_2, M_3) of $\mathfrak{su}(1,1)$. It is known that $\mathrm{SU}(1,1)$ is connected.

(a) Show that, for every $g \in \mathrm{SU}(1,1)$, $\psi(g) \in \mathrm{O}(2,1)$.

(b) Show that ψ is a Lie group morphism of $\mathrm{SU}(1,1)$ into $\mathrm{SO}_0(2,1)$. What is the kernel of ψ?

(c) Find the differential of ψ. Show that ψ is surjective from $\mathrm{SU}(1,1)$ onto $\mathrm{SO}_0(2,1)$.

Solutions

1. The condition $AJ\,{}^t\bar{A} = J$ can be written

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

which is equivalent to $a\bar{a} - b\bar{b} = 1$, $c\bar{c} - d\bar{d} = -1$, $a\bar{c} - b\bar{d} = 0$, $\bar{a}c - \bar{b}d = 0$. We deduce that $ac\bar{c} - bc\bar{d} = 0$. Because $ad - bc = 1$, we have $ac\bar{c} - ad\bar{d} + \bar{d} = 0$, whence $d = \bar{a}$ and $c = \bar{b}$. Thus $A = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix}$ and the condition $\det A = 1$ is equivalent to $|a|^2 - |b|^2 = 1$.

To show that the Lie group $\mathrm{SU}(1,1)$ is not compact, it suffices to show that the set $\{(a,b) \in \mathbb{C}^2 \mid \ |a|^2 - |b|^2 = 1\}$ is not bounded; it contains, for example, all pairs of positive reals of the form $(\sqrt{x^2+1}, x)$.

2. If $\tau = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$, then $\tau^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$. Let $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R})$. Then $\mu(g)$ is the matrix

$$\tau g \tau^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \alpha+\delta-i(\beta-\gamma) & \beta+\gamma-i(\alpha-\delta) \\ \beta+\gamma+i(\alpha-\delta) & \alpha+\delta+i(\beta-\gamma) \end{pmatrix}.$$

We verify that $d = \bar{a},\ c = \bar{b}$ and $|a|^2 - |b|^2 = \alpha\delta - \beta\gamma = 1$ because $g \in \mathrm{SL}(2,\mathbb{R})$. Furthermore, $\det \mu(g) = \det g = 1$. Thus the image of μ is contained in $\mathrm{SU}(1,1)$.

On the other hand if $h = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SU}(1,1)$, we have

$$\tau^{-1} h \tau = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \frac{1}{2}\begin{pmatrix} a+d+i(b-c) & b+c+i(a-d) \\ b+c-i(a-d) & a+d-i(b-c) \end{pmatrix}.$$

We see that, under the hypothesis $d = \bar{a}$ and $c = \bar{b}$, the numbers α, β, γ and δ are real. Furthermore, because $h \in \mathrm{SU}(1,1)$,

$$\det(\tau^{-1} h \tau) = \det h = 1.$$

Thus μ is a bijection of $\mathrm{SL}(2,\mathbb{R})$ onto $\mathrm{SU}(1,1)$. Furthermore, μ is a group morphism since $\mu(gg') = \mu(g)\mu(g')$ for every g and $g' \in \mathrm{SL}(2,\mathbb{R})$. Finally the coefficients of $\mu(g)$ are linear functions and thus are smooth functions of the coefficients of g. Thus μ is a Lie group isomorphism of $\mathrm{SL}(2,\mathbb{R})$ onto $\mathrm{SU}(1,1)$.

3. Let M be a matrix of $\mathfrak{gl}(2,\mathbb{C})$. By differentiating both sides of the equation

$$\exp(sM)J\exp(s\ {}^t\overline{M}) = J$$

with respect to $s \in \mathbb{R}$, we obtain

$$\exp(sM)(MJ + J\ {}^t\overline{M})\exp(s\ {}^t\overline{M}) = 0.$$

By evaluating at $s = 0$, we see that in order for $\exp(sM)$ to belong to $\mathrm{SU}(1,1)$ for every $s \in \mathbb{R}$, we must have $MJ + J\ {}^t\overline{M} = 0$. And conversely, if the condition $MJ + J\ {}^t\overline{M} = 0$ is satisfied, the matrix $\exp(sM)J\exp(s\ {}^t\overline{M})$ is constant and equal to its initial value, which is J. On the other hand, the condition $\mathrm{Tr}M = 0$ is equivalent to $\mathrm{Det}(\exp sM) = 1$, for every s. Hence the Lie algebra of $\mathrm{SU}(1,1)$ is

$$\mathfrak{su}(1,1) = \{M \in \mathfrak{sl}(2,\mathbb{C}) \mid MJ + J\ {}^t\overline{M} = 0\}.$$

The matrices of $\mathfrak{su}(1,1)$ are of the form $\begin{pmatrix} u & v \\ \bar{v} & -u \end{pmatrix}$ where u is pure imaginary. A basis of $\mathfrak{su}(1,1)$ is thus

$$\frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -i\xi_1 = M_1, \quad \frac{1}{2}\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} = -i\xi_2 = M_2, \quad \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \xi_3 = M_3.$$

The commutation relations are

$$[M_1, M_2] = -M_3,\ [M_2, M_3] = M_1,\ [M_3, M_1] = M_2.$$

4. (a) If $x = \sum_{k=1}^{3} x_k e_k$, then $\alpha(x) = \sum_{k=1}^{3} x_k M_k = \frac{1}{2}\begin{pmatrix} ix_3 & x_1 + ix_2 \\ x_1 - ix_2 & -ix_3 \end{pmatrix}$ whose determinant is $\frac{1}{4}(-(x_1^2 + x_2^2) + x_3^2) = -\frac{1}{4}Q(x)$.

(b) By definition $K(X,Y) = \mathrm{Tr}(\mathrm{ad}_X \mathrm{ad}_Y)$. We have, in the basis (M_k),

$$\mathrm{ad}_{M_1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}, \quad \mathrm{ad}_{M_2} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathrm{ad}_{M_3} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Set $X = \alpha(x) = \sum_{k=1}^{3} x_k M_k$ and set $Y = \alpha(y) = \sum_{k=1}^{3} y_k M_k$. Then

$$\mathrm{ad}_X = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ x_2 & -x_1 & 0 \end{pmatrix},$$

whence $\mathrm{Tr}(\mathrm{ad}_X \mathrm{ad}_Y) = 2(x_1y_1 + x_2y_2 - x_3y_3)$. Thus $K(\alpha(x), \alpha(x)) = 2Q(x)$.

5. The group $\mathrm{O}(2,1)$ is the group of transformations of $\mathbb{R}^3$ preserving the quadratic form Q which is of signature (2,1), and therefore not positive definite. In fact, if X is the column vector of components x_1, x_2, x_3,

$$Q(x) = {}^tX J_{21} X,$$

and if $A \in \mathrm{GL}(3,\mathbb{R})$,

$$Q(Ax) = {}^tX\ {}^tA\ J_{21}\ A\ X,$$

then $Q(Ax) = Q(x)$ for every x is equivalent to ${}^tAJ_{21}A = J_{21}$. Now, because $(J_{21})^2 = I$, the condition ${}^tA \in \mathrm{O}(2,1)$ is equivalent to $A \in \mathrm{O}(2,1)$. In fact, ${}^tAJ_{21}A = J_{21}$ implies $AJ_{21}\,{}^tAJ_{21}A = A$, whence $AJ_{21}\,{}^tA = J_{21}$.

6. Using the same argument as in Question 3, we see that if $M \in \mathfrak{gl}(3,\mathbb{R})$, the condition $\exp(sM)J_{21}\exp(s\ {}^tM) = J_{21}$, for every $s \in \mathbb{R}$, is equivalent to the condition $MJ_{21} + J_{21}\ {}^tM = 0$. The matrices M of $\mathfrak{gl}(3,\mathbb{R})$ satisfying the condition $MJ_{21} + J_{21}\,{}^tM = 0$ are of the form $\begin{pmatrix} 0 & u & v \\ -u & 0 & w \\ v & w & 0 \end{pmatrix}$, where $u, v, w \in \mathbb{R}$. Thus $\mathrm{Tr}M = 0$. The Lie algebra of $\mathrm{SO}(2,1)$ is therefore

$$\mathfrak{so}(2,1) = \{M \in \mathfrak{sl}(3,\mathbb{R}) \mid MJ_{21} + J_{21}\,{}^tM = 0\}.$$

The matrices

$$\zeta_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix},\ \zeta_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},\ \zeta_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

form a basis of $\mathfrak{so}(2,1)$.

The commutation relations are

$$[\zeta_1, \zeta_2] = -\zeta_3\ ,\ [\zeta_2, \zeta_3] = \zeta_1,\ [\zeta_3, \zeta_1] = \zeta_2.$$

It follows from the remark above on the vanishing of the trace that the Lie algebra of $\mathrm{O}(2,1)$ is equal to $\mathfrak{so}(2,1)$.

7. The linear mapping of $\mathfrak{su}(1,1)$ on $\mathfrak{so}(2,1)$ defined by $M_k \mapsto \zeta_k$, for $k = 1,2,3$, is an isomorphism of Lie algebras since in these bases the commutation relations are the same in the two Lie algebras.

On the other hand, one can deduce from the result of Question 2 that the Lie algebras of $\mathrm{SU}(1,1)$ and $\mathrm{SL}(2,\mathbb{R})$ are isomorphic. Thus the Lie algebras $\mathfrak{so}(2,1), \mathfrak{su}(1,1)$ and $\mathfrak{sl}(2,\mathbb{R})$ are isomorphic.

Observe that, because $\mathrm{Tr}((\mathrm{ad}\xi_k)^2) = -2$, the Killing form of $\mathfrak{su}(2)$ is negative definite. The Lie algebras $\mathfrak{su}(1,1)$ and $\mathfrak{su}(2)$ are not isomorphic since their Killing forms do not have the same signature.

8. (a) By definition $\alpha \circ \psi(g) = \mathrm{Ad}_g \circ \alpha$, for every $g \in \mathrm{SU}(1,1)$. The linear mapping Ad_g preserves the determinants, and hence $\psi(g)$ preserves Q, because $\det \alpha(x) = -\frac{1}{4}Q(x)$, for every $x \in \mathbb{R}^3$. Thus $\psi(g) \in \mathrm{O}(2,1)$.

(b) Because $\mathrm{SU}(1,1)$ is connected, the image of ψ is contained in the connected component of the identity of $\mathrm{O}(2,1)$ which is the component $\mathrm{SO}_0(2,1)$ of its subgroup $\mathrm{SO}(2,1)$.

It is clear that ψ is a Lie group morphism. The kernel of ψ is the subgroup of $\mathrm{SU}(1,1)$,

$$\{g \in \mathrm{SU}(1,1) \mid \mathrm{Ad}_g = \mathrm{Id}_{\mathfrak{su}(1,1)}\} = \{g \in \mathrm{SU}(1,1) \mid \forall M \in \mathfrak{su}(1,1), gM = Mg\}.$$

A calculation shows that $g = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix}$ satisfies this condition if and only if $b = 0$, $a = \bar{a}$, whence $a = \pm 1$ and $\mathrm{Ker}\,\psi = \{I, -I\}$.

(c) The differential of ψ is the morphism of Lie algebras that associates to $X \in \mathfrak{su}(1,1)$ the matrix of ad_X in the basis (M_k). This morphism is the isomorphism of $\mathfrak{su}(1,1)$ onto $\mathfrak{so}(2,1)$ constructed in Question 7, because the matrix of ad_{M_k} is ζ_k, for $k = 1, 2, 3$, as we have seen in the proof of Questions 4. (b) and 6.

On the other hand, each element of $\mathrm{SO}_0(2,1)$ is a product of exponentials. Thus to show that ψ is surjective, it suffices to show that, for every $M \in \mathfrak{so}(2,1)$, there is an $N \in \mathfrak{su}(1,1)$ such that $\psi(\exp N) = \exp M$. Because $\psi(\exp N) = \exp((D\psi)N)$, and because $D\psi$ is an isomorphism, it suffices to take $N = (D\psi)^{-1}(M)$.

The morphism ψ is a double cover of $\mathrm{SO}_0(2,1)$. But $\mathrm{SU}(1,1)$ is not simply connected. The universal cover of $\mathrm{SO}_0(2,1)$ is an example of a Lie group that is not a linear Lie group.

We can define the analogue of Euler angles for $\mathrm{SU}(1,1)$ and express the Haar integral in this parametrization. Since the Lie group $\mathrm{SU}(1,1)$ is not compact, it has infinite-dimensional irreducible unitary representations. One can realize the representations of $\mathrm{SU}(1,1)$ in spaces of homogeneous functions of a complex variable, study the special functions associated to the matrix coefficients of these representations, etc. See Vilenkin (1968) or Vilenkin and Klimyk (1991).

In physics, the non-compact Lie group $\mathrm{SO}_0(3,2)$ is called the *anti-de Sitter group*, while $\mathrm{SO}_0(4,1)$ is called the *de Sitter group*. The importance of the anti-de Sitter group is due to the fact that the Poincaré group can be obtained from it by contraction. The group $\mathrm{O}(3,2)$ is the symmetry group of a one-sheeted hyperboloid in $\mathbb{R}^5$, called the *anti-de Sitter space-time*, AdS_5.

The case of $\mathrm{O}(2,1)$ corresponding to the two-dimensional space-time AdS_3 has been studied intensively. See, for example, the anti-de Sitter version of

Hamiltonian mechanics of J.-P. Gazeau, "On two analytic elementary systems in quantum mechanics", in *Géométrie complexe*, F. Norguet, S. Ofman and J.-J. Szczeciniarz, eds., Hermann, 1996, p. 175–199. We quote from an article by S. Detournay, D. Orlando, P. Petropoulos and P. Spindel, "Three-dimensional black holes from deformed anti-de Sitter", *JHEP* **07** (2005) "Three-dimensional anti-de Sitter space provides a good laboratory for studying many aspects of gravity and strings, including black-holes physics." The geometry of the anti-de Sitter space-time as well as the representations of the groups $\mathrm{SU}(1,1)$ and $\mathrm{SL}(2,\mathbb{R})$ play an essential role in these theories. In the preprint list <http://fr.arxiv.org/find/hep-th>, the number of papers containing "anti-de Sitter" or "AdS" in the title was 77 in 2005, 100 in 2006, 123 in 2007, and 176 in 2008. In 2021, it is well over 2000.

6 Irreducible Representations of $\mathbf{SU(2) \times SU(2)}$

Let $\mathbb{H}$ be the associative algebra of quaternions, a real vector space of dimension 4 with basis $(1, i, j, k)$ and multiplication defined by

$$i^2 = j^2 = k^2 = -1,\ ij = -ji = k,\ jk = -kj = i,\ ki = -ik = j.$$

For $q = \alpha_0 1 + \alpha_1 i + \alpha_2 j + \alpha_3 k$, we set $\bar{q} = \alpha_0 1 - \alpha_1 i - \alpha_2 j - \alpha_3 k$ and $\|q\|^2 = q\bar{q}$. It is known that $\|q\|^2 = \sum_{p=0}^{3} (\alpha_p)^2$ and that, for all $q, q' \in \mathbb{H}$,

$$\|qq'\| = \|q\|\|q'\|.$$

Denote by $\mathbb{S}$ the group of quaternions of unit norm,

$$\mathbb{S} = \{q \in \mathbb{H} \mid \|q\| = 1\}.$$

Consider the real vector space $\mathcal{H}$ generated by the matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\ \mathcal{I} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},\ \mathcal{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},\ \mathcal{K} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

Observe that $\mathrm{SU}(2)$ and $\mathfrak{su}(2)$ are contained in $\mathcal{H}$.

Part I

1. We write $M \mapsto q_M$ for the isomorphism of real vector spaces from $\mathcal{H}$ onto $\mathbb{H}$ defined by $I \mapsto 1$, $\mathcal{I} \mapsto i$, $\mathcal{J} \mapsto j$, $\mathcal{K} \mapsto k$.

(a) Show that the restriction to $\mathrm{SU}(2)$ of this map is an isomorphism of the group $\mathrm{SU}(2)$ onto the group $\mathbb{S}$.

(b) Show that, for every $X \in \mathfrak{su}(2)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}(q_{\exp(tX)})\Big|_{t=0} = q_X.$$

2. Let $(A, B) \in \mathrm{SU}(2) \times \mathrm{SU}(2)$. We consider the map

$$\sigma(A, B) \ : \mathbb{H} \to \mathbb{H}$$

defined by

$$\sigma(A, B)q = q_A \ q \ (q_B)^{-1}, \quad \forall q \in \mathbb{H}.$$

We identify $\mathbb{H}$ and $\mathbb{R}^4$ by means of the $\mathbb{R}$-linear map defined by $1 \mapsto e_0$, $i \mapsto e_1$, $j \mapsto e_2$, $k \mapsto e_3$, where (e_0, e_1, e_2, e_3) is the canonical basis of $\mathbb{R}^4$.

(a) Show that $\sigma(A, B)$ can be identified with an element of O(4).

(b) Show that σ is a group morphism of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ into SO(4).

(c) Find the kernel of the morphism σ.

3. (a) Find the differential $D\sigma : \mathfrak{su}(2) \oplus \mathfrak{su}(2) \to \mathfrak{so}(4)$ of σ.

(b) Show that $D\sigma$ is an isomorphism of Lie algebras.

(c) Show that the image of σ is SO(4).

Part II

Let (E_1, ρ_1) and (E_2, ρ_2) be finite-dimensional representations of compact Lie groups G_1 and G_2, respectively. We define a representation ρ of $G_1 \times G_2$ in $E_1 \otimes E_2$ by

$$\rho(g_1, g_2)(v_1 \otimes v_2) = \rho_1(g_1)v_1 \otimes \rho_2(g_2)v_2,$$

for $g_1 \in G_1, \ g_2 \in G_2, \ v_1 \in E_1, \ v_2 \in E_2$. We denote this representation by $\rho = \rho_1 \times \rho_2$.

We state without proof that each irreducible representation of $G_1 \times G_2$ is of the form $(E_1 \otimes E_2, \rho_1 \times \rho_2)$, where (E_i, ρ_i) is an irreducible representation of G_i, $i = 1, 2$.

4. Find the irreducible representations of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ and those of SO(4).

5. Let $\mathfrak{g}_1$ (resp., $\mathfrak{g}_2$) be the Lie algebra of G_1 (resp., G_2). Let R be the differential of $\rho = \rho_1 \times \rho_2$. Evaluate $R(X_1, X_2)(v_1 \otimes v_2)$, for $X_1 \in \mathfrak{g}_1$, $X_2 \in \mathfrak{g}_2, \ v_1 \in E_1, \ v_2 \in E_2$, by means of $D\rho_1$ and $D\rho_2$.

We denote the representation R of $g_1 \oplus \mathfrak{g}_2$, and also the corresponding representation of the complexified Lie algebra $g_1^{\mathbb{C}} \oplus \mathfrak{g}_2^{\mathbb{C}}$, by $D\rho_1 \times D\rho_2$.

6. Let $j_1, j_2 \in \frac{1}{2}\mathbb{N}$. Consider, with the notation of Question 5, the representation $D^{(j_1 j_2)} = D^{j_1} \times D^{j_2}$ of $\mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$ on $V^{j_1} \otimes V^{j_2}$. Set

$$J_1 = -\frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ J_2 = -\frac{i}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \ J_3 = -\frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Let $|j,m\rangle$ be the usual basis of the space V^j. Choose a basis of $V^{j_1}\otimes V^{j_2}$ and write the matrices of $D^{(j_1j_2)}(X,0)$ and of $D^{(j_1j_2)}(0,X)$ in this basis,

(a) for $X=J_1,\ J_2,\ J_3$, when $j_1=j_2=\frac{1}{2}$,

(b) for $X=J_3$, when $j_1=1$ and $j_2=\frac{1}{2}$,

(c) for $X=J_3$, when $j_1=\frac{1}{2}$ and $j_2=1$.

7. Are the representations $D^{(j_1j_2)}$ and $D^{(j_2j_1)}$ equivalent? Why?

8. For $X\in\mathfrak{sl}(2,\mathbb{C})$, set $D_1^{(j_1j_2)}(X^2)=D^{(j_1j_2)}(X,0)\circ D^{(j_1j_2)}(X,0)$ and $D_2^{(j_1j_2)}(X^2)=D^{(j_1j_2)}(0,X)\circ D^{(j_1j_2)}(0,X)$. Set $C_i^{(j_1j_2)}=\sum_{\ell=1}^3 D_i^{(j_1j_2)}(J_\ell^2)$, $i=1,\ 2$. Evaluate the operators $C_1^{(j_1j_2)}$ and $C_2^{(j_1j_2)}$.

9. We continue to use the notation of the preceding questions. We want to evaluate the multiplicity of the negative-energy bound states of a hydrogen atom. Let e,m and $\hbar$ be constants. We define the Rydberg constant,

$$R_H=\frac{me^4}{2\hbar^2}.$$

Let R_ℓ and l_ℓ, $\ell=1,2,3$, be elements of $\mathfrak{sl}(2,\mathbb{C})$. We set $R^2=\sum_{\ell=1}^3 D_1^{(jj)}(R_\ell^2)$, and $l^2=\sum_{\ell=1}^3 D_2^{(jj)}(l_\ell^2)$. Assume that

$$R^2=e^4+2E\frac{l^2+\hbar^2}{m},$$

and that

$$-\frac{m}{2E}R^2+l^2=2\hbar^2(C_1^{(jj)}+C_2^{(jj)}).$$

Find the value of the energy E as a function of the constant R_H and of the integer $n=2j+1$.

Solutions

Part I

1. (a) The map $M\mapsto q_M$ respects multiplication since

$$\mathcal{I}^2=\mathcal{J}^2=\mathcal{K}^2=-I,$$

$$\mathcal{I}\mathcal{J}=-\mathcal{J}\mathcal{I}=\mathcal{K},\ \mathcal{J}\mathcal{K}=-\mathcal{K}\mathcal{J}=\mathcal{I},\ \mathcal{K}\mathcal{I}=-\mathcal{I}\mathcal{K}=\mathcal{J}.$$

The restriction of this map is thus a group morphism. The image of SU(2) is $\mathbb{S}$ since, if $A\in\mathrm{SU}(2)$, $A=\alpha_0 I+\alpha_1\mathcal{I}+\alpha_2\mathcal{J}+\alpha_3\mathcal{K}$ with $\sum_{p=0}^3(\alpha_p)^2=1$, hence $q_A\in\mathbb{S}$, and, conversely, each quaternion of norm 1 is the image of a matrix A belonging to SU(2). Thus $A\in\mathrm{SU}(2)\mapsto q_A\in\mathbb{S}$ is a group isomorphism.

(b) The vector space $\mathcal{H}$ is in fact a subalgebra of the associative algebra of complex 2×2 matrices (with matrix multiplication). If $X \in \mathfrak{su}(2)$, then $\frac{X^k}{k!} \in \mathcal{H}$ for each $k \in \mathbb{N}$, and one has, in $\mathbb{H}$,

$$q_{\exp(tX)} = q_{I+tX+\mathrm{O}(t^2)} = 1 + tq_X + \mathrm{O}(t^2), \text{ for } t \to 0.$$

Hence $\frac{\mathrm{d}}{\mathrm{d}t} q_{\exp(tX)}\big|_{t=0} = q_X$.

One can also use the equation $\exp(tX) = \cos t\ I + \sin t\ X$, which holds for $X \in \mathfrak{su}(2)$ of determinant 1, whence, for every $X \in \mathfrak{su}(2)$, $X \neq 0$,

$$\exp(tX) = \cos(t\|X\|)I + \sin(t\|X\|)\frac{X}{\|X\|},$$

whence $q_{\exp(tX)} = \cos(t\|X\|)1 + \frac{1}{\|X\|}\sin(t\|X\|)q_X$, and therefore $\frac{\mathrm{d}}{\mathrm{d}t} q_{\exp(tX)}\big|_{t=0} = q_X$.

2. (a) In the identification of $\mathbb{H}$ with $\mathbb{R}^4$, the quaternion norm can be identified with the Euclidean norm. The map $\sigma(A, B)$ preserves the quaternion norm because

$$\|q_A q (q_B)^{-1}\| = \|q_A\|\|q\|\|(q_B)^{-1}\|,$$

which is equal to $\|q\|$, since $\|q_A\| = \|q_B\| = \|(q_B)^{-1}\| = 1$, for $A, B \in \mathrm{SU}(2)$. Thus σ can be identified with an element of $\mathrm{O}(4)$.

(b) Let (A, B) and (A', B') be elements of $\mathrm{SU}(2) \times \mathrm{SU}(2)$. Then

$$\sigma(A, B) \circ \sigma(A', B')\ q = \sigma(A, B)(q_{A'} q (q_{B'})^{-1}) = q_A q_{A'} q (q_{B'})^{-1} (q_B)^{-1}.$$

Now $A \mapsto q_A$ is a group morphism, and hence

$$q_A q_{A'} = q_{AA'} \quad \text{and} \quad (q_{B'})^{-1}(q_B)^{-1} = (q_B q_{B'})^{-1} = (q_{BB'})^{-1}.$$

Hence

$$\sigma(A, B) \circ \sigma(A', B') = \sigma(AA', BB').$$

The image of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ is contained in $\mathrm{SO}(4)$ because $\mathrm{SU}(2) \times \mathrm{SU}(2)$ is connected.

(c) The kernel of σ is the set of ordered pairs (A, B) such that for every $q \in \mathbb{H}$, $q_A q = q q_B$. Taking $q = 1$, we see that $q_A = q_B$. We set

$$q_A = \alpha_0 1 + \alpha_1 i + \alpha_2 j + \alpha_3 k.$$

The condition $q_A i = i q_A$ implies

$$\alpha_0 i - \alpha_1 - \alpha_2 k + \alpha_3 j = \alpha_0 i - \alpha_1 + \alpha_2 k - \alpha_3 j,$$

whence

$$\alpha_2 = \alpha_3 = 0.$$

The condition $q_A j = j q_B$ implies $\alpha_1 = 0$. One thus obtains $q_A = q_B = \alpha_0 1$. Because $\|q_A\| = 1$, the solutions are $q_A = q_B = 1$, and $q_A = q_B = -1$. The kernel of σ is thus the set

$$\{(I, I), (-I, -I)\} \subset \mathrm{SU}(2) \times \mathrm{SU}(2).$$

3. (a) We have

$$(D\sigma)(X, Y)q = \frac{\mathrm{d}}{\mathrm{d}t}\sigma(\exp(tX), \exp(tY))q\big|_{t=0}$$

$$= \frac{\mathrm{d}}{\mathrm{d}t}(q_{\exp(tX)}\ q\ q_{\exp(-tY)})\big|_{t=0} = q_X q - q q_Y,$$

by the equation $\frac{\mathrm{d}}{\mathrm{d}t} q_{\exp(tX)}\big|_{t=0} = q_X$.

(b) We know that $D\sigma : \mathfrak{su}(2) \oplus \mathfrak{su}(2) \to \mathfrak{so}(4)$ and that $D\sigma$ is a morphism of Lie algebras.

The kernel of $D\sigma$ is the set of pairs $(X, Y) \in \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ such that $q_X q = q q_Y$, for every $q \in \mathbb{H}$. This implies $q_X = q_Y = \alpha_0 1$. Because X and Y belong to $\mathfrak{su}(2)$, q_X and q_Y are zero. Hence $\operatorname{Ker} D\sigma = (0, 0)$. Because $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ and $\mathfrak{so}(4)$ have the same dimension, 6, this injective morphism is an isomorphism.

(c) Every element of SO(4) can be written $\exp Z_1 \dots \exp Z_k$, where $Z_\ell \in \mathfrak{so}(4)$. Because $D\sigma$ is bijective, $Z_\ell = D\sigma(X_\ell, Y_\ell)$ for some pair $(X_\ell, Y_\ell) \in \mathfrak{su}(2) \times \mathfrak{su}(2)$. So $\exp Z_\ell = \exp D\sigma(X_\ell, Y_\ell) = \sigma(\exp X_\ell, \exp Y_\ell)$, thus $\exp Z_\ell \in \operatorname{Im}\sigma$. We deduce that $\exp Z_1 \dots \exp Z_k \in \operatorname{Im}\sigma$, and thus $\operatorname{Im}\sigma = \mathrm{SO}(4)$.

Part II

4. The irreducible representations of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ are $\mathcal{D}^{j_1} \times \mathcal{D}^{j_2}$, $(j_1, j_2) \in \frac{1}{2}\mathbb{N} \times \frac{1}{2}\mathbb{N}$. We set

$$\mathcal{D}^{(j_1 j_2)} = \mathcal{D}^{j_1} \times \mathcal{D}^{j_2}.$$

The irreducible representations of SO(4) are the $\mathcal{D}^{(j_1 j_2)}$ that can be factored through the projection σ of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ onto SO(4), that is, that satisfy $\mathcal{D}^{(j_1 j_2)}(-I, -I) = \mathrm{Id}_{V^{j_1} \otimes V^{j_2}}$. Now,

$$\mathcal{D}^{(j_1 j_2)}(-I, -I)(v_1 \otimes v_2) = \mathcal{D}^{j_1}(-I)v_1 \otimes \mathcal{D}^{j_2}(-I)v_2 = (-1)^{2(j_1 + j_2)} v_1 \otimes v_2.$$

Therefore the irreducible representations of SO(4) are $\mathcal{D}^{(j_1 j_2)}$, $(j_1, j_2) \in \frac{1}{2}\mathbb{N} \times \frac{1}{2}\mathbb{N}$, such that $j_1 + j_2 \in \mathbb{N}$.

The irreducible representations of SU(2) × SU(2) and their dimensions are thus

$\mathcal{D}^{(00)}$	$\mathcal{D}^{(0\frac{1}{2})}$	$\mathcal{D}^{(01)}$	$\mathcal{D}^{(0\frac{3}{2})}$	$\mathcal{D}^{(02)}$	$\dots$	$\mathcal{D}^{(0j_2)}$	$\dots$
1	2	3	4	5		$2j_2+1$	
$\mathcal{D}^{(\frac{1}{2}0)}$	$\mathcal{D}^{(\frac{1}{2}\frac{1}{2})}$	$\mathcal{D}^{(\frac{1}{2}1)}$	$\mathcal{D}^{(\frac{1}{2}\frac{3}{2})}$	$\mathcal{D}^{(\frac{1}{2}2)}$	$\dots$	$\mathcal{D}^{(\frac{1}{2}j_2)}$	$\dots$
2	4	6	8	10		$2(2j_2+1)$	$\cdots$
$\mathcal{D}^{(10)}$	$\mathcal{D}^{(1\frac{1}{2})}$	$\mathcal{D}^{(11)}$	$\mathcal{D}^{(1\frac{3}{2})}$	$\mathcal{D}^{(12)}$	$\dots$	$\mathcal{D}^{(1j_2)}$	$\dots$
3	6	9	12	15		$3(2j_2+1)$	$\cdots$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
$\mathcal{D}^{(j_10)}$	$\mathcal{D}^{(j_1\frac{1}{2})}$	$\mathcal{D}^{(j_11)}$	$\mathcal{D}^{(j_1\frac{3}{2})}$	$\mathcal{D}^{(j_12)}$	$\dots$	$\mathcal{D}^{(j_1j_2)}$	$\dots$
$2j_1+1$	$2(2j_1+1)$	$3(2j_1+1)$	$4(2j_1+1)$	$5(2j_1+1)$		$(2j_1+1)(2j_2+1)$	$\cdots$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$

The irreducible representations of SO(4) are

$\mathcal{D}^{(00)}$	$\mathcal{D}^{(01)}$	$\mathcal{D}^{(02)}$	$\dots$	$\mathcal{D}^{(0j_2)}$	j_2 integer
$\mathcal{D}^{(\frac{1}{2}\frac{1}{2})}$	$\mathcal{D}^{(\frac{1}{2}\frac{3}{2})}$		$\dots$	$\mathcal{D}^{(\frac{1}{2}j_2)}$	j_2 half-integer
$\mathcal{D}^{(10)}$	$\mathcal{D}^{(11)}$		$\dots$	$\mathcal{D}^{(1j_2)}$	j_2 integer
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
$\dots$	$\dots$	$\dots$	$\dots$	$\mathcal{D}^{(j_1j_2)}$	j_1 and j_2 integers or j_1 and j_2 half-integers.

5. By definition, we have

$$\begin{aligned} &D(\rho_1\times\rho_2)(X_1,X_2)(v_1\otimes v_2)\\ &=\frac{\mathrm{d}}{\mathrm{d}t}\rho_1(\exp(tX_1))v_1\otimes\rho_2(\exp(tX_2))v_2\big|_{t=0}\\ &=D\rho_1(X_1)v_1\otimes v_2+v_1\otimes D\rho_2(X_2)v_2. \end{aligned}$$

6. In V^j, we use the basis $(|j,-j\rangle,\ |j,-j+1\rangle,\ \dots,|j,j-1\rangle,\ |j,j\rangle)$ in this order. The matrix of $D^j(J_3)$ in this basis is

$$\begin{pmatrix} -j & 0 & & 0\\ 0 & -j+1 & & 0\\ & & \ddots & \\ 0 & 0 & & j \end{pmatrix}.$$

On the other hand $J_\pm = J_1 \pm iJ_2$, whence $J_1 = \frac{1}{2}(J_+ + J_-)$ and $J_2 = -\frac{i}{2}(J_+ - J_-)$, whence

$$D^j(J_1)=\frac{1}{2}\begin{pmatrix} 0 & \sqrt{2j} & 0 & \cdot\\ \sqrt{2j} & 0 & \sqrt{2(2j-1)} & \cdot\\ 0 & \sqrt{2(2j-1)} & 0 & \cdot\\ \cdots & \cdots & \cdots & 0 \end{pmatrix},$$

$$D^j(J_2) = \frac{1}{2}\begin{pmatrix} 0 & i\sqrt{2j} & 0 & \cdot \\ -i\sqrt{2j} & 0 & i\sqrt{2(2j-1)} & \cdot \\ 0 & -i\sqrt{2(2j-1)} & 0 & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdots & \cdots & \cdots & 0 \end{pmatrix}.$$

In the basis $(|\frac{1}{2}, -\frac{1}{2}\rangle,\ |\frac{1}{2}, \frac{1}{2}\rangle)$ of $V^{\frac{1}{2}}$,

$$D^{\frac{1}{2}}(J_3) = \frac{1}{2}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = J_3,\ D^{\frac{1}{2}}(J_1) = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -J_1,\ D^{\frac{1}{2}}(J_2) = \frac{1}{2}\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} = -J_2.$$

In the basis $(|1, -1\rangle,\ |1, 0\rangle,\ |1, 1\rangle)$ of V^1,

$$D^1(J_3) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and

$$D^1(J_1) = \frac{1}{2}\begin{pmatrix} 0 & \sqrt{2} & 0 \\ \sqrt{2} & 0 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix},\ D^1(J_2) = \frac{1}{2}\begin{pmatrix} 0 & i\sqrt{2} & 0 \\ -i\sqrt{2} & 0 & i\sqrt{2} \\ 0 & -i\sqrt{2} & 0 \end{pmatrix}.$$

By definition,

$$D^{(j_1 j_2)}(X, 0) = D^{j_1}(X) \otimes \mathrm{Id}_{V^{j_2}},\quad D^{(j_1 j_2)}(0, X) = \mathrm{Id}_{V^{j_1}} \otimes D^{j_2}(X).$$

We set, for fixed j_1 and j_2, $|j_1 m_1\rangle \otimes |j_2 m_2\rangle = |m_1; m_2\rangle$ and we use the basis

$$(|-j_1\,; -j_2\rangle, |-j_1\,; -j_2+1\rangle, \ldots, |-j_1\,; j_2\rangle, |-j_1+1\,; -j_2\rangle, \ldots, |-j_1+1\,; j_2\rangle, \ldots, |j_1\,; j_2\rangle).$$

a) Let $j_1 = j_2 = \frac{1}{2}$. Then,

$$D^{(\frac{1}{2}\frac{1}{2})}(J_3, 0) = \frac{1}{2}\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},\ D^{(\frac{1}{2}\frac{1}{2})}(0, J_3) = \frac{1}{2}\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$D^{(\frac{1}{2}\frac{1}{2})}(J_1, 0) = \frac{1}{2}\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},\ D^{(\frac{1}{2}\frac{1}{2})}(0, J_1) = \frac{1}{2}\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

and

$$D^{(\frac{1}{2}\frac{1}{2})}(J_2, 0) = \frac{1}{2}\begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & i \\ -i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix},\ D^{(\frac{1}{2}\frac{1}{2})}(0, J_2) = \frac{1}{2}\begin{pmatrix} 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \end{pmatrix}.$$

(b) Set $j_1 = 1$ and $j_2 = \frac{1}{2}$. Then,

$$D^{(1\frac{1}{2})}(J_3,0) = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, D^{(1\frac{1}{2})}(0,J_3) = \frac{1}{2}\begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

(c) Let $j_1 = \frac{1}{2}$ and $j_2 = 1$. Then,

$$D^{(\frac{1}{2}1)}(J_3,0) = \frac{1}{2}\begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, D^{(\frac{1}{2}1)}(0,J_3) = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

7. If $j_1 > j_2$, then j_1 is an eigenvalue of $D^{(j_1 j_2)}(J_3,0)$ and not of $D^{(j_2 j_1)}(J_3,0)$. Thus the representation $D^{(j_1 j_2)}$ is equivalent to $D^{(j_2 j_1)}$ if and only if $j_1 = j_2$.

8. We have

$$\sum_{\ell=1}^{3} D_1^{(j_1 j_2)}(J_\ell^2)(v_1 \otimes v_2) = \sum_{\ell=1}^{3} D^{j_1}(J_\ell^2)v_1 \otimes v_2 = j_1(j_1+1)v_1 \otimes v_2,$$

for $v_1 \in V^{j_1}$ and $v_2 \in V^{j_2}$, and a similar result for $\sum_{\ell=1}^{3} D_2^{(j_1 j_2)}(J_\ell^2)$. Hence

$$C_i^{(j_1 j_2)} = j_i(j_i+1)\mathrm{Id}_{V^{j_1}\otimes V^{j_2}},\ i = 1,2.$$

9. From $R^2 = e^4 + 2E\frac{l^2+\hbar^2}{m}$, we find that

$$-\frac{m}{2E}R^2 + l^2 + \hbar^2 = -\frac{me^4}{2E},$$

whence, by replacement from $-\frac{m}{2E}R^2 + l^2 = 2\hbar^2(C_1^{(jj)} + C_2^{(jj)})$,

$$\hbar^2(2(C_1^{(jj)} + C_2^{(jj)}) + 1) = -\frac{me^4}{2E}.$$

Because $C_1^{(jj)} = C_2^{(jj)} = j(j+1)$, and since we set $\frac{me^4}{2\hbar^2} = R_H$, we obtain

$$E = -\frac{R_H}{(2j+1)^2},$$

or

$$E = -\frac{R_H}{n^2}.$$

The degeneracy of the energy level $E = -\frac{R_H}{n^2}$ of the hydrogen atom is the dimension $(2j+1)^2 = n^2$ of the corresponding eigenspace of the Hamiltonian operator which is the support of the irreducible representation $D^{(jj)}$ of SO(4). This degeneracy, which has been observed experimentally, can therefore be explained by the fact that the full symmetry group of the hydrogen atom is the group SO(4), a symmetry group larger than the obvious symmetry group SO(3). This is what is called the *hidden symmetry of the hydrogen atom*. See, for example, Singer (2005) or Gilmore (2008). The operator (R_1, R_2, R_3) corresponds to the Laplace vector $\overrightarrow{R}$ in classical mechanics. The fact that it involves a representation with $j_1 = j_2$ is a consequence of the fact that the Laplace vector $\overrightarrow{R}$ is orthogonal to the angular momentum $\overrightarrow{\ell}$. In other publications, the constant called the Rydberg constant is $R'_H = \dfrac{R_H}{h} = \dfrac{R_H}{2\pi\hbar}$. The values of these constants are $R_H = 13.61$ eV and $R'_H = 3.29 \times 10^{15}$ s^{-1}.

7 Projection Operators

Part I

Let $G = \mathfrak{S}_3$ be the symmetric group on three elements. We denote by e the identity, by c the cyclic permutation $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ and by t the transposition $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$. We denote by ρ_0 the representation of $\mathfrak{S}_3$ in $\mathbb{C}^2$ such that

$$\rho_0(c) = \begin{pmatrix} j & 0 \\ 0 & j^2 \end{pmatrix}, \quad \rho_0(t) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

where j is the zero of $x^2 + x + 1$ with positive imaginary part.

Let $E = \mathcal{F}(G) = \mathbb{C}[G]$, the vector space of functions on G taking values in $\mathbb{C}$, with basis $(\epsilon_g)_{g\in G}$. Let (E, R) be the regular representation of G.

1. Write the decomposition of R as a direct sum of irreducible representations.

We propose finding a vector subspace of E invariant under R and such that the restriction of R to this subspace is equivalent to ρ_0.

We denote by $\rho_{\alpha\beta}$ the matrix coefficients of ρ_0 $(1 \le \alpha \le 2,\ 1 \le \beta \le 2)$. We set

$$P_{\alpha\beta} = \frac{1}{3} \sum_{g\in G} \rho_{\beta\alpha}(g^{-1}) R(g) \in \text{ End } (E).$$

2. (a) Find the matrices $\rho_0(e), \rho_0(c^2), \rho_0(tc), \rho_0(tc^2)$.

(b) Express P_{11}, P_{22}, P_{21} and P_{12} explicitly as linear combinations of $R(g)$, $g \in G$.

(c) Calculate $P_{11}(\epsilon_h)$, for every $h \in G$. What is the dimension of the vector subspace $P_{11}(E)$ of E? Is it invariant under R?

(d) Same questions as in (c) replacing P_{11} by P_{22}.

3. Choose a basis (u_1, u_1') of $P_{11}(E)$.

(a) Calculate $P_{21}(u_1)$ and show that $P_{21}(u_1) \in P_{22}(E)$.

(b) Let $E_{0,1}$ be the vector subspace of E generated by u_1 and $P_{21}(u_1)$. Show that $E_{0,1}$ is invariant under R and that the restriction of R to $E_{0,1}$ is equivalent to ρ_0.

(c) What can be said about $R\big|_{E'_{0,1}}$, where $E'_{0,1}$ is the vector subspace of E generated by u_1' and $P_{21}(u_1')$?

4. Show that $P_{11}+P_{22}$ is the projection P_0 of E onto the isotypic component of type ρ_0 of E, and that $P_0(E) = E_{0,1} \oplus E'_{0,1}$.

Part II

Let G be a finite group, and let $(E^{(i)}, \rho^{(i)}), i = 1, \ldots, N$, be the inequivalent irreducible representations of G. We set $\dim E^{(i)} = d_i$ and we denote by $\rho^{(i)}_{\alpha\beta}$, for $1 \leq \alpha \leq d_i, 1 \leq \beta \leq d_i$, the matrix coefficients of the representation $\rho^{(i)}$ in a basis $(e^{(i)}_\alpha)$ of $E^{(i)}$. We recall that, for all integers $\alpha, \beta, \lambda, \mu$ between 1 and d_i,

$$\sum_{g\in G} \rho^{(i)}_{\beta\alpha}(g^{-1})\rho^{(j)}_{\lambda\mu}(g) = \begin{cases} 0 & \text{if } i \neq j, \\ \frac{|G|}{d_i}\delta_{\alpha\lambda}\delta_{\beta\mu} & \text{if } i = j. \end{cases} \tag{1}$$

Let (E, ρ) be a representation of G. We set $E = \bigoplus_{i=1}^N V^{(i)}$, with $V^{(i)} = m_i E^{(i)}$, where m_i is the multiplicity of $E^{(i)}$ in E. We fix the index $i, 1 \leq i \leq N$, and for $1 \leq \alpha \leq d_i, 1 \leq \beta \leq d_i$, we consider

$$P^{(i)}_{\alpha\beta} = \frac{d_i}{|G|} \sum_{g\in G} \rho^{(i)}_{\beta\alpha}(g^{-1})\rho(g) \in \text{ End } (E).$$

5. Compare $\sum_{\alpha=1}^{d_i} P^{(i)}_{\alpha\alpha}$ to the projection $P^{(i)}$ of E onto the isotypic component $V^{(i)}$ of E.

6. Set $1 \leq i \leq N,\ 1 \leq j \leq N$. Show that $P^{(i)}_{\alpha\beta}(V^{(j)}) = \{0\}$ if $j \neq i$, and that

$$P^{(i)}_{\alpha\beta}(e^{(i)}_\gamma) = \delta_{\beta\gamma} e^{(i)}_\alpha, \text{ for } 1 \leq \gamma \leq d_i.$$

7. (a) Let $\alpha, \beta, \lambda, \mu$ be integers between 1 and d_i. Show that $P^{(i)}_{\alpha\beta} \circ P^{(j)}_{\lambda\mu} = 0$ if $i \neq j$, and that

$$P^{(i)}_{\alpha\beta} \circ P^{(i)}_{\lambda\mu} = \delta_{\beta\lambda} P^{(i)}_{\alpha\mu}.$$

(b) Conclude that $P^{(i)}_{\alpha\beta}$ vanishes on $P^{(i)}_{\gamma\gamma}(E)$ if $\gamma \neq \beta$, and that $P^{(i)}_{\alpha\beta}$ defines an isomorphism of $P^{(i)}_{\beta\beta}(E)$ onto $P^{(i)}_{\alpha\alpha}(E)$.

8. Show that $\rho(g) \circ P^{(i)}_{\alpha\beta} = \sum_{\gamma=1}^{d_i} \rho^{(i)}_{\gamma\alpha}(g) P^{(i)}_{\gamma\beta}$.

9. Let $x \in P^{(i)}_{11}(E)$, $x \neq 0$. For $\beta = 1, \ldots, d_i$, set $x_\beta = P^{(i)}_{\beta 1}(x)$. Let $E^{(i)}_1(x)$ be the vector subspace of E generated by $(x_1, \ldots, x_{d_i})$. Show that the restriction of ρ to $E^{(i)}_1(x)$ is equivalent to $\rho^{(i)}$.

10. Construct m_i vector subspaces of E whose direct sum is $V^{(i)}$ and such that the restriction of ρ to each is equivalent to $\rho^{(i)}$.

11. Define projection operators $P^{(i)}_{\alpha\beta}$ in order to apply an analogous procedure to the decomposition of a representation of a compact group.

Solutions

Part I

1. The irreducible representations of $G = \mathfrak{S}_3$ are the trivial representation $\underline{1}$ and the signature representation ϵ, both one-dimensional, and the two-dimensional representation ρ_0. Hence

$$R = \underline{1} \oplus \epsilon \oplus 2\rho_0.$$

2. (a) By calculating matrix products, we obtain

$$\rho_0(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \rho_0(c) = \begin{pmatrix} j & 0 \\ 0 & j^2 \end{pmatrix}, \rho_0(c^2) = \begin{pmatrix} j^2 & 0 \\ 0 & j \end{pmatrix},$$

$$\rho_0(t) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \rho_0(tc) = \begin{pmatrix} 0 & j^2 \\ j & 0 \end{pmatrix}, \rho_0(tc^2) = \begin{pmatrix} 0 & j \\ j^2 & 0 \end{pmatrix}.$$

We shall make use of the equations $c^3 = e, t^2 = e, ct = tc^2, c^2t = tc$ and $ctc = t$.

(b) We note that $c^{-1} = c^2, (c^2)^{-1} = c$. By definition

$$P_{11} = \frac{1}{3}(R(e) + j^2 R(c) + jR(c^2)),$$

$$P_{22} = \frac{1}{3}(R(e) + jR(c) + j^2 R(c^2)).$$

Because t, tc and tc^2 are equal to their inverses, and since e, c and c^2 are diagonal, we obtain

$$P_{12} = \frac{1}{3} \sum_{g \in G} \rho_{21}(g^{-1}) R(g) = \frac{1}{3}(R(t) + jR(tc) + j^2 R(tc^2)),$$

$$P_{21} = \frac{1}{3}\sum_{g\in G}\rho_{12}(g^{-1})R(g) = \frac{1}{3}(R(t) + j^2R(tc) + jR(tc^2)).$$

(c) We know that $R(g)\epsilon_h = \epsilon_{gh}$, for $g, h \in G$. Thus

$$P_{11}(\epsilon_h) = \frac{1}{3}(\epsilon_h + j^2\epsilon_{ch} + j\epsilon_{c^2h}).$$

Hence

$$P_{11}(\epsilon_e) = \frac{1}{3}(\epsilon_e + j^2\epsilon_c + j\epsilon_{c^2}),$$

$$P_{11}(\epsilon_t) = \frac{1}{3}(\epsilon_t + j^2\epsilon_{ct} + j\epsilon_{c^2t}) = \frac{1}{3}(\epsilon_t + j\epsilon_{tc} + j^2\epsilon_{tc^2}).$$

We set

$$P_{11}(\epsilon_e) = u_1, \quad P_{11}(\epsilon_t) = u_1',$$

which are linearly independent. Then,

$$P_{11}(\epsilon_c) = ju_1,\ P_{11}(\epsilon_{c^2}) = j^2u_1,$$

$$P_{11}(\epsilon_{tc}) = j^2u_1'\ ,\ P_{11}(\epsilon_{tc^2}) = ju_1'.$$

Thus $P_{11}(E)$ is two-dimensional.

We have $R(t)u_1 = \frac{1}{3}(\epsilon_t + j^2\epsilon_{tc} + j\epsilon_{tc^2})$, which does not lie in $P_{11}(E)$. Thus $P_{11}(E)$ is not invariant under R.

(d) By (b) and the equation $R(g)\epsilon_h = \epsilon_{gh}$, we obtain

$$P_{22}(\epsilon_h) = \frac{1}{3}(\epsilon_h + j\epsilon_{ch} + j^2\epsilon_{c^2h}).$$

Hence

$$P_{22}(\epsilon_e) = \frac{1}{3}(\epsilon_e + j\epsilon_c + j^2\epsilon_{c^2}),$$

$$P_{22}(\epsilon_t) = \frac{1}{3}(\epsilon_t + j\epsilon_{ct} + j^2\epsilon_{c^2t}) = \frac{1}{3}(\epsilon_t + j^2\epsilon_{tc} + j\epsilon_{tc^2}).$$

We set

$$P_{22}(\epsilon_t) = v_1, \quad P_{22}(\epsilon_e) = v_1',$$

which are linearly independent. We have

$$P_{22}(\epsilon_c) = j^2v_1',\ P_{22}(\epsilon_{c^2}) = jv_1',$$

$$P_{22}(\epsilon_{tc}) = jv_1,\ P_{22}(\epsilon_{tc^2}) = j^2v_1.$$

Thus $P_{22}(E)$ is two-dimensional.

We have $R(t)v_1 = \frac{1}{3}(\epsilon_e + j^2\epsilon_c + j\epsilon_{c^2})$ which does not lie in $P_{22}(E)$. Thus $P_{22}(E)$ is not invariant under R.

3. (a) We calculate $P_{21}(u_1) = \frac{1}{3}(R(t)u_1 + j^2R(tc)u_1 + jR(tc^2)u_1)$. We have

$$R(t)u_1 = \frac{1}{3}(\epsilon_t + j^2\epsilon_{tc} + j\epsilon_{tc^2}) = v_1,$$

$$R(tc)u_1 = \frac{1}{3}(\epsilon_{tc} + j^2\epsilon_{tc^2} + j\epsilon_t) = jv_1,$$

$$R(tc^2)u_1 = \frac{1}{3}(\epsilon_{tc^2} + j^2\epsilon_t + j\epsilon_{tc}) = j^2v_1.$$

Hence $P_{21}(u_1) = v_1$, which belongs to $P_{22}(E)$. Furthermore u_1 and v_1 are linearly independent.

(b) We have $R(c)u_1 = \frac{1}{3}(\epsilon_c + j^2\epsilon_{c^2} + j\epsilon_e) = ju_1$ and $R(t)u_1 = v_1$, and thus u_1 and $P_{21}(u_1)$ generate a two-dimensional vector subspace $E_{0,1}$ of E, invariant under R. We have $R(c)v_1 = \frac{1}{3}(\epsilon_{tc^2} + j\epsilon_{tc} + j^2\epsilon_t) = j^2v_1$ and $R(t)v_1 = \frac{1}{3}(\epsilon_e + j\epsilon_{c^2} + j^2\epsilon_c) = u_1$. Thus the matrices of $R\big|_{E_{0,1}}$ in the basis (u_1, v_1) are

$$C = \begin{pmatrix} j & 0 \\ 0 & j^2 \end{pmatrix},\ T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thus $R\big|_{E_{0,1}}$ is equivalent to ρ_0.

(c) Similarly $R\big|_{E'_{0,1}}$ is equivalent to ρ_0.

4. By definition,

$$\begin{aligned} P_{11} + P_{22} &= \tfrac{1}{3}\textstyle\sum_{g\in G}(\rho_{11}(g^{-1}) + \rho_{22}(g^{-1}))R(g) \\ &= \tfrac{1}{3}\textstyle\sum_{g\in G}\overline{\chi_{\rho_0}(g)}R(g), \end{aligned}$$

where χ_{ρ_0} is the character of the representation ρ_0. We know that this endomorphism of E is the projection P_0 onto the isotypic component of type ρ_0, which here has multiplicity two. Because $E_{0,1}$ and $E'_{0,1}$ are of type ρ_0 and have trivial intersection $\{0\}$, we have

$$P_0(E) = E_{0,1} \oplus E'_{0,1}.$$

Part II

5. We know that $\sum_{\alpha=1}^{d_i} \rho^{(i)}_{\alpha\alpha}(g^{-1}) = \chi_{\rho^{(i)}}(g^{-1}) = \overline{\chi_{\rho^{(i)}}(g)}$. Thus, by the definitions,

$$\sum_{\alpha=1}^{d_i} P^{(i)}_{\alpha\alpha} = P^{(i)}.$$

6. Let (φ_λ), $\lambda=1,\ldots,d_j$, be a basis of $E^{(j)}$. Since $\rho(g)(\varphi_\mu)=\sum_{\lambda=1}^{d_j}\rho_{\lambda\mu}^{(j)}(g)\varphi_\lambda$, it follows from (1) that, if $i\neq j$,

$$P_{\alpha\beta}^{(i)}(\varphi_\mu)=\frac{d_i}{|G|}\sum_{\lambda=1}^{d_j}\left(\sum_{g\in G}\rho_{\beta\alpha}^{(i)}(g^{-1})\rho_{\lambda\mu}^{(j)}(g)\right)\varphi_\lambda=0.$$

Because $V^{(j)}=m_jE^{(j)}$, the operator $P_{\alpha\beta}^{(i)}$ vanishes on $V^{(j)}$.

On the other hand, using (1) for $i=j$,

$$P_{\alpha\beta}^{(i)}(e_\gamma^{(i)})=\frac{d_i}{|G|}\sum_{\lambda=1}^{d_i}\left(\sum_{g\in G}\rho_{\beta\alpha}^{(i)}(g^{-1})\rho_{\lambda\gamma}^{(i)}(g)\right)e_\lambda^{(i)}=\sum_{\lambda=1}^{d_i}\delta_{\alpha\lambda}\delta_{\beta\gamma}e_\lambda^{(i)}=\delta_{\beta\gamma}e_\alpha^{(i)}.$$

7. (a) We calculate

$$P_{\alpha\beta}^{(i)}\circ P_{\lambda\mu}^{(j)}=\frac{d_id_j}{|G|^2}\left(\sum_{g\in G}\rho_{\beta\alpha}^{(i)}(g^{-1})\rho(g)\right)\circ\left(\sum_{h\in G}\rho_{\mu\lambda}^{(j)}(h^{-1})\rho(h)\right)$$

$$=\frac{d_id_j}{|G|^2}\sum_{g\in G}\sum_{h\in G}\rho_{\beta\alpha}^{(i)}(g^{-1})\rho_{\mu\lambda}^{(j)}(h^{-1})\rho(gh).$$

Setting $gh=k$, whence $g^{-1}=hk^{-1}$, we obtain

$$P_{\alpha\beta}^{(i)}\circ P_{\lambda\mu}^{(j)}=\frac{d_id_j}{|G|^2}\sum_{k\in G}\left(\sum_{h\in G}\rho_{\beta\alpha}^{(i)}(hk^{-1})\rho_{\mu\lambda}^{(j)}(h^{-1})\right)\rho(k).$$

We use $\rho_{\beta\alpha}^{(i)}(hk^{-1})=\sum_{\gamma=1}^{d_i}\rho_{\beta\gamma}^{(i)}(h)\rho_{\gamma\alpha}^{(i)}(k^{-1})$, along with equation (1) and we obtain

$$P_{\alpha\beta}^{(i)}\circ P_{\lambda\mu}^{(j)}=\frac{d_id_j}{|G|^2}\sum_{k\in G}\sum_{\gamma=1}^{d_i}\left(\sum_{h\in G}\rho_{\beta\gamma}^{(i)}(h)\rho_{\mu\lambda}^{(j)}(h^{-1})\right)\rho_{\gamma\alpha}^{(i)}(k^{-1})\rho(k)$$

$$=\delta_{ij}\frac{d_i}{|G|}\sum_{k\in G}\sum_{\gamma=1}^{d_i}\delta_{\gamma\mu}\delta_{\beta\lambda}\rho_{\gamma\alpha}^{(i)}(k^{-1})\rho(k)=\delta_{ij}\delta_{\beta\lambda}\frac{d_i}{|G|}\sum_{k\in G}\rho_{\mu\alpha}^{(i)}(k^{-1})\rho(k)=\delta_{ij}\delta_{\beta\lambda}P_{\alpha\mu}^{(i)},$$

which proves the desired results.

(b) In particular, $P_{\alpha\alpha}^{(i)}\circ P_{\beta\beta}^{(j)}=0$, if $i\neq j$, and $P_{\alpha\beta}^{(i)}\circ P_{\gamma\gamma}^{(i)}=0$, if $\gamma\neq\beta$. Furthermore,

$$P_{\alpha\alpha}^{(i)}\circ P_{\beta\beta}^{(i)}=0,\ \text{if}\ \alpha\neq\beta,\quad P_{\alpha\alpha}^{(i)}\circ P_{\alpha\alpha}^{(i)}=P_{\alpha\alpha}^{(i)}.$$

Therefore, the $P_{\alpha\alpha}^{(i)}$ are projection operators onto subspaces of $V^{(i)}$ such that their images $P_{\alpha\alpha}^{(i)}(E)$ and $P_{\beta\beta}^{(i)}(E)$ have trivial intersection if $\alpha\neq\beta$.

We now show, by means of the result of Question (a), that $P^{(i)}_{\alpha\beta}$ is an isomorphism of $P^{(i)}_{\beta\beta}(E)$ onto $P^{(i)}_{\alpha\alpha}(E)$. We use

$$P^{(i)}_{\alpha\beta} \circ P^{(i)}_{\beta\beta} = P^{(i)}_{\alpha\beta} = P^{(i)}_{\alpha\alpha} \circ P^{(i)}_{\alpha\beta},$$

to show that $P^{(i)}_{\alpha\beta}(P^{(i)}_{\beta\beta}(E)) \subset P^{(i)}_{\alpha\alpha}(E)$. Let us show that the restriction of $P^{(i)}_{\alpha\beta}$ to $P^{(i)}_{\beta\beta}(E)$ is bijective. It is injective since if $u \in E$ satisfies $P^{(i)}_{\alpha\beta}P^{(i)}_{\beta\beta}(u) = 0$, then $0 = P^{(i)}_{\beta\alpha}P^{(i)}_{\alpha\beta}P^{(i)}_{\beta\beta}(u) = P^{(i)}_{\beta\beta}P^{(i)}_{\beta\beta}(u) = P^{(i)}_{\beta\beta}(u)$. It is also surjective onto $P^{(i)}_{\alpha\alpha}(E)$, because for every $u \in E$, and for every $\beta = 1, \ldots, d_i$,

$$P^{(i)}_{\alpha\alpha}(u) = P^{(i)}_{\alpha\beta}P^{(i)}_{\beta\alpha}(u) = P^{(i)}_{\alpha\beta}P^{(i)}_{\beta\beta}P^{(i)}_{\beta\alpha}(u),$$

and thus $P^{(i)}_{\alpha\alpha}(u)$ belongs to the image under $P^{(i)}_{\alpha\beta}$ of $P^{(i)}_{\beta\beta}(E)$.

8. We calculate $\rho(g) \circ P^{(i)}_{\alpha\beta} = \frac{d_i}{|G|} \sum_{h\in G} \rho^{(i)}_{\beta\alpha}(h^{-1})\rho(g)\rho(h)$. We set $gh = k$, whence $h = g^{-1}k$ and $h^{-1} = k^{-1}g$. We obtain

$$\begin{aligned}\rho(g) \circ P^{(i)}_{\alpha\beta} &= \tfrac{d_i}{|G|} \textstyle\sum_{k\in G} \rho^{(i)}_{\beta\alpha}(k^{-1}g)\rho(k) \\ &= \tfrac{d_i}{|G|} \textstyle\sum_{k\in G} \sum_{\gamma=1}^{d_i} \rho^{(i)}_{\beta\gamma}(k^{-1})\rho^{(i)}_{\gamma\alpha}(g)\rho(k) \\ &= \textstyle\sum_{\gamma=1}^{d_i} \rho^{(i)}_{\gamma\alpha}(g)P^{(i)}_{\gamma\beta}.\end{aligned}$$

9. We consider the family $(x_\beta = P^{(i)}_{\beta 1}(x))$, $\beta = 1, \ldots, d_i$, where $x \in P^{(i)}_{11}(E)$ and $x \neq 0$. Let us show that $(x_1, \ldots, x_\beta, \ldots, x_{d_i})$ are linearly independent. We remark first that $x_1 = x$, since $(P^{(i)}_{11})^2 = P^{(i)}_{11}$. Assume that there are complex numbers, $\lambda_1, \ldots, \lambda_{d_i}$, such that $\sum_{\beta=1}^{d_i} \lambda_\beta P^{(i)}_{\beta 1}(x) = 0$. Then, for every $\alpha = 1, \ldots, d_i$,

$$\sum_{\beta=1}^{d_i} \lambda_\beta P^{(i)}_{1\alpha}P^{(i)}_{\beta 1}(x) = \sum_{\beta=1}^{d_i} \lambda_\beta \delta_{\alpha\beta} P^{(i)}_{11}(x) = \lambda_\alpha x.$$

Because $x \neq 0$, we see that $\lambda_\alpha = 0$, for each α. The vector space $E^{(i)}_1(x)$ generated by $(x_1, \ldots, x_{d_i})$ is thus of dimension d_i. We also recall that $P^{(i)}_{\beta 1}$ is an isomorphism of $P^{(i)}_{11}(E)$ onto $P^{(i)}_{\beta\beta}(E)$ and that the $(P^{(i)}_{\beta\beta}(E))$, $\beta = 1, \ldots, d_i$, form a direct sum decomposition of $V^{(i)} = m_i E^{(i)}$.

By the equation proved in Question 8,

$$\begin{aligned}\rho(g)x_\beta &= \rho(g)(P^{(i)}_{\beta 1}(x)) = \textstyle\sum_{\gamma=1}^{d_i} \rho^{(i)}_{\gamma\beta}(g)P^{(i)}_{\gamma 1}(x) \\ &= \textstyle\sum_{\gamma=1}^{d_i} \rho^{(i)}_{\gamma\beta}(g)x_\gamma,\end{aligned}$$

which shows that $\rho\big|_{E^{(i)}_1(x)}$ is equivalent to $\rho^{(i)}$.

10. In order to obtain m_i vector subspaces of E whose direct sum is $V^{(i)}$ and on each of which the group acts by the representation $\rho^{(i)}$, it suffices to apply the procedure of Question 9 successively to the vectors of a basis of $P_{11}^{(i)}(E)$, as we did in Part I. In fact, we have seen that for fixed x in $P_{11}^{(i)}(E)$, the corresponding vector subspace $E_1^{(i)}(x)$ is invariant under G. If x and y are linearly independent, $E_1^{(i)}(x) \cap E_1^{(i)}(y)$ reduces to $\{0\}$ by the irreducibility of the representation $\rho^{(i)}$.

We have already noted that the $(P_{\alpha\alpha}^{(i)}(E))$, $\alpha = 1, \ldots, d_i$, form a direct sum. Since $V^{(i)} = P^{(i)}(E) = \left(\sum_{\alpha=1}^{d_i} P_{\alpha\alpha}^{(i)}\right)(E)$, we obtain

$$V^{(i)} = \bigoplus_{\alpha=1}^{d_i} P_{\alpha\alpha}^{(i)}(E).$$

Because the $P_{\alpha\alpha}^{(i)}(E), \alpha = 1, \ldots, d_i$, are isomorphic, we have $\dim P_{\alpha\alpha}^{(i)}(E) = m_i$. Thus by choosing m_i basis vectors $(x^1, \ldots, x^{m_i})$ in $P_{11}^{(i)}(E)$ and by subjecting them to the process of Question 9 we obtain m_i vector subspaces $E_1^{(i)}(x^A)$, $A = 1, \ldots, m_i$, such that the restriction of ρ to $E_1^{(i)}(x^A)$ is equivalent to $\rho^{(i)}$ for every A and such that $V^{(i)} = \bigoplus_{A=1}^{m_i} E_1^{(i)}(x^A)$. This is the desired decomposition.

Clearly, Part I of the problem consisted of the direct study of a particular case of this process, where $\rho^{(i)} = \rho_0$, one has chosen the basis (u_1, u_1') of $P_{11}^{(i)}(E) = P_{11}(E)$ and one has set $E_1^{(i)}(u^1) = E_{0,1}, E_1^{(i)}(u_1') = E_{0,1}'$.

11. For a compact group, we use the Haar integral $\int_G f(g)dg$ to define

$$P_{\alpha\beta}^{(i)} = d_i \int_G \rho_{\beta\alpha}^{(i)}(g^{-1})\rho(g)dg,$$

and we proceed as in the case of a finite group.

This method is useful for determining the Clebsch–Gordan coefficients, and has many applications in quantum mechanics. See for example the study of the distribution of isotopic-spin states for a collection of free particles in Barut and Raczka (1986). In that book, the authors write, "The technique of the projection operators is extremely useful, elegant and effective in the solution of various practical problems in representation theory and quantum physics."

8 Symmetries of Fullerene Molecules

We propose to study certain properties of the icosahedron group, which is the symmetry group of rotations of "soccer-ball molecules", C_{60}, also called *fullerenes*.

Part I: Character table of $\mathfrak{S}_5$

We give below, without proof, the list of conjugacy classes of the symmetric group $\mathfrak{S}_5$ on five elements and the number of elements of each class. We denote by

e the class of the identity, with one element,

a the class of the permutation (12) : (1,2,3,4,5) $\mapsto$ (2,1,3,4,5), with 10 elements,

α the class of (123) : (1,2,3,4,5) $\mapsto$ (2,3,1,4,5), with 20 elements,

b the class of (1234) : (1,2,3,4,5) $\mapsto$ (2,3,4,1,5), with 30 elements,

γ the class of (12345) : (1,2,3,4,5) $\mapsto$ (2,3,4,5,1), with 24 elements,

β the class of (12)(34) : (1,2,3,4,5) $\mapsto$ (2,1,4,3,5), with 15 elements,

c the class of (12)(345) : (1,2,3,4,5) $\mapsto$ (2,1,4,5,3), with 20 elements.

1. The permutation representation of $\mathfrak{S}_5$ decomposes as the direct sum of the trivial representation ρ_1 and a four-dimensional representation ρ_4. Find their characters. Find the character of the signature representation, ρ_1', as well as the character of the representation ρ_4' obtained as the tensor product of ρ_4 with ρ_1'. Verify that ρ_4 and ρ_4' are irreducible.

2. Show that for each finite-dimensional representation (V, ρ) of a finite group G of character χ_ρ, the character of the representation $\wedge^2\rho$ satisfies

$$\forall g \in G, \ \ \chi_{\wedge^2\rho}(g) = (1/2)((\chi_\rho(g))^2 - \chi_\rho(g^2)),$$

and that $\chi_{\mathrm{S}^2\rho} = (\chi_\rho)^2 - \chi_{\wedge^2\rho}$. (Here $\wedge$ denotes the exterior product and S the symmetric product.)

3. Calculate the characters of the representations $\wedge^2(\rho_4)$ and $\mathrm{S}^2(\rho_4)$. [Hint: Use the equations $\alpha^2 = \alpha, b^2 = b, \gamma^2 = \gamma, c^2 = \alpha$.] Show that $\wedge^2(\rho_4)$ is irreducible. Show that $\mathrm{S}^2(\rho_4)$ is a direct sum of three irreducible representations, and then show that $\mathrm{S}^2(\rho_4) = \rho_1 \oplus \rho_4 \oplus \rho_5$, where ρ_5 is a five-dimensional irreducible representation.

4. Give the character table of $\mathfrak{S}_5$.

Part II: Character table of $\mathfrak{A}_5$

Here is the list of conjugacy classes of the alternating group $\mathfrak{A}_5$ with the number of elements of each class:

e the class of the identity, with one element,

α the class of (123), with 20 elements,

β the class of (12)(34), with 15 elements,

γ the class of (12345), with 12 elements,

γ^2 the class of (23451) : (1,2,3,4,5) $\mapsto$ (3,4,5,1,2), with 12 elements.

5. Show that the representations ρ_1, ρ_4 and ρ_5 stay irreducible when restricted to $\mathfrak{A}_5$. We denote these also by ρ_1, ρ_4, ρ_5.

Show that the restriction to $\mathfrak{A}_5$ of $\wedge^2(\rho_4)$ is the direct sum of two inequivalent irreducible representations, ρ_3 and ρ_3', and that each of them is three-dimensional.

6. Let $\tau = \dfrac{1+\sqrt{5}}{2}$ be the golden mean. Show that the values of the characters of the representations ρ_3 and ρ_3' of $\mathfrak{A}_5$ on the classes $(e, \alpha, \beta, \gamma, \gamma^2)$ are $(3, 0, -1, \tau, 1-\tau)$ and $(3, 0, -1, 1-\tau, \tau)$.

7. Find the character table of $\mathfrak{A}_5$.

Part III: Some representations of the icosahedron group

It is known that the group $\mathfrak{A}_5$ can be identified with the subgroup $\mathcal{I}$ of the group of rotations of $\mathbb{R}^3$ leaving a regular icosahedron invariant. In this identification, the elements of the conjugacy classes $(e, \alpha, \beta, \gamma, \gamma^2)$ of $\mathfrak{A}_5$ become rotations through angles $(0, \dfrac{2\pi}{3}, \pi, \dfrac{2\pi}{5}, \dfrac{4\pi}{5})$, respectively. We recall that $2\cos\dfrac{\pi}{5} = \dfrac{1+\sqrt{5}}{2} = \tau$.

8. Denote by φ the covering morphism of SU(2) onto SO(3). What is the image under φ of the element $g_\alpha = \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{pmatrix}$ of SU(2)?

For $j \in \mathbb{N}$, we denote by $\underline{\mathcal{D}}^j$ the irreducible representation of dimension $2j+1$ of SO(3) obtained from the representation $\mathcal{D}^j$ of SU(2) by passing to the quotient. Show that the value of the character $\chi_{\underline{\mathcal{D}}^j}$ of $\underline{\mathcal{D}}^j$ on a rotation through an angle $\theta \in [0, \pi]$ is

$$\chi_{\underline{\mathcal{D}}^j}(\theta) = \sum_{m=-j}^{j} e^{im\theta}.$$

For each $j \in \mathbb{N}$, we consider the restriction to the subgroup $\mathcal{I}$ of the representation $\underline{\mathcal{D}}^j$. We denote it by R^j and its character by χ^j.

9. Calculate χ^j for $j \in \mathbb{N}$. Show that for each element $g \in \mathcal{I}$, $g \neq e$, and for every integer $\ell \in \mathbb{N}$,

$$\chi^{30\ell+j}(g) = \chi^j(g).$$

10. Show that the representations R^0, R^1 and R^2 of the group $\mathcal{I}$ are irreducible. Decompose the representation R^3 into a direct sum of irreducible representations, and then decompose the representations $R^{30\ell+3}$, for every integer $\ell \geq 1$.

11. Decompose the representation $R^2 \otimes R^3$ of the group $\mathcal{I}$ into a direct sum of irreducible representations.

Epilogue: Representations of $G = \varphi^{-1}(\mathcal{I}) \subset \mathrm{SU}(2)$

Similarly, we can study the representations of the double cover of the icosahedron group, the inverse image of the group $\mathcal{I}$ under the morphism φ. It is a subgroup G of SU(2) with 120 elements. It has nine conjugacy classes, whose representatives are g_α, with

$$\alpha = 0, \pi, \frac{\pi}{2}, \frac{\pi}{3}, \frac{4\pi}{3}, \frac{\pi}{5}, \frac{7\pi}{5}, \frac{6\pi}{5}, \frac{2\pi}{5}.$$

These classes have $1, 1, 30, 20, 20, 12, 12, 12, 12$ elements respectively. Can you find the character table of G? Can you show that, restricted to G, the representation $\mathcal{D}^j$ of SU(2) is irreducible for $0 \leq j \leq \frac{5}{2}$? Can you find the decomposition of the restrictions to G of the representations $\mathcal{D}^j, j \in \frac{1}{2}\mathbb{N}$, into a direct sum of irreducible representations?

Solutions

Part I: Character table of $\mathfrak{S}_5$

1. Let ρ_1 be the one-dimensional trivial representation. We know that $\chi_{\rho_1}(g) = 1$, for every g. In the permutation representation, $\chi(g)$ is the number of elements that are invariant under g. Because $\chi_{\rho_1} + \chi_{\rho_4} = \chi$, we find the following character table.

	1 e	10 a	20 α	30 b	24 γ	15 β	20 c
ρ_1	1	1	1	1	1	1	1
ρ_4	4	2	1	0	-1	0	-1
ρ_1'	1	-1	1	-1	1	1	-1
ρ_4'	4	-2	1	0	-1	0	1

We calculate $(\chi_{\rho_4}|\chi_{\rho_4}) = (\chi_{\rho_4'}|\chi_{\rho_4'}) = \frac{1}{120}(16 + 40 + 20 + 24 + 20) = 1$, and thus ρ_4 and ρ_4' are irreducible.

2. The representation ρ is unitarizable, so for every $g \in G$, $\rho(g)$ is diagonizable. Let (e_i) a basis of V such that $\rho(g)e_i = \lambda_i(g)e_i$, where $\lambda_i(g) \in \mathbb{C}$. Then

$$(\rho(g) \otimes \rho(g))(e_i \otimes e_j - e_j \otimes e_i) = \lambda_i(g)\lambda_j(g)(e_i \otimes e_j - e_j \otimes e_i),$$

whence

$$\begin{aligned} \chi_{\wedge^2\rho}(g) &= \sum_{i<j} \lambda_i(g)\lambda_j(g) = \frac{1}{2}\Big((\sum_i \lambda_i(g))^2 - \sum_i (\lambda_i(g))^2\Big) \\ &= \frac{1}{2}\left((\chi_\rho(g))^2 - \chi_\rho(g^2)\right). \end{aligned}$$

Because $\rho \otimes \rho = \wedge^2\rho \oplus \mathrm{S}^2\rho$, we have $\chi_{\mathrm{S}^2\rho} = (\chi_\rho)^2 - \chi_{\wedge^2\rho}$. Therefore

$$\chi_{\mathrm{S}^2\rho}(g) = \frac{1}{2}\left((\chi_\rho(g))^2 + \chi_\rho(g^2)\right).$$

3. The calculation yields

$$\chi_{\wedge^2(\rho_4)} = (6, 0, 0, 0, 1, -2, 0) \quad \text{and} \quad \chi_{\mathrm{S}^2(\rho_4)} = (10, 4, 1, 0, 0, 2, 1).$$

We have $(\chi_{\wedge^2(\rho_4)}|\chi_{\wedge^2(\rho_4)}) = \dfrac{1}{120}(36 + 24 + 60) = 1$, and thus $\wedge^2(\rho_4)$ is irreducible. On the other hand,

$$(\chi_{\mathrm{S}^2(\rho_4)}|\chi_{\mathrm{S}^2(\rho_4)}) = \frac{1}{120}(100 + 160 + 20 + 60 + 20) = 3.$$

We cannot write $3 = \sum m_i^2$, where the m_i are integers, unless three of them are equal to 1, and the others are zero. We have

$$(\chi_{\rho_1}|\chi_{\mathrm{S}^2(\rho_4)}) = 1, \ (\chi_{\rho_4}|\chi_{\mathrm{S}^2(\rho_4)}) = 1,$$

thus $\mathrm{S}^2(\rho_4) = \rho_1 \oplus \rho_4 \oplus \rho_5$, where $\dim \rho_5 = 5$, and $\chi_{\rho_5} = (5, 1, -1, -1, 0, 1, 1)$, thus $(\chi_{\rho_5}|\chi_{\rho_5}) = 1$, and ρ_5 is therefore irreducible.

4. We denote by $\rho_6 = \wedge^2(\rho_4)$ the six-dimensional irreducible representation of $\mathfrak{S}_5$. We find the character table of $\mathfrak{S}_5$.

	1 e	10 a	20 α	30 b	24 γ	15 β	20 c
ρ_1	1	1	1	1	1	1	1
ρ_1'	1	-1	1	-1	1	1	-1
ρ_4	4	2	1	0	-1	0	-1
ρ_4'	4	-2	1	0	-1	0	1
ρ_5	5	1	-1	-1	0	1	1
ρ_5'	5	-1	-1	1	0	1	-1
ρ_6	6	0	0	0	1	-2	0

We verify the orthogonality of the columns, and of the rows weighted by the number of elements of each class.

Part II: Character table of $\mathfrak{A}_5$

5. Using the preceding table, we obtain the following characters of $\mathfrak{A}_5$.

	1 e	20 α	15 β	12 γ	12 γ^2
ρ_1	1	1	1	1	1
ρ_4	4	1	0	-1	-1
ρ_5	5	-1	1	0	0
ρ_6	6	0	-2	1	1

We see that

$$(\chi_{\rho_1}|\chi_{\rho_1}) = 1, \quad (\chi_{\rho_4}|\chi_{\rho_4}) = 1, \quad (\chi_{\rho_5}|\chi_{\rho_5}) = 1, \quad (\chi_{\rho_6}|\chi_{\rho_6}) = 2.$$

Thus ρ_1, ρ_4, ρ_5 are irreducible. Writing $60 - (1 + 16 + 25) = 18$ as a sum of squares shows that $\rho_6 = \rho_3 \oplus \rho_3'$ with

$$\dim \rho_3 = \dim \rho_3' = 3.$$

6. Let $(3, A, B, C, D)$ and $(3, A', B', C', D')$ be the last two rows of the character table of $\mathfrak{A}_5$. Using the orthogonality of columns and rows, we obtain

$A = A' = 0$, $B = B' = -1$, $C + D = 1$, $C' + D' = 1$, $C'^2 - D'^2 = 0$. Because $D' = -C$ is impossible, we have $D' = C$ and $C' = D$, whence $C^2 - C - 1 = 0$. We can choose $C = \tau$ and then $C' = \tau - 1, D = 1 - \tau, D' = \tau$. If we had chosen $C = 1 - \tau$, we would only have inverted the last two rows of the character table.

7. Hence we obtain the character table of $\mathfrak{A}_5$.

	1 e	20 α	15 β	12 γ	12 γ^2
ρ_1	1	1	1	1	1
ρ_4	4	1	0	-1	-1
ρ_5	5	-1	1	0	0
ρ_3	3	0	-1	τ	$1-\tau$
ρ'_3	3	0	-1	$1-\tau$	τ

Part III: Some representations of the icosahedron group

08. In the projection $\varphi : \mathrm{SU}(2) \to \mathrm{SO}(3)$, the element $g_\alpha = \exp(2\alpha\xi_3) = \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{pmatrix}$ projects to the rotation through an angle 2α,

$$\mathrm{Rot}(e_3, 2\alpha) = \exp(2\alpha\eta_3) = \begin{pmatrix} \cos 2\alpha & -\sin 2\alpha & 0 \\ \sin 2\alpha & \cos 2\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

We know that $\chi_{\mathcal{D}^j}(g_\alpha) = \sum_{m=-j}^{j} e^{2im\alpha}$, whence

$$\chi_{\underline{\mathcal{D}}^j}(\mathrm{Rot}(e_3, \theta)) = \chi_{\mathcal{D}^j}(g_{\frac{\theta}{2}}) = \sum_{m=-j}^{j} e^{im\theta}.$$

9. We calculate $\chi^j(\theta) = \sum_{m=-j}^{j} e^{im\theta} = \dfrac{\sin(j + \frac{1}{2})\theta}{\sin\frac{\theta}{2}}$, for $\theta \neq 0$.

We find $\chi^j(0) = 2j + 1$ and for $\theta = \dfrac{2\pi}{3}, \pi, \dfrac{2\pi}{5}, \dfrac{4\pi}{5}$,

$$\chi^j\left(\frac{2\pi}{3}\right) = \begin{cases} 1 & \text{if } j \equiv 0 \,(\mathrm{mod}\ 3) \\ 0 & \text{if } j \equiv 1 \,(\mathrm{mod}\ 3) \\ -1 & \text{if } j \equiv 2 \,(\mathrm{mod}\ 3) \end{cases}$$

$$\chi^j(\pi) = \begin{cases} 1 & \text{if } j \equiv 0 \,(\mathrm{mod}\ 2) \\ -1 & \text{if } j \equiv 1 \,(\mathrm{mod}\ 2) \end{cases}$$

$$\chi^j\left(\frac{2\pi}{5}\right) = \begin{cases} 1 & \text{if } j \equiv 0 \,(\mathrm{mod}\ 5) \\ 2\cos\dfrac{\pi}{5} = \tau & \text{if } j \equiv 1 \,(\mathrm{mod}\ 5) \\ 0 & \text{if } j \equiv 2 \,(\mathrm{mod}\ 5) \\ -2\cos\dfrac{\pi}{5} = -\tau & \text{if } j \equiv 3 \,(\mathrm{mod}\ 5) \\ -1 & \text{if } j \equiv 4 \,(\mathrm{mod}\ 5) \end{cases}$$

$$\chi^j\left(\frac{4\pi}{5}\right) = \begin{cases} 1 & \text{if } j \equiv 0 \,(\mathrm{mod}\ 5) \\ 1-\tau & \text{if } j \equiv 1 \,(\mathrm{mod}\ 5) \\ 0 & \text{if } j \equiv 2 \,(\mathrm{mod}\ 5) \\ \tau - 1 & \text{if } j \equiv 3 \,(\mathrm{mod}\ 5) \\ -1 & \text{if } j \equiv 4 \,(\mathrm{mod}\ 5) \end{cases}$$

It is thus clear that, for every integer $\ell \in \mathbb{N}$ and each rotation $g \in \mathcal{I}$ different from the identity,

$$\chi^{j+30\ell}(g) = \chi^j(g).$$

We can find the table of all characters χ^j, for $0 \le j \le 29$. We write the first rows.

	j	1 0	20 $\frac{2\pi}{3}$	15 π	12 $\frac{2\pi}{5}$	12 $\frac{4\pi}{5}$	
R^0	0	1	1	1	1	1	χ^0
R^1	1	3	0	-1	τ	$1-\tau$	χ^1
R^2	2	5	-1	1	0	0	χ^2
R^3	3	7	1	-1	$-\tau$	$\tau-1$	χ^3
R^4	4	9	0	1	-1	-1	χ^4
R^5	5	11	-1	-1	1	1	χ^5
R^6	6	13	1	1	τ	$1-\tau$	χ^6
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

10. We see immediately that $R^0 = \rho_1$ (the trivial representation), $R^1 = \rho_3$, $R^2 = \rho_5$ and are thus irreducibles. We denote by $\chi_1, \chi_3, \chi_{3'}, \chi_4, \chi_5$ the characters of $\rho_1, \rho_3, \rho_3', \rho_4, \rho_5$. We have

$$(\chi_1|\chi^3) = 0,\ (\chi_3|\chi^3) = 0,\ (\chi_{3'}|\chi^3) = 1,\ (\chi_4|\chi^3) = 1,\ (\chi_5|\chi^3) = 0.$$

Thus

$$R^3 = \rho_3' \oplus \rho_4.$$

We set, for $m = 1, 3, 3', 4, 5$,

$$(\chi_m|\chi^j) = \frac{1}{60}((2j+1)\underline{m} + A_m^j),$$

where $\underline{m} = m$ for $m = 1, 3, 4, 5$ and $\underline{m} = 3$ for $m = 3'$. Then

$$\begin{aligned}(\chi_m|\chi^{30\ell+j}) &= \frac{1}{60}((2j+1+60\ell)\underline{m} + A_m^j)\\ &= \ell\underline{m} + (\chi_m|\chi^j).\end{aligned}$$

Thus if $(\chi_m|\chi^j) = \alpha_1^j, \alpha_3^j, \alpha_{3'}^j, \alpha_4^j, \alpha_5^j$, then

$$(\chi_m|\chi^{30\ell+j}) = \alpha_1^j + \ell, \alpha_3^j + 3\ell, \alpha_{3'}^j + 3\ell, \alpha_4^j + 4\ell, \alpha_5^j + 5\ell.$$

In particular

$$(\chi_m|\chi^{30\ell+3}) = \ell, 3\ell, 3\ell+1, 4\ell+1, 5\ell,$$

whence

$$R^{30\ell+3} = \ell\rho_1 \oplus 3\ell\rho_3 \oplus (3\ell+1)\rho_3' \oplus (4\ell+1)\rho_4 \oplus 5\ell\rho_5.$$

We verify that $2(30\ell+3)+1 = \ell + 9\ell + 9\ell + 3 + 16\ell + 4 + 25\ell$.

11. In order to decompose $R^2 \otimes R^3$, we can use the Clebsch–Gordan formula

$$\mathcal{D}^2 \otimes \mathcal{D}^3 = \mathcal{D}^1 \oplus \mathcal{D}^2 \oplus \mathcal{D}^3 \oplus \mathcal{D}^4 \oplus \mathcal{D}^5$$

which implies the same relationship between the $\underline{\mathcal{D}}^j$ and thus between the R^j.

The decompositions of R^4 and of R^5 are

$$R^4 = \rho_4 \oplus \rho_5, \quad R^5 = \rho_3 \oplus \rho_3' \oplus \rho_5.$$

Hence

$$R^2 \otimes R^3 = 2\rho_3 \oplus 2\rho_3' \oplus 2\rho_4 \oplus 3\rho_5.$$

We verify that $5 \times 7 = 6 + 6 + 8 + 15$.

One can also write

$$R^2 \otimes R^3 = \rho_5 \otimes (\rho_3' \oplus \rho_4) = (\rho_5 \otimes \rho_3') \oplus (\rho_5 \otimes \rho_4).$$

The character of $\rho_5 \otimes \rho_3'$ is $(15, 0, -1, 0, 0)$, and thus

$$\rho_5 \otimes \rho_3' = \rho_3 \oplus \rho_3' \oplus \rho_4 \oplus \rho_5.$$

The character of $\rho_5 \otimes \rho_4$ is $(20, -1, 0, 0, 0)$, and hence

$$\rho_5 \otimes \rho_4 = \rho_3 \oplus \rho_3' \oplus \rho_4 \oplus 2\rho_5.$$

We recover

$$R^2 \otimes R^3 = 2\rho_3 \oplus 2\rho_3' \oplus 2\rho_4 \oplus 3\rho_5.$$

Epilogue: Representations of $G = \varphi^{-1}(\mathcal{I}) \subset \mathrm{SU2})$

Each conjugacy class of G has a representative g_α, $0 \leq \alpha < 2\pi$.

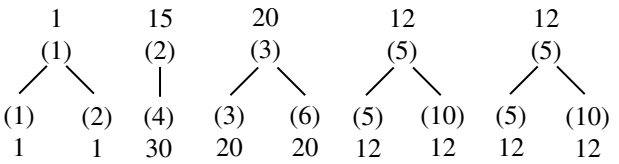
order	1	2	4	6	3	10	10	5	5
cardinality	1	1	30	20	20	12	12	12	12
α	0	π	$\frac{\pi}{2}$	$\frac{\pi}{3}$	$\frac{4\pi}{3}$	$\frac{\pi}{5}$	$\frac{7\pi}{5}$	$\frac{6\pi}{5}$	$\frac{2\pi}{5}$

The class of $e, \theta = 0$, of order 1, yields $\alpha = 0$ (order 1) and $\alpha = \pi$ (order 2).

The class with 15 elements, $\theta = \pi$, of order 2, yields a class with 30 elements of order 4, $\alpha = \dfrac{\pi}{2}$.

The class with 20 elements, $\theta = \dfrac{2\pi}{3}$, of order 3, yields a class with 20 elements of order 3, $\alpha = \dfrac{4\pi}{3}$ and a class with 20 elements of order 6, $\alpha = \dfrac{\pi}{3}$.

Each class with 12 elements, $\theta = \dfrac{2\pi}{5}$ (resp., $\theta = \dfrac{4\pi}{5}$), of order 5, yields a class with 12 elements of order 5, $\alpha = \dfrac{6\pi}{5}$ (resp., $\alpha = \dfrac{2\pi}{5}$) and a class with 12 elements of order 10, $\alpha = \dfrac{\pi}{5}$ (resp., $\alpha = \dfrac{7\pi}{5}$).



We can easily write the characters of the representations $\rho_1, \rho_3, \rho_3', \rho_4, \rho_5$. The representations $\rho_1, \rho_3, \rho_4, \rho_5$ are the restrictions of $\mathcal{D}^0, \mathcal{D}^1, \mathcal{D}^{3/2}, \mathcal{D}^2$. In addition, for $j = \dfrac{1}{2}$, there is a representation ρ_2 obtained by restriction of $\mathcal{D}^{1/2}$ and a representation ρ_2'. Hence we obtain the character table of $G = \varphi^{-1}(\mathcal{I})$.

	order	1	2	4	6	3	10	10	5	5
	cardinality	1	1	30	20	20	12	12	12	12
		0	π	$\frac{\pi}{2}$	$\frac{\pi}{3}$	$\frac{4\pi}{3}$	$\frac{\pi}{5}$	$\frac{7\pi}{5}$	$\frac{6\pi}{5}$	$\frac{2\pi}{5}$
$\mathcal{D}^0$	$\mathbf{1}$	1	1	1	1	1	1	1	1	1
$\mathcal{D}^{1/2}$	$\mathbf{2}$	2	-2	0	1	-1	τ	$1-\tau$	$-\tau$	$\tau-1$
$\mathcal{D}^1$	$\mathbf{3}$	3	3	-1	0	0	τ	$1-\tau$	τ	$1-\tau$
$\mathcal{D}^{3/2}$	$\mathbf{4}$	4	-4	0	-1	1	1	1	-1	-1
$\mathcal{D}^2$	$\mathbf{5}$	5	5	1	-1	-1	0	0	0	0
$\mathcal{D}^{5/2}$	$\mathbf{6}$	6	-6	0	0	0	-1	-1	1	1
	$\mathbf{4}'$	4	4	0	1	1	-1	-1	-1	-1
	$\mathbf{3}'$	3	3	-1	0	0	$1-\tau$	τ	$1-\tau$	τ
	$\mathbf{2}'$	2	-2	0	1	-1	$1-\tau$	τ	$\tau-1$	$-\tau$

The first six rows are irreducible characters, as shown by the calculation of the norm squared. Since, for $j \geq \frac{1}{2}$, $\mathcal{D}^{\frac{1}{2}} \otimes \mathcal{D}^j = \mathcal{D}^{j-\frac{1}{2}} \oplus \mathcal{D}^{j+\frac{1}{2}}$, we obtain, for $k \geq 2$,

$$\mathbf{2} \otimes \mathbf{k} = (\mathbf{k-1}) \oplus (\mathbf{k+1}).$$

Furthermore,

$$\begin{aligned} \mathbf{2} \otimes \mathbf{2}' &= \mathbf{4}', \\ \mathbf{2} \otimes \mathbf{3}' &= \mathbf{6} = \mathbf{2}' \otimes \mathbf{3}, \\ \mathbf{2} \otimes \mathbf{4}' &= \mathbf{2}' \oplus \mathbf{6}. \end{aligned}$$

We can show that

$$\begin{aligned} \mathbf{3} \oplus \mathbf{1} &= \mathbf{2}^{\otimes \mathbf{2}} \\ \mathbf{4} \oplus 2\mathbf{2} &= \mathbf{2}^{\otimes \mathbf{3}} \\ \mathbf{5} \oplus 3\mathbf{2}^{\otimes \mathbf{2}} &= \mathbf{2}^{\otimes \mathbf{4}} \otimes \mathbf{1} \\ \mathbf{6} \oplus 4\mathbf{2}^{\otimes \mathbf{3}} &= \mathbf{2}^{\otimes \mathbf{5}} \otimes 3\mathbf{2} \end{aligned}$$

and similarly $\mathbf{4}', \mathbf{3}', \mathbf{2}'$ can be written in terms of linear combinations of tensor powers of $\mathbf{2}$.

A reference for the properties proved in this problem is F. R. K. Chung, B. Kostant, S. Sternberg, "Groups and the buckyball", in *Lie Theory and Geometry*, J.-L. Brylinski, R. Brylinski, V. Guillemin, V. Kac, eds., Birkhäuser, 1994,

p. 97–126. It is proved there that the multiplicities $a_{k,r}$ of the representations $r = \mathbf{1}, \mathbf{2}, \ldots, \mathbf{2}'$ in the restriction to G of the representations $\mathcal{D}^{k/2}, k \in \mathbb{N}$, are the coefficients of the series expansion of a rational function, where only the numerator depends on r.

9 Matrix Coefficients and Spherical Harmonics

Let j be a non-negative integer or half-integer. We denote by V^j the complex vector space of homogeneous polynomials in two variables, (z_1, z_2), with complex coefficients, of degree $2j$.

1. To each matrix $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ belonging to $\mathrm{SL}(2, \mathbb{C})$ and each polynomial $f \in V^j$, we associate the polynomial $\rho_V^j(g)f$ defined by

$$(\rho_V^j(g)f)(z_1, z_2) = f(az_1 + cz_2, bz_1 + dz_2).$$

Show that we have thus defined a representation ρ_V^j of $\mathrm{SL}(2, \mathbb{C})$ in V^j.

We denote by P^j the complex vector space of polynomials with complex coefficients in one variable x of degree $\leq 2j$.

2. Show that the map Φ of V^j into P^j defined by

$$(\Phi(f))(x) = f(x, 1),$$

for $f \in V^j$, is an isomorphism of vector spaces.

3. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{C})$ and $\varphi \in P^j$, we define $\rho^j(g)(\varphi) \in P^j$ by

$$(\rho^j(g)\varphi)(x) = (bx + d)^{2j} \varphi\left(\frac{ax + c}{bx + d}\right).$$

Show that we have thus defined a representation, ρ^j, of $\mathrm{SL}(2, \mathbb{C})$ in P^j, equivalent to the representation (V^j, ρ_V^j).

4. We denote by r^j the differential of the representation ρ^j. Find the differential operator $r^j(X)$ for $X = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathfrak{sl}(2, \mathbb{C})$.

For $-j \leq m \leq j$ and $j - m$ integer, we set

$$\varphi_m^j(x) = \frac{x^{j-m}}{\sqrt{(j-m)!(j+m)!}}.$$

5. We consider the basis $J_3 = \frac{1}{2}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, $J_+ = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}$, $J_- = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$ of $\mathfrak{sl}(2, \mathbb{C})$. Find the images under each of the operators $r^j(J_3)$, $r^j(J_+)$, $r^j(J_-)$ of the polynomials φ_m^j, for $-j \leq m \leq j$ and $j - m$ integer.

6. Show that the representation ρ^j of $\mathrm{SL}(2,\mathbb{C})$ on P^j is irreducible.

7. Show that there is a unique scalar product on P^j invariant under the restriction to $\mathrm{SU}(2)$ of the representation ρ^j of $\mathrm{SL}(2,\mathbb{C})$ and such that the polynomials φ^j_m form an orthonormal basis of P^j. Is the matrix of $\rho^j(g)$ in the basis (φ^j_m) unitary for every $g \in \mathrm{SL}(2,\mathbb{C})$?

We write $(\ |\ )$ for the scalar product defined above. The matrix coefficients, $\rho^j_{m,n}$, of the representation ρ^j in the basis (φ^j_m) are defined by

$$\rho^j_{m,n}(g) = (\varphi^j_m | \rho^j(g)\varphi^j_n),$$

for $g \in \mathrm{SL}(2,\mathbb{C})$.

8. Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{C})$. Use Taylor's formula for the function $(ax + c)^{j-n}(bx + d)^{j+n}$, along with a change of variable, to show that, if $a \neq 0$ and $b \neq 0$,

$$\rho^j_{m,n}(g) = C(j,m,n)\frac{b^{n-m}}{a^{n+m}}\frac{\mathrm{d}^{j-m}}{\mathrm{d}x^{j-m}}((1-x)^{j-n}(1+x)^{j+n})\big|_{x=ad+bc},$$

where $C(j,m,n)$ is a constant to be determined.

9. (a) Let $\xi_1 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$. We set $\rho^j(\theta) = \rho^j(\exp(\theta\xi_1))$, $0 \leq \theta \leq \pi$. Find $\rho^j(\pi)$. Show that $\rho^j_{m,-n}(\theta) = \rho^j_{-m,n}(\theta)$.

(b) We assume that j and m are integers and that $m \geq 0$. We recall that the Legendre function $P_{j,m}$ is defined, for $-1 \leq x \leq 1$, by

$$P_{j,m}(x) = \frac{(-1)^{j+m}}{2^j j!}(1-x^2)^{\frac{m}{2}}\frac{\mathrm{d}^{j+m}}{\mathrm{d}x^{j+m}}(1-x^2)^j.$$

Show that $\rho^j_{m,0}(\theta) = \mathcal{P}^j_m(\cos\theta)$, where $\mathcal{P}^j_m$ is a function of one variable proportional to $P_{j,m}$, and find the coefficient of proportionality.

For $-j \leq m \leq j$, what is the relation between the function $\rho^j_{m,0}(\theta)e^{im\phi}$ and the spherical harmonic $Y^j_m(\theta,\phi)$ $(0 \leq \theta \leq \pi, 0 \leq \phi < 2\pi)$?

10. Prove the orthonormality property of the system of spherical harmonics from the orthogonality relations for the matrix coefficients of irreducible representations of $\mathrm{SU}(2)$.

Solutions

1. For every matrix $g \in \mathrm{SL}(2,\mathbb{C})$ and for every $f \in V^j$, the polynomial $\rho^j_V(g)f$ is homogeneous of degree $2j$, and its coefficients depend continuously on the

coefficients of the matrix g. The definition can be written $\rho_V^j(g)f = f \circ {}^t g$; hence, if g_1 and $g_2 \in \mathrm{SL}(2,\mathbb{C})$,

$$\rho_V^j(g_1 g_2)f = f \circ {}^t g_2 \, {}^t g_1 = \rho_V^j(g_1)(\rho_V^j(g_2)f).$$

Thus ρ_V^j is a representation of $\mathrm{SL}(2,\mathbb{C})$ on V^j.

2. It is clear that Φ is linear. The vector spaces V^j and P^j have the same dimension, $2j+1$. Furthermore, Φ is injective. In fact, if $\Phi(f) = 0$, all the coefficients of the homogeneous polynomial f are zero, and hence it is zero. Thus Φ is an isomorphism. The inverse Ψ of Φ is given by

$$(\Psi(\varphi))(z_1, z_2) = z_2^{2j}\varphi(\frac{z_1}{z_2}), \quad \varphi \in P^j.$$

3. When we identify V^j with P^j by means of the isomorphism Φ, the transformation $\rho_V^j(g)$ can be identified with the transformation $\rho^j(g)$. In fact, for $f \in V^j$,

$$\rho^j(g)(\Phi(f))(x) = (bx+d)^{2j}(\Phi(f))(\frac{ax+c}{bx+d}) = (bx+d)^{2j} f(\frac{ax+c}{bx+d}, 1)$$

$$= f(ax+c, bx+d) = (\rho_V^j(g)f)(x,1) = \Phi(\rho_V^j(g)f)(x).$$

We deduce that ρ^j is a representation of $\mathrm{SL}(2,\mathbb{C})$ in P^j, equivalent to ρ_V^j. The isomorphism Φ intertwines the representations ρ_V^j and ρ^j.

4. Let $X = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathfrak{sl}(2,\mathbb{C})$. Then $\exp(tX) = \begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix}$ is a matrix of $\mathrm{SL}(2,\mathbb{C})$ such that $a(0) = d(0) = 1$, $b(0) = c(0) = 0$ and $a'(0) = \alpha$, $b'(0) = \beta$, $c'(0) = \gamma$, $d'(0) = \delta$. One thus obtains, for $\varphi \in P^j$,

$$(r^j(X)\varphi)(x) = \frac{\mathrm{d}}{\mathrm{d}t}(\rho^j(\exp tX)\varphi)(x)\big|_{t=0} = \frac{\mathrm{d}}{\mathrm{d}t}(b(t)x+d(t))^{2j}\varphi\left(\frac{a(t)x+c(t)}{b(t)x+d(t)}\right)\Big|_{t=0}$$

$$= 2j(\beta x+\delta)\varphi(x) + (-\beta x^2 + (\alpha-\delta)x + \gamma)\frac{\mathrm{d}\varphi}{\mathrm{d}x}(x),$$

that is,

$$r^j(X) = 2j(\beta x+\delta) + (-\beta x^2 + (\alpha-\delta)x+\gamma)\frac{\mathrm{d}}{\mathrm{d}x}.$$

5. In particular,

$$r^j(J_3) = j - x\frac{\mathrm{d}}{\mathrm{d}x}, \quad r^j(J_+) = -\frac{\mathrm{d}}{\mathrm{d}x}, \quad r^j(J_-) = -2jx + x^2\frac{\mathrm{d}}{\mathrm{d}x},$$

and hence

$$r^j(J_3)(\varphi_m^j) = m\varphi_m^j,$$

$$r^j(J_+)(\varphi_m^j) = -\sqrt{(j-m)(j+m+1)}\,\varphi_{m+1}^j,$$

$$r^j(J_-)(\varphi_m^j) = -\sqrt{(j+m)(j-m+1)}\varphi_{m-1}^j.$$

6. The preceding equations show that the representation r^j of $\mathfrak{sl}(2,\mathbb{C})$ is equivalent to the representation D^j, the basis $(-1)^{j-m}\varphi_m^j$ corresponding to the standard basis $|j,m\rangle$. We know that this representation is irreducible. We deduce immediately that the representation ρ^j of $\mathrm{SL}(2,\mathbb{C})$ is irreducible.

7. There exists a unique scalar product on P^j such that the basis (φ_m^j) is orthonormal. Let us show that this scalar product is invariant under $\rho^j\big|_{\mathrm{SU}(2)}$, that is, that the restriction of ρ^j to $\mathrm{SU}(2)$ is unitary for this scalar product. Because $\mathrm{SU}(2)$ is connected, it suffices to show that the matrices in the basis (φ_m^j) of the operators $r^j(\xi_k)$, $k=1,2,3$, are antihermitian or, equivalently, that the matrix of $r^j(J_3)$ is Hermitian and that the matrices of $r^j(J_+)$ and $r^j(J_-)$ are conjugate transposes of one another. Because the operator $r^j(J_3)$ is real and diagonal, it is Hermitian. We have

$$(r^j(J_+)\varphi_m^j|\varphi_n^j) = -\sqrt{(j-m)(j+m+1)}\delta_{m+1,n},$$

$$\begin{aligned}(\varphi_m^j|r^j(J_-)\varphi_n^j) &= -\sqrt{(j+n)(j-n+1)}\delta_{m,n-1}\\ &= -\sqrt{(j+m+1)(j-m)}\delta_{m+1,n},\end{aligned}$$

which shows that the matrices of $r^j(J_+)$ and $r^j(J_-)$ are real and are transposes of one another. The operators $\rho^j(g)$ are thus unitary for $g\in\mathrm{SU}(2)$. But clearly this is not true for an arbitrary element of $\mathrm{SL}(2,\mathbb{C})$: the operators $r^j(X)$ are antihermitian for $X\in\mathfrak{su}(2)$ and Hermitian for $X\in i\mathfrak{su}(2)$.

8. The coefficient of x^{j-m} in $(ax+c)^{j-n}(bx+d)^{j+n}$ is

$$\frac{1}{(j-m)!}\frac{\mathrm{d}^{j-m}}{\mathrm{d}x^{j-m}}((ax+c)^{j-n}(bx+d)^{j+n})\big|_{x=0},$$

and hence

$$\begin{aligned}\rho_{m,n}^j(g) &= \Big(\frac{x^{j-m}}{\sqrt{(j-m)!(j+m)!}}\Big|\frac{(ax+c)^{j-n}(bx+d)^{j+n}}{\sqrt{(j-n)!(j+n)!}}\Big)\\ &= \frac{\sqrt{(j+m)!}}{\sqrt{(j-n)!(j+n)!(j-m)!}}\frac{\mathrm{d}^{j-m}}{\mathrm{d}x^{j-m}}((ax+c)^{j-n}(bx+d)^{j+n})\big|_{x=0}.\end{aligned}$$

We set $\frac{y+1}{2} = a(bx+d)$. Taking the equation $ad-bc=1$ into account, we have $\frac{y-1}{2} = b(ax+c)$. We assume $a\neq 0$ and $b\neq 0$. Because the variable y is a linear function of x, we find

$$\begin{aligned}&\rho_{m,n}^j(g)\\ &= \frac{\sqrt{(j+m)!}}{\sqrt{(j-n)!(j+n)!(j-m)!}}\frac{(-1)^{j-n}}{2^{j+m}}\frac{b^{n-m}}{a^{n+m}}\frac{\mathrm{d}^{j-m}}{\mathrm{d}y^{j-m}}((1-y)^{j-n}(1+y)^{j+n})\big|_{y=ad+bc}.\end{aligned}$$

9. (a) We know that $\exp(\theta\xi_1) = \begin{pmatrix} \cos\frac{\theta}{2} & i\sin\frac{\theta}{2} \\ i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$. In particular, $\exp(\pi\xi_1) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$. Now, $(\rho^j(\pi)\varphi^j_m)(x) = \frac{1}{\sqrt{(j-m)!(j+m)!}} i^{2j} x^{j+m} = i^{2j}\varphi^j_{-m}(x)$, and thus

$$\rho^j(\pi)\varphi^j_m = i^{2j}\varphi^j_{-m}, \qquad \rho^j_{m,n}(\pi) = i^{2j}\delta_{m,-n}.$$

From the equation $\rho^j(\theta)\rho^j(\pi) = \rho^j(\pi)\rho^j(\theta)$, we deduce that

$$\rho^j_{m,-n}(\theta) = \rho^j_{-m,n}(\theta).$$

In particular, $\rho^j_{m,0}(\theta) = \rho^j_{-m,0}(\theta)$.

(b) For $0 < \theta < \pi$, we calculate $\rho^j_{m,0}(\theta) = \rho^j_{-m,0}(\theta)$ by the general formula obtained above. We find

$$\rho^j_{m,0}(\theta) = \frac{(-1)^j}{2^j j!}\sqrt{\frac{(j-m)!}{(j+m)!}}\, i^m \sin^m\theta \frac{\mathrm{d}^{j+m}}{\mathrm{d}x^{j+m}}(1-x^2)^j\big|_{x=\cos\theta}.$$

When $0 \le \theta \le \pi$, the equation $\sin\theta = (1-\cos^2\theta)^{\frac{1}{2}}$ is valid. Consequently, for $0 < \theta < \pi$, $\rho^j_{m,0}(\theta) = \mathcal{P}^j_m(\cos\theta)$, where the function $\mathcal{P}^j_m$ is defined for $-1 \le x \le 1$ by

$$\mathcal{P}^j_m(x) = \frac{(-1)^j}{2^j j!}\sqrt{\frac{(j-m)!}{(j+m)!}}\, i^m (1-x^2)^{\frac{m}{2}} \frac{\mathrm{d}^{j+m}}{\mathrm{d}x^{j+m}}(1-x^2)^j.$$

We verify that, for j and m integers, $m \ge 0$, $\mathcal{P}^j_m(1) = \delta_{m,0}$ and $\mathcal{P}^j_m(-1) = (-1)^j\delta_{m,0}$, which extends the result to the case $\theta = 0$ and $\theta = \pi$. If we use $P_{j,m}$ to denote the Legendre function, we obtain an equation valid for $0 \le \theta \le \pi$,

$$\rho^j_{m,0}(\theta) = \sqrt{\frac{(j-m)!}{(j+m)!}}\, i^{-m} P_{j,m}(\cos\theta).$$

We know that, for $0 \le m \le j$, the spherical harmonics are the functions

$$Y^j_m(\theta,\phi) = \sqrt{\frac{2j+1}{4\pi}}$$

We deduce that, in this case,

$$Y^j_m(\theta,\phi) = \sqrt{\frac{2j+1}{4\pi}}\, i^m \rho^j_{m,0}(\theta) e^{im\phi}.$$

This equation is valid for $-j \le m < 0$ as well. In fact, by definition, $Y^j_m = (-1)^m\overline{Y^j_{-m}}$, therefore $Y^j_m = (-1)^m\sqrt{\frac{2j+1}{4\pi}}\, i^m \overline{\rho^j_{-m,0}(\theta)} e^{im\phi}$. Since $\rho^j_{-m,0}(\theta) = \rho^j_{m,0}(\theta)$ and $\overline{\rho^j_{m,0}(\theta)} = (-1)^m\rho^j_{m,0}(\theta)$, the result follows.

10. We know that a dense open set of SU(2) is parametrized by the Euler angles. For a matrix $g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \in \mathrm{SU}(2)$ such that $\mathrm{Im}(a) \neq 0$, $\mathrm{Re}(b) \neq 0$, there are unique parameters $\phi \in (0, 2\pi)$, $\theta \in (0, \pi)$, $\psi \in (-2\pi, 2\pi)$ such that

$$g = g(\phi, \theta, \psi) = \begin{pmatrix} \cos\frac{\theta}{2} e^{i\frac{\phi+\psi}{2}} & i\sin\frac{\theta}{2} e^{i\frac{\phi-\psi}{2}} \\ i\sin\frac{\theta}{2} e^{-i\frac{\phi-\psi}{2}} & \cos\frac{\theta}{2} e^{-i\frac{\phi+\psi}{2}} \end{pmatrix} = \exp(\phi\xi_3)\exp(\theta\xi_1)\exp(\psi\xi_3).$$

In this parametrization, the Haar integral on SU(2) can be written

$$I(f) = \frac{1}{16\pi^2} \int_0^{2\pi} \int_0^{\pi} \int_{-2\pi}^{2\pi} f(g(\phi, \theta, \psi)) \sin\theta \, \mathrm{d}\phi \, \mathrm{d}\theta \, \mathrm{d}\psi.$$

Now consider the matrix coefficients of the representation ρ^j of SU(2). Because $g(\phi, 0, 0) = \exp(\phi\xi_3) = \begin{pmatrix} e^{i\frac{\phi}{2}} & 0 \\ 0 & e^{-i\frac{\phi}{2}} \end{pmatrix}$, for each function $\varphi \in P^j$, we find $(\rho^j(g(\phi, 0, 0))\varphi)(x) = e^{-ij\phi}\varphi(e^{i\phi}x)$, whence

$$\rho^j(g(\phi, 0, 0))x^{j-m} = e^{-im\phi}x^{j-m}.$$

The matrix of $\rho^j(g(\phi, 0, 0))$ in the basis φ^j_m is thus diagonal, with elements $e^{-im\phi}$, $-j \leq m \leq j$. Similarly, the matrix of $\rho^j(g(0, 0, \psi))$ is diagonal, with elements $e^{-im\psi}$.

We have set $\rho^j(\theta) = \rho^j(\exp(\theta\xi_1))$, *i.e.*, $\rho^j(\theta) = \rho^j(g(0, \theta, 0))$. By writing

$$\rho^j_{m,n}(g(\phi, \theta, \psi)) = \sum_{p,q} \rho^j_{m,p}(g(\phi, 0, 0))\rho^j_{pq}(\theta)\rho^j_{qn}(g(0, 0, \psi)),$$

we obtain

$$\rho^j_{m,n}(g(\phi, \theta, \psi)) = e^{-i(m\phi+n\psi)}\rho^j_{m,n}(\theta).$$

In particular,

$$\rho^j_{m,0}(g(\phi, \theta, \psi)) = e^{-im\phi}\rho^j_{m,0}(\theta).$$

The orthogonality relations for the matrix coefficients of unitary irreducible representations ρ^j of SU(2) imply

$$\frac{1}{16\pi^2} \int_0^{2\pi} \int_0^{\pi} \int_{-2\pi}^{2\pi} \overline{\rho^j_{m,0}(g(\phi, \theta, \psi))}\rho^{j'}_{m',0}(g(\phi, \theta, \psi)) \sin\theta \, \mathrm{d}\phi \, \mathrm{d}\theta \, \mathrm{d}\psi$$

$$= \frac{1}{2j+1}\delta_{m,m'}\delta_{j,j'},$$

whence, because $\rho^j_{m,0}$ does not depend on ψ,

$$\frac{2j+1}{4\pi} \int_0^{2\pi} \int_0^{\pi} \overline{\rho^j_{m,0}(g(\phi, \theta, 0))}\rho^{j'}_{m',0}(g(\phi, \theta, 0)) \sin\theta \, \mathrm{d}\phi \, \mathrm{d}\theta = \delta_{m,m'}\delta_{j,j'}.$$

The elements corresponding to $\psi = 0$ can be identified with the points of the sphere S^2 : to $\exp(\phi\xi_3)\exp(\theta\xi_1)$ corresponds the point $\exp(\phi\eta_3)\exp(\theta\eta_1)e_3$, with spherical coordinates $(\theta, \phi - \frac{\pi}{2})$.

Now, for $j \in \mathbb{N}$, $-j \leq m \leq j$,

$$Y_m^j(\theta, \phi) = \sqrt{\frac{2j+1}{4\pi}} i^m e^{im\phi} \rho_{m,0}^j(\theta).$$

Furthermore, we know that $\rho_{m,0}^j(\theta)$ is either real or pure imaginary. This equation means that, up to a coefficient, the spherical harmonic $Y_m^j(\theta, \phi)$ is the conjugate of the matrix coefficient $\rho_{m,0}^j(g(\phi, \theta, 0))$. Taking into account the value of the modulus of this coefficient, the orthogonality relation above can be written

$$\int_0^{\pi} \int_0^{2\pi} \overline{Y_m^j(\theta, \phi)}\, Y_{m'}^{j'}(\theta, \phi) \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \delta_{m,m'} \delta_{j,j'},$$

which proves that the spherical harmonics (Y_m^j), $j \in \mathbb{N}$, $-j \leq m \leq j$, form an orthonormal set in $L^2(S^2)$ equipped with the unnormalized scalar product.

From the properties of the matrix coefficients of the irreducible representations of the group SU(2), one can prove the orthogonality properties, the addition theorems and many others for the Legendre polynomials, Legendre functions, and the spherical harmonics.

***

Endnote

The topics discussed at the end of the preceding chapters do not exhaust the list of famous mathematicians and physicists whose name appears in various contexts in this book. I could, of course, drown the mathematics in a sea of anecdotes, but this is not my objective. To make amends for the neglect of some very important scientists who should have appeared in the preceding pages, I shall add at least a few more as briefly as possible.

Cartesian coordinates are named after the French philosopher, mathematician and physicist, René Descartes (1596–1650) whose name was also spelled Des Cartes.

The name "Fourier analysis" encompasses the study of the theory and innumerable applications of the Fourier series and Fourier integrals, named after Joseph Fourier (1768–1830) who lived through the French Revolution, was appointed professor at the newly created École Polytechnique, took leave to participate in Napoleon's expedition to Egypt from 1798 to 1801, and then returned to a very important career in science.

The contributions of the Irish mathematician William Rowan Hamilton (1805–1865) to mechanics, physics and astronomy were equally fundamental. In an 1847 paper in the *Proceedings of the Irish Royal Academy*, he applied the theory of quaternions to the study of rotations in space. The quaternions are not his only claim to fame and he is known to have been very conscious of the importance of that discovery.

The German mathematician Karl Weierstrass (1815–1897) was the analyst of great importance who proved "the Weierstrass theorem" in his "On the possibility of giving an analytic representation to an arbitrary function of a real variable", published in 1885 and reprinted with additions in his collected works in 1903.

The Lorentz group bears the name of the Dutch theoretical physicist, Hendrik Lorentz (1853–1928), who was awarded the Nobel prize in 1902 for his "masterly theory of the electron". He introduced the notion of local time as early as 1892. It was Poincaré who, in 1906, gave the name of Lorentz to the invariance group of the theory of relativity.

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Planck's constant is named after the German theoretical physicist Max Planck (1858–1947), who introduced it in his study of heat radiation, in a communication to the Berlin Academy of 14 December 1900 that would become fundamental to the development of quantum theory.

"Minkowski space-time" is named after Hermann Minkowski who was born in Russia in 1864, studied in Königsberg, and held positions in Bonn, Zurich and Göttingen. He died in 1909 after making important contributions to relativity and geometry.

The German physicist Werner Heisenberg (1901–1976) was awarded the Nobel prize in physics in 1932 "for the creation of quantum mechanics".

Bertram Kostant (1928-2017) was an American mathematician, born in Brooklyn, who made key contributions to many aspects of Lie group theory, and in particular to representation theory.

For further information on these scientists and the others who are cited elsewhere in this book, the interested student will find a wealth of documents online, some of which are reliable and some much less so, and in books, often available online on the site of university libraries, and you may even find your way to a brick-and-mortar library well stocked with BOOKS!



Hermann Weyl (1885–1955) wrote his doctoral dissertation in 1908 at the University of Göttingen under the direction of David Hilbert, taught in Zurich from 1913 to 1930, succeeded Hilbert as a professor in Göttingen and left Germany in 1933 for the United States where he accepted a position at the newly formed Institute for Advanced Study in Princeton. His most important contributions to group theory date from the years 1920–1940.

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[1] The literature on finite groups, Lie algebras, Lie groups and representation theory is immense. We present a list containing

- classical and recent works which seem most useful for a first approach to the topics dealt with in this text, and we mark them with the symbol (*),
- books for further study, and
- texts of historical significance.

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